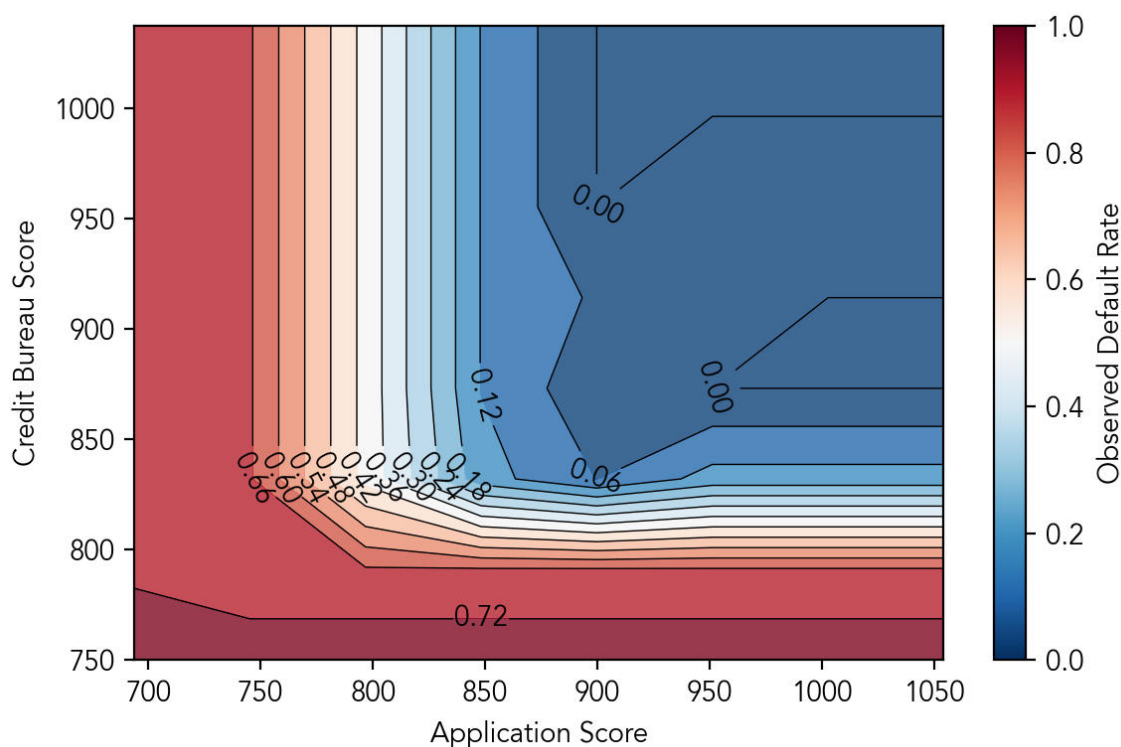




Machine Learning in Credit Risk

A collection of white papers and research artifacts



Contents:

- Machine Learning in Credit Risk Modeling (James AI)
 - Machine Learning Approach for Credit Scoring
 - Consumer Credit Risk Models via ML Algorithms
 - Explainable ML Models of Consumer Credit Risk
- Predicting Consumer Default: a DL Approach (NBER)
 - ML Applications in Assessing Credit Risk (IMF)

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WHITE PAPER

Machine Learning in Credit Risk Modeling

Efficiency should not come at
the expense of Explainability



JAMES

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Executive Summary

Nowadays, and though it has been around for decades, Machine Learning (ML) is the talk of the town. Yet, it is only a combination of theoretical breakthroughs and growing computing power at an affordable cost that made it possible for companies like [James](#) (The Artificial Intelligence (AI) for Credit Risk) to include it in their product.

The question is: as it is disrupting many sectors of our societies, from Medicine to the Transportation Industry, why does ML make Credit Risk estimation more efficient? Also, and most importantly, why doesn't this Efficiency come at the cost of Explainability?

In the present paper, we answer these questions thoroughly by applying ML techniques to build non-linear and non-parametric forecasting credit risk models. With the latter, we are able to significantly improve the prediction of defaults with gains in Gini reaching 27%. Further on, we explain the main models used in James in order to fight the *black box myth* around ML.

Results

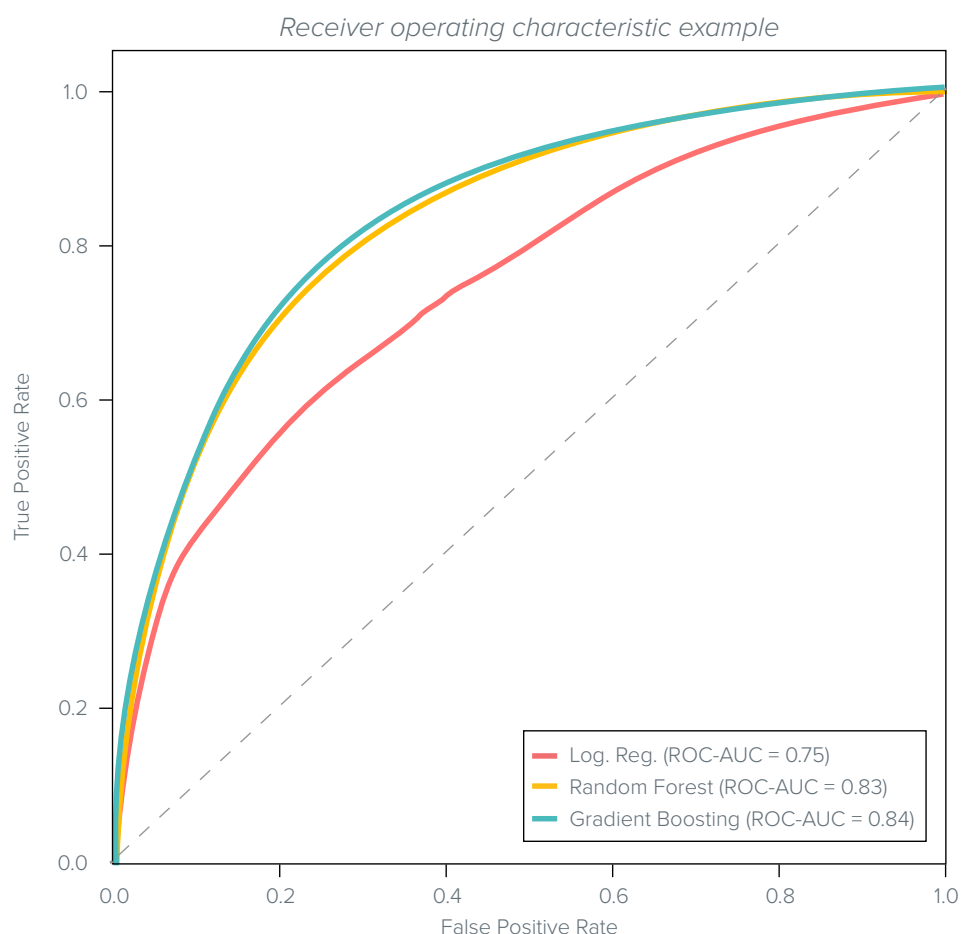
Presentation of the results

In order to prove that ML is an efficient tool when it comes to Credit Risk estimation, we work with a typical Credit Risk dataset of approximately 150,000 observations and 12 features, including the default label.

After identifying the label, we preprocess the data in James, eliminating collinear features and [binning the features](#) when needed. We then use [cross-validation](#) to train our models on one part of the dataset (80%) and test it on the remaining 20% sample. For more information on the way James works and processes the data, feel free to check the Finovate 2017 James demo on [our website](#).

Here are the results for three selected models: the regular **Logistic Regression** and two state-of-the-art Machine Learning algorithms: **Random Forest (RF)** and **Gradient Boosting (GB)**.

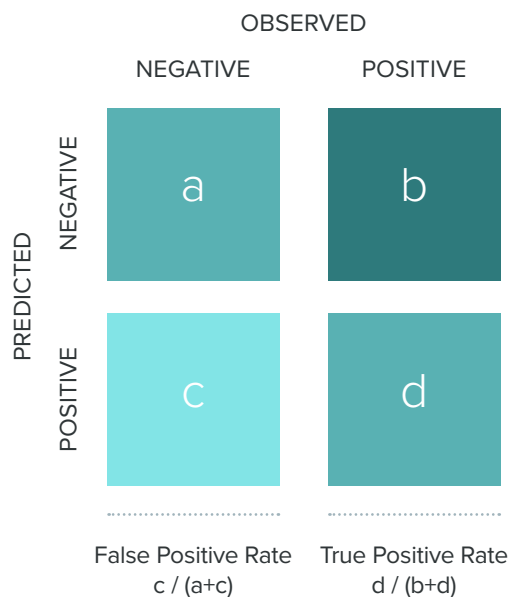
The ROC-AUC score (presented in the following section) equals to **0.75** for the Logit model, reaches **0.83** for RF and goes up to **0.84** for GB. The gain in prediction quality is obvious with **a gain of up to 9% in ROC-AUC score**. In parallel, the same happens with the Gini score with **a gain of up to 27%** from Logit to GB.



Scoring a model

We need to define the performance metrics that are often used in ML applied to credit risk. The most common one is the **AUC** (Area under curve) that derives from the **ROC** curve (Receiving Operator Characteristic). Let us consider a two-class prediction problem ([binary classification](#)), in which the outcomes are labeled either as default (D) or safe (S). In Credit Scoring, we consider defaults as positives. There are four possible outcomes from a binary classifier. If the outcome from a prediction is D and the actual value is also D, then it is called a *true positive* (TP); however if the actual value is S then it is said to be a *false positive* (FP). Conversely, a *true negative* (TN) has occurred when both the prediction outcome and the actual value are S, and *false negative* (FN) is when the prediction outcome is S while the actual value is D.

The four outcomes can be summarized in a 2x2 *confusion matrix*, as follows:



The contingency table can derive several evaluation “metrics”. To draw an ROC curve, the **true positive rate** (TPR) and **false positive rate** (FPR) are needed (as functions of some classifier parameter). The TPR defines how many correct positive results occur among all positive samples available during the test. The FPR, on the other hand, defines how many incorrect positive results occur among all negative samples available during the test.

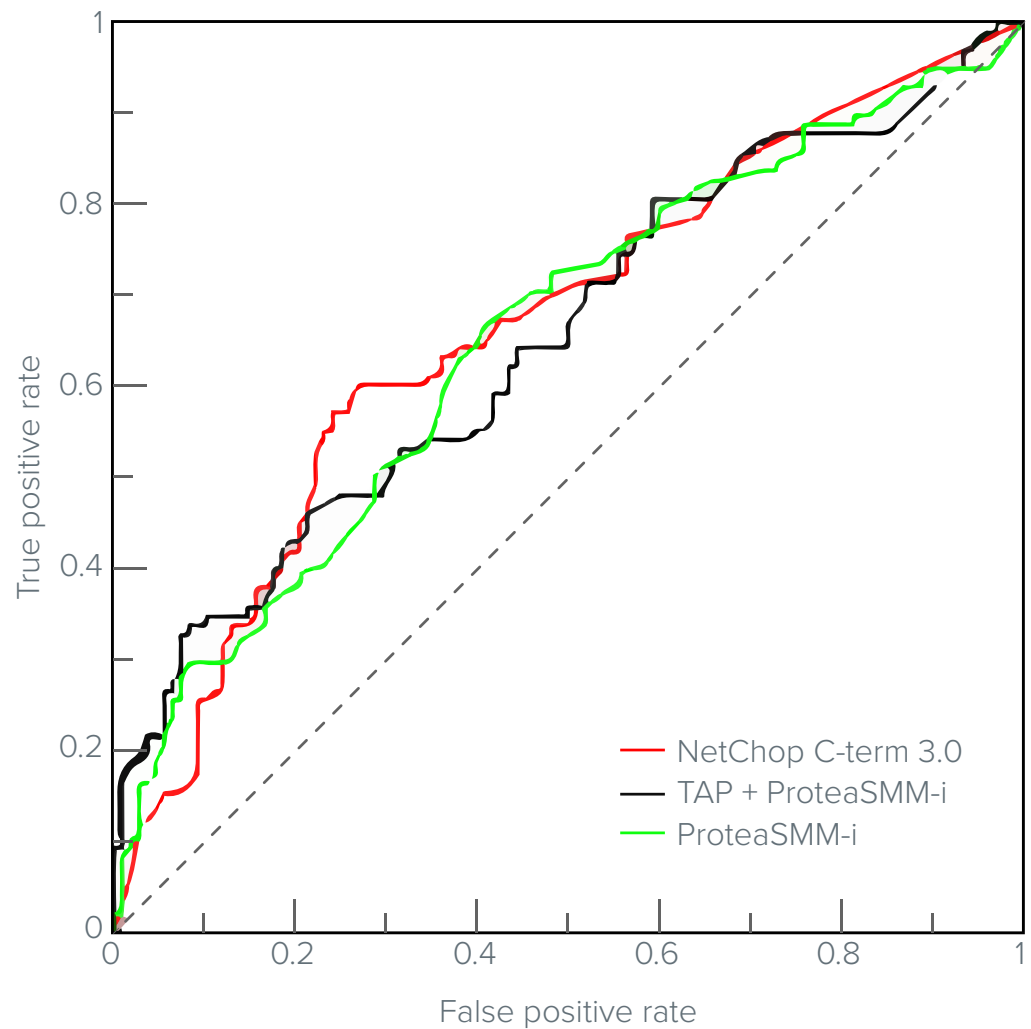
In binary classification, the class prediction for each instance is often made based on a continuous random variable X , which is a “score” computed for the instance (e.g. estimated probability in logistic regression). Given a threshold parameter T , the instance is classified as “positive” if $X > T$ and “negative” otherwise. X follows a probability density $f_1(x)$ if the instance actually belongs to class “positive”, and $f_0(x)$ if otherwise. Therefore, the true positive rate is given by $TPR(T) = \int_T^{\infty} f_1(x).dx$ and the false positive

rate is given by $FPR(T) = \int_T^\infty f_0(x).dx$. The ROC curve plots parametrically $TPR(T)$ versus $FPR(T)$ with T as the varying parameter.

Here is what the result looks like:

Example of ROC curve from 3 different predictive models.

(source: [Wikipedia](#))



The further the curve is from the diagonal (random picking, often called the *no-discrimination line*), the better the classifier is. That's why it makes sense to calculate the AUC in order to determine the quality of a classifier. The AUC is related to the **Gini coefficient** G_I ; $G_I = 2AUC - 1$

Models and parameters: how does this work?

Introduction to Machine Learning

What is Machine Learning?

ML is a subfield of computer science that “gives the computers the ability to learn without being explicitly programmed” (Arthur Samuel, 1959). More precisely, the main idea is to build algorithms that **can learn from and make predictions on** data.

There are two main types of machine learning:

- the first type of learning is called **supervised learning**. We have a dataset with an input describing the characteristics of given individuals and an output with the **labelling** of these individuals given by a “teacher” (a doctor for instance who defines a tumor as cancerous/non-cancerous) or by reality and experience (in reinforcement learning, a robot might “fall” and learn from its falls). The algorithm learns from the association between the label and the characteristics and is then able to predict the label of an individual given his characteristics.
- the second type of learning is called **unsupervised learning**. This time, no labels are given to the algorithm who has to find structure in its input on its own.

In the case of Credit Risk, we focus on supervised learning since the loans are **labelled** (in this case, “default/safe” or “good/bad”) and we are trying to predict whether one given customer is likely to default or not, given their characteristics.

Supervised Learning

Explanation

All algorithms used by James are based on a Machine Learning field called supervised learning.

Example of data file with history (each row represents a loan)

# Loan	Age	Income	Debts	Purpose	Amount	...	Defaulted
1	28	\$1000	\$20k	Car	\$4000		Yes
2	49	\$2300	-	Edu	\$2000		No
3	55	\$600	-	Car	\$4000		No
4	67	\$4000	\$1k	Home	\$500		Yes
5	35	\$1100	-	Edu	\$1000		Yes

Given a **history** of loans, we want to understand which variables (e.g. Age, Income, etc..) will help to predict if the client defaulted on the loan. With supervised learning, we will create a model which, given the characteristics of the loan (input variables), will estimate its **probability of default**.

Objective Function

In Supervised Learning, a fundamental function is the **objective function** defined as follows:

$$Obj(\Theta) = L(\Theta) + \Omega(\Theta)$$

$L(\Theta)$ is the **loss function**, measuring how well our model fits to our training data: it measures the quality of the prediction and therefore the **bias** of the latter. We can write the function this way:

$$L(\Theta) = \sum_{i=1}^n l(y_i, \hat{y}_i)$$

n is the number of individuals, y_i is the label of individual i and $\hat{y}_i$ is the prediction of individual i 's label.

Several loss functions can be used depending on the cases:

- the Squared Loss: $l(y_i, \hat{y}_i) = (y_i - \hat{y}_i)^2$
- the Absolute Loss: $l(y_i, \hat{y}_i) = |y_i - \hat{y}_i|$
- the Logistic Loss: $l(y_i, \hat{y}_i) = y_i \ln(1 + e^{-\hat{y}_i}) + (1 - y_i) \ln(1 + e^{\hat{y}_i})$

One advantage of the Absolute Loss over the Squared Loss is that it is **more robust to outliers** since it gives the same weight to all observations. On the other hand, the Squared Loss is more useful if we are concerned about large errors whose consequences are much bigger than equivalent smaller ones.

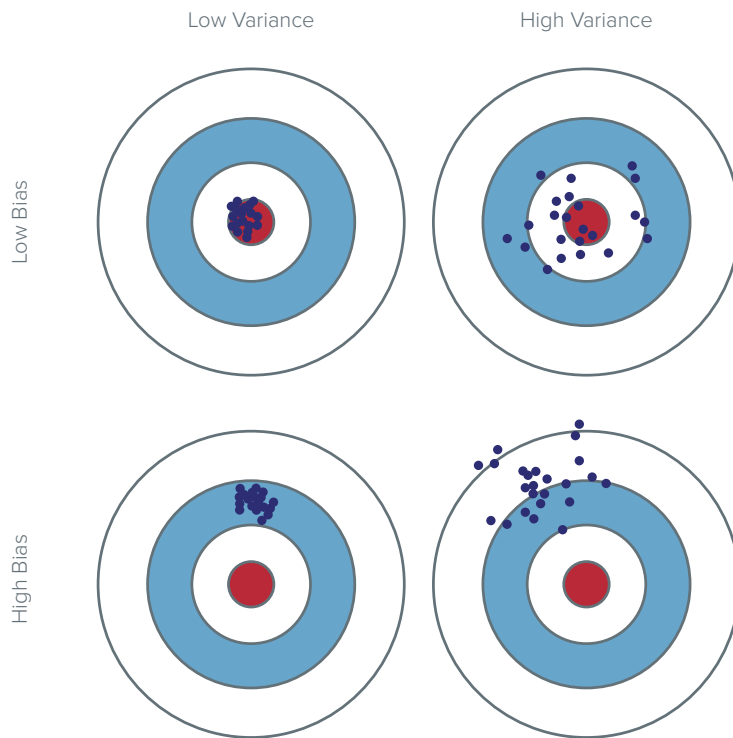
If the problem is a binary classification (i.e. labels are -1 and +1), Logistic loss is convenient although Squared loss may work as well.

$\Omega(\Theta)$ is the **regularization term**, calculating the complexity of the model and therefore its [variance](#).

For regularization terms, we use different **norms** depending on the purposes:

- L1 norm: $\Omega(\beta) = \alpha \|\beta\|_1$
- L2 norm: $\Omega(\beta) = \alpha \|\beta\|^2$

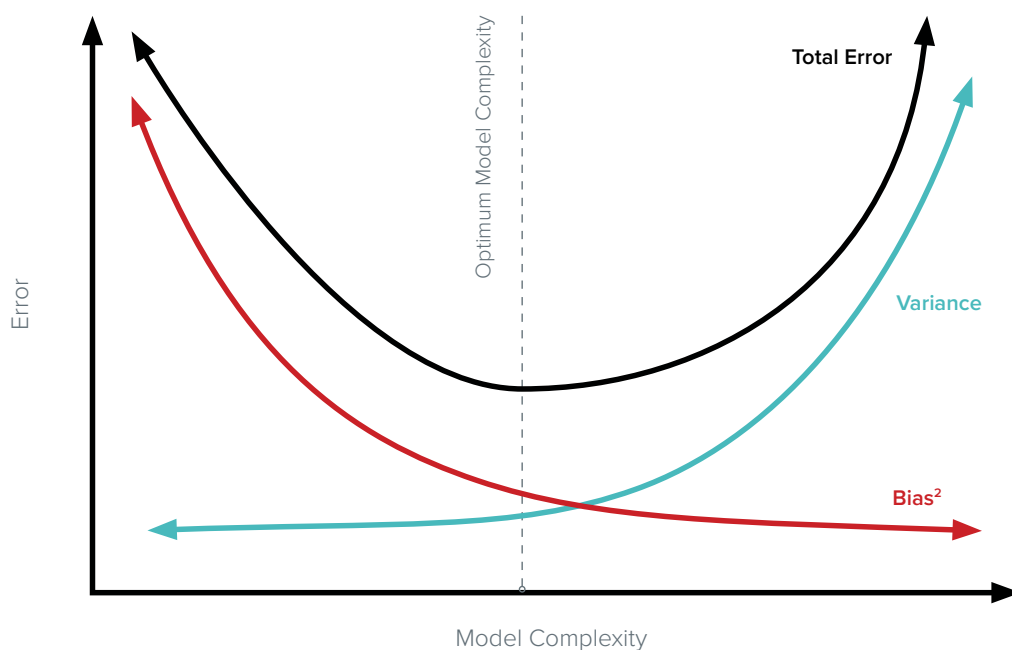
In supervised learning, we face a **bias-variance tradeoff**: we want our model to be **predictive (low bias)** without **being too complex (high variance)**.



Graphical illustration of bias and variance.
(source: [Scott Fortmann–Roe](#))

Decreasing the [bias](#) may lead to high variance and **overfitting**¹. We therefore have to find a **tradeoff** between the two terms of the objective function we are minimizing in order to reach the optimum model complexity.

¹ Overfitting – “learn by heart” the training sample, building a very complex model that will not make efficient predictions on unseen data.



Bias and variance contributing to total error.
(source: [Scott Fortmann–Roe](#))

Model, Variables, Parameters and Hyperparameters

We refer to the relations which supposedly describe a certain physical situation, as a **model**. Typically, a model consists of one or more equations. The quantities appearing in the equations are classified into **variables** and **parameters**. The distinction between these is not always clear cut, and it frequently depends on the context in which the variables appear. Usually, a model is designed to explain the relationships that exist among quantities which can be measured independently in an experiment; these are the **variables** of the model. To formulate these relationships, however, one frequently introduces “constants” which stand for inherent properties of nature (or of the materials and equipment used in a given experiment). These are the **parameters**.

We use the term **hyperparameter** in order to distinguish from standard model **parameters** that are obtained by fitting the model with the data.

Hyperparameters are the kind of parameters that cannot be directly learned from the regular training process. These parameters express “higher-level” properties of the model, such as its complexity or how fast it should learn, and are usually either fixed before the actual process begins or learned by hyperparameter optimizers such as Grid Search. For non-parametric models (RF, Boosting, ...), parameters and hyperparameters are the same thing (though “hyper” is more proper).

Linear models

When we talk about linear models, there are a [lot](#) of models that come to mind. In this paper, we will focus on linear and logistic regression. We won't spend much time on **linear** regressions: feel free to check [this Yale course](#) which introduces this model quickly and efficiently.

The baseline to start from in a **binary classification problem** is the binary **logistic regression**. This model, like the ones to follow, is used to estimate the probability of a binary response based on one or more predictors.

Why is logistic regression more successful than linear regression in our case?

Let Y be our target variable with X being a vector including the observed values of all the features.

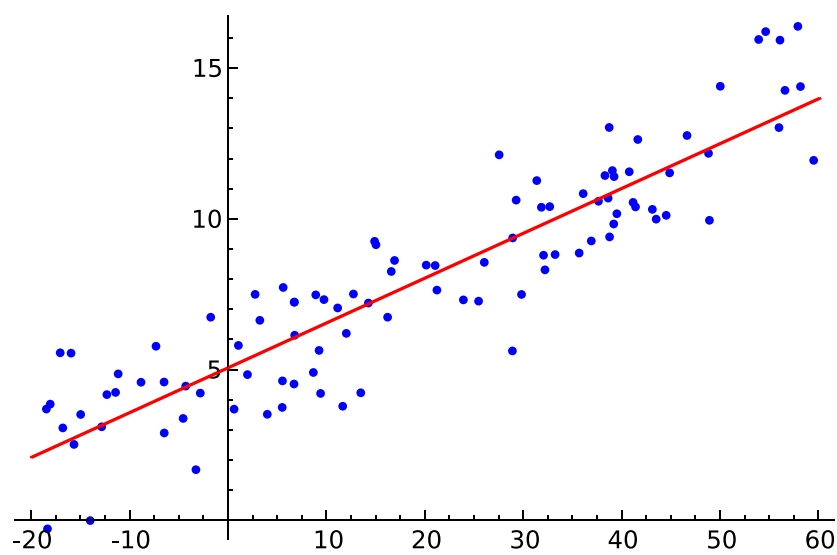
Like in any statistical problem, we assume that the observations are a partial description of a wider phenomenon that we want to know more about.

In a **linear regression**, this phenomenon will be shaped by a straight line whose parameters are found by minimizing the square error (an example is shown in the next figure).

$$Y = \beta_0 + \beta_1 X + \epsilon$$

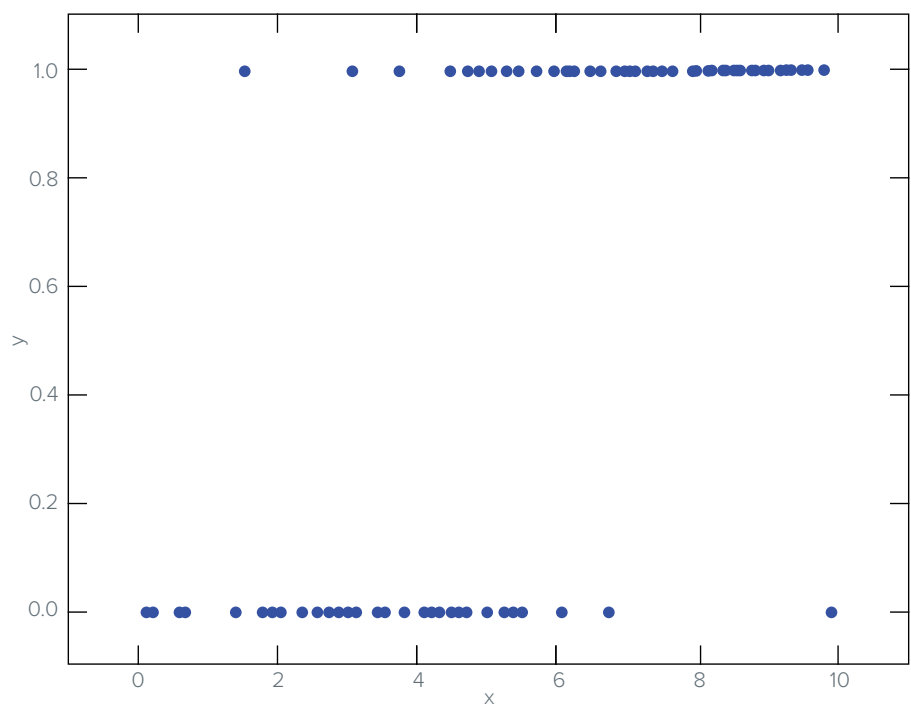
with β_0 as the intercept (5 on the graphic below) and β_1 as the slope.

The graphic below represents the observations (in blue) and the regression line (in red) on a (X,Y) axis.



(source: [Wikipedia](#))

Here, with a binary variable as our target, drawing a straight line would not make much sense. Indeed, our observations look more like this:



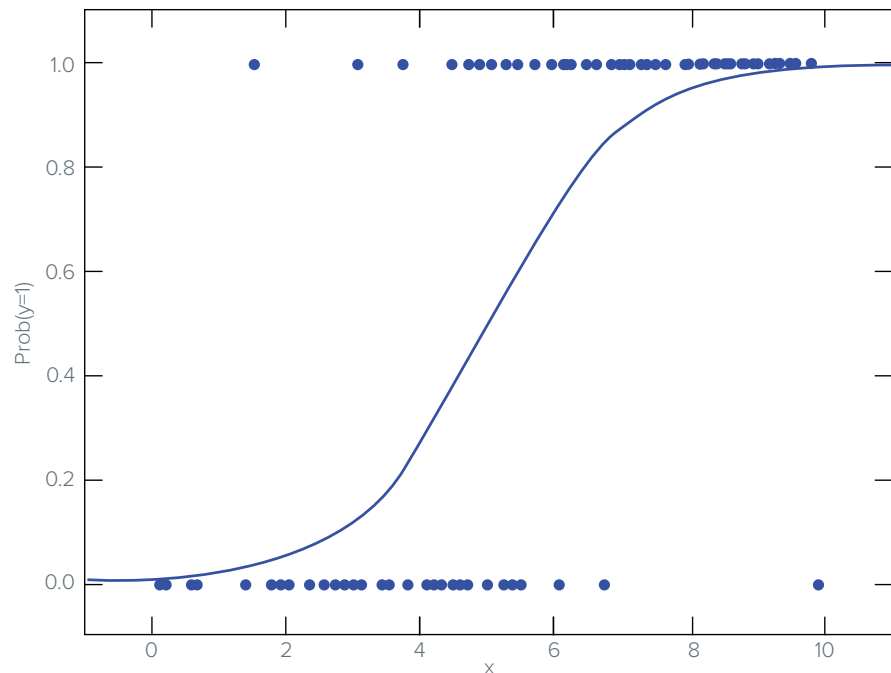
How does logit work?

The way to go in this case is to make an assumption about the probability law that defines the probability of Y being 1 given the characteristics of an individual (otherwise written $P(Y = 1 \mid X = x)$). In the case of a **logistic regression**, we assume that this probability follows a **logistic law** defined by the following cumulative distribution function:

$$F(x) = \frac{1}{1 + exp^{-x}}$$

Let's look at a graphic showing our observations and the logistic cumulative distribution function.

(source: [Analytics Vidhya](#))



Notice that the Y axis is no longer defined by the dependent variable but by the probability that this dependent variable equals to 1 (for instance, it could be the probability that a customer defaults). This probability is what we want to estimate for each customer. Let's get back to the model to see how we will do this:

$$P(Y = 1 | X) = \frac{1}{1 + \exp^{-\beta^T X}}$$

Indeed, we made the assumption that our model follows a logistic law but we still don't know what the parameters of this law are. Therefore, the parameter we need to estimate here is β of the logistic law.

The way to go about this is to find the parameter β that maximizes the likelihood² of Y . Many statistical softwares do this calculation for us and with the estimated parameters, we're able to define **the probability for each individual to default given its characteristics**.

From the probability we found, we build a **decision boundary** defined as follows: if a customer has a probability higher than a particular value (our threshold, 0.5 by default) to default, the model will predict that this customer will default. Otherwise, the model will predict that it is a safe customer. This is why the logistic regression is actually not a regression model but a **classification** model and a **linear classifier** to be more precise.

To illustrate this with an example, let's assume that we only have two regressors X_1 and X_2 . The model is therefore:

²Likelihood – the joint probability of having every Y_i equal to its observed value, either 0 or 1, given the characteristics X .

$$P(Y = 1 | X) = \frac{1}{1 + \exp^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2)}}$$

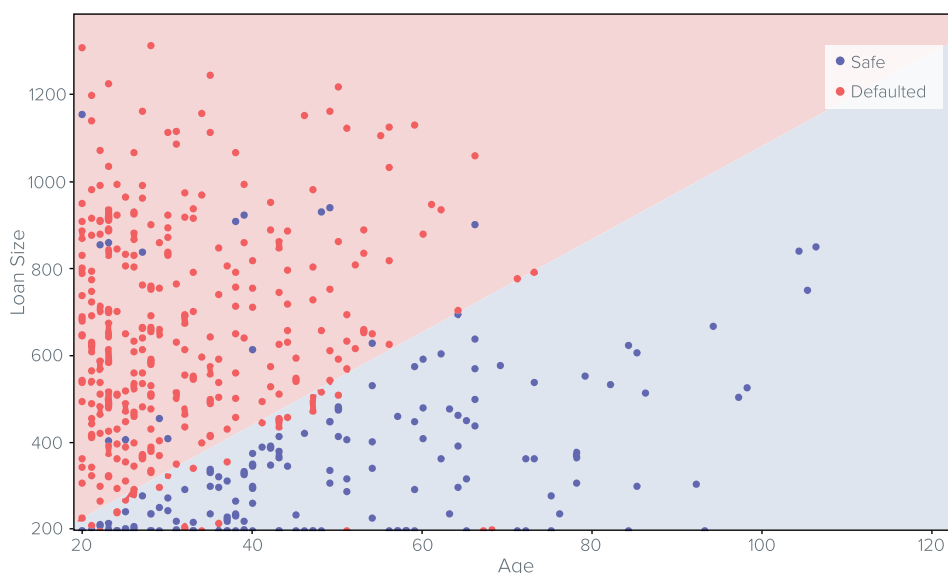
We then estimate β_0 , β_1 and β_2 and in order to build the decision boundary line, we choose: $P(Y = 1 | X) = 0.5$ which implies that $\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$

We then find the equation of the decision boundary:

$$X_2 = -\frac{\beta_1}{\beta_2} X_1$$

Results overview

If we take the loan size and the age as regressors and an example dataset, the result looks like the following graphic. Each point is an individual located by his age (X axis) and loan size (Y axis). The color defines the individual's situation: a blue dot refers to a safe individual and a red dot refers to an individual that defaulted. The background color refers to the algorithm's prediction: a red background implies that the algorithm predicted a default for the individuals located in this zone and vice versa.

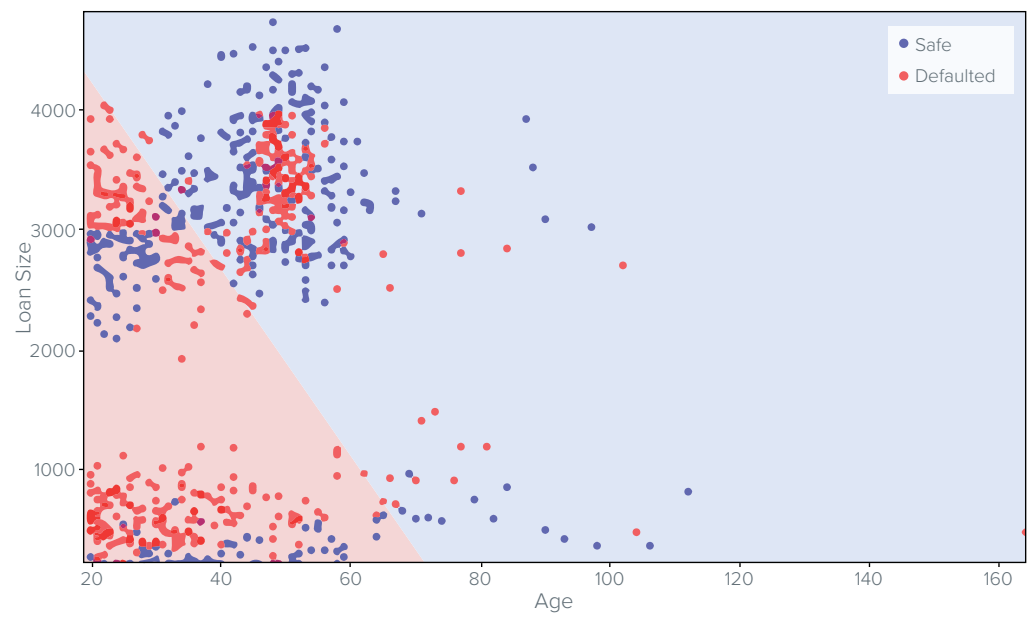


The result looks fairly good on this dataset and will look fairly good in general as long as you expect **your features to be roughly linear** and **the problem to be linearly separable** as it is more or less here.

Logistic regression can also be useful in its **Least Absolute Shrinkage and Selection Operator (LASSO)** and/or **Ridge** version(s). The idea is to create models in the presence of a great number of features. As model complexity increases, the number of coefficients goes up very fast. This is why putting a constraint on the magnitude of coefficients is a good way to reduce the **model's complexity**. It therefore allows one to reduce overfitting and to reduce the probable computational challenges of big datasets (complexity and speed). This is basically what **Ridge** does

by adding a penalization term $\lambda \|x\|_1$ (L2 norm defined earlier) to the log-likelihood in the maximization part. The particularity of **LASSO** is that it also allows one to do a **feature selection** by putting down to zero the least important features after a certain value of α (use of L1 norm defined earlier).

Nevertheless, logistic regression has disadvantages: for instance, high correlation between features can become a problem above a certain threshold ([multicollinearity](#)). Also and more simply, what if our features are not linear?



We observe in the above graphics that in this case, the logistic regression is not an efficient tool to separate our data.

Conclusion on linear models

- Model: how to make prediction $\hat{y}_i$ given x_i .
 - Linear model: $\hat{y}_i = \sum_j \beta_j x_{ij}$ (include linear/logistic regression);
 - The prediction score $\hat{y}_i$ can have different interpretations depending on the task:
 - In linear regression, $\hat{y}_i$ is the predicted score;
 - In logistic regression, $1/(1 + e^{-\hat{y}_i})$ is the probability of the instance being positive.
- Parameters:
 - Linear model: $\Theta = \{\beta_j \mid j = 1, \dots, d\}$, with d being the number of regressors.
- [Hyperparameters:](#)
 - `C (classifier__C)` - Strength of regularization, a smaller value implies stronger regularization;
 - Class weight (`classifier__class_weight`) - If null, default and non-default observations have entered the model with the same weight; otherwise it indicates the respective weights;
 - Fit intercept (`classifier__fit_intercept`) - If True, a constant intercept was added to the logistic function;
 - Intercept scaling (`classifier__intercept_scaling`) - The value of the intercept, if one was used;
 - Regularization type (`classifier__penalty`) - L1 or L2 regularization.
- **Advantages:**
 - Good results if you expect your features to be linear and your problem to be linearly separable;
 - Explainability.
- **Disadvantages:**
 - Does not handle categorical features very well;
 - High correlation between variables create issues;
 - Does not work well if the features are not linear;
 - Prone to underfitting.

Decision Trees

Parametric models such as logits make certain assumptions on the data structure that may not be fully correct. This is when other **nonparametric** models come in handy. **Nonparametric** means that we do not make any assumptions as to the form or parameters of a frequency distribution.

Decision Trees are one of them and are an important type of algorithm in Machine Learning.

The classical decision tree algorithm has been around for decades and ensemble models which are based on decision trees, such as Random Forest, are among the most powerful techniques available today.

The goal of a decision tree is to create a model **that predicts the value of a target variable by learning simple decision rules inferred from the historical data.**

In other words, a decision tree is a set of rules used to classify data into categories. It looks at the variables in a data set, determines which are most important, and then comes up with a tree of decisions which best partitions the data. The tree is created by splitting data up by one feature at each step and then counting to see how many are in each bucket after each split.

The splitting rule is explained in the next section and is based on

$$E(T) = - \sum_{i=1}^d p_i \log(p_i)$$

which variable (e.g., Income) will best divide the population in 2 new segments (ex: Salary above and below 1000).

Splitting rules and examples

There are several possible splitting criterion in decision trees. **Entropy** is one of them and it is the one we will present here. Gini or node impurity is another such possible criteria.

There are two kinds of entropy we must calculate at each step in order to build a decision tree:

- **the target entropy:** we look at our whole data set and define the number of individuals in each class, which allows us to calculate the p_i (percentage of total individuals in class i) and then the target entropy:

$$E(T, X) = \sum_{c \in X} P(c) E(c)$$

d is the number of classes, in our case $d=2$ (Safe and Default).

- **each feature's entropy:** same principle but different mode of calculation: for each feature X , we use its frequency table.

As an example, if “Play golf” is the target and “Outlook” one feature:

(source: [Saed Sayad](#))

		Play golf	
		YES	NO
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	2	3
		14	

$$\begin{aligned} E(\text{PlayGolf}, \text{Outlook}) &= P(\text{Sunny}) \cdot E(3,2) + P(\text{Overcast}) \cdot E(4,0) + P(\text{Rainy}) \cdot E(2,3) \\ &= (5/14) \cdot 0.971 + (4/14) \cdot 0.0 + (5/14) \cdot 0.971 \\ &= 0.693 \end{aligned}$$

So we end up with the target entropy and the entropy of each feature. The **information gain** is defined as follows (with the target T and a feature X):

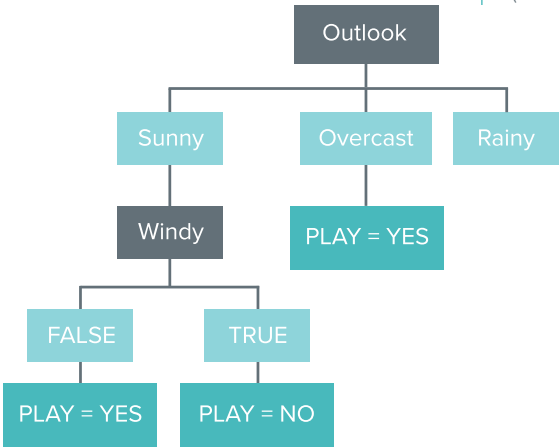
$$Gain(T, X) = E(T) - E(T, X)$$

We choose the feature corresponding to the highest information gain and we use it to make the **split**. In case there is a branch that only has individuals belonging to a unique class (entropy=0), this branch is a **leaf node**, i.e., does not need more splitting.

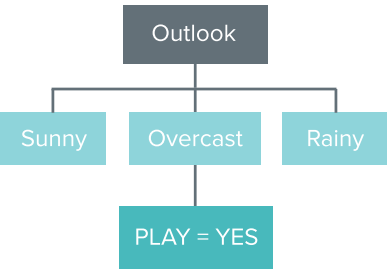
As long as the entropy is strictly higher than 0, the branch benefits from additional splitting.

(source: [Saed Sayad](#))

Temp	Humidity	Windy	Play Golf
Mild	High	FALSE	Yes
Cool	Normal	FALSE	Yes
Mild	Normal	FALSE	Yes
Cool	Normal	TRUE	No
Mild	High	TRUE	No



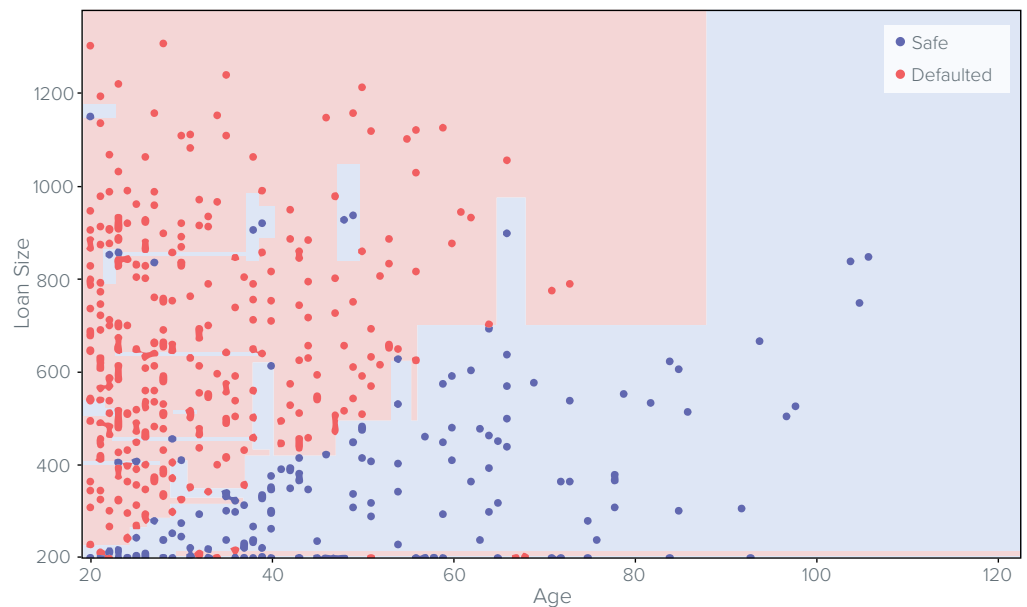
Temp	Humidity	Windy	Play Golf
Hot	High	FALSE	Yes
Cool	Normal	TRUE	Yes
Mild	High	TRUE	Yes
Hot	Normal	FALSE	Yes



To make predictions on new data, the algorithm will apply the set of rules to the new observations. This will tell us to which “endpoint” and which class the observation belongs to.

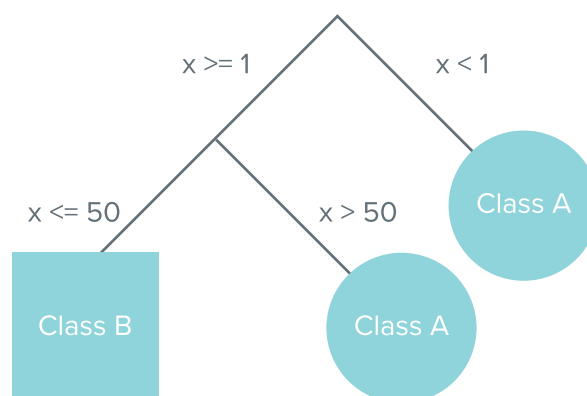
Results overview

Here is an example of a classification made by a decision tree, with the same dataset that was used earlier for the logistic regression example.



³Outliers – individuals with very different characteristics from the rest of the population.

We can see that the classification is already more precise, partly because **it does not expect the features to be linear**. **Outliers³** are also handled well by decision trees: the trees will easily take care of the cases where you have class A at the low end of feature X , class B in the mid-range of feature X , and A again at the high end.



In general, decision trees are easy to interpret and to explain but one of their main disadvantages is that they tend to **overfit**. Indeed, the trees are grown to the maximum possible extent so that they reach the lowest possible **bias** but it also implies a big dependence on the training set's characteristics, hence a very **complex** model and therefore a high **variance**.

Conclusion on decision trees

- Model:
 - For *regression*, output is a real figure;
 - For *classification* (which is our case here); output: class that the individuals belong to. If the trees are not grown to their fully extent, the output of each leaf node is the proportion of individuals that belong to the most represented class. In this case, by default, an individual ending up in one leaf node will be classified in the class that is most represented in this node.
- Parameters:
 - Criterion: function used to measure the quality of a split (we presented *entropy* here; [Gini impurity](#) may alternatively be used);
 - Max depth of the tree: if None, then nodes are expanded until all leaves are pure or until all leaves contain less than min_samples_split samples;
 - min_samples_split: the minimum number of samples to make a split. (default = 2).
- **Advantages:**
 - Explainability;
 - It does not expect the features to be linear;
 - Handles outliers reasonably well.
- **Disadvantages:**
 - Tends to overfit: poor predictive performance.

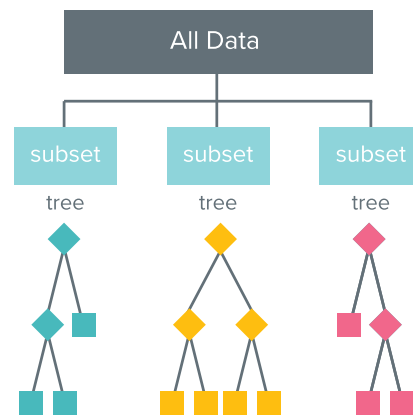
Ensemble Models

To handle the overfitting tendency of the decision trees and reduce the overall variance, ensemble models come in handy. We will present here the two main ensemble models: **Random Forest** and **Gradient Boosting**.

Random Forest

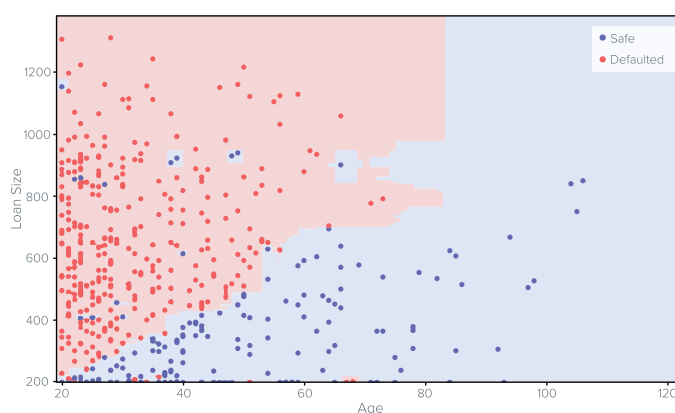
The Random Forest algorithm builds decision trees out of **random samples of the data** and a **random selection of features**. The production phase of random samples is called **bootstrap aggregating** (also known as **bagging**): the idea is to produce m new training sets D_i of size n out of one initial training set D of size n by sampling from D uniformly and with replacement. With the selected population, the tree is built with a different splitting method: **each split is made based on a random subset of features**. Each tree outputs a class for each individual. In the end, a majority vote takes place: we calculate the proportion of trees that classify one individual in each class and the biggest proportion defines the predicted class membership

(source: [Mapr](#))



Results

With the same example dataset, the results look like this:



In many scenarios, Random Forest gives much more accurate predictions when compared to simple decision trees or linear models. This is because this technique implies the creation of different decision trees that have **small bias and very (very) high variance**. Thus, taking the average prediction of all these different classifiers **lowers the variance by a lot** while **keeping a fairly small bias**.

Conclusion on Random Forest

- Model:
 - Use bagging to sample the data for each tree;
 - Solve each split using a random subset of features;
 - Majority vote in order to average the prediction.
- (Hyper)parameters:
 - Bootstrap (*classifier__bootstrap*) - indicates if the samples used for building the trees were drawn with the bootstrapping method;
 - Class weight (*classifier__class_weight*) - if 'None', default and non-default observations have entered the model with the same weight; otherwise it indicates the respective weights. 'Balanced' indicates that weights are automatically adjusted to match the proportion of defaults in the training data set;
 - Criterion (*classifier__criterion*) - see Decision Trees;
 - Max depth (*classifier__max_depth*) - maximum depth of the decision trees;
 - n_estimators: the number of trees in the forest;
 - Max features (*classifier__max_features*) - the number of features to use when looking for the best split in decision tree nodes;
 - Min samples leaf (*classifier__min_samples_leaf*) - the minimum number of samples required at each leaf node (as fraction of total number of samples).
- **Advantages:**
 - It is one of the most accurate learning algorithms available. For many data sets, it produces a highly accurate classifier.
 - It runs efficiently on large databases.
 - It can handle thousands of input variables without variable deletion.
 - It gives estimates of which variables are important in the classification.
 - It generates an internal unbiased estimate of the generalization error as the forest building progresses.
 - It has an effective method for estimating missing data and maintains accuracy when a large proportion of the data is missing.
 - It has methods for balancing error in class population unbalanced data sets.
 - Prototypes are computed that give information about the

relation between the variables and the classification.

- It computes proximities between pairs of cases that can be used in clustering, locating outliers, or (by scaling) give interesting views of the data.
 - The capabilities of the above can be extended to unlabeled data, leading to unsupervised clustering, data views and outlier detection.
 - It offers an experimental method for detecting variable interactions.
 - Does not require feature scaling.
- **Disadvantages:**
 - Random forest has been observed to overfit for some datasets with noisy classification/regression data;
 - Harder to explain relatively to Decision Trees and Logit;
 - Hard to calibrate.

Gradient Boosting

Gradient Boosting is an ensemble technique that is rooted in the concept of **Gradient descent**. The latter is a first-order optimization algorithm that is usually used to calculate a function’s local minimum.

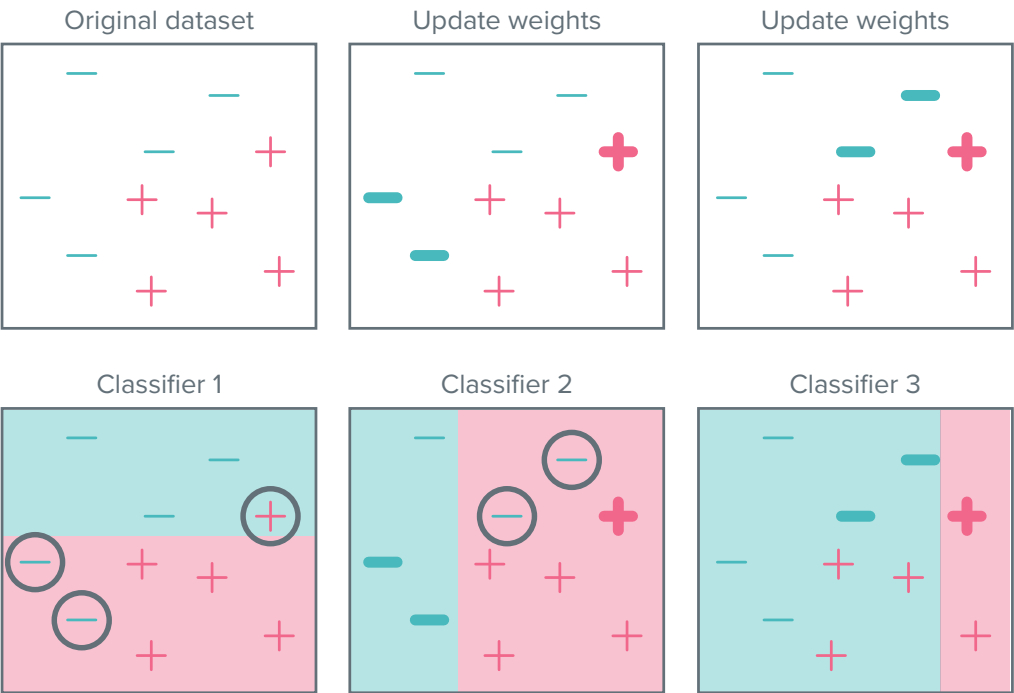
The idea of boosting came from the idea about whether a **weak learner**⁴ can be modified to become better. The classifiers are built in a sequential manner and each member of the ensemble is an expert on the errors of its predecessors.

⁴Weak learner – classifier that does slightly better than randomly picking the class.

Boosting: Adaboost

Before explaining **Gradient Boosting** itself, let’s focus on an application of Boosting to decision trees: **Adaboost**.

Adaboost is the first practical Boosting algorithm and was introduced by **Y. Freund** and **R. Schapire** in 1996. As you can see on the image below, we first build one classifier h_1 out of our training data. This classifier will perform a good prediction for some individuals and a worse prediction for others (errors circled in black in the image below). The error rate is calculated with a **loss function** defined earlier in the process. From this classifier, we want to iteratively build other classifiers that will do better than the weak learner h_1 . To do so, we will give a **higher weight to the individuals that the first classifier misclassified in the training sample of the second classifier** so that this one will **pay more attention to these individuals and classify them in a better way**. The evolution in weights will influence the loss function minimization phase and emphasize the importance of predicting the *heavier* individuals well.



(source: “Ensemble Classifiers”, Roberto Henriques)

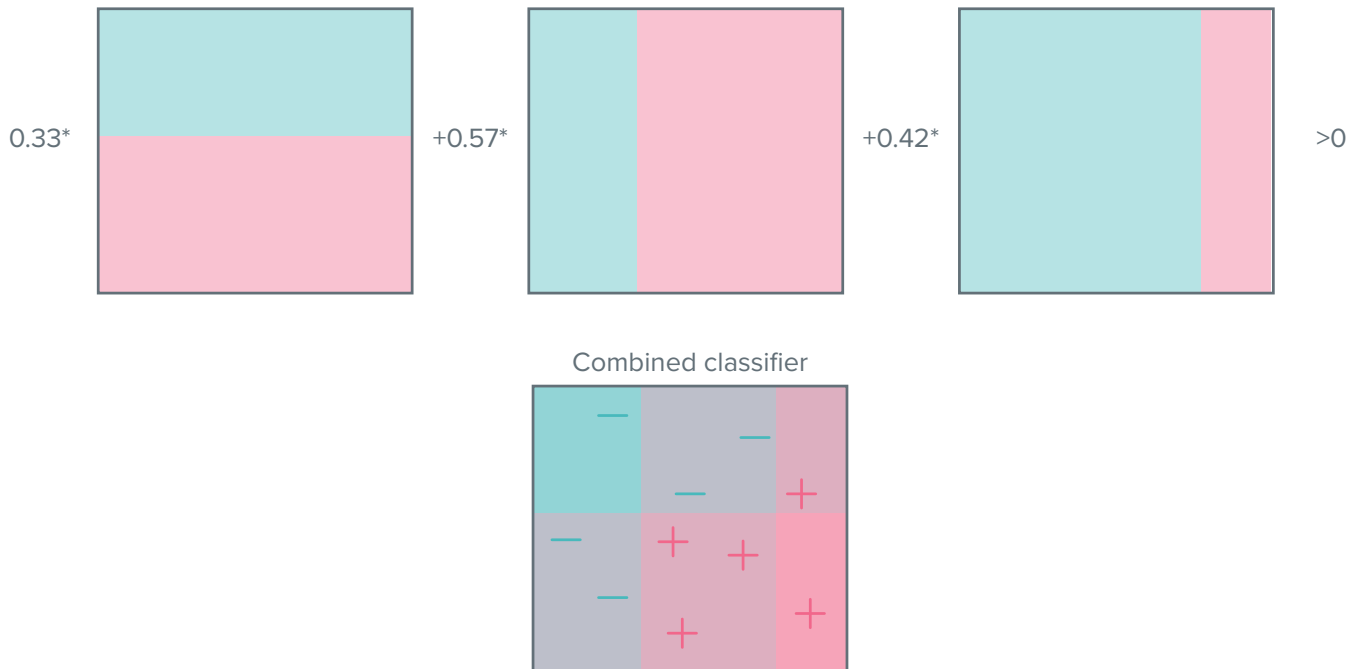
In the end, we have M classifiers that equal to the number of iterations we chose when launching the algorithm. We then evaluate the weight α_t of each of them, based on their error rate ϵ_t . The final classifier is made from the weighted sum of all our classifiers.

- Weigh each classifier:

Negative logit function multiplied by 0.5.

$$\alpha_t = \frac{1}{2} \cdot \ln \left(\frac{1 - \epsilon_t}{\epsilon_t} \right)$$

- Weigh each classifier and combine them:



Adaboost had a lot of success, in [face recognition](#) for instance, and Freund and Schapire won the Gödel Prize in 2003 for their work. This raised the attention of the statistics community on Boosting methods and [Jérôme Friedman came up in 1999](#) with **Gradient Boosting** which is a **generalization of Boosting to arbitrary loss functions**.

Introduction to Gradient Boosting: Regression Method (Ben Gorman)

Based on the idea of boosting, we present the intuition of Gradient Boosting in **pseudo-code**:

1. Fit the model to the data: $F_1(x) = y$
2. Fit a model to the residuals: $h_1(x) = F_1(x) - y$
3. Create a new model $F_2(x) = F_1(x) + h_1(x)$

Repeating this model iteratively will keep on improving our classifier.

$$F(x) = F_1(x) \rightarrow F_2(x) = F_1(x) + h_1(x) \rightarrow \dots \rightarrow F_M(x) = F_{M-1}(x) + h_{M-1}(x)$$

Note that h_M is just a model and **nothing in our definition requires it to be a tree-based model**. This is one of the broader concepts and advantages of Gradient Boosting: the latter is a framework for iteratively improving **any weak learner**. In theory, a Gradient Boosting module would **allow to plug in various classes of weak learners**. In practice however, h_M is almost always a regression tree. Also, note that the number of iterations M is found by testing different values via **cross-validation**.

In order to present an algorithm that is conform to most gradient boosting implementations, we initialize the model with a single prediction value:

$$F_0(x) = \underset{\gamma}{\operatorname{argmin}} \sum_{x=1}^n L(y_i, \gamma)$$

Our first classifier minimizes the total loss. Note that this works with any loss function. For instance, we can define L as the **squared error**: this is what the regression tree does by default. In this case, making the prediction one unit closer to its target will reduce the error a lot more if the prediction is far from the target (check the “Objective Function” part in “Supervised Learning” for more details). Therefore, the tree will focus on improving the predictions that are far from their respective target. On the contrary, the **absolute error** does not discriminate predictions since the improvement of one unit of a predictor towards its target reduces the error by one, no matter the quality of prediction.

With this in mind, suppose that instead of training h_0 on the residuals, we instead train h_0 on **the gradient of the loss function** $L(y, F_0(x))$ with respect to the prediction values produced by $F_0(x)$. **Essentially, we will train h_0 on the cost reduction for each sample if the predicted value were to become one unit closer to the observed value.** In the case of absolute error, h_m will simply consider the sign of every F_m residual (as opposed to squared error which would consider the magnitude of every residual). After samples in h_m are grouped into leaves, an average gradient can be calculated and then scaled by some factor, γ , so that $F_m + \gamma h_m$ minimizes the loss function for the samples in each leaf. (Note that in practice, a different factor is chosen for each leaf.)

Gradient Descent (GD) - a short introduction (Abishek Ghose)

To go on in our explanation, we need to understand the concept of **Gradient Descent**.

Let $f(x)$ be the function you want to minimize. Assume x to be a scalar. One way to **iteratively** minimize, and find a (possibly local) minimizer x , is to follow this update rule at the i -th iteration:

$$x_{(i)} = x_{(i-1)} - \eta \frac{\delta f(x_{(i-1)})}{\delta(i-1)}$$

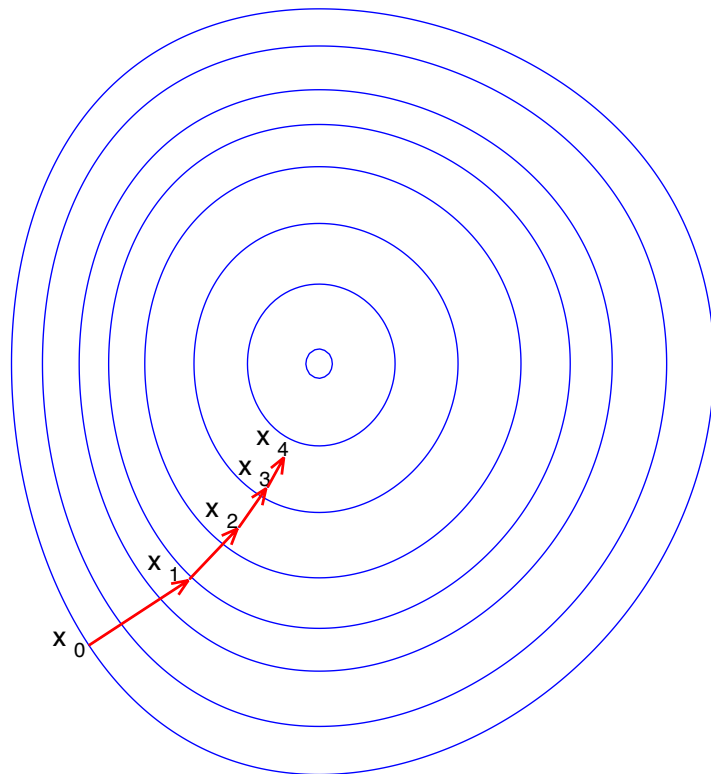
The term that multiplies η is called the **gradient**. Graphically, for a single-variable function, it represents its steepness. It is positive if f is rising and negative if f is decreasing. η is called the **step magnitude** and is a positive constant.

The next value of x , $x(i+1)$ is the current value $x(i)$ plus a small step against the gradient, that is, towards the direction of steepest descent. That is why it will reach a (local) minima. We stop when $x(i) = x(i-1)$. We may start with an arbitrary value for $x(0)$.

The following figure shows how this works:

Illustration of Gradient Descent on a series of level sets.

(source: [Wikipedia](https://en.wikipedia.org/wiki/Gradient_descent))



In the case where x is a vector, the principle remains the same. Now, we adjust every *individual dimension* of x based on the slope along that direction. For the i -th iteration and the j -th dimension, this is what the update rule looks like:

$$x_j^{(i)} = x_j^{(i-1)} - \eta \frac{\delta f(x^{(i-1)})}{\delta x_j^{i-1}}$$

At each iteration *all* dimensions are adjusted. The idea is, we want to move the vector x , as a whole, in a direction where *each individual component minimizes $f(x)$* . This is **important** - this form will show up in the subsequent discussion.

Leveraging Gradient Descent: Gradient Boosting for regression

We can now use gradient descent for our gradient boosting model. The objective function we want to minimize is L . Our starting point is $F_0(x)$. For iteration $m = 1$, we compute the gradient of L with respect to $F_0(x)$. Then we fit a weak learner to the gradient components. In the case of a regression tree, leaf nodes produce an average gradient among samples with similar features. For each leaf, we step in the direction of the average gradient (using line search to determine the step magnitude). The result is F_1 . Then we can repeat the process until we have F_m .

We modified our gradient boosting algorithm so that it works with any differentiable loss function. Let's clean up the ideas above and reformulate our gradient boosting model once again.

Initialize the model with a constant value:

$$F_0(x) = \underset{\gamma}{\operatorname{argmin}} \sum_{i=1}^n L(y_i, \gamma)$$

For $m = 1$ to M :

- Compute pseudo residuals,

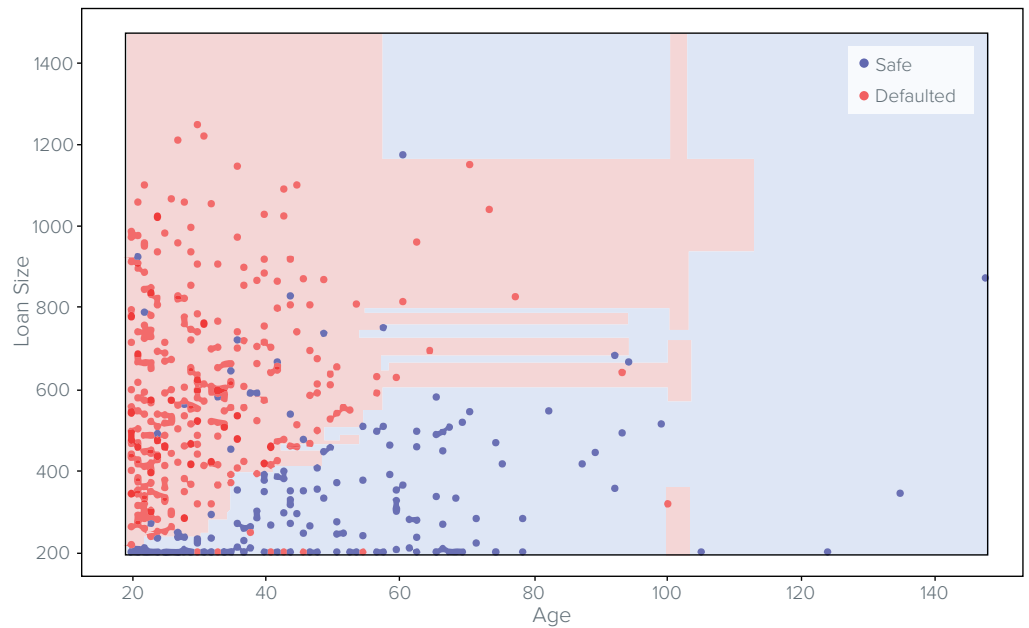
$$r_{im} = - \left[\frac{\delta L(y_i, F(x_i))}{\delta F(x_i)} \right]_{F(x)=F_{m-1}(x)} \text{ for } i = 1, \dots, n.$$

- Fit base learner, $h_m(x)$ to pseudo residuals
- Compute step magnitude multiplier γ_m . (In the case of tree models, compute a different γ_m for every leaf.)
- Update $F_m(x) = F_{m-1}(x) + \gamma_m h_m(x)$

For a classification problem, the main difference is that we **don't improve just one model but as many models as the number of classes**. Feel free to read Cheng Li's ["Gentle Introduction to Gradient Boosting"](#) for more information on GB for classification.

Results

The results are the following and look similar to Random Forest in this case:



After presenting these two ensemble methods, one could rightfully ask themselves which works the best. In 2005, **Caruana & al.** made an empirical comparison of supervised learning algorithms. They included Random Forest and Boosted decision trees and concluded that: “*With excellent performance on all eight metrics, **calibrated boosted trees** were the best learning algorithm overall. Random Forest are close second.*”

Conclusion on Gradient Boosting

- Model:
 - Many weak learners together make a strong one;
 - Trees built iteratively relative to the former ones;
 - Each new tree aims at doing better than the former one in the areas it was failing.
- [\(Hyper\)parameters:](#)
 - Learning rate (*classifier__learning_rate*) - degree of contribution of each new decision tree;
 - Loss (*classifier__loss*) - loss function to be optimized by the gradient descent algorithm;
 - n_estimators: number of boosting stages to perform (basically, number of trees);
 - Subsampling (*classifier__subsample*) - Fraction of the number of samples to be used in creating the decision trees;
 - Max depth - see Random Forest;
 - Max features - see Random Forest;
 - min samples leaf - see Random Forest.
- **Advantages:**
 - Handle heterogeneous data very well (features measured on different scale);
 - Support different loss functions;
 - Automatically detects (non-linear) feature interactions.
- **Disadvantages:**
 - Longer training (since it's iterative);
 - Requires careful tuning;
 - Prone to overfitting (however there are strategies to avoid it: good tuning of parameters and a big number of boosting stages);
 - Cannot be used to extrapolate.

General conclusion

Overview of algorithms:

This table summarizes everything that has been said on the binary classifiers.

	Logistic Regression	Random Forest	Gradient Boosting
Output	PDs	PDs	PDs
Accuracy	Low	High	Very High
Base Classifier	Log. Reg.	Decision Trees	Decision Trees
Data assumptions	Linear Structure	No assumptions	No assumptions
Main advantage	Traditionally considered more intuitive	High performance with reduced overfitting	High performance with reduced overfitting
Main disadvantage	Low accuracy	Cannot be described as an equation	Cannot be described as an equation, slow to train

Final words

In this paper, we presented the results of the main Machine Learning algorithms applied to Credit Risk estimation. With gains in Gini of **up to 27%**, **Random Forest** and especially **Gradient Boosting** perform a lot better than **Logistic Regression**.

The aim of this paper was also **to show our customers that the models we use are not black-boxes** and can be interpreted just as a score-card or a logistic regression can be.

Finally, not only are the results better with ML algorithms but most importantly, **James allows you to do in a few hours what is usually done in several weeks.**

This is why our product is **such a valuable tool for financial institutions who wish to get the most out of the growing and promising AI sector.**

We hope you enjoyed learning more about Machine Learning and Credit Risk: if you wish to know more about us, **feel free to contact us at james@james.finance.**

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Machine Learning approach for Credit Scoring

A. R. Provenzano*, D. Trifirò, A. Datteo, L. Giada, N. Jean, A. Riciputi,
G. Le Pera, M. Spadaccino, L. Massaron and C. Nordio



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Working paper¹

Abstract

In this work we build a stack of machine learning models aimed at composing a state-of-the-art credit rating and default prediction system, obtaining excellent out-of-sample performances. Our approach is an excursion through the most recent ML / AI concepts, starting from natural language processes (NLP) applied to economic sectors' (textual) descriptions using embedding and autoencoders (AE), going through the classification of defaultable firms on the base of a wide range of economic features using gradient boosting machines (GBM) and calibrating their probabilities paying due attention to the treatment of unbalanced samples. Finally we assign credit ratings through genetic algorithms (differential evolution, DE). Model interpretability is achieved by implementing recent techniques such as SHAP and LIME, which explain predictions locally in features' space.

JEL Classification codes: C45, C55, G24, G32, G33 **AMS** Classification codes: 62M45, 68T01, 68T50, 91G40

Keywords: Artificial Intelligence, Machine Learning, Explainable AI, Autoencoders, Embedding, LightGBM, Differential Evolution, SHAP, LIME, Credit Risk, Rating Model, Default, Probability of Default, Classification

Introduction

In the aftermath of the economic crisis, the probability of default (PD) has become a topical theme in the field of financial research. Indeed, given its usage in the risk management, in the valuation of the credit derivatives, in the estimation of the creditworthiness of a borrower and in the calculation of economic or regulatory capital for banking institution (under Basel II), incorrect PD prediction can lead to false valuation of risk, unreasonable rating and incorrect pricing of financial instruments. In the last decades, a growing number of approaches has been

*corresponding author: angela.provenzano@illimity.com

¹This paper reflects the authors' opinions and not necessarily those of their employers.

developed to model the credit quality of a company, by exploring statistical techniques. Several works have employed probit models[1] or linear and logistic regression to estimate company ratings using the main financial indicators as model input. However, these models suffer from their clear inability to capture non-linear dynamics, which are prevalent in financial ratio data[2]. New statistical techniques, especially from the field of machine learning, have gained a worldwide reputation thanks to their ability to efficiently capture information from big dataset by recognizing non-linear patterns and temporal dependencies among data. Zhao et al. (2015) [3] employed feed forward neural networks in credit corporate rating determination. Petropoulos et al.[4] explore two state of the art techniques namely Extreme Gradient Boosting (XGBoost) and deep learning neural networks in order to estimate loan PD and calibrate an internal rating system, useful both for internal usage and regulatory scope. Addo et al. (2018)[5] built binary classifiers based on machine and deep learning models on real data to predict loan probability of default. They observed that the tree-based models are more stable than ones based on multilayer artificial neural networks.

Starting from these studies, we propose a sophisticated framework of machine learning models which, on the basis of company annual (end-of-year) financial statements coupled with relevant macroeconomic indicators, attempts to classify the status of a company (performing - “in-bonis” - or defaulted) and to build a robust rating system in which each rating class will be matched to an internally calibrated default probability. In this regard, here the target variable is different from a previous work by some of the authors [6], where the goal was to predict the credit rating that Moody’s would assign, according to an approach commonly called “shadow rating”. The novelty of our approach lies in the combination of data preprocessing algorithms, responsible for feature engineering and feature selection, and a core model architecture made of a concatenation of a Boosted Tree default classifier, a probability calibrator and a rating attribution system based on genetic algorithm. Great attention is then given to model interpretability, as we propose two intuitive approaches to interpret the model output by exploring the property of local explainability. In details, the article is composed of the following sections: Section 1 is devoted to describe the input dataset and the preprocessing phase; Section 2 in which the core model architecture is explained; Section 3 which collects results from the core model structure (i.e. default classifier, PD calibrator and rating clustering); finally Section 4 is left to model explainability.

1 Dataset description

Data used for model training have been collected from the Credit Research Database (CRD) provided by Moody’s, and consist of 919,636 annual (end of year) financial statements of 157,986 Italian companies belonging to different sectors (e.g. automotive, construction, consumer goods durable and not-durable, energy, high-tech industries, media, etc. to the exclusion of FIRE sector, i.e. finance, insurance, and real estate sectors). The dependent variable in our dataset, i.e. the *target* of the proposed default prediction model, is a binary indicator with the value of 1 flagging a default event (i.e. a bankruptcy occurrence over a one-year horizon), 0 otherwise. In accordance to the above-defined target variable, input variables of our model have been selected to be consistent with factors that can affect the companies capacity to service external debt (a full explanation of the input model’s features is reported in Appendix A). In particular they consist of balance-sheet indexes and ratios, and *Key Performance Indicators* (KPI) calculated from CRDs financial reports[7]. The latter include indicators for efficiency (i.e. measures of operating performance), liquidity (i.e. ratios used to determine how quickly a company can turn its assets into cash if it is experiencing financial distress or impending bankruptcy),

solvency (i.e. ratios that depict how much a company relies upon its debt to fund operations) and profitability (i.e. measures that demonstrate how profitable a company is). Since business cycles can have great impact on a firm profitability and influence its risk profile, we joined original information with more general macro variables (2 years lagged historical data) addressing the surrounding climate in which companies operate. Among the wide range of macroeconomic indicators provided by Oxford Economics [8], a subset of the most influential ones has been selected as explanatory variables². Some of them are country-specific, others are common to the whole Eurozone³. The combined dataset of balance-sheet indexes, financial ratios and macro variables along with data transformations and feature selection (better described hereafter in Section 1.1), led to a set of 179 features and covers the period 2011 – 2017.

As fully described hereafter, the obtained dataset was split into three parts: an out-of-time dataset which includes the data referred to year 2017 (marked in light-blue in Figure 1), used to test model performance; a stratified pair of train/test dataset (constituting each the 80% and the 20% of the input dataset) which covers the period 2011 – 2016, employed for model development and calibration.

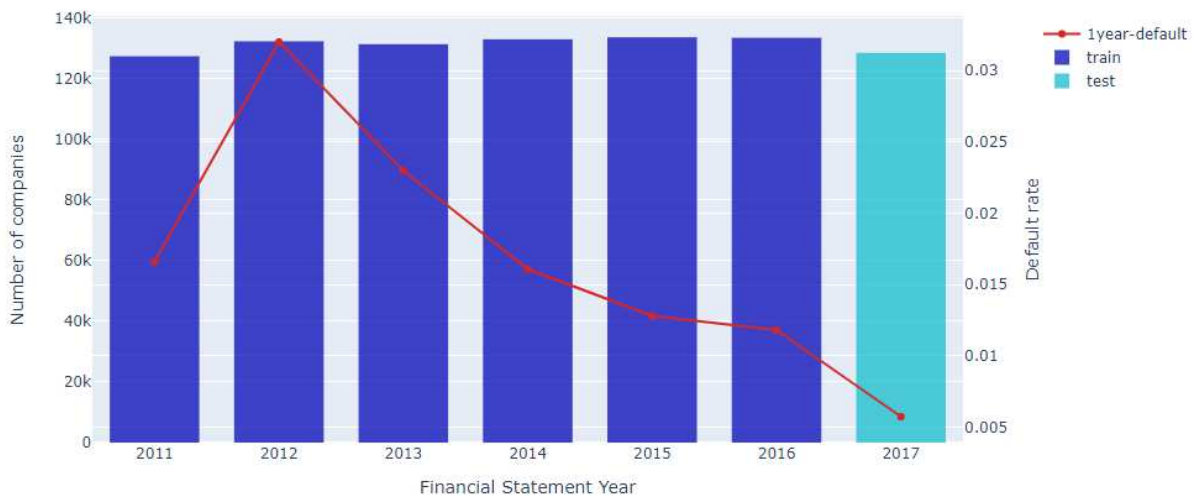


Figure 1: Number of balance-sheets per financial statement year and the corresponding 1-year default rate. The default rate shows an increasing trend in the 2011 – 2012 period, and then decreases till 2017.

1.1 Feature engineering and feature selection

A preliminary step for building a machine learning model consists in generating a set of features suitable for model training. This task involves data manipulation processes like transformation of categorical features, missing values treatment, infinite values handling, outliers detection, data leakage avoidance. In particular, categorical, non-ordinal variables are one of the main issue that must be tackled in order to feed any machine learning model[9].

Different *encoding* technique can be used to make categorical data legible for a machine learning algorithm. The most common way to deal with categories is to simply map each category with

²The list of selected indicators are reported in Section A.3

³Regional aggregate *Eurozone* includes the following countries: Austria, Belgium, Cyprus, Estonia, Finland, France, Germany, Greece, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Portugal, Slovenia, Slovakia and Spain

a number. By applying such transformation, known as *Label Encoding*, a model would treat categories as ordered integers, which would imply non-existent ordinal relationships between data, that could be misleading for model training. Another simple way to handle categorical data is the *One-Hot Encoding* technique, which consists in transforming each categorical feature into fixed-size sparse vector of all zeros but 1 in the cell used uniquely to identify specific realization of that variable. The main drawback of this technique relies on the fact that categories with an high number of possible realizations would generate large dimension datasets, which makes it a memory-inefficient encoder. Moreover this sparse representation does not preserve similarity between feature values. An alternative approach to overcome these issues is represented by *Categorical Embedding*, which consists in mapping via Deep Neural Network (DNN) each possible discrete value of a given categorical variable in a low-dimensional, learned, continuous vector representation. This method allows to place each categorical feature in a Euclidean space, keeping coherent relationship with other realization of the same variable. The extension of categorical embedding approach to words and document representation is known as *Word Embedding*[10]. In particular, *Sentence Embedding* is an application of word embedding aiming at representing a full sentence into a vector space. In this study, we applied sentence embedding to represent industry sectors descriptions associated to each "NACE code"⁴. In order to guarantee the "*semantics*" of the original data and work in a low dimensional space, we propose a framework of embedding with autoencoder regularization, in which the original data are embedded into low dimension vectors. The obtained embeddings maintain local similarity and can be easily reverted to their original forms. The encoding of the NACE is a novel way to overcome the NACE1-NACE2 mapping conundrum. In our dataset both NACE are used and as already stated in many papers the two encoding systems are not fully compatible[11]. Moreover the NACE encoding allows for proper industry segment description of multi sector firms that cannot be easily described by a single NACE code, further extending the predictive power of the economic sector category.

A different encoding method is the *Target encoding*, in which categorical features are replaced with the mean target value for samples having that category. This allows to encode an arbitrary number of features without increasing data dimensionality. However, as a drawback, a naive application of this type of encoding can allow data leakage, leading to model overfitting and poor predictive performance. A target encoding algorithm developed for preventing data leakage is known as *James-Stein estimator*, and is the one used in our model. In more details, it transforms each categorical feature with a weighted average of the mean target value for the observed feature value and the mean target value computed regardless of the feature realization.

As described above, some feature transformations could result in a general increase of input data dimensionality, which makes it urgent to implement a robust and independent feature selection framework. In fact, training a machine learning model to a huge number of independent variables is doomed to suffer from the so-called *curse of dimensionality*[12], i.e. the problem of exponential increase in volume associated with adding extra dimensions to a vector space. We employed a voting ensemble of models to independently assign importance to the available features and efficiently select those features which will contribute most to model prediction.

Hereafter in this section we will look into the implementation of satellite models aiming at: performing sentence embedding of the industry sectors description; reducing embeddings dimensionality via stacked autoencoder; selecting relevant features via voting approach.

⁴The "*Statistical Classification of Economic Activities in the European Community*", commonly referred to as NACE, is the industry standard classification system used in the European Union

Sentence Embedding of sector descriptions A common practice in Natural Language Processing (NLP) is the use of pre-trained embeddings to represent words or sentence in a document. Following this common practice, we use the pre-trained models built in SpaCy NLP library for embedding the sequence of NACE sector textual description. In particular, we performed sentence embedding i.e. we transform each description into a 300-dimensional real-value vector. Each sentence embedding is automatically constructed by SpaCy averaging the 300-dimensional real-value pre-trained vectors which map each word in that sentence.

Here’s a glimpse at how spaCy processes textual data. It first segments text into words, punctuations, symbols and others by applying specific rules to each language (i.e. it tokenizes the text). Then it performs Part-of-speech (POS) tagging to understand the grammatical properties of each word by means of built in statistical model. A model consists of binary data trained on a dataset large enough to allow the system to make predictions that generalize across the language. A key assumption to the word embedding approach is the idea of using for each word a dense distributed representation learned based on the usage of words [13]. This allows words that are used in similar ways to have similar representations, naturally capturing their meaning[14]. Given the high importance industry sector has in financial literature as default prediction driver, embedding NACE industry descriptions improves the overall model performance in application by helping the model to generalize better and to smoothly handle unseen elements.

Dimensionality reduction via stacked autoencoder The aforementioned word embedding models are powerful way to represent categorical variables which preserve relationship between data, but at the cost of increase in dimensionality. In order to reduce the number of dimensions of the output embeddings from 300 to 5, a *stacked autoencoder* (SAE) of 6 layers⁵ has been developed via tensorflow[15].

In details, *autoencoders* (AE) are a family of neural networks in which input and output coincide. They work by compressing the input into a latent-space representation and then reconstruct the output by means of this representation. They consist of two principal component: the *encoder* which takes the input and compresses it into a representation with less dimensions, and the *decoder* which tries to reconstruct the input. Among AEs, stacked autoencoders are deep neural networks in which the output of each hidden layer is connected to the input of the successive hidden layer. All hidden layers are trained by an unsupervised algorithm and then fine-tuned by a supervised method aimed at minimize the cost function. Since they can learn even non-linear transformations, unlike PCA, by using a non-linear activation function and a multiple layer structure, autoencoders are efficient tools for dimensionality reduction. Moreover, in our application, SAE exhibited a low reconstruction loss⁶ (around 6% of MSE), contrary to a low fraction of variance explained by the PCA.

Voting approach for feature selection Feature selection is a key component when building machine learning models. We can either demand this task to the main model or use a set of lighter models in a preparatory task so that the required effort for further feature selections will be reduced when training the main model. This is particularly useful for multi parameters models like Light-GBM where the training phase involves also the calibration of a set of hyperparameters usually spanning very wide ranges. Neglecting the expert based component, algorithmic feature selection methods are usually divided into three classes: *filter* methods, *wrapper* methods and

⁵a 3-layer encoder and 3-layer decoder

⁶The *reconstruction loss* is the loss function (usually either the mean-squared error or cross-entropy between the reconstructed output and the input) which penalizes the network for creating outputs different from the original input

embedded methods. Filter-based methods apply a statistical measure to assign a score to each feature; variables of the starting dataset are then ranked according to their scores and either selected to be kept or removed. Wrapper-based methods consider the selection of a set of features as a "search problem", where different combinations are prepared, evaluated and compared to other combinations. In details, a predictive model is used to evaluate a combination of features and assign a score based on model accuracy. Embedded-based methods learn which features best contribute to the accuracy of the model while the model is being created.

We combined a set of 6 different models for feature selection, stacking each algorithm into a hard voting framework where features which receive the highest number of votes among all the models have been selected. In particular, after having transformed categorical features via target encoding (by means of James-Stein encoder), each feature in the dataset has been ranked on the basis of the following models:

- **Pearson criterion.** It is a *filter-based* method which consists in checking the absolute value of the Pearson correlation between the target and features in the input dataset and keeping the top n features based on this score.
- **Chi-squared criterion.** It is another *filter-based* method in which we calculate the chi-squared metric between each feature and the target and select the desired number of features which exhibit the best chi-squared scores. The underlying intuition is that if a feature is independent to the target it is uninformative for classifying information.
- **Recursive Feature Elimination (RFE).** This is a *wrapper-based* method whose goal is to select features by recursively considering smaller and smaller sets of features. First, the estimator is trained on the initial set of features and the importance of each feature is computed. In our specific case the estimator used is a Logistic Regression. Then, the least important features are pruned from current set of features. That procedure is recursively repeated on the pruned set until the desired number of features to select is eventually reached.
- **Random Forest Classifier (RF⁷).** This is a *wrapper-based* method that uses a built-in algorithm for feature selection. In particular, variables are selected accordingly to feature importance, obtained by averaging of all decision tree feature importance.
- **Logistic Lasso Regression** It is an *embedded-based* method which uses the built-in feature selection algorithm embedded into the Logistic Regression with L1 regularization model.
- **Light-GBM[16] (LGBM⁸).** It is a *wrapper-based* method analogous to the above-mentioned RF classifier.

2 Model architecture

Moving beyond the satellite models described in Section 1.1, and used in the preprocessor phase, in this section we will present the core model architecture. It consists of a concatenation of three

⁷RF is an *ensemble* of Decision Trees generally trained via *bagging method*: this approach consists in using the same training algorithm for every predictor, but training them on different random subsets of the Train set. Once all predictors are trained, the ensemble can make a prediction for a new instance by simply aggregating the predictions of all predictors

⁸LGBM is a fast, high-performance gradient boosting framework based on decision tree algorithm

machine learning models aiming at building a robust and reliable framework whose purpose is not only to classify the company status (as in bonis or defaulted), but also to construct an internally calibrated rating system in which each rating class will correspond to a self-consistent default probability.

In details, we developed a Boosted Tree algorithm in order to classify the status of a company over a one-year horizon. This is a classical binary classification problem whose target variable is 1 in case of default event or 0 otherwise. Hyper-parameters tuning has been performed via an extension of cross validation for time-series that will be further described hereafter.

The purpose of the second model is to fit the output score of the binary classifier to the actual default rate. We built a calibrator which consists of a Logistic Regression trained on the leaf assignments of the Boosted Tree.

Finally, we calibrated our own rating system by splitting the refitted default probability into 9 clusters via genetic algorithm.

2.1 Light-GBM for default classification

In order to leverage the availability of a large scale dataset, enriched with a high number of features, we developed a robust machine learning approach based on *Gradient Boosting* decision trees known as *Light-GBM*.

Gradient Boosting trees model[17] is one method of combining a group of "weak learners" (specifically decision trees) in order to form a "strong predictor" model, by reducing both variance and bias. Differently from other tree methods like Random Forest, Boosted trees work by sequentially adding predictors to an ensemble, each one corrects its predecessor by trying to fit the new predictor to the residuals of the previous one. These residuals are the gradient of the loss functional being minimized, with respect to the model values at each training data point evaluated at the current step. Specifically, at each iteration a sub-sample of the training data is drawn at random (without replacement) from the full training data-set. This randomly selected sub-sample is then used in place of the full sample to fit the "weak learner" and compute the model update for the current iteration.

In particular *Light GBM* (LGBM) is a fast, high-performance gradient boosting framework based on decision tree algorithm, which has proved to be highly effective in classification and regression models when applied on tabular, structured data, such as the ones we are dealing with. The model hyper-parameters have been tuned via *out-of-time cross-validation* procedure based on a custom extension of the F_β -measure where the balance of specificity⁹ (also called "the true negative rate") and recall¹⁰ (also called "the true positive rate") in the calculation of the harmonic mean is controlled by a coefficient called β as follows:

$$F_\beta = (1 + \beta^2) \frac{\text{specificity} \cdot \text{recall}}{\beta^2 \cdot \text{specificity} + \text{recall}} \quad (1)$$

In this procedure, each test sets consist of a single year of future observation, while the corresponding training sets are made up of the observations that occurred prior to the observation that forms the test set. In this way, the model is optimized in predicting what will happen in the future using only information available up to the the present day. The objective function used for the classification problem was the *log-loss*, which measures the distance between each predicted probability and the actual class output value by means of a logarithmic penalty. Due

⁹The specificity is defined as the number of true negatives over the number of true negatives plus the number of false negatives

¹⁰The recall is defined as the number of true positives over the number of true positives plus the number of false negatives

to the high unbalance between 0 and 1 target flags, we have used the modified unbalanced log-loss, by setting the *scale_pos_weight* parameter of the Light-GBM equal to the ratio between the number of 0s and the number of 1s. Other objective functions we have used like Focal Loss[18] and custom weighted log losses have not given any specific advantage compared to the unbalanced log loss.

2.2 Probability calibration for tree-base model

A natural extension of the issue of classification of corporate default forecast consists in predicting the probability of default. Complex non-linear machine learning algorithms can provide poor estimates of the class probabilities, especially in case the target variable is highly unbalanced, so that the distribution and behaviour of the probabilities may not be reflective of the true underlying probability of the sample. The unbalanced log loss objective chosen for the classification task creates a custom metric in the default probability space that is reflected in distorted class probabilities. The perfect classifier would have only 0 and 1 probabilities, but these would not be able to match historical default rates. They simply represent the probability of belonging to a class with the switch threshold at 0.5, they are not predicted default rates. Fortunately, it is possible to adjust the probability distribution in order to better match the actual distribution observed in the data, without losing predictive power. This adjustment is referred to as *calibration*. In particular, calibrating a classifier consists in fitting a regressor (known as *calibrator*) which maps the output of the classifier f_i to a calibrated probability $p(y_i = 1|f_i)$ in $[0, 1]$.

Taking inspiration from the hybrid model structure proposed by Xinran He et al [19], based on the concatenation of boosted decision trees and of a probabilistic sparse classifier, we calibrated the output probabilities of the LGBM classifier by fitting a L2-Regularized Logistic Regression (LR) (where the inverse of regularization strength, i.e. the c parameter, has been optimized on the out-of-time sample of the training-set) on its one-hot encoded leaf assignments.

We first fitted the LGBM on the stratified test-set we left aside for the classification task, as the sample used to train the calibrator should not be used to train the target classifier. We treated the output of each individual tree of the LGBM classifier as a categorical feature that takes as value the index of the leaf an instance ends up falling in. We applied one-hot encoding to have dummies indicating leaf assignments on which fitting the LR model. Finally we tested the calibrator on the train-set previously used for LGBM training phase. This methodology of using an intermediate result and change the output from classification to regression is analogous to what is currently known as Transfer Learning in the Deep Neural Network world, where the final neural net layer is removed and substituted with a novel output. The main advantage of this method is in preserving all the inner complex feature engineering that the system learned in the original training task and transferring it to a different problem, in our specific case the prediction of actual default rates.

2.3 Rating attribution via genetic algorithm

A robust default classification system, able to meet both supervisory requirements and internal banking usage, provides a way to represent the internally calibrated probability of default to a rating system, in which each PD bucket is matched to a rating grade. In order to calibrate our own rating system, the refitted default probability has been split into 9 groups (corresponding to 9 different rating classes) by means of a genetic algorithm known *Differential Evolution*[20]. The algorithmic task of calibrating a rating system can be stated as an optimization problem, as it tries to minimize the Brier Score, maximise the similarities among elements of the same groups (the so called *cohesion* i.e. the items in a cluster should be as similar as possible), minimise the

dissimilarities between different groups (the so called *separation* i.e. any two clusters should be as distinct as possible in terms of similarity of items), ensure both PD monotonicity (i.e. lower default rates have to correspond to low rating grades and vice versa) and have an acceptable cluster size (i.e. each cluster has to include a fraction of the total population that has to be roughly homogeneous among them). Among the partitioning clustering algorithms, Genetic Algorithms (GA) are stochastic search heuristic inspired by the concepts of Darwinian evolution and genetics. They are based on the idea of creating a population of candidate solutions to an optimization problem, which is iteratively refined by alteration (mutation) and selection of good solutions for the next iteration. Candidate solutions are selected according to a so-called fitness function, which evaluates their quality in respect to the optimization problem. In the case of Differential Evolution (DE) algorithms the candidate solutions are linear combinations of existing solutions. In the end, the best individual of the population is returned. This individual represents the best solution discovered by the algorithm.

3 Results

The metric used to evaluate the model performance is the AUROC or AUC-ROC score (*Area Under the Receiver Operating Characteristics*). In particular, ROC is a probability curve and AUC represents the degree of separability. This measure tells how much a model is capable of distinguishing between classes: for an excellent model it is near to 1, for a poor model it near to 0. The ROC curve is constructed by evaluating the fraction of "true positives" (*tpr* or *True Positive Rate*) and "false positives" (*fpr* or *False Positive Rate*) for different threshold values. In details, *tpr*, also known as *Recall* or *Sensitivity*, is defined in Equation (2) as the number of items correctly identified as positive out of total true positives:

$$tpr = \frac{TP}{TP + FN} \quad (2)$$

where TP is the number of true positives and FN is the number of false negatives. The *fpr*, also known as *Type I Error*, is defined in Equation (3) as the number of items wrongly identified as positive out of total true negatives:

$$fpr = \frac{FP}{FP + TN} \quad (3)$$

where FP is the number of false positive and TN is the number of true negatives.

Prediction results are then summarized into a *confusion matrix* which counts the number of correct and incorrect predictions made by the classifier. A threshold is applied to the cut-off point in probability between the positive and negative classes, which for the default classifier has been set at 0.5. However, a trade-off exists between *tpr* and *fpr*, such that changing the threshold of classification will change the balance of predictions towards improving the *True Positive Rate* at the expense of *False Positive Rate*, or vice versa.

The metric used to evaluate the performance of internally calibrated PD prediction is the *Brier score* (BS), i.e. a way to verify the accuracy of a probability forecast in term of distance between the actual results. The most common formulation of the *Brier score* is *mean squared error*:

$$BS = \frac{1}{N} \sum_{t=1}^N (f_t - o_t)^2 \quad (4)$$

in which f_t is the forecast probability, o_t the actual outcome of the event at instance t and N is the number of forecasting instances. The best possible Brier score is 0, for total accuracy, the lowest possible score is 1, which mean the forecast was wholly inaccurate.

Note that, all the metrics described so far have been calculated on the *Test-set* obtained by splitting the dataset along the financial statement year¹¹.

3.1 Default Classification

We obtained an high performance corresponding to an AUROC = 95.0% (see Figure 2) and summarized in the normalized confusion matrix of Figure 3. In highly unbalanced datasets, usually the confusion matrix is skewed towards predicting well only the majority class, producing unsatisfactory performances in the minority class, even if the event of misclassification for the minority class is the event business usually try to minimize the most. The distortion in the default probability space and an accurate choice of feature selection and hyperparameters created a system able to effectively discriminate events in the minority class, reducing the occurrence of false negatives.

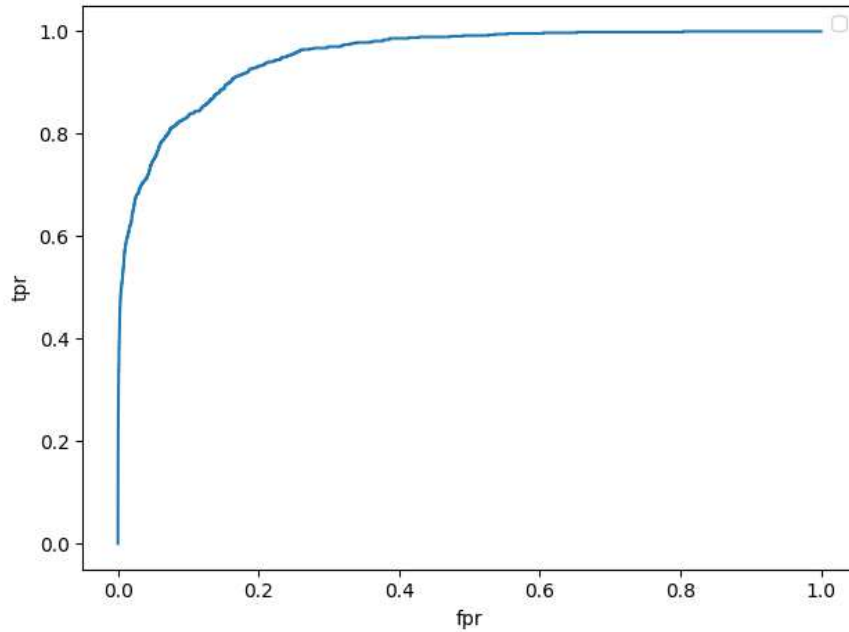


Figure 2: ROC curve for Light-GBM classifier

¹¹Train-set spans from 2011 to 2016, Test-set covers 2017

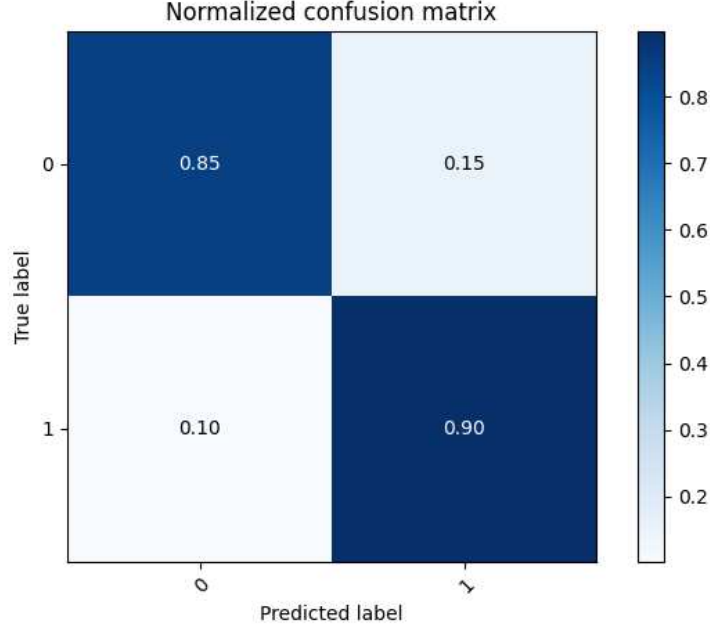


Figure 3: Normalized confusion matrix with 50% threshold for Light-GBM classifier

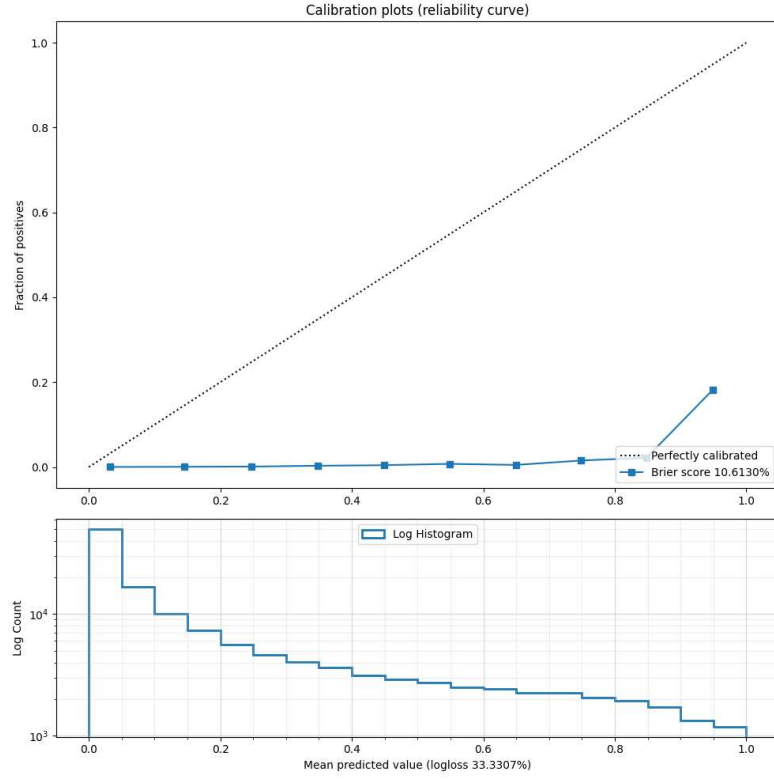
3.2 PD refitting

Default probabilities forecast before and after refitting procedure are summarized in the *calibration plots* (also called *reliability curves*) of Figure 4, which allow checking if the predicted probabilities produced by the model are well calibrated. Specifically, a calibration plot consists of a line plot of the relative observed frequency (y-axis) versus the predicted probabilities (x-axis)¹². A perfect classifier would produce only 0 and 1 predictions but would not be able to forecast actual default rates. A perfect actual default rate model would produce reliability diagrams as close as possible to the main diagonal from the bottom left to the top right of the plot. The refitting procedure maps the perfect classifier to a reliable default rate predictor.

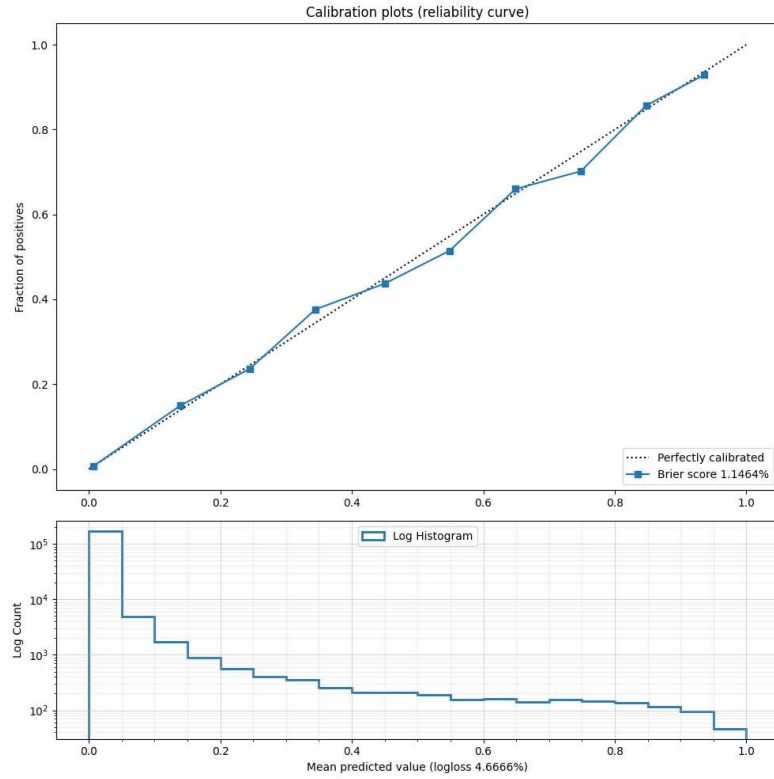
The refitting procedure left AUROC score of the model unchanged (AUROC = 95.0%); the calibration performance is evaluated with the Brier score ($BS = 1.2\%$ ¹³); classification results are summarized in the normalized confusion matrix reported in Figure 6, where the threshold to the cut-off point between the positive and negative classes has been optimized on the Figure 5.

¹²In details, the predicted probabilities are divided up into a fixed number of buckets along the x-axis. The number of target events (i.e. the occurrence of 1-year default) are then counted for each bin (i.e. the relative observed frequency). Finally, the counts are normalized and the results are plotted as a line plot

¹³The closer the Brier score is to zero the better is the forecast of default probabilities



(a)



(b)

Figure 4: Calibration plots and log-scaled histograms of forecast probability before (Figure 8a) and after (Figure 8b) refitting. Accuracy of predicted probabilities is expressed in term of *log loss* measure

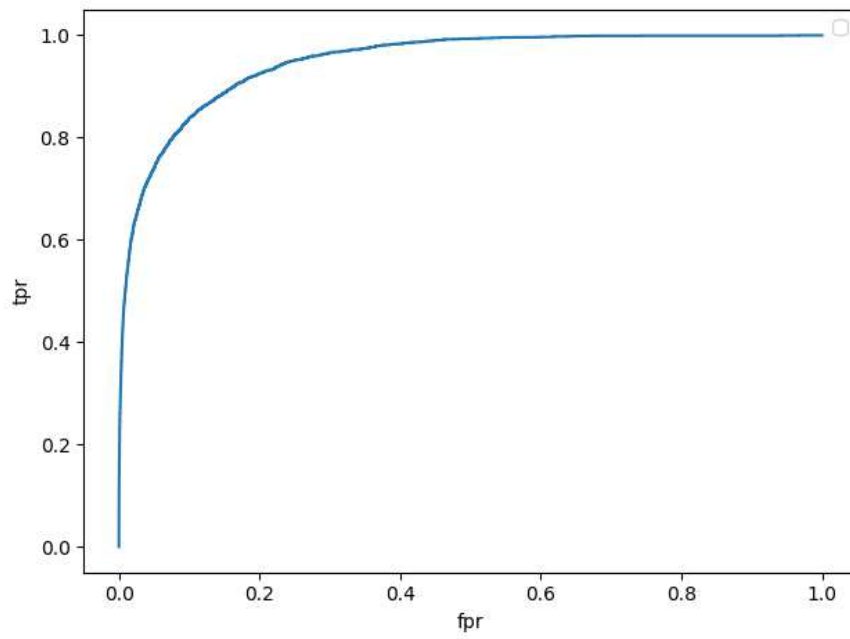


Figure 5: ROC curve for the calibrated classifier

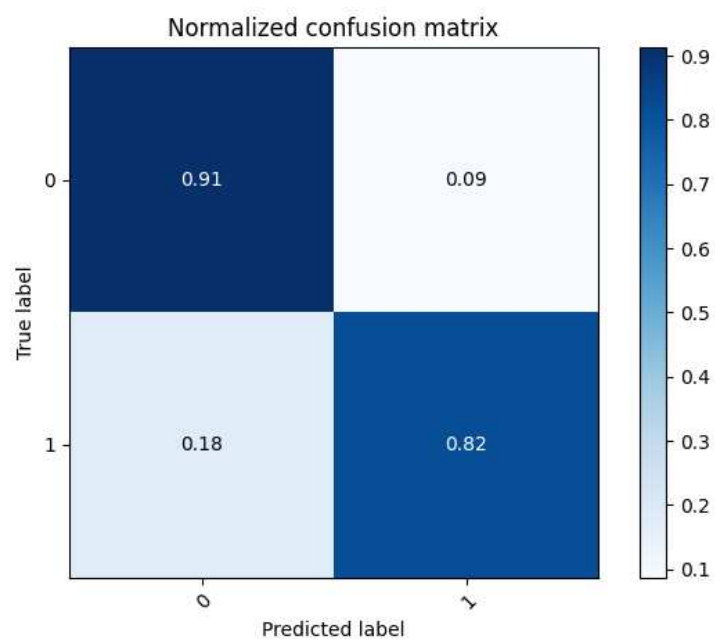


Figure 6: Normalized confusion matrix with optimized threshold for the calibrated classifier

3.3 PD clustering

Among the several common statistical tests that can be performed to validate the assignment of a probability of default to a certain rating grade, two approaches have been used: the *one-sided Binomial test* and the *Extended Traffic-Light Approach*.

The *Binomial Test* is one of the most popular single-grade single-period¹⁴ test performed for rating system validation. For a certain rating grade $k \in \{1, \dots, K\}$, being K the number of rating classes¹⁵, we made the assumptions that default events are independent within the grade k and could be modelled as a binomially distributed random variable X with size parameter N_k and "success" probability PD_k . Thus, we can assess the correctness of the PD forecast by testing the null hypothesis H_0 , where:

- H_0 : the actual default rate is less than or equal to the forecast default rate given by the PD;

The null hypothesis H_0 is rejected at a confidence level α in case the number of observed defaults d per rating grade is greater than or equal to the critical value, as reported in Equation 5:

$$d_\alpha = \min \left\{ d : \sum_{j=d}^{N_k} \binom{N_k}{j} PD_k^j (1 - PD_k)^{N_k-j} \leq 1 - \alpha \right\} \quad (5)$$

The *Extended Traffic Light Approach* is a novel technique for default probability validation, first adopted by Tasche (2003)[21]. The implementation used in this section refers to a heuristic approach proposed by Blochwitz et al. (2005) [22] which is based on the estimation of a relative distance between observed default rates and forecast probabilities of default, under the key assumption of binomially distributed default events. Four coloured zones, *Green*, *Yellow*, *Orange*, *Red*, are established to analyse the deviation of forecasts and actual realizations. In details: if the result of the validation assessment lies in the *Green* zone there is no obvious contradiction between forecast and realized default rate; the *Yellow* and *Orange* lights indicate that the realized default rate is not compatible with the PD forecast, however, the difference of realized rate and forecast is still in the range of usual statistical fluctuations; and last *red* traffic light indicates a wrong forecast of default probability. The boundaries between the afore mentioned light-zones are summarized in Equation 6:

$$\begin{cases} \text{Green} & p_k < PD_k \\ \text{Yellow} & PD_k \leq p_k < PD_k + K^y \sigma(PD_k, N_k) \\ \text{Orange} & PD_k + K^y \sigma(PD_k, N_k) \leq p_k < PD_k + K^0 \sigma(PD_k, N_k) \\ \text{Red} & PD_k + K^0 \sigma(PD_k, N_k) \leq p_k \end{cases} \quad (6)$$

where $\sigma(PD_k, N_k) = \sqrt{PD_k(1 - PD_k)/N_k}$. The parameters K^y and K^0 play a major role in the validation assessment, so have to be tuned carefully. A proper choice based on practical considerations is setting $K^y = 0.84$ and $K^0 = 1.44$, which corresponds to a probability of observing green of 0.5, observing yellow with 0.3, orange with 0.15 and red with 0.05.

The results of the application of the above mentioned statistical test are summarized in Table 1.

¹⁴usually one year

¹⁵9 in our case

Rating class	PD Bins (%)	Rating Class PD (%)	Out-of-sample Default Rate (%)	One-sided Binomial Test	Extended Traffic Light Approach
AAA	[0.00, 0.005)	0.03	0.00	Passed	Green
AA	[0.05, 0.42)	0.24	0.03	Passed	Green
A	[0.42, 0.55)	0.48	0.08	Passed	Green
BBB	[0.55, 0.74)	0.64	0.21	Passed	Green
BB	[0.74, 1.00)	0.87	0.40	Passed	Green
B	[1.00, 1.42)	1.21	0.83	Passed	Green
CCC	[1.42, 2.12)	1.77	1.29	Passed	Green
CC	[2.12, 9.03)	5.57	5.06	Passed	Green
C	[9.03,100)	54.52	33.77	Passed	Green

Table 1: Internally calibrated PD clustering into 9 rating classes. Despite being borrowed from S&P rating scales, the labels are assigned to a PD calibrated on an internal dataset (the one used during the training-phase) and does not correspond to any rating agencies PD

4 Model explainability

Machine learning models which operate in higher dimensions than cannot be directly visualized by human mind are often referred as "*black boxes*", in the sense that high model performance is often achieved on the detriment of output explainability, leading users not to understand the logic behind model predictions. Even greater attention to model interpretability has led to the development of several methods to provide an explanation to machine learning outputs, both in term of global and local interpretability. In the first case, the goal is being able to explain and understand model decisions based on conditional interactions between the dependent variable (i.e. target) and the independent features on the entire dataset. In the latter case, the aim is to understand model output for a single prediction by looking at a local subregion in the feature space around that instance.

Two popular approaches described hereafter in this section are SHAP and LIME, which explore and leverage the property of *local explainability* to build surrogate models which are able to interpret the output of any machine learning models. The technique upon which these algorithms are based is slightly tweaking the input and modelling the changes in prediction by means of surrogate agnostic models. In particular, SHAP measures how much each feature in our model contributes, either positively or negatively, to each prediction, in term of difference between the actual prediction and its expected value. LIME builds sparse linear models around each prediction to explain how the black box model works in that local vicinity.

4.1 SHAP

SHAP, which stands for (*SH*apley Additive exPlanation) [23], is a novel approach for model explainability which exploits the idea of *Shapley regression value*¹⁶ to model feature influence scoring. SHAP values quantify the magnitude and direction (positive or negative) of a feature's effect on a prediction via an additive feature attribution method. In simple words, SHAP builds model explanations by asking, for each prediction i and feature j , how i changes when j is removed from the model. Since SHAP considers all possible predictions for an instance using all

¹⁶The technical definition of Shapley value is the average marginal contribution of a feature value over all possible coalitions

possible combinations of feature inputs, it can guarantee both consistency and local accuracy. More in details, SHAP method computes Shapley values from *coalitional game theory*. The feature values of a data instance act as players¹⁷ in a coalition: Shapley values suggest how to fairly distribute the *payout* (i.e. the prediction) among the features.

SHAP summary plot As reported in Figure 10, it combines feature importance with feature effects to measure the global impact of features on the model. For each feature shown in the y-axis, and ordered according to their importance, each point on the plot represents the Shapley value (reported along the x-axis) for a given prediction. The color of each point represents the impact of the feature on model output from low (i.e. blue) to high (i.e. red). Overlapping points are littered in y-axis direction, so we get a sense of the distribution of the Shapley values per feature.

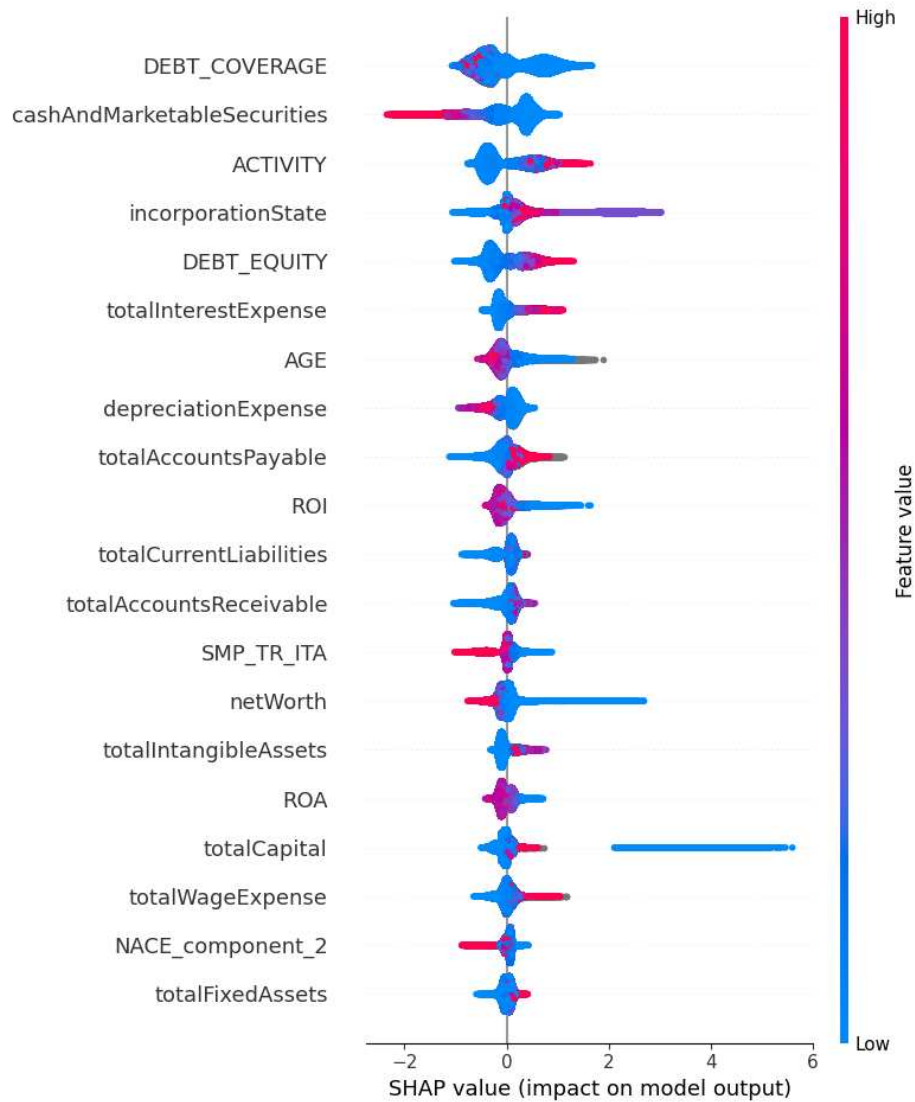


Figure 7: SHAP summary plot for Light-GBM classifier. The details of model's feature description are reported in Appendix A

¹⁷note that a player can be an individual feature value or a group of feature values

SHAP dependence plot It is a scatter plot that shows the effect a single feature has on the model predictions. In particular, each dot represents a single prediction where the feature value is on the x-axis and its SHAP value, representing how much knowing that feature's value changes the output of the model for that sample's prediction, on the y-axis. The color corresponds to a second feature that may have an interaction effect with the plotted feature. If an interaction effect is present it will show up as a distinct vertical pattern of colouring.

SHAP waterfall plot The waterfall plot reported in Figure 9 is designed to display how the SHAP values of each feature move the model output from our prior expectation under the background data distribution ($E[f(X)]$), to the final model prediction ($f(X)$) given the evidence of all the features. Features are sorted by the magnitude of their SHAP values with the smallest magnitude features grouped together at the bottom of the plot. The color of each row represents the impact of the feature on model output from low (i.e. blue) to high (i.e. red).

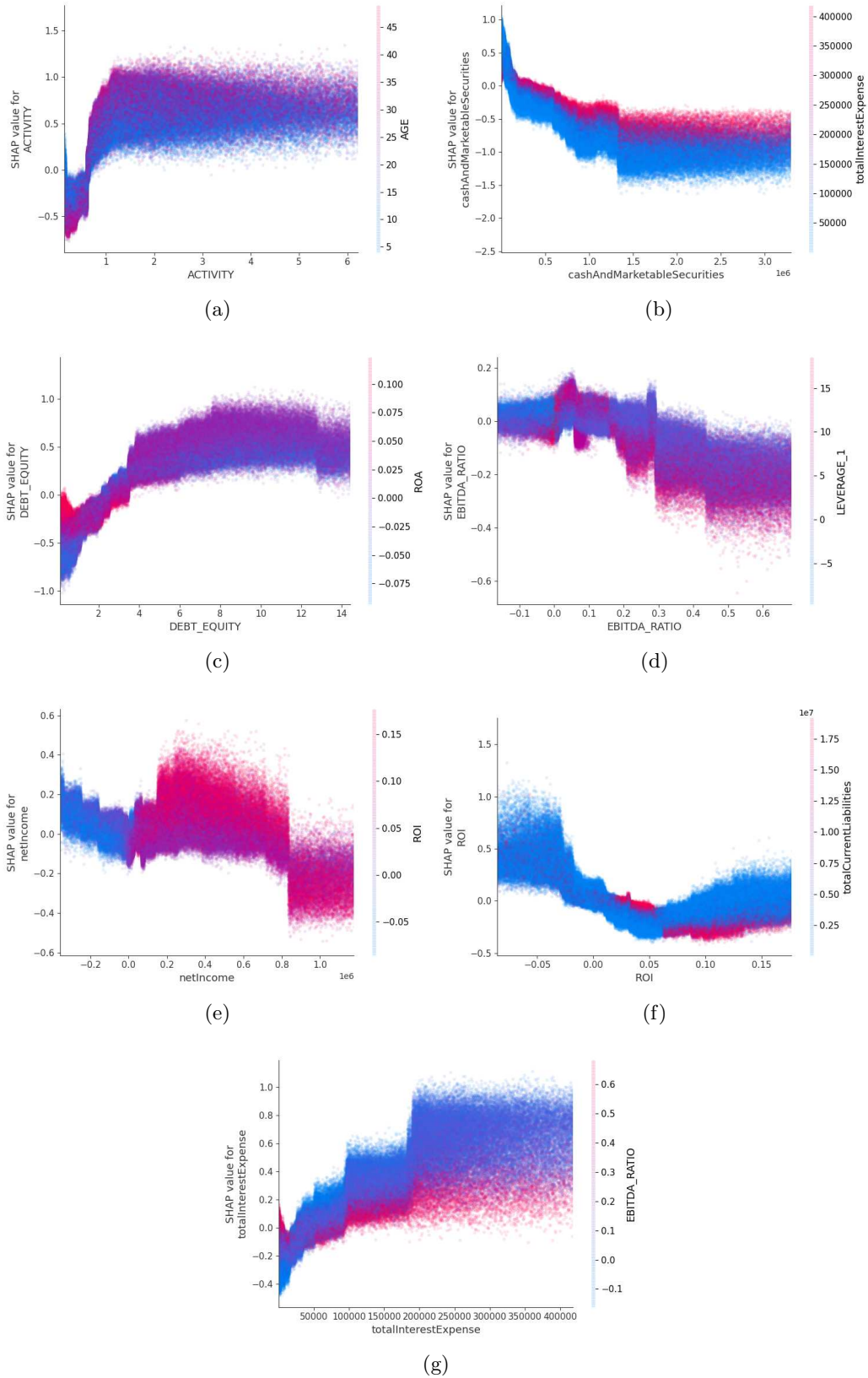


Figure 8: SHAP dependency plots for ACTIVITY in Figure 8a, cashAndMarketableSecurities in Figure 8b, DEBT_EQUITY in Figure 8c, EBITDA_RATIO in Figure 8d, netIncome in Figure 8e, ROI in Figure 8f, totalInterestExpense in Figure 8g. The details of model's feature description are reported in Appendix A

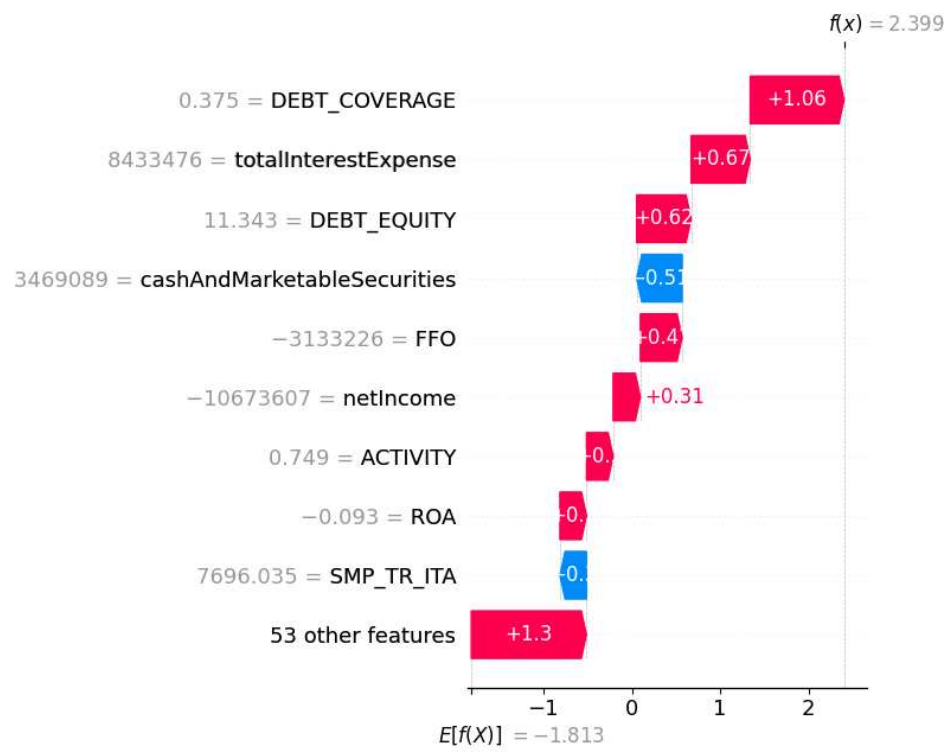


Figure 9: SHAP waterfall plot The details of model's feature description are reported in Appendix A

4.2 Lime

LIME, *Local Interpretable Model-agnostic Explanations*, is a novel technique that explains the predictions of any classifier in an interpretable and faithful manner, by learning an interpretable model locally around the prediction[24]. Behind the workings of LIME lies the assumption that every complex model is linear on a local scale, so it is possible to fit a simple model around a single observation that will mimic how the global model behaves at that locality. The output of LIME is a list of explanations, reflecting the contribution of each feature to the prediction of a data sample, allowing to determine which feature changes will have most impact on the prediction. Note that, LIME has the desirable property of additivity, i.e. the sum of the individual impact is equal to the total impact. Results for a prediction are summarized in Figure 10.

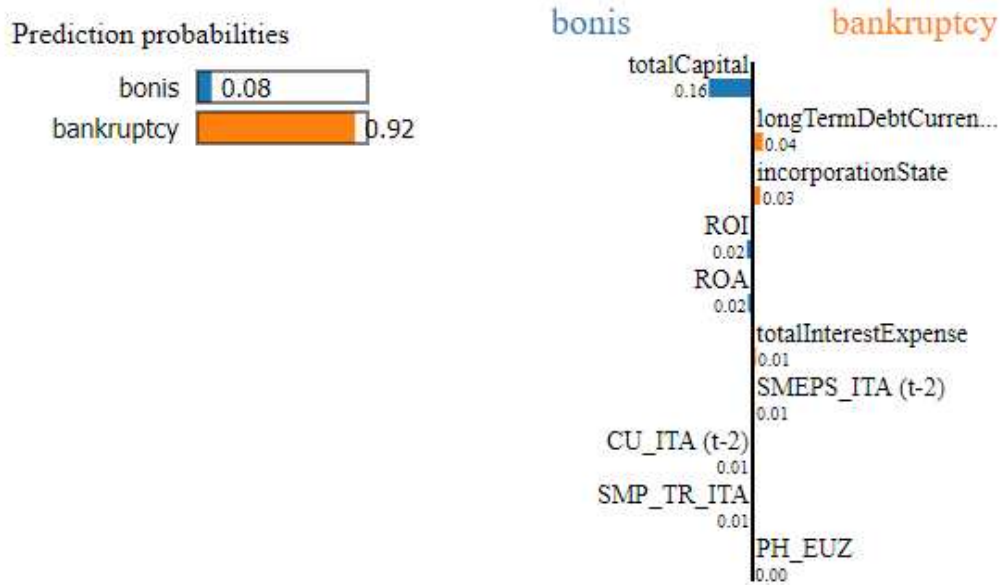


Figure 10: Lime local explanation for a prediction from Light-GBM classifier. The details of model’s feature description are reported in Appendix A

5 Conclusions

Starting from Moody’s dataset of historical balancesheets, bankruptcy statuses and macroeconomic variables we have built three models: a classifier, a default probability model and a rating system. By leveraging on modern techniques in both data processing and parameter calibration we have reached state of the art results. The three models show stunning out of sample performances allowing for an intensive usage in risk averse businesses where the occurrence of false negatives can dramatically harm the firm itself. The explainability layers via Shap and Lime give a set of extra tools to increase the confidence in the model and help in understanding the main features determining a specific result. These information can be leveraged by the analyst to understand how to reduce the bankruptcy probability of a specific firm or to get insight in which balance sheet fields need to be improved to increase the rating, therefore providing a business instrument to actively manage clients and structured finance deals.

Acknowledgements

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Appendices

A Model's Features descriptions

In this section the details of selected features, upon which the model have been trained, are reported.

A.1 Balancesheet index descriptions

Code	Definition
cashAndMarketableSecurities	Cash and marketable securities.
depreciationExpense	The depreciation expense for current period.
ebitda	Earnings before interest, taxes, depreciation and amortization before extraordinary items.
entityConsolidation-Type	For companies with subsidiaries, i
incorporationRegion	The entity's incorporation region.
incorporationState	The entity's incorporation province or administrative division where the entity has a legal representation.
longTermDebtCurrent-Maturities	The current maturities of long-term debt, principal payments due within 12 months.
netIncome	Net income is the total period-end earnings.
netWorth	Net worth is the sum of all equity items, including retained earnings and other equity.
payableToTrade	Accounts payable to regular trade accounts.
receivableFromTrade	Accounts receivable from trade.
retainedEarnings	Retained earnings.
tangibleNetWorth	Defined as the difference between netWorth and totalIntangibleAssets.
totalAccountsPayable	The sum of accounts payable.
totalAccountsReceivable	The total accounts receivable, net of any provision from loss.
totalAmortizationAnd-Depreciaton	The sum of amortization and depreciation expense for current period.
totalAssets	The total assets of the borrower which is the sum of the current assets and non-current assets.
totalCapital	Total subscribed and share capital.
totalCapital_and_totalLiabilities	Defined as the sum of totalLiabilities and totalCapital
totalCurrentAssets	The sum of all current assets.
totalCurrentLiabilities	The sum of all current liabilities.
totalFixedAssets	Total fixed assets are the Gross Fixed Assets less Accumulated Depreciation.
totalIntangibleAssets	Total intangible assets.

Code	Definition
totalInterestExpense	The total interest expense is any gross interest expense generated from short-term, long-term, subordinated or related debt.
totalInventory	The sum of all the inventories.
totalLiabilities	The sum of Total Current Liabilities and Total non-current liabilities.
totalLongTermDebt	The amount due to financial and other institutions after 12 months.
totalOperatingExpense	The sum of all operating expenses.
totalOperatingProfit	The Gross Profit less Total Operating Expense
totalProvisions	Total provisions for pensions, taxes, etc.
totalSales	Total sales.
totalWageExpense	The total wage expense.
workingCapital	Defined as the sum of receivableFromTrade, totalAccountsReceivable and totalInventory minus payableToTrade and totalAccountsPayable.

A.2 KPIs descriptions

$$ACID = \frac{\text{cashAndMarketableSecurities} + \text{totalAccountsReceivable}}{\text{totalCurrentLiabilities}} \quad (7)$$

$$ACTIVITY = \frac{\text{totalCurrentLiabilities}}{\text{totalSales}} \quad (8)$$

$$AGE = \frac{\text{financialStatementDate} - \text{incorporationDate}}{365.0} \quad (9)$$

being "financialStatementDate" the date of the financial statement and "incorporationDate" the date in which the entity was incorporated

$$ASSET_TURNOVER = \frac{\text{totalSales}}{\text{totalAssets}} \quad (10)$$

$$CURRENT_RATIO = \frac{\text{totalCurrentAssets}}{\text{totalCurrentLiabilities}} \quad (11)$$

$$DEBT_COVERAGE = \frac{\text{ebitda}}{\text{totalInterestExpense}} \quad (12)$$

$$DEBT_EQUITY = \frac{\text{totalLiabilities}}{\text{netWorth}} \quad (13)$$

$$EBITDA_RATIO = \frac{\text{ebitda}}{\text{totalSales}} \quad (14)$$

$$FFO = \text{netIncome} + \text{totalAmortizationAndDepreciaton} + \text{depreciationExpense} \quad (15)$$

$$IND_ROTA = \frac{\text{workingCapital}}{\text{totalSales}} \quad (16)$$

$$\text{IND_STRUTT} = \frac{\text{PFN}}{\text{netWorth}} \quad (17)$$

$$\text{INVENTORY_TURNOVER} = \frac{\text{totalInventory}}{\text{totalSales}} \quad (18)$$

$$\text{LEVERAGE_1} = \frac{\text{PFN}}{\text{ebitda}} \quad (19)$$

$$\text{LEVERAGE_2} = \frac{\text{FFO}}{\text{PFN}} \quad (20)$$

$$\text{LONG-TERM-DEBT_EQUITY} = \frac{\text{totalLongTermDebt}}{\text{netWorth}} \quad (21)$$

$$\text{NETINCOME_RATIO} = \frac{\text{netIncome}}{\text{totalSales}} \quad (22)$$

$$\text{PFN} = \text{totalLongTermDebt} + \text{longTermDebtCurrentMaturities} - \text{cashAndMarketableSecurities} \quad (23)$$

$$\text{ROA} = \frac{\text{netIncome}}{\text{totalAssets}} \quad (24)$$

$$\text{ROE} = \frac{\text{netIncome}}{\text{netWorth}} \quad (25)$$

$$\text{ROI} = \frac{\text{totalOperatingProfit}}{\text{totalAssets}} \quad (26)$$

$$\text{SHORT-TERM-DEBT_EQUITY} = \frac{\text{totalCurrentLiabilities}}{\text{netWorth}} \quad (27)$$

A.3 Macro-economic factors descriptions

Country specific indicators

Code	Name	Definition
C	Consumption, private, real	The volume of goods and services consumed by households and non-profit institutions serving households.
CD	Durable goods	The volume of real personal consumption expenditures.
CREDR	Credit rating, average	The sovereign risk rating, based on the average of the sovereign ratings provided by Moodys, S&P and Fitch.
CU	Capacity utilisation	A measure of the extent to which the productive capacity of a business is being used.

Country specific indicators

Code	Name	Definition
DOMD	Domestic demand, real	The volume of consumption, investment, stockbuilding and government consumption expressed in local currency and at prices of the country's base year.
EE	Employees in employment	Employees in employment.
ET	Employment, total	Employment, total.
GC	Consumption, government, real	The volume of government spending on goods and services.
GDP	GDP, real	The volume of all final goods and services produced within a country in a given period of time.
GDPHEAD	GDP per capita, real, US\$, constant prices	GDP per capita, real, US\$, constant prices.
IF	Investment, total fixed investment, real	Investment, total fixed investment, real.
IP	Industrial production index	The volume of investment in tangible and intangible capital goods, including machinery and equipment, software, and construction.
IPNR	Investment, private sector business, real	The volume of investment in private sector business.
IPRD	Investment, private dwellings, real	The volume of investment in private dwellings.
IS	Stockbuilding, real	The volume of stocks of outputs that are still held by the units that produced them and stocks of products acquired from other units that are intended to be used for intermediate consumption or for resale.
M	Imports, goods & services, real	The volume of goods and services imports.
MG	Imports, goods, real	The volume of goods imports.
MS	Imports, services, real	The volume of services imports.
PEWFP	GDP, compensation of employees, total, nominal	The values of wages and salaries of employees as a component of GDP.
PH	House price index	Index of house prices.
POIL\$	Oil price US\$ per toe	Oil price US\$ per toe.
RCB	Interest rate, central bank policy	The rate that is used by central bank to implement or signal its monetary policy stance (expressed as an average).

Country specific indicators

Code	Name	Definition
RCORP_SPREADEOP	Credit spreads, end of period	The difference in yield between two bonds of similar maturity but different credit quality, expressed as end of period value.
RLG	Interest rate, long-term government bond yields	Interest rate, long-term government bond yields.
RS	Retail Sales volume index, excluding automotive	Volume index for retail sales excluding automotive.
RSH	Interest rate, short-term	The 3-month interbank rate.
RSHEOP	Interest rate, short-term, end of period	The 3-month interbank rate for the end of period.
SMEPS	Stockmarket earnings per share	Stockmarket earnings per share calculated as Stockmarket earnings, LCU * 1000 / Stockmarket shares outstanding.
SMP_TR	Share price total return index	Share price total return index.
TFE	Total final expenditure, real	The sum of volumes of consumption, investment, stockbuilding, government consumption and exports.
U	Unemployment	The total number of people without a job, but actively searching for one.
UP	Unemployment rate	The percentage of the labour force that is unemployed at a given date.
X	Exports, goods & services, real	The volume of goods and services exports expressed in local currency and at the country's base year.
XG	Exports, goods, real	The volume of goods exports expressed in local currency.
XS	Exports, services, real	The volume of services exports expressed in local currency.

Eurozone indicators

Code	Name	Definition
PH	House price index	Index of house prices.
RCB	Interest rate, central bank policy	The rate that is used by central bank to implement or signal its monetary policy stance (expressed as an average).

Eurozone indicators

Code	Name	Definition
RCBEOP	Interest rate, central bank policy, end of period	The rate that is used by central bank to implement or signal its monetary policy stance (expressed as an end of period value).
REONIA	Interest rate, EONIA	1-day interbank interest rate for the Eurozone.
RLG	Interest rate, long-term government bond yields	Interest rate, long-term government bond yields.
RSH	Interest rate, short-term	The 3-month interbank rate.
RSH6M	Interest rate, 6-month	The 6-month interbank rate.
RSHEOP	Interest rate, short-term, end of period	The 3-month interbank rate for the end of period.
RSWAP2YR	Interest Rate Swap, 2-year	Swap par-rate for 2 years tenor.
RSWAP5YR	Interest Rate Swap, 5-year	Swap par-rate for 5 years tenor.
RSWAP10YR	Interest Rate Swap, 10-year	Swap par-rate for 10 years tenor.
RSWAP30YR	Interest Rate Swap, 30-year	Swap par-rate for 30 years tenor.

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Consumer Credit Risk Models via Machine-Learning Algorithms*

Amir E. Khandani[†], Adlar J. Kim[‡], and Andrew W. Lo[§]

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Abstract

We apply machine-learning techniques to construct nonlinear nonparametric forecasting models of consumer credit risk. By combining customer transactions and credit bureau data from January 2005 to April 2009 for a sample of a major commercial bank's customers, we are able to construct out-of-sample forecasts that significantly improve the classification rates of credit-card-holder delinquencies and defaults, with linear regression R^2 's of forecasted/realized delinquencies of 85%. Using conservative assumptions for the costs and benefits of cutting credit lines based on machine-learning forecasts, we estimate the cost savings to range from 6% to 25% of total losses. Moreover, the time-series patterns of estimated delinquency rates from this model over the course of the recent financial crisis suggests that aggregated consumer-credit risk analytics may have important applications in forecasting systemic risk.

Keywords: Household Behavior; Consumer Credit Risk; Credit Card Borrowing; Machine Learning; Nonparametric Estimation

JEL Classification: G21, G33, G32, G17, G01, D14

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[†]Post Doctoral Associate, MIT Sloan School of Management and Laboratory for Financial Engineering.

[‡]Post Doctoral Associate, MIT Sloan School of Management and Laboratory for Financial Engineering.

[§]Harris & Harris Group Professor, MIT Sloan School of Management; director, Laboratory for Financial Engineering.

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1 Introduction

One of the most important drivers of macroeconomic conditions and systemic risk is consumer spending, which accounted for over two thirds of U.S. gross domestic product as of October 2008. With \$13.63 trillion of consumer credit outstanding as of the fourth quarter of 2008 (\$10.47 trillion in mortgages \$2.59 trillion in other consumer debt), the opportunities and risk exposures in consumer lending are equally outsized.¹ For example, as a result of the recent financial crisis, the overall charge-off rate in all revolving consumer credit across all U.S. lending institutions reached 10.1% in the third quarter of 2009, far exceeding the average charge-off rate of 4.72% during 2003 to 2007.² With a total of \$874 billion of revolving consumer credit outstanding in the U.S. economy as of November 2009,³ and with 46.1% of all families carrying a positive credit-card balance in 2007,⁴ the potential for further systemic dislocation in this sector has made the economic behavior of consumers a topic of vital national interest.

The large number of decisions involved in the consumer lending business makes it necessary to rely on models and algorithms rather than human discretion, and to base such algorithmic decisions on “hard” information, e.g., characteristics contained in consumer credit files collected by credit bureau agencies. Models are typically used to generate numerical “scores” that summarize the creditworthiness of consumers.⁵ In addition, it is common for lending institutions and credit bureaus to create their own customized risk models based on private information about borrowers. The type of private information typically consist of both “within-account” as well as “across-account” data regarding customers’ past behavior.⁶ However, while such models are generally able to produce reasonably accurate ordinal mea-

¹U.S. Federal Reserve Flow of Funds data, June 11, 2009 release.

²Data available from the Federal Reserve Board at <http://www.federalreserve.gov/releases/chargeoff/>.

³See the latest release of Consumer Credit Report published by the Federal Reserve Board, available at <http://www.federalreserve.gov/releases/g19/Current/>

⁴See the *Survey of Consumer Finances, 2009* (SCF), released in February 2009 and available at <http://www.federalreserve.gov/pubs/bulletin/2009/pdf/scf09.pdf>. This report shows that the median balance for those carrying a non-zero balance was \$3,000, while the mean was \$7,300. These values have risen 25% and 30.4%, respectively, since the previous version of the SCF conducted three year earlier. The SCF also reports that the median balance has risen strongly for most demographic groups, particularly for higher-income groups.

⁵See Hand and Henley (1997) and Thomas (2009) for reviews of traditional and more recent statistical modeling approaches to credit scoring.

⁶The impact of such relationship information in facilitating banking engagement has been studied extensively in the area of corporate and small business lending (see, for example, Boot 2000) and, more recently, in consumer lending (see Agarwal, Chomsisengphet, Liu and Souleles, 2009).

asures, i.e., rankings, of consumer creditworthiness, these measures adjust only slowly over time and are relatively insensitive to changes in market conditions. Given the apparent speed with which consumer credit can deteriorate, there is a clear need for more timely cardinal measures of credit risk by banks and regulators.

In this paper, we propose a cardinal measure of consumer credit risk that combines traditional credit factors such as debt-to-income ratios with consumer banking transactions, which greatly enhances the predictive power of our model. Using a proprietary dataset from a major commercial bank (which we shall refer to as the “Bank” throughout this paper to preserve confidentiality) from January 2005 to April 2009, we show that conditioning on certain changes in a consumer’s bank-account activity can lead to considerably more accurate forecasts of credit-card delinquencies in the future. For example, in our sample, the unconditional probability of customers falling 90-days-or-more delinquent on their payments over any given 6-month period is 5.3%, but customers experiencing a recent decline in income—as measured by sharp drops in direct deposits—have a 10.8% probability of 90-days-or-more delinquency over the subsequent 6 months. Such conditioning variables are statistically reliable throughout the entire sample period, and our approach is able to generate many variants of these transactions-based predictors and combine them in nonlinear ways with credit-bureau data to yield even more powerful forecasts. By analyzing patterns in consumer expenditures, savings, and debt payments, we are able to identify subtle nonlinear relationships that are difficult to detect in these massive datasets using standard consumer credit-default models such as logit, discriminant analysis, or credit scores.

We use an approach known as “machine learning” in the computer science literature, which refers to a set of algorithms specifically designed to tackle computationally intensive pattern-recognition problems in extremely large datasets. These techniques include radial basis functions, tree-based classifiers, and support-vector machines, and are ideally suited for consumer credit-risk analytics because of the large sample sizes and the complexity of the possible relationships among consumer transactions and characteristics.⁷ The extraordinary speed-up in computing in recent years, coupled with significant theoretical advances in machine-learning algorithms, have created a renaissance in computational modeling, of which our consumer credit-risk model is just one of many recent examples.

⁷See, for example, Li, Shiue, and Huang (2006) and Bellotti and Crook (2009) for applications of machine learning based model to consumer credit.

One measure of the forecast power of our approach is to compare the machine-learning model’s forecasted scores of those customers who eventually default during the forecast period with the forecasted scores of those who do not. Significant differences between the forecasts of the two populations is an indication that the forecasts have genuine discriminating power. Over the sample period from May 2008 to April 2009, the average forecasted score among individuals who do become 90-days-or-more delinquent during the 6-month forecast period is 61.9, while the average score across all customers is 2.1. The practical value-added of such forecasts can be estimated by summing the cost savings from credit reductions to high-risk borrowers and the lost revenues from “false positives”, and under a conservative set of assumptions, we estimate the potential net benefits of these forecasts to be 6% to 25% of total losses.

More importantly, by aggregating individual forecasts, it is possible to construct a measure of systemic risk in the consumer-lending sector, which accounts for one of the largest components of U.S. economic activity. We show that the time-series properties of our machine-learning forecasts are highly correlated with realized delinquency rates (linear regression R^2 ’s of 85%), implying that a considerable portion of the consumer credit cycle can be forecasted 6 to 12 months in advance. This has obvious implications for macroprudential risk management.

In Section 2, we describe our dataset, discuss the security issues surrounding it, and document some simple but profound empirical trends. Section 3 outlines our approach to constructing useful variables or feature vectors that will serve as inputs to the machine-learning algorithms we employ. In Section 4, we describe the machine-learning framework for combining multiple predictors to create more powerful forecast models, and present our empirical results. Using these results, we provide two applications in Section 5, one involving model-based credit-line reductions and the other focusing on systemic risk measures. We conclude in Section 6.

2 The Data

In this study, we use a unique dataset consisting of transaction-level, credit-bureau, and account-balance data for individual consumers. This data is obtained for a subset of the Bank’s customer base for the period from January 2005 to April 2009. Integrating transaction, credit-bureau, and account-balance data allows us to compute and update measures

of consumer credit-risk much more frequently than the slower-moving credit-scoring models currently being employed in the industry and by regulators. This data was assembled from a number of different sources which we review in Section 2.1. Given the sensitive nature of the data and consumer privacy protection laws, all individual identification data elements such as names, addresses, and social security numbers were removed by the bank before sharing the data with us, and all computations were performed on machines physically located at the Bank’s office and within the bank’s electronic “firewalls”. We review these security protocols in Section 2.2. In Section 2.3, we document some trends in debt balances and payments over the sample period that clearly point to substantial but gradual deterioration in the financial health of consumers over this period.

2.1 Data Sources

The raw data was provided to us in flat-file format. A set of programs were then used to pre-process the data, producing higher-level variables or “feature vectors” and time-aggregated to yield monthly observations that were stored in a database (see Figure 1). The following is a description of each of these components.

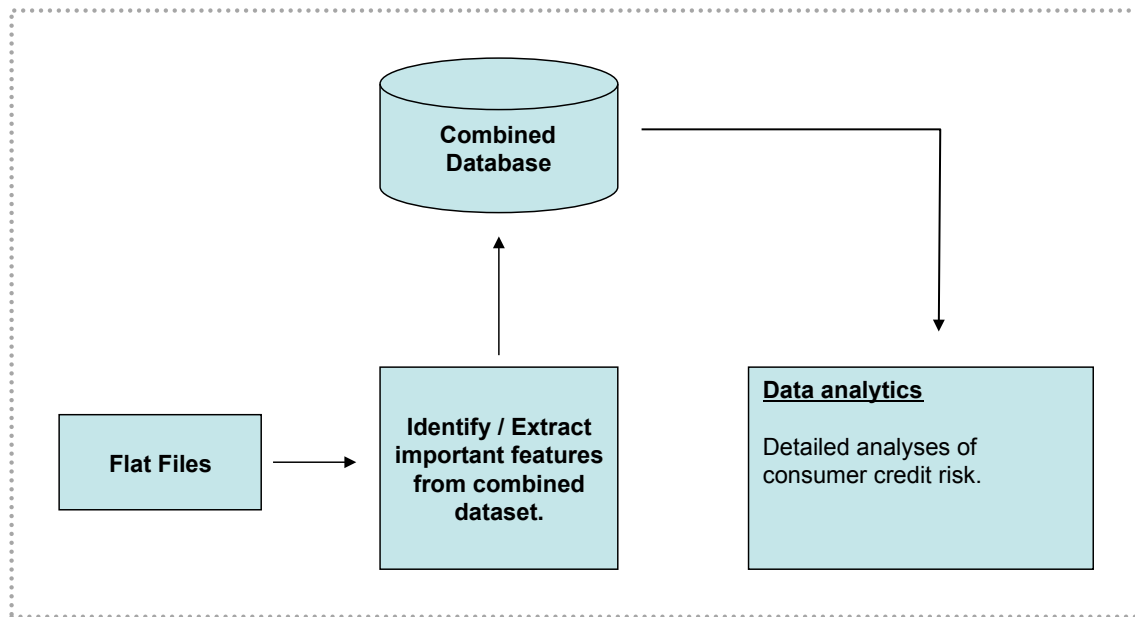


Figure 1: Construction of an integrated database of transactions and credit bureau statistics used in machine-learning model of consumer credit risk.

Transactions Data

The Bank provided us with account-level transactions data for a subset of its customer base. In addition to identifying each transaction’s amount and direction (inflow or outflow), this dataset contains two other classification fields with each transaction: *channel* and *category*. The channel refers to the medium through which the transaction took place; a complete list of possible channels are: Automated Clearing House (ACH), Adjustments to Account (ADJ), Automated Teller Machine (ATM), Online Bill Payment (BPY), Credit Card (CC), Check (CHK), Debit Card (DC), Account Fees (FEE), Interest on Account (INT), Wire Transfer (WIR), Internal Transfer (XFR) and Miscellaneous Deposit or Withdrawal (not in another channel) (MSC). There was also very rare occurrences of an Unidentifiable Channel (BAD).

The category classification is much broader and meant to capture the nature of the transaction. Examples of transaction categories include: restaurant expenditure, bar expenditure, grocery, food, etc. In the raw data, we observed 138 categories, which were produced by the Bank’s data team based on the information that was associated with each transaction. Naturally, the level of accuracy and coverage varied greatly across channels. In particular, transactions taking place via Credit Card (CC) or Debit Card (DC) channels had much more accurate categories associated with them than those associated with ATM or Check transaction channels.

The sheer size and scope of this dataset proved to be a significant challenge, even for computing basic summary statistics. To develop some basic intuition for the data, we began by computing the total number of transactions per month, total inflows, and total outflows across all transactions for each individual in the database, and then constructed the same metrics for each of the following channels separately: Automated Clearing House (ACH), Automated Teller Machine (ATM), Credit Card (CC), Debit Card (DC), Interest on Account (INT), Online Bill Payment (BPY) and Wire Transfer (WIR). These channels were chosen because they represent the vast majority of the entire sample.

We also attempted to distill the most important information contained in the 138 categories by selecting a smaller set of data elements that broadly represent the key aspects of consumers’ daily expenditures. In particular, Table 1 lists the subset of data items we selected from the transactions level data.⁸ It should be emphasized that this data represents

⁸Note that not all of these items were used in creating our risk models because of legal restrictions (see

an incomplete picture of a typical consumer’s expenditures. For example, the ultimate purposes of transactions that take place via ATM and Check channels are not typically known. Also, while employment-related payments (e.g., salary, bonus, etc.) are captured if they are directed via ACH channel, these payments are missed and not classified if an individual deposits a check or cash into his or her account. Similarly, any expense paid in cash is not captured in this data. Finally, an individual may have banking relationships with multiple institutions, hence the data from the Bank may be an incomplete view of the individual’s financial activities. However, to the extent that such levels of incompleteness are stable through time, the trends observed in the data may still be used to make inferences and predictions about the changing behavior of the population. While a certain degree of caution is warranted because of these issues, the apparent success of the statistical models created from this data (see Section 4.4 and 5) suggests that there is considerable incremental value for consumer credit risk management even in this limited dataset.

Data from the Credit Bureaus

The Bank complements its account transactions data with credit file data provided by one of the credit bureaus for their customers. This data consists of variables listed under the “Individual Level” data items in Table 2. The generic credit score provided in this data, which we shall refer to as CScore, is similar in terms of statistical properties and predictive power to scores that are standard technique for measuring consumer credit quality.⁹ We will use this credit score as a benchmark against which the performance of machine-learning models of credit risk can be compared.

Figure 2 depicts the relationship between CScore for the months of January of 2005, 2006, 2007 and 2008 and the realized default frequency in the following 6 months. In each case, we consider the initial CScore for individuals and calculate the default frequency in the subsequent 6 months for each level of the score. In addition, we plot second-order-polynomial-fitted curves in each graph, and Figure 2 shows that the CScore is remarkably successful in rank-ordering future default frequencies. In Figure 3, we plot the four polynomial approximations to compare the performance across different market conditions. As economic conditions began deteriorating in 2007 and 2008, the curves started to shift to the

Section 2.2 for further discussion). The list of data items used in our analysis is given in Section 4.2 and Table 4.

⁹See Avery et al. (2003) for an excellent review of consumer data.

right, indicating a higher absolute default rate for each level of the CScore. This is to be expected since the distribution of the CScore is static overtime as shown in Figure 4.

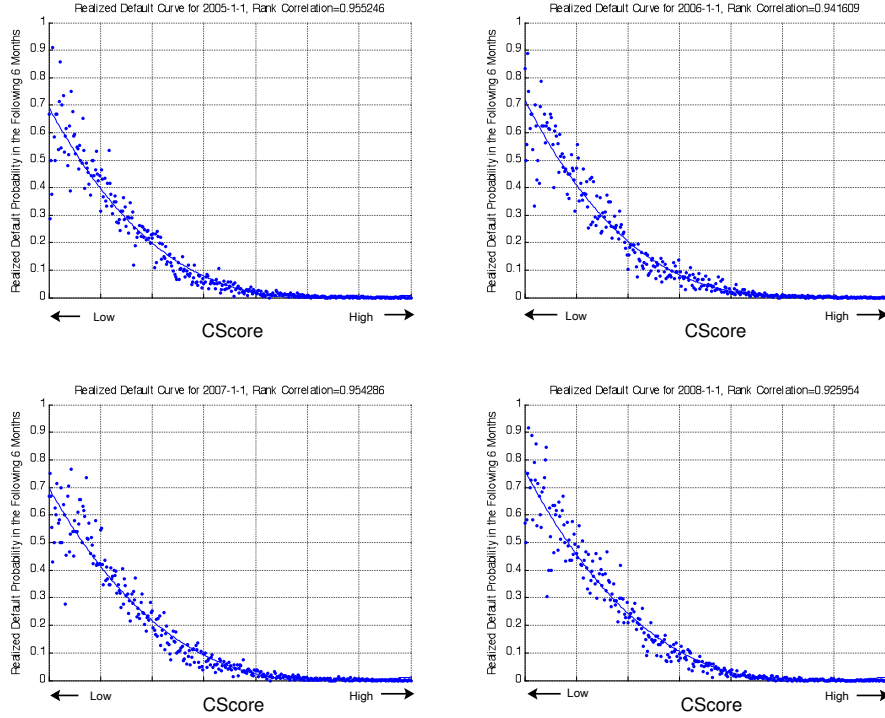


Figure 2: Relationship between CScore and subsequent default frequency (over the next 6 months) for January of 2005, 2006, 2007, 2008. Second-order polynomial approximations are also shown (solid lines).

In addition, this data contains information regarding all credit or loan facilities, also referred to as “Trade Line”, that each customer has across all financial institutions. For example, for a customer with a checking account with the Bank and a mortgage with another lender, this dataset would contain the mortgage balance, age, payment history and other relevant information without specifying the name of identity of the lender. Trade lines are divided into the following six types: Auto (AUT), Mortgage (MTG), Credit Card (CCA), Home Line of Credit (HLC), Home Loan (HLN), Other Loans (ILA), Other Lines of Credit (LOC), and Speciality type (SPC). The data collected for each Trade Line are listed in Table 2 under the “Trade Line Level” column.

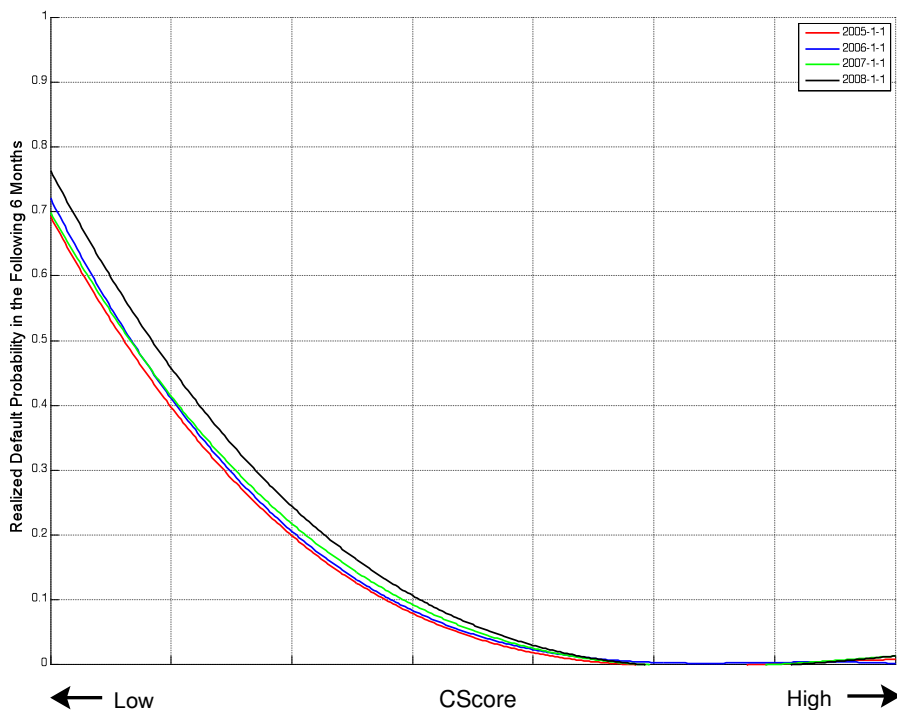


Figure 3: Second-order polynomial approximation of the relationship between CScore and subsequent default frequency (over the next 6 months) for January of 2005, 2006, 2007, 2008.

Account Balance Data

We also match the transaction-level and credit-bureau data with information about account balances from checking accounts and CDs that a customer has with the Bank. These items are listed in Table 3.

2.2 Data Security

Given the sensitivity of consumer data, a number of steps were taken to preserve the confidentiality of this data. First, the Bank de-identified all records—deleting all identification information such as names, addresses, social security numbers, and customer age—before they were transferred into databases to which we were given access. All computations were performed on dedicated MIT computers installed on site at the Bank’s office, and all external network connections for these computers were disabled. At no point during this project was any raw consumer data transferred outside of the Bank’s electronic firewalls.

Moreover, several data items were excluded from our models due to legal restrictions such

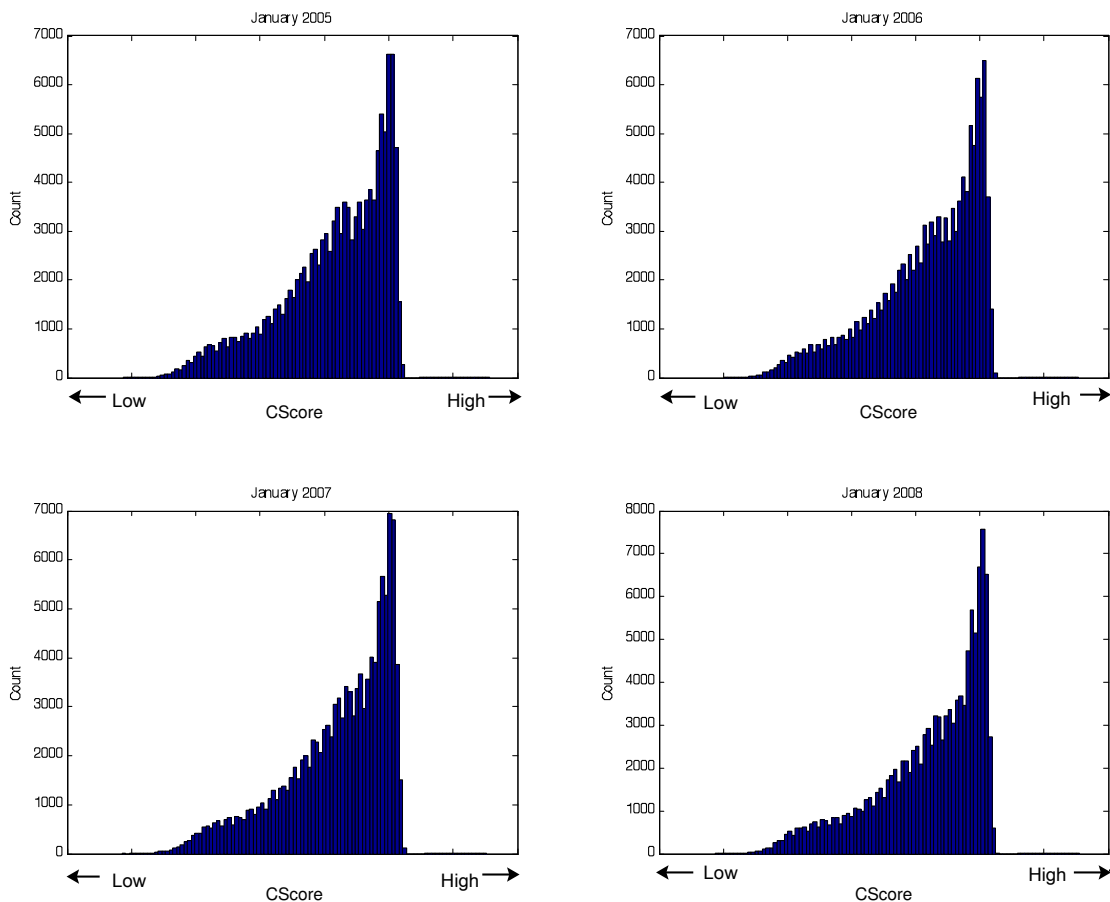


Figure 4: Histograms of CScore for January of 2005, 2006, 2007, 2008.

as the Fair Credit Reporting Act. In particular, all healthcare, insurance, unemployment, government, treasury, account and file age information were excluded from our analysis (see Section 4.2 and Table 4 for the list of data items used in our analysis).

2.3 Data Trends from 2005 to 2008

To provide some intuition for the empirical characteristics of the data, in this section we document several trends in the data over the period from January 2005 to December 2008. Since the main focus of our study is predictive models for credit-card defaults and delinquencies, we confine our attention in this section to observed trends in credit-card balances and payments.

Figure 5(a) shows the median, 75th, 90th, and 95th percentiles of credit-card balances, based on the total balance across all credit cards. Because the data used in this calculation are taken from the matched sample obtained from credit bureaus (see Table 2), the balance represents the total balance across credit cards provided by all financial institutions, and is an accurate representation of the total credit-card debt of individuals. This data shows that the top 5% of the sample had credit-card debt of approximately \$20,190 at the beginning of the sample, which increased to \$27,612 by the end of the sample. The pattern of increasing credit-card indebtedness is clear, and even more extreme for other percentiles of the distribution as shown in Figure 5(a). For example, the median credit-card balance almost doubled from \$647 to \$1,259 over this period.

While increasing credit-card debt probably does not bode well for systemic risk, it need not represent a decline in the overall financial health of the consumer if the increased debt was accompanied by a comparable increase in income. For example, if increasing credit-card indebtedness is concurrent with a change in the usage among customers such that those consumers with high incomes—and who had not previously carried positive credit-card balances—started borrowing with their credit cards, this scenario may not represent a substantial change in systemic risk. If, on the other hand, customers increase their credit-card borrowing without any change in their income, the result would be an increase in systemic risk. To attempt to distinguish between these two scenarios, we plot the ratio of credit-card balances to monthly income in Figure 5(b).¹⁰ This figure shows that in the first half of the sample, from January 2005 to January 2006, the increase in credit-card debt documented in Figure 5(a) was concurrent with increased monthly income or a shift in usage such that there was little or no change in the ratio of credit-card balances to income. However, starting in January 2007, a different pattern emerges—the ratio of the balances to monthly income starts to increase steadily. For example, for the top 5%, this ratio was approximately 11 for the first half of the sample but increased by about 2.5 to reach 13.5 over the period from January 2007 to December 2008.

While the trends shown in Figure 5(a) and 5(b) point to an increasing level of credit-card debt, both in absolute terms and relative to income, over this period, this by itself may be the result of a substitution effect where credit-card debt replaced another form of credit over

¹⁰We have used the average of monthly income, as measured by the “Employment Inflow” data shown in Table 1, over the precessing 6 months, or as many month as available, as the denominator.

this period. To explore this possibility, in Figure 5(c) we depict the same statistics for credit-card payments during this period.¹¹ We see that credit-card payments exhibited a positive trend from January 2005 to July 2007, but the increase stopped after July 2007. Finally, Figure 5(d) contains the ratio of credit-card payments to monthly income over this period, which shows that while credit-card payments were increasing as a percentage of monthly income for period of January 2005 to July 2007, this trend broke after July 2007.

However, this pattern must be interpreted with some care due to certain data issues. In particular, as discussed above, the monthly income data is only captured by the bank’s transaction categorization engine if it comes through ACH (such as direct deposit), and there is no assurance that this system will capture an individual’s entire income, even if all of it is deposited with the Bank (for example, income in the form of deposited checks would not be categorized as income). Furthermore, there may be some outliers, as in the case where individuals use their credit cards to purchase big-ticket items such as vacations or major appliances, in which case their credit-card balances can exceed their monthly income. Nevertheless, we believe that the patterns in Figures 5(a)–5(d) are still revealing with respect to the deteriorating financial condition of consumers in months leading up to the financial crisis.

Figures 6(a) and 6(b) contain other trends that confirm the developing problems in consumer credit during this period. Figure 6(a) shows the ratio of monthly payments for credit cards, auto loans, student loans, and mortgages to monthly income, and the pattern is consistent with Figure 5(d); monthly payments across all debt instruments were increasing as a percentage of monthly income (but again, we may not have identified all sources of an individual’s income). To develop a better sense of these trends, Figure 6(b) plots the ratio of debt payment to total monthly outflow over this period.¹² The total outflow provides a useful robustness check since the direction of each transaction (inflow or outflow) is known with certainty, so the total monthly outflow is a complete picture of all expenditures from all Bank accounts over the course of the month. The pattern that emerges from this analysis, see 6(b), supports the trends in the previous figures.

The patterns reported in this section are a very small subset of the many trends that

¹¹In particular, we use average credit-card payments over the preceding 6 months, or as many month as are available.

¹²Total monthly outflow is measured as the average monthly outflow over the preceding 6 months, or as many month as available.

are readily observed in this extremely rich data set. Even based on this very limited set of metrics, it is clear that credit-card balances were increasing, both in absolute and relative terms, long before the credit crisis hit, and similar trends were apparent among a broader set of debt instruments. Such patterns suggest that predictive systemic risk indicators in the consumer lending business are feasible.

3 Constructing Feature Vectors

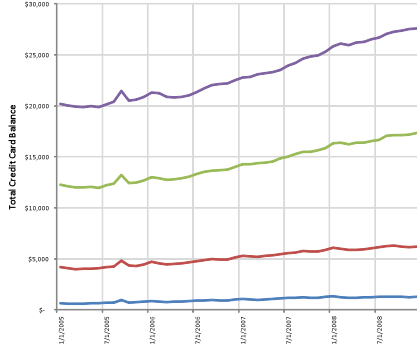
The objective of any machine-learning model is the identification of statistically reliable relationships between certain features of the input data and the target variable or outcome. In the models that we construct in later sections, the features we use include data items such as total inflow, total income, credit-card balance, etc. and the target variable is a binary outcome that indicates whether an account is delinquent by 90 days or more within the subsequent 3-, 6-, or 12-month window. In this section, we provide two illustrative examples of the relationship between certain features and subsequent delinquencies to develop intuition for both the data and our machine-learning algorithms. In Section 3.1, we consider high levels of the ratio of credit-card balances to income, and in Section 3.2, we explore the impact of sharp income declines.

3.1 High Balance-to-Income Ratio

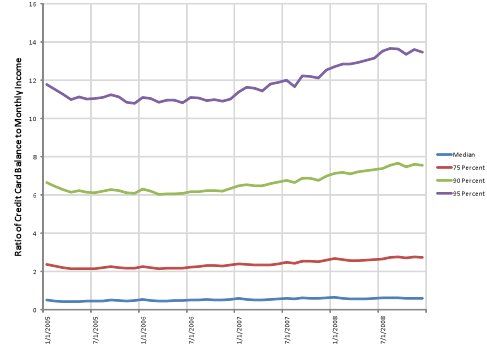
For each month in our sample (January 2005 to December 2008), we estimate the probability of 90-days-or-more delinquencies on any of the credit-card accounts in the following 3 and 6 months for all customers. We then compute the same estimates for customers with credit-card-balance-to-income ratios greater than 10.¹³ This threshold is chosen arbitrarily, and is meant to identify consumers experiencing great financial distress. As shown in Figure 5(b), the top 5% of the sample have credit-card-balance-to-income ratios exceeding 10.

Figures 7(a) and 7(b) contain the outcome of this stratification for both 3-month and 6-month evaluation periods. Observe that the last month shown on the horizontal axis is June or Sept 2008 since we need 6 or 3 months, respectively, to calculate subsequent realized delinquency rates, and our sample ends in December 2008. Each axis also begins in June

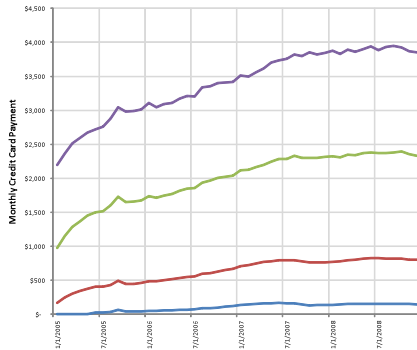
¹³We define “credit-card balance” as the total credit-card balances across all financial institutions obtained from credit bureau data. Also, we measure the rate of 90-days-or-more delinquencies on any credit-card account across all financial institutions.



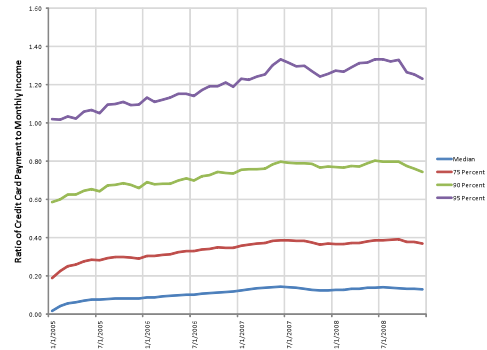
(a) Historical Monthly Credit-Card Balance



(b) Historical Monthly Credit-Card Balance to Income Ratio

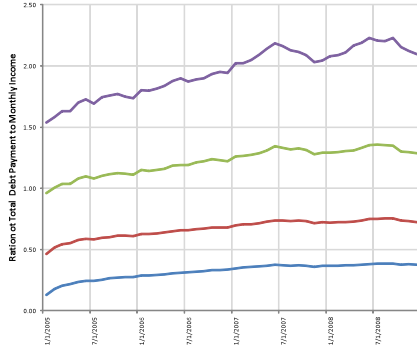


(c) Historical Monthly Credit-Card Payment

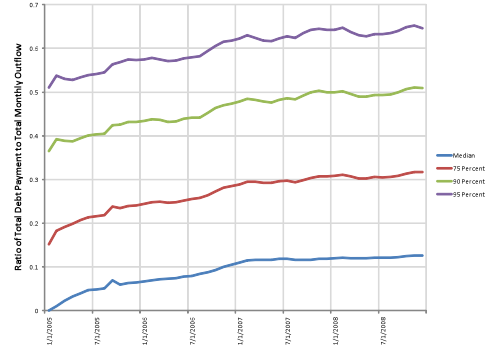


(d) Historical Monthly Credit-Card Payment to Income Ratio

Figure 5: Characteristics of credit-card balances, payments and their relationship to monthly income, based on data from January 2005 to December 2008. In Panel (a), the credit-card balance, which is the total balance across all credit cards of the customer in the month ending shown on the horizontal axis, is shown. In Panel (b), the ratio of credit-card balance to monthly income is shown. Panel (c) shows credit-card payments over this period while Panel (d) shows the ratio of credit-card payments to monthly income. For this calculation, the average income and credit-card payment over 6 months preceding the date on the horizontal axis or as many months as available, is used for the calculation. See Section 2 for further details.



(a) Historical Monthly Ratio of All Debt Payments to Income



(b) Historical Monthly Ratio of All Debt Payments to Total Outflow

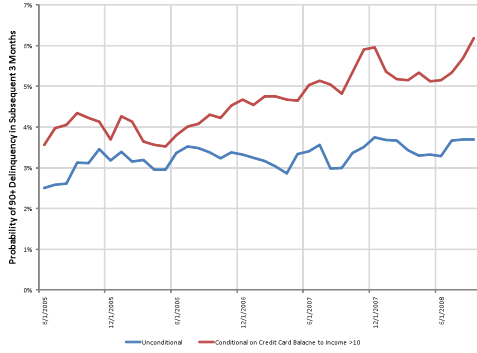
Figure 6: Characteristics of payments of all forms of debt to monthly income and total monthly outflow based on data from January 2005 to December 2008. Payments for the following debt instruments are included in this calculation: auto loans, student loans, banks loans, mortgages, and credit cards. The average of monthly figures over 6 months proceeding the date on the horizontal axis or as many months as available, is used for this calculation. See Section 2 for further details.

2005 since we use the average income level over the prior 6 months for the denominator, hence we lose the first 6 months of the sample. This is consistent with the analysis presented in Figure 5(a)–5(d) and Figure 6(a)–6(b).

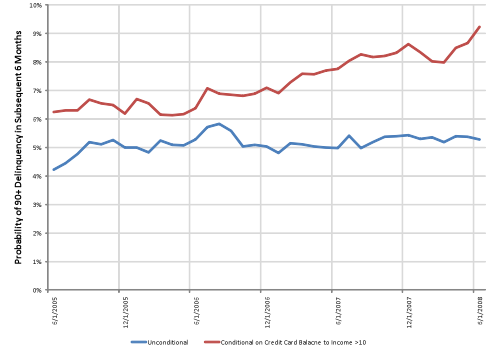
As shown in Figures 7(a) and 7(b), customers with very high credit-card-balance-to-income ratios are much more likely to experience delinquencies in subsequent 3- and 6-month periods. Furthermore, the separation has become wider in the latter part of the sample.

To evaluate the reliability of the relationship identified in Figures 7(a) and 7(b), in Figures 8(a) and 8(b) we report the same statistics but on the limited sample of customers who were current on *all* their credit-card accounts as of the conditioning date. The rate of future delinquency is lower (compare the blue lines in Figures 7(a) and 7(b) with those in Figures 8(a) and 8(b)), but the gap between delinquency rates for typical customers and for highly indebted customers is still clearly apparent.

This example illustrates the presence of statistically stable relationships that can be exploited in creating a statistical model for predicting future delinquencies and defaults. However, since the gap between the future delinquency rates of average customers versus highly indebted customers has become wider in more recent periods, it is clear that the data contain important non-stationarities. Therefore, any machine-learning model of delinquen-

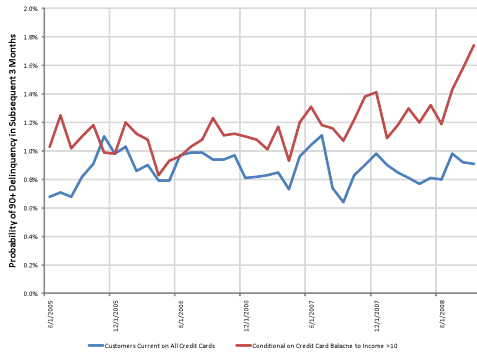


(a) Impact of High Balance-to-Income Ratios on Future 3-Month Delinquency Rates

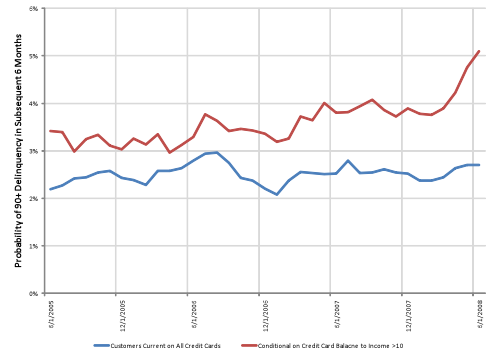


(b) Impact of High Balance-to-Income Ratios on Future 6-Month Delinquency Rates

Figure 7: Delinquency rates of customers with high credit-card-payment-to-income ratios. For each month, the rates of 90-days-or-more delinquencies during the following 3- or 6-month window for all customers as well as customers with total-credit-card-debt-to-income ratios greater than 10 are computed. The data covers the period from January 2005 to December 2008. A 6-month moving-average of income is the denominator used in calculating the ratio of credit-card balance to income, hence the horizontal axis starts in June 2005 and ends in June or Sept 2008 depending on whether 6- or 3-month forecast windows apply.



(a) Impact of High Balance-to-Income Ratios on Future 3-Month Delinquency Rates



(b) Impact of High Balance-to-Income Ratios on Future 6-Month Delinquency Rates

Figure 8: Delinquency rates for customers with high credit-card-payment-to-income ratios. For each month, the rates of 90-days-or-more delinquencies during the following 3- or 6-month window for customers who are current on all their credit-card accounts are computed, as well as customers who are current and have total-credit-card-debt-to-income ratios greater than 10. The data covers the period from January 2005 to December 2008. A 6-month moving-average of income is the denominator used in calculating the ratio of credit-card balance to income, hence the horizontal axis starts in June 2005, and ends in June or Sept 2008 depending on whether 6- or 3-month forecast windows apply.

cies and defaults must be re-calibrated frequently enough to capture such changes in a timely manner. We will discuss this issue in more detail in Section 4.

3.2 Negative Income Shocks

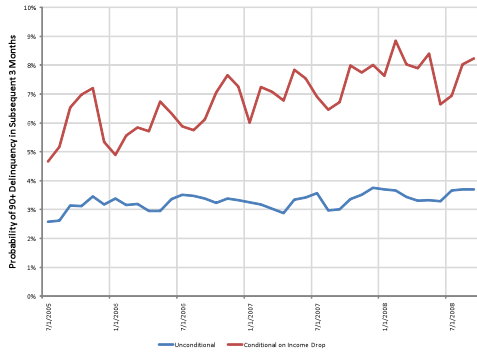
A sharp and sudden drop in income—most likely due to unemployment—is a plausible potential predictor of future delinquency and default, and we proceed in a fashion similar to Section 3.1 to evaluate this possibility. The challenge is to identify income shocks that are “significant”, i.e., large relative to his/her historical levels and variability of income. To that end, in each month, we construct an income-shock variable by taking the difference between the current month’s income and the 6-month moving-average of the income, and then divide this difference by the standard deviation of income over the same trailing 6-month window. By dividing the income drop by the estimated volatility, we are able to account for those individuals with jobs that produce more volatile income streams. We can stratify the sample according to this income-shock variable, and compare the 3- and 6-month subsequent 90-days-or-more delinquency rates for individuals with and without significant income shocks during each month in our sample (January 2005 to December 2008).

Figure 9(a) and 9(b) plots the 90-days-or-more delinquency rates for all customers (blue) and for those customers with income-shock variables less than -2 (red). The gap between the delinquency rates of these two groups is clear. Furthermore, like the balances-to-income ratio of Section 3.1, this gap has also become wider over time, consistent with the patterns in Figures 7(a)–7(b) and Figures 8(a)–8(b).

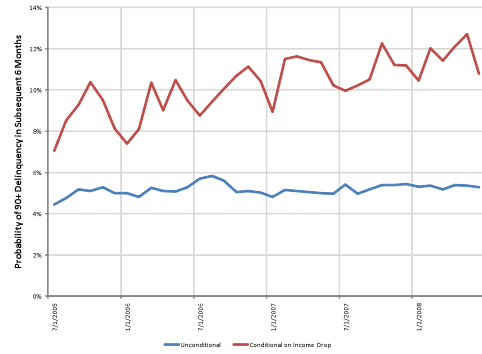
4 Modeling Methodology

In this section, we describe the machine-learning algorithms we use to construct delinquency forecast models for the Bank’s consumer credit and transactions data from January 2005 to April 2009.¹⁴ This challenge is well suited to be formulated as a supervised learning problem, which is one of the most widely used techniques in the machine-learning literature. In the supervised learning framework, a learner is presented with input/output pairs from past data, in which the input data represent pre-identified attributes to be used to determine the output value. Such input data are commonly represented as a vector and, depending on

¹⁴There is an extensive literature on machine learning, and we refer readers to Bishop (2006), Duda et al. (2000), and Vapnik (1998) for excellent overviews of this large and still-growing literature.



(a) Impact of Income Ratio Drop on Future 3-Month Default Rates



(b) Impact of Income Ratio Drop on Future 6-Month Default Rates

Figure 9: Delinquency rates for customers experiencing negative income shocks. For each month and customer, the income-shock variable is defined as the difference between the current month's income and the 6-month moving-average of income over the prior 6 months, divided by the standard deviation of income over that period. For each month, the rates of 90-days-or-more delinquencies during the following 3- or 6-month window for all customers as well as customers with income shocks less than -2 are calculated. The data covers the period from January 2005 to December 2008. Because of the 6-month moving-average for income, the horizontal axis starts in June 2005, and ends in June or Sept 2008 depending on whether 6- or 3-month forecast windows apply.

learning algorithm, can consist of continuous and/or discrete values with or without missing data. The supervised learning problem is referred as a “regression problem” when the output is continuous, and as a “classification problem” when the output is discrete.

Once such input/output data are presented, the learner’s task is to find a function that correctly maps such input vectors to the output values. One brute force way of accomplishing this task is to “memorize” all previous values of input/output pairs. Although this correctly maps the input/output pairs in the training data set, it is unlikely to be successful in forecasting the output values if the input values are different from the ones in the training data set, or when the training data set contains noise. Therefore, the challenge of supervised learning is to find a function that generalizes beyond the training set, so that the resulting function also accurately maps out-of-sample inputs to out-of-sample outcomes.

Using this machine-learning approach, we build a credit-risk forecast model that predicts delinquencies and defaults for individual consumers. For example, an output of our model is a continuous variable between 0 and 1 that can be interpreted (under certain conditions) as an estimate of the probability of 90-days-or-more delinquency sometime within the next 3 months of a particular credit-card account, given specific input variables from that account. Using this model, we address two questions:

1. How can this model be used to improve decisions about credit-line cuts?
2. What is the rate of credit-card delinquencies in the aggregate economy?

In addition, we need to take model accuracy into account in addressing these two questions, and more detailed discussion of this and other modeling issues are provided in Section 5. For both questions, selecting a parsimonious set of input variables that are relevant to credit risk is critical, and requires intuition for the underlying drivers of consumer credit, which cannot be resolved by the learning algorithm alone. The approach we described in Section 3 can be used to identify such variables, and Figure 10 summarizes our supervised learning framework.

4.1 Machine-Learning Framework

We use generalized classification and regression trees (CART) (Breiman et al. 1984) to construct our forecast models. CART is a widely used statistical technique in which a dependent

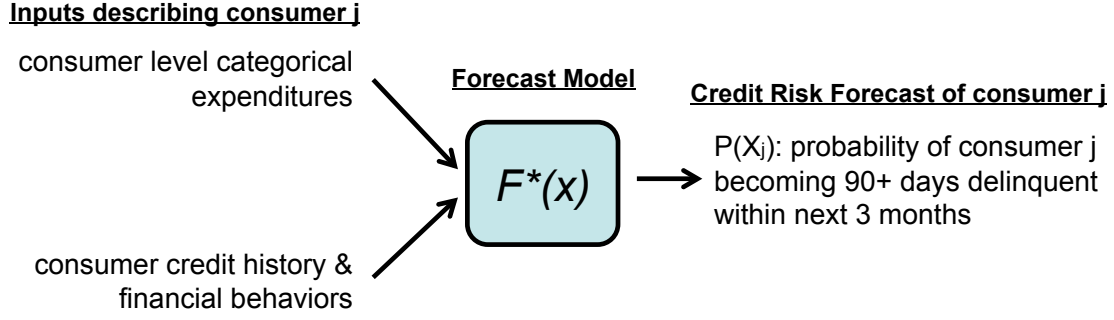


Figure 10: A summary of the machine-learning algorithm used to construct the consumer credit risk model.

or “output” variable (either continuous or discrete) is related to a set of independent or “input” variables through a recursive sequence of simple binary relations (hence the reference to a “tree”). The collection of recursive relations partitions the multi-dimensional space of independent variables into distinct regions in which the dependent variable is typically assumed to be constant (classification tree) or linearly related (regression tree) to the independent variables with parameters unique to each region.

Figure 11 provides an illustration of a CART model with two non-negative independent variables, $\{x_1, x_2\}$, also known as a “feature vector”, and a discrete dependent variable that takes on two possible values, *good* and *bad*. The sequence of recursive binary relations represented by the tree in Figure 11 partitions the domain of $\{x_1, x_2\}$ into five distinct regions determined by the parameters $L_1, \dots, L_4$. In particular, this model implies that all realizations of $\{x_1, x_2\}$ in which $x_1 < L_1$ and $x_2 < L_2$ is associated with a *bad* outcome, and all realizations of $\{x_1, x_2\}$ in which $x_1 < L_1$ and $x_2 \geq L_2$ is associated with a *good* outcome. The parameters $\{L_j\}$ are chosen to minimize the distance between the dependent variable and the fitted values of the tree using an appropriate distance metric (typically mean-squared error), and forecasts may be readily constructed from new realizations of the feature vector $\{x'_1, x'_2\}$ by traversing the tree from its top-most node to the bottom.

The growing popularity of the CART model stems from the fact that it overcomes the limitations of standard models such as logit and probit in which the dependent variable is forced to fit a single linear model throughout the entire input space. In contrast, CART is capable of detecting nonlinear interactions between input variables, which dramatically

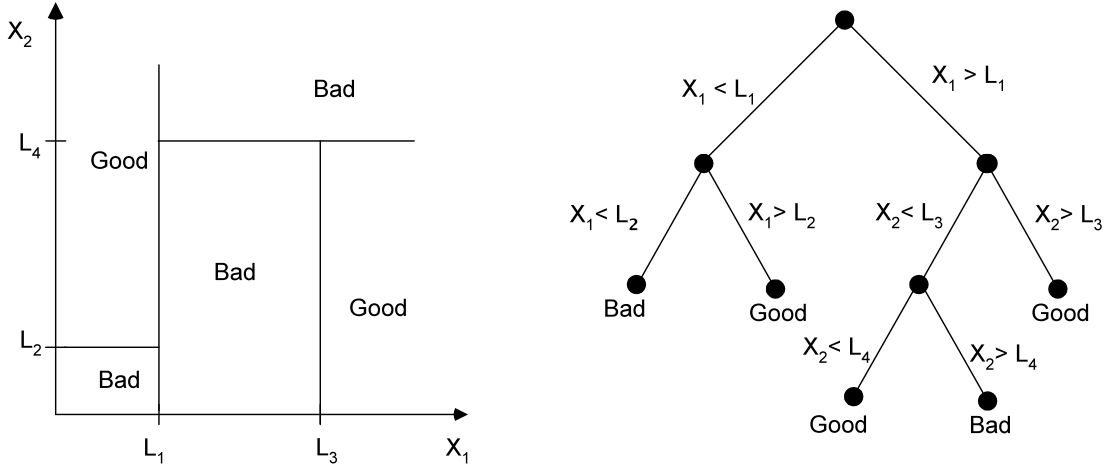


Figure 11: An example of a CART model for a discrete dependent variable with two outcomes, *good* and *bad*, and two independent variables $\{x_1, x_2\}$.

increases the types of relations that can be captured and the number of independent variables that can be used. Moreover, CART models produce easily interpretable decision rules whose logic is clearly laid out in the tree. This aspect is particularly relevant for applications in the banking sector in which “black-box” models are viewed with suspicion and skepticism.

CART models can easily be applied to problems with high-dimensional feature spaces. Suppose we have N observations of the dependent variable $\{y_1, \dots, y_N\}$ and its corresponding D -dimensional feature vectors $\{x_1, \dots, x_N\}$. We estimate the parameters of the CART model on the training dataset by recursively selecting features from $x \in \{x_1, \dots, x_D\}$ and parameters $\{L_j\}$ that minimize the residual sum-of-squared errors. Of course, we must impose a “pruning criterion” for stopping the expansion of the tree so as to avoid overfitting the training data. One of the most widely used measures for pruning is the *Gini* measure:

$$G(\tau) \equiv \sum_{k=1}^K P_{\tau}(k)(1 - P_{\tau}(k)) \quad (1)$$

where τ refers to a leaf node of a CART model and $P_{\tau}(k)$ refers to the proportion of training data assigned to class k at leaf node τ . Then the pruning criterion for CART model T is

defined as,

$$C(T) \equiv \sum_{\tau=1}^{|T|} G(\tau) + \lambda|T| \quad (2)$$

where $|T|$ refers to a number of leaf nodes in CART model T , and λ refers to a regularization parameter chosen by cross-validation. Once the pruning criterion reaches the minimum, the CART algorithm will stop expanding the tree.

An issue that is common among credit-default data is highly skewed proportion of *bad* and *good* realizations (credit defaults takes up of only 2% of our data set). There are several ways to improve the forecast power of CART models in these cases, one of which is a technique called “boosting”. Instead of equally weighting all the observations in the training set, we can weight the scarcer observations more heavily than the more populous ones (see Freund, and Shapire, 1996). This “adaptive boosting” technique trains the model on examples with pre-assigned weights (initially equally weighted), and the weights are adjusted recursively as a function of the goodness-of-fit. In particular, the weight for observation i at the n^{th} iteration is given by:

$$w_i^{(n)} = w_i^{(n-1)} \exp [\alpha_{n-1} I(f_{n-1}(\mathbf{x}_i) \neq y_i)] \quad (3)$$

and the data re-weighting coefficient α_{n-1} is defined as,

$$\alpha_{n-1} \equiv \ln \left(\frac{1 - \epsilon_{n-1}}{\epsilon_{n-1}} \right) \quad (4)$$

where $I(\cdot)$ is an indicator function that indicates whether the model has correctly predicted the outcome y_i given the input vector $\mathbf{x}_i$, and ϵ_{n-1} is the weighted average error of the model from the $(n-1)^{\text{th}}$ iteration.

Related approaches have been used by Atiya (2001), Shin et al. (2005), and Min et al. (2005), in which artificial neural networks (ANN) and support vector machines (SVM) were applied to the problem of predicting corporate bankruptcies. Ong et al. (2005) conclude that genetic-algorithms-based consumer credit-scoring models outperform models based on ANNs, decision trees, rough sets, and logistic regression. Martens et al. (2007) introduce rule-extraction methods for SVMs that can generate consumer-credit models with human-

interpretable rules and little loss in accuracy. Galindo and Tamayo (2000) test CART decision-tree algorithms on mortgage-loan data to identify defaults, and also compare their results to k -nearest neighbor (KNN), ANNs, and probit models. Huang et al. (2004) provide an excellent survey of these and other related studies.

In the remainder of this section, we describe the attributes we selected for our inputs and discuss the model’s performance. To identify the most powerful set of attributes, we combine bank transaction, consumer credit, and account balance databases and process them in parallel to construct our feature vectors (see Figure 1).

4.2 Model Inputs

Table 4 lists the data elements from the three sources discussed in Section 2.1 that were used as inputs to our delinquency prediction model. As discussed in Section 2.2 not all data elements can be used because of legal restrictions. We conducted extensive exploratory data analysis, similar to illustrative examples in Section 3, and finalized the set of input variables to be used in constructing our feature vectors listed in Table 4. For all transaction-related items, their average values over the prior 6 months, or as many months as available, are used.

4.3 Comparison to CScore

The empirical results of Section 2.1 show that the CScore is indeed useful in rank-ordering customers by their delinquency and default rates, hence it should serve as a reasonable benchmark for the machine-learning forecasts.

Figure 12 compares the machine-learning forecasts of 90-days-or-more delinquencies of the Bank’s credit-card accounts to CScore for December 2008. The machine-learning forecasts are constructed by training the model on delinquency data from October 2008 to December 2008 and using input data from September 2008 to construct the feature vectors, and then computing “fitted values” using features in December 2008, which yields December-2008 forecasts of delinquencies in the 3-month period from January 2009 to March 2009. CScore scores from December 2008 are plotted on the x-axis and the corresponding December-2008 machine-learning forecasts are plotted on the y-axis.

This figure shows that machine-learning forecasts differ substantially from CScore scores. In particular, there are a handful of accounts with relatively high CScore scores (note that

higher CScore scores indicate higher credit quality or lower credit risk) that have high forecasted delinquency and default risk according to the machine-learning model. To provide further intuition for these conflicts, in Figure 13 we reproduce the same plot but have included color-coding for status of each account status as of the prediction date (December 2008)—green for accounts that are current (no late payments), blue for accounts that are 30-days delinquent, yellow for accounts 60-days delinquent, and red for accounts that are 90-days-or-more delinquent. Individuals with high CScore and high credit-risk forecasts are those already delinquent in their payments as of the prediction date (in some cases 90-days-or-more). However, those with low CScore but low forecasted credit risk are individuals that are current in their payments. While these patterns are not meant to be a conclusive comparison of the predictive ability of the two measures, Figure 13 does suggest that machine-learning forecasts contain different information than traditional credit scores like CScore. We provide a more quantitative assessment in the next section.

Transaction Data	
Transaction Count	By Category (Cont.)
Total Inflow	
Total Outflow	Hotel expenses
	Travel expenses
By Channel:	Recreation
	Department Stores Expenses
ACH (Count, Inflow and Outflow)	Retail Stores Expenses
ATM (Count, Inflow and Outflow)	Clothing expenses
BPY (Count, Inflow and Outflow)	Discount Store Expenses
CC (Count, Inflow and Outflow)	Big Box Store Expenses
DC (Count, Inflow and Outflow)	Education Expenses
INT (Count, Inflow and Outflow)	Total Food Expenses
WIR (Count, Inflow and Outflow)	Grocery Expenses
	Restaurant Expenses
By Category	Bar Expenses
	Fast Food Expenses
Employment Inflow	Total Rest/Bars/Fast-Food
Mortgage payment	Healthcare related expenses
Credit car payment	Fitness expenses
Auto loan payment	Health insurance
Student loan payment	Gas stations expenses
All other types of loan payment	Vehicle expenses
Other line of credit payments	Car and other insurance
Brokerage net flow	Drug stores expenses
Dividends net flow	Government
Utilities Payments	Treasury
TV	Pension Inflow
Phone	Collection Agencies
Internet	Unemployment Inflow

Table 1: Inflows and outflows into accounts are collected by the Bank. Direction, amount, channel, and category information is associated with each transaction. The elements outlined in this table are collected and aggregated from a much larger set of channel and category data collected by the the Bank. Not all fields are used because of legal restrictions, e.g., insurance, healthcare related, social security, collection agencies, unemployment and several more were excluded.

Individual Level	Account Level
Credit Score File Age Bankruptcy (Date & Code) MSA & Zip Code	Age of Account Open/Closed Flag & Date of Closure Type (CC, MTG, AUT, etc) Balance Limit if applicable Payment Status 48-Month Payment Status History

Table 2: Data items extracted from credit bureaus. Not all fields extracted were used because of legal restrictions, e.g., MSA, zip code, file age, and account age data were not used.

Deposit Data	
Checking Account Balance Brokerage Account Balance Saving Account Balance	CD Account Balance IRA Account Balance

Table 3: Balance data for accounts held with the Bank. Not all fields extracted were used due to legal restrictions.

Model Inputs	
Credit Bureau Data Total Number of Trade Lines Number of Open Trade Lines Number of closed trade lines Number and balance of auto loans Number and balance of credit cards Number and balance of home line of credits Number and balance of home loans Number and balance of all other loans Number and balance of all other lines of credit Number and balance of all mortgages Balance of all auto loans to total debt Balance of all credit cards to total debt Balance of all home line of credit to total debt Balance of all home loans to total debt Balance of all other loans to total debt Balance of all other lines of credit to total debt Ratio of total mortgage balance to total debt Total credit-card balance to limits Total home line of credit balances to limits Total balance on all other lines of credit to limits Transaction Data Number of Transactions Total inflow Total outflow Total pay inflow Total all food related expenses Total grocery expenses Total restaurant expenses Total fast food expenses Total bar expenses	Transaction Data (Cont.) Total expenses at discount stores Total expenses at big-box stores Total recreation expenses Total clothing stores expenses Total department store expenses Total other retail stores expenses Total utilities expenses Total cable TV & Internet expenses Total telephone expenses Total net flow from brokerage account Total net flow from dividends and annuities Total gas station expenses Total vehicle related expenses Total logging expenses Total travel expenses Total credit-card payments Total mortgage payments Total outflow to car and student loan payments Total education related expenses Deposit Data Savings account balance Checking account balance CD account balance Brokerage account balance

Table 4: The final inputs used in the machine-learning model of consumer credit risk.

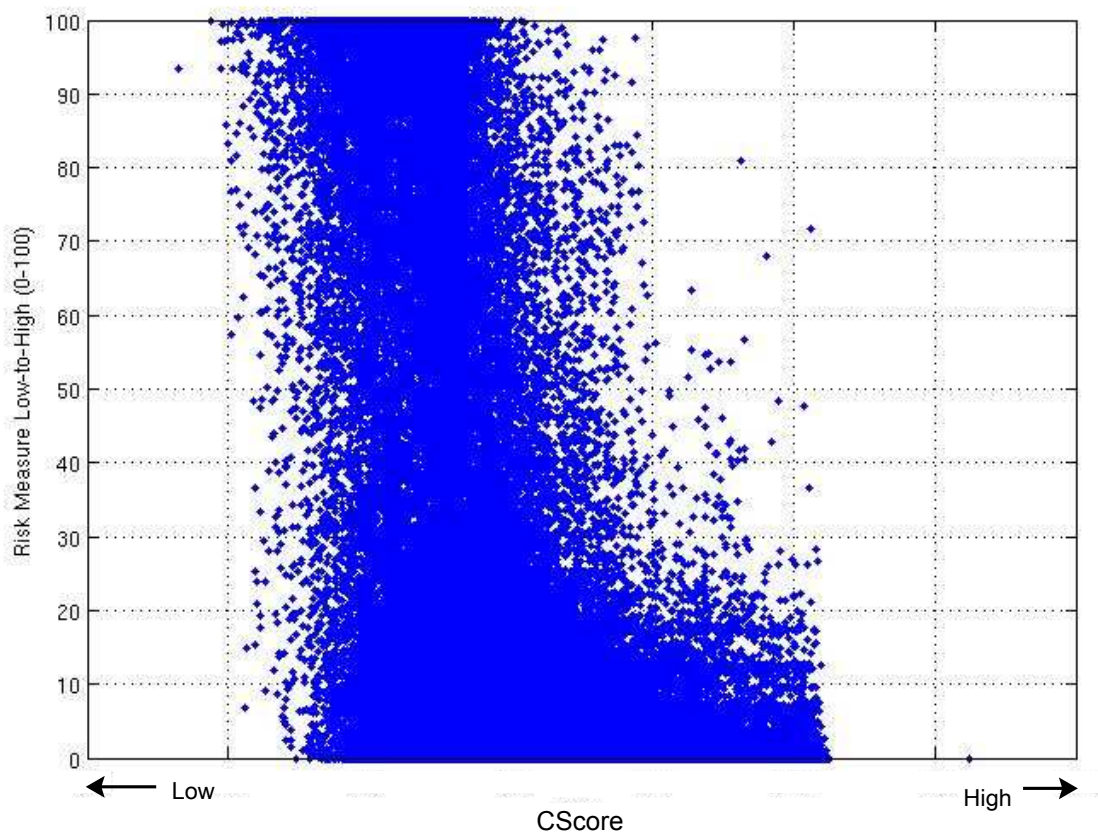


Figure 12: Comparison of December 2008 CScore (x-axis, and where higher values are associated with lower credit risk) and machine-learning forecasts of 90-days-or-more delinquency rates over subsequent 3-month windows using realized delinquency events from October to December 2008 and September 2008 feature vectors. Once calibrated, the machine-learning model is applied to the December 2008 feature vector, and the resulting “fitted values” are plotted (y-axis) against the December 2008 CScores.

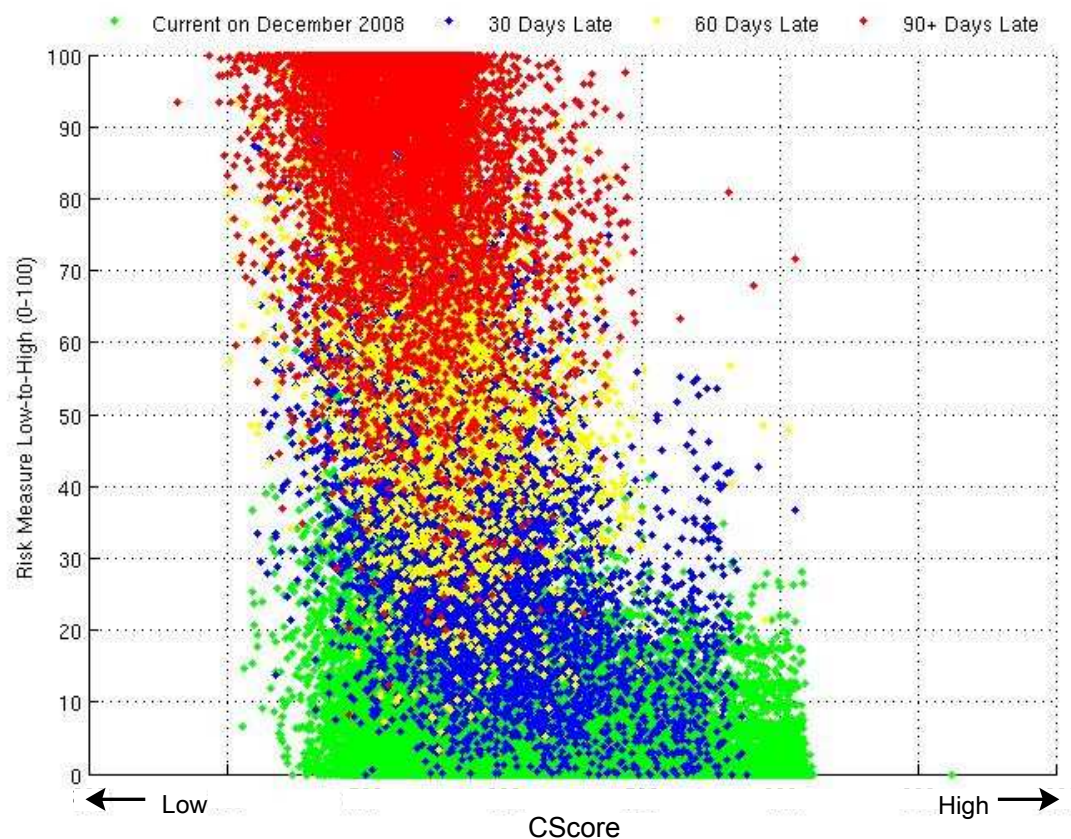


Figure 13: Color-coded comparison of December 2008 CScore (x-axis, and where higher values are associated with lower credit risk) and machine-learning forecasts of 90-days-or-more delinquency rates over subsequent 3-month windows using realized delinquency events from October to December 2008 and September 2008 feature vectors. Once calibrated, the machine-learning model is applied to the December 2008 feature vector, and the resulting “fitted values” are plotted (y-axis) against the December 2008 CScores. The color coding indicates whether an account is current (green), 30-days delinquent (blue), 60-days delinquent (yellow), or 90-days-or-more delinquent as of December 2008.

Input	Training Period			Evaluation Period		
	Date	Delinquency Period		Prediction Date	Delinquency Period	
		Start Date	End Date		Start Date	End Date
Jan-08		Feb-08	Apr-08	Apr-08	May-08	Jul-08
Feb-08		Mar-08	May-08	May-08	Jun-08	Aug-08
Mar-08		Apr-08	Jun-08	Jun-08	Jul-08	Sep-08
Apr-08		May-08	Jul-08	Jul-08	Aug-08	Oct-08
May-08		Jun-08	Aug-08	Aug-08	Sep-08	Nov-08
Jun-08		Jul-08	Sep-08	Sep-08	Oct-08	Dec-08
Jul-08		Aug-08	Oct-08	Oct-08	Nov-08	Jan-09
Aug-08		Sep-08	Nov-08	Nov-08	Dec-08	Feb-09
Sep-08		Oct-08	Dec-08	Dec-08	Jan-09	Mar-09
Oct-08		Nov-08	Jan-09	Jan-09	Feb-09	Apr-09

Table 5: The 10 calibration and testing periods used in evaluating the performance of machine-learning forecasts of consumer credit risk.

4.4 Model Evaluation

We now turn to a more rigorous evaluation of our models. We will focus on the model used to produce forecasts of 90-days-or-more delinquencies over subsequent 3-month windows for the period from January 2008 to April 2009. We choose this period because it includes the most severe deterioration in consumer credit quality in recent history, providing a natural laboratory for gauging the performance of machine-learning models during periods of financial dislocation. To minimize the effects of look-ahead bias, we only train the model based on delinquencies over 3-month windows that were observable at the time of the forecast. For example, the first model was calibrated using input data from January 2008 and realized delinquencies over the 3-month window from February to April 2008, and then the parameter estimates were applied to the input data in April 2008 to produce forecasts of delinquencies over the 3-month window from May to July 2008. This rolling-window approach yields 10 calibration and testing periods listed in Table 5, and Table 6 reports sample sizes for each of the 10 prediction periods. The percentage of 90-days-or-more delinquent accounts varies from 2.0% to 2.5%.

The output of the machine-learning model is a continuous variable that, under certain conditions, can be interpreted as an estimate of the probability of an account becoming 90-days-or-more delinquent during the subsequent 3-month forecasting window. One measure of the model’s success is the difference between the forecasts of those accounts that did

3 Months Starting Date	3 Months Ending Date	Customers going 90+ days delinquent (%)	Customers NOT going 90+ days delinquent (%)
May-08	Jul-08	2.4	97.6
Jun-08	Aug-08	2.2	97.8
Jul-08	Sep-08	2.1	97.9
Aug-08	Oct-08	2.0	98.0
Sep-08	Nov-08	2.1	97.9
Oct-08	Dec-08	2.1	97.9
Nov-08	Jan-09	2.1	97.9
Dec-08	Feb-09	2.3	97.7
Jan-09	Mar-09	2.6	97.4
Feb-09	Apr-09	2.5	97.5

Table 6: Sample sizes of accounts and 90-days-or-more delinquencies for each of the 10 3-month evaluation windows from May 2008 to April 2009. This data is based on Bank-issued credit cards only.

become delinquent and those that did not; if accounts that subsequently became delinquent had the same forecasts as those that did not, the forecasts have no value. Table 7 reports the difference between the average forecast associated with accounts that did and did not fall into the 90-days-or-more delinquency category over the 10 evaluation periods. For example, during the period from May to July 2008, the model’s average forecast among the 2.4% of accounts (see Table 6) that did become 90-days-or-more delinquent was 61.2, whereas the average forecast among the 97.6% of accounts that did not fall into this category was only 1.0. We should emphasize that these forecasts are truly out-of-sample, being based on input data available at the end of April 2008, and calibrated using data from February to April 2008 (see Table 5). This implies significant forecast power in distinguishing between accounts that will and will not become delinquent over subsequent 3-month periods. Moreover, this forecast power does not diminish over the 10 calibration and evaluation periods, likely due to the frequent re-calibration of the model that captures some of the changing dynamics of consumer behavior documented in Section 3.

Another interesting challenge for the machine-learning model is the identification of “straight-rollers”, accounts that are current as of the forecast date but become 90-days-or-more delinquent within the subsequent 3-month window. Such accounts typically do not

become current after short delinquencies, but usually “roll straight” into the most extreme delinquency level that is commonly used as an indication of likely default and a charge-off loss for that bank. Recall from the examples of Section 3 (see, in particular, Figures 8(a) and 8(b)) that certain input variables do seem to be related to future delinquencies, even among customers with current accounts as of the forecast date.

To evaluate the machine-learning model’s ability to identify straight-rollers, we compare the model’s average forecast among customers who were current on their accounts but became 90-days-or-more delinquent with the average forecast among customers who were current and did not become delinquent. Since this is a harder learning problem, we expect the model’s performance to be less impressive than the values reported in Table 7. Nevertheless the values reported in Table 8 still indicate that the model is remarkably powerful, and is clearly able to separating the two populations. For example, using input date from April 2008, the average model forecast among customers who were current on their account and did not become 90-days-or-more delinquent from May to July 2008 was 0.7, while the average forecast for straight-rollers was 10.3. And as before, the degree of separation is remarkably consistent across the 10 evaluation periods listed in Table 8.

Training Window		Prediction Date	Evaluation Window		Average Predicted Probability of Going 90+ Delinquent		
Start	End		Start	End	Among All	Among Customers Going 90+ Days Delinquent	Among Customers NOT Going 90+ Days Delinquent
Feb-08	Apr-08	Apr-08	May-08	Jul-08	2.5	61.2	1.0
Mar-08	May-08	May-08	Jun-08	Aug-08	2.0	62.1	0.6
Apr-08	Jun-08	Jun-08	Jul-08	Sep-08	1.9	60.4	0.7
May-08	Jul-08	Jul-08	Aug-08	Oct-08	1.9	62.5	0.6
Jun-08	Aug-08	Aug-08	Sep-08	Nov-08	2.0	62.4	0.7
Jul-08	Sep-08	Sep-08	Oct-08	Dec-08	2.1	63.6	0.8
Aug-08	Oct-08	Oct-08	Nov-08	Jan-09	2.1	62.5	0.8
Sep-08	Nov-08	Nov-08	Dec-08	Feb-09	2.2	59.8	0.8
Oct-08	Dec-08	Dec-08	Jan-09	Mar-09	2.4	60.8	0.8
Nov-08	Jan-09	Jan-09	Feb-09	Apr-09	2.4	62.8	0.9

Table 7: Performance metrics for machine-learning forecasts over 10 3-month evaluation windows from May 2008 to April 2009. For each evaluation period, the model is calibrated on credit-card delinquency data over the 3-month period specified in the *Training Window* columns, and predictions are based on the data available as of the date in the *Prediction Date* column. For example, the first row reports the performance of the model calibrated using input data available in January 2008 and credit-card delinquency data from February to April 2008, and applied to the April 2008 feature vector to generate forecasts of delinquencies from May to July 2008. Average model forecasts over all customers, and customers that (ex-post) did and did not become 90-days-or-more delinquent over the evaluation period are also reported.

Training Window		Prediction Date	Evaluation Window		Average Predicted Probability of Going 90+ Delinquent		
Start	End		Start	End	Among All	Among Customers Going 90+ Days Delinquent	Among Customers NOT Going 90+ Days Delinquent
Feb-08	Apr-08	Apr-08	May-08	Jul-08	0.7	10.3	0.7
Mar-08	May-08	May-08	Jun-08	Aug-08	0.4	8.0	0.4
Apr-08	Jun-08	Jun-08	Jul-08	Sep-08	0.5	9.7	0.4
May-08	Jul-08	Jul-08	Aug-08	Oct-08	0.4	8.6	0.3
Jun-08	Aug-08	Aug-08	Sep-08	Nov-08	0.4	9.6	0.4
Jul-08	Sep-08	Sep-08	Oct-08	Dec-08	0.5	9.5	0.4
Aug-08	Oct-08	Oct-08	Nov-08	Jan-09	0.5	10.1	0.4
Sep-08	Nov-08	Nov-08	Dec-08	Feb-09	0.5	10.6	0.5
Oct-08	Dec-08	Dec-08	Jan-09	Mar-09	0.5	8.6	0.5
Nov-08	Jan-09	Jan-09	Feb-09	Apr-09	0.5	10.2	0.5

Table 8: Performance metrics for machine-learning forecasts over 10 3-month evaluation windows from May 2008 to April 2009 for “straight-rollers”, customers who are current as of the forecast date but who become 90-days-or-more delinquent in following 3-month window. For each evaluation period, the model is calibrated on credit-card delinquency data over the 3-month period specified in the *Training Window* columns and predictions are based on the data available as of the date in the *Prediction Date* column, using only customers who are current as of the forecast date. For example, the first row shows the performance of the model calibrated using input data available in January 2008 and credit-card delinquency data from February to April 2008, and applied to the April 2008 feature vector to generate forecasts about delinquencies from May to July 2008.

5 Applications

In this section we apply the models and methods of Sections 3 and 4 to two specific challenges in consumer-credit risk management: deciding when and how much to cut individual-account credit lines, and forecasting aggregate consumer-credit delinquencies for the purpose of enterprise-wide and macroprudential risk management.

The former application is of particular interest from the start of 2008 through early 2009 as banks and other financial institutions responded to the developing crisis with massive risk reductions. In Section 5.1, we use input data from December 2008 to make forecasts about individual-account delinquencies from January to March 2009, which are then compared with realized delinquencies during that period to evaluate the out-of-sample predictive power of our machine-learning model.

In Section 5.2, we investigate the robustness of machine-learning forecasts by comparing their performance across subsets of the data stratified by the number of features available for each account. We find that delinquencies among accounts with more features are significantly easier to predict than those with few features. This difference underscores the potential value of the conditioning information we have constructed from the combined data items in Table 4.

In Section 5.3, we combine individual-account forecasts to produce an aggregate forecast of consumer-credit delinquencies, which we propose as a new measure of systemic risk. Despite the fact that our data represents a small subset of the Bank’s customer base, the machine-learning model is surprisingly effective in capturing the changing risk characteristics of the population. In fact, our results indicate near-monotonic increases in aggregate consumer-credit risk from January 2005 to April 2009, suggesting that consumer-based measures of financial distress may play an important role in macroprudential risk management.

5.1 Credit-Line Risk Management

Once a line of credit is extended to a customer by a bank, the bank actively monitors the creditworthiness of the customer. The credit line may be increased or the interest rate reduced in the case of improving creditworthiness, while the opposite may occur if there is a decline in the customer’s credit profile. Therefore, the particular application considered in this section is the primary focus of any consumer-credit business. For the business to

be successful, credit-line decisions must be based on a model that captures changes in a customer’s risk profile in a timely and accurate manner. In fact, during the sample period of our dataset, the major decision facing banks and other consumer lenders was identifying high-risk customers and taking the appropriate actions with respect to their credit lines, i.e., lowering credit limits, increasing interest rates, or some combination of both. However, to simplify our analysis, we assume that once a customer is identified as high-risk, his/her remaining (unused) credit line will be reduced to zero.

As noted in Section 4, the type of problem in which the target variable or output is binary is known as a *classification* problem in the machine-learning literature. Although the forecasts produced by the model described in Section 4 can be interpreted, under certain assumptions, as estimated delinquency probabilities, these numerical scores can be easily converted to binary decisions by comparing the forecast to a specified threshold and classifying customers with scores exceeding that threshold as high-risk. Setting the appropriate level for this threshold involves a tradeoff. A very low threshold results in many customers being classified as high-risk, and while it may correctly identify customer who are genuinely high-risk and about to fall behind on their payments, this choice may also result in many low-risk customers incorrectly classified as high-risk. On the other hand, a high threshold may allow too many high-risk customers to be classified as low-risk, yielding greater subsequent delinquencies and losses.

This type of trade-off is present in any classification problem, and involves the well-known statistical balancing act between Type-I (false positives) and Type-II (false negatives) error in classical hypothesis testing contexts. In the more practical context of credit-risk management, the trade-off can be made explicitly by a cost/benefit analysis of false positives vs. false negatives, and selecting the threshold that will optimize some criterion function in which such costs and benefits are inputs. Before turning to this cost/benefit analysis, we wish to develop further intuition for the statistical properties of this tradeoff.

To understand the statistical behavior of any classification algorithm, one commonly used performance metric in the machine-learning and statistics literatures is a 2×2 contingency table often called the “confusion matrix” for reasons that will become obvious once we define its entries (see the left side of Figure 14).¹⁵ The two rows correspond to the two types of

¹⁵A confusion matrix is typically used in supervised learning to describe actual and predicted classifications by the learning model. See Kohavi and Provost (1998) for a formal definition of this and other machine-learning terminology.

customers in our sample, low-risk or “Good” and high-risk or “Bad”, or *ex post* realizations of delinquency. In our application, we define Good customers as those who did not become 90-days-or-more delinquent during the forecast period, and Bad customers of those who did. The two columns correspond *ex ante* classifications of the customers into these same two categories, Good and Bad. When any forecast model is applied to a given set of customers, each customer falls into one of these four cells, hence the performance of the model can be gauged by the relative frequencies of the entries (see Figure 14). In the traditional Neyman-Pearson hypothesis-testing framework, the lower-left entry is defined as Type-I error and the upper right as Type-II error, and the objective of the classical statistician is to find testing procedures to reduce Type-II error (or maximize “power”) subject to a fixed level of Type-I error (or “size”).

The right side of Figure 14 contains the confusion matrix for our machine-learning model for December 2008 using a 3-month forecasting horizon from January to March 2009 and a threshold of 20%, i.e., accounts with estimated delinquency probabilities greater than 20% are classified as “Bad” and 20% or below as “Good”. For this month, the machine-learning model classified $96.37\% + 0.44\% = 96.81\%$ of the accounts as good, of which 96.37% were, in fact, good and 0.44% of which were bad, i.e., were 90-days-or-more delinquent sometime during January to March 2009. Similarly, of the $0.44\% + 2.13\% = 2.57\%$ that became 90-days-or-more delinquent during the three-month forecast period, the model correctly identified 2.13% of the accounts, a correct-classification rate of 82.9%. Of course this success comes at the cost of labeling 1.06% of customers who ended up not falling into the 90-days-or-more delinquency state as bad.

Four specific performance metrics are typically computed from the entries of the confusion matrix, and their formulas are given in Figure 14. “Precision” refers to the model’s accuracy in instances that the model classified an account as bad, while “Recall” refers to the number of bad accounts identified by the model divided by the actual number of bad accounts. “True positive rate” and “false positive rate” are self-explanatory. Ideally, we would like our model to have very high precision and recall, or, equivalently, to have a high “true positive rate” and a low “false positive rate”. The role of the classification threshold is now clear: if a lender wishes to aggressively identify bad accounts by using a lower threshold, such aggressiveness will result in more incorrect classifications of good accounts as bad, leading to the opportunity cost of lost interest income from cutting the credit lines of low-risk customers. Figure 15

			Example		
Actual Outcome	Model Prediction		Actual Outcome	Model Prediction	
	Good	Bad		Good	Bad
	Good	True Positive		False Negative	96.37%
	Bad	False Positive	True Negative	0.44%	2.13%
Performance Metrics			Example Performance Metrics		
Precision =		$TN/(TN+FN)$	Precision =		66.78%
Recall =		$TN/(TN+FP)$	Recall =		82.84%
True Positive Rate =		$TP/(TP+FN)$	True Positive Rate =		98.91%
False Positive Rate=		$FP/(FP+TN)$	False Positive Rate=		17.16%

Figure 14: Confusion matrix framework, model performance metrics, and numerical example based on December 2008 machine-learning forecasts of 90-days-or-more delinquencies over the 3-month forecast horizon from January to March 2009. Rows correspond to actual customer types (“Bad” is defined as 90-days-or more delinquent) and columns correspond to forecasted types. TN: True Negative, FN: False Negative, FP: False Positive; TN: True Negative.

illustrates such tradeoffs for four different threshold levels.

By applying our forecast model with varying levels of the classification threshold, we can trace out explicitly the trade-off between true and false positives, and this trade-off is plotted in Figure 16. The blue line, known as the Receiver Operating Characteristic (ROC) curve in the machine-learning literature, is the pairwise plot of true and false positive rates for various classification thresholds (green line), and as the threshold increases, Figure 16 shows that the true positive rate increases, but the false positive rate increases as well.

The ROC curve shows that the trade-offs are not linear, meaning that increase in true positive rates is not always commensurate with the increase false positive rates. If the cost of false positives is equal to the gain of true positives, the optimal threshold will correspond to the tangent point of the ROC curve with the 45° line. If the cost/benefit trade-off is not one-for-one, the optimal threshold is different, but can easily be determined from the ROC curve. We have provided a simple optimization framework in the appendix for making such trade-offs.

Table 9 presents a set of standard performance statistics widely used in machine learning literature (see Appendix A.1 for their definitions) for each of the 12 models discussed in Section 4 (see Table 5).¹⁶ Our models exhibit strong predictive power across the various

¹⁶The reported results are for classifiers based on the ROC-tangency classification threshold. In particular,



Figure 15: Confusion matrices of machine-learning forecasts of 90-days-or-more delinquencies for four classification thresholds. Rows correspond to actual customer types (“Bad” is defined as 90-days-or more delinquent). The numerical example is based on the December 2008 model forecast for the 3-month forecast horizon from January to March 2009.

performance metrics. The two simplest measures reported are the percentage of correctly and incorrectly classified observations, and the reported values show that our models routinely achieve approximately 99% correct classification rates. However, the correct-classification rate is not necessarily a sufficient measure of a model’s predictive power when the distribution of the dependent variable is highly skewed. Therefore, we report several more-sophisticated performance measures to capture other aspects of our model’s predictive power.

The kappa statistic is a well-known relative measure of predictive success in classification problems, where success is measured relative to purely random classifications. According to Landis and Koch (1977), a kappa statistic between 0.6 and 0.8 represents “substantial”, and a value greater than 0.8 implies “almost perfect” agreement. Table 9 shows that our model’s kappa statistic ranges from 0.72 and 0.83, indicating strong agreement between real and forecasted values, even after correcting for agreement due to chance. Both mean-absolute-error and root-mean-squared-error measures also indicate significant predictive power, as do the F -measures which are the harmonic means of the model’s precision and recall (see Figure 14), which are 0.73 or greater, implying a good balance between high precision and

for each model we constructed the ROC curve, determined the threshold that corresponded to the 45-degree tangency line, and used that threshold to classify observations into the two possible classes.

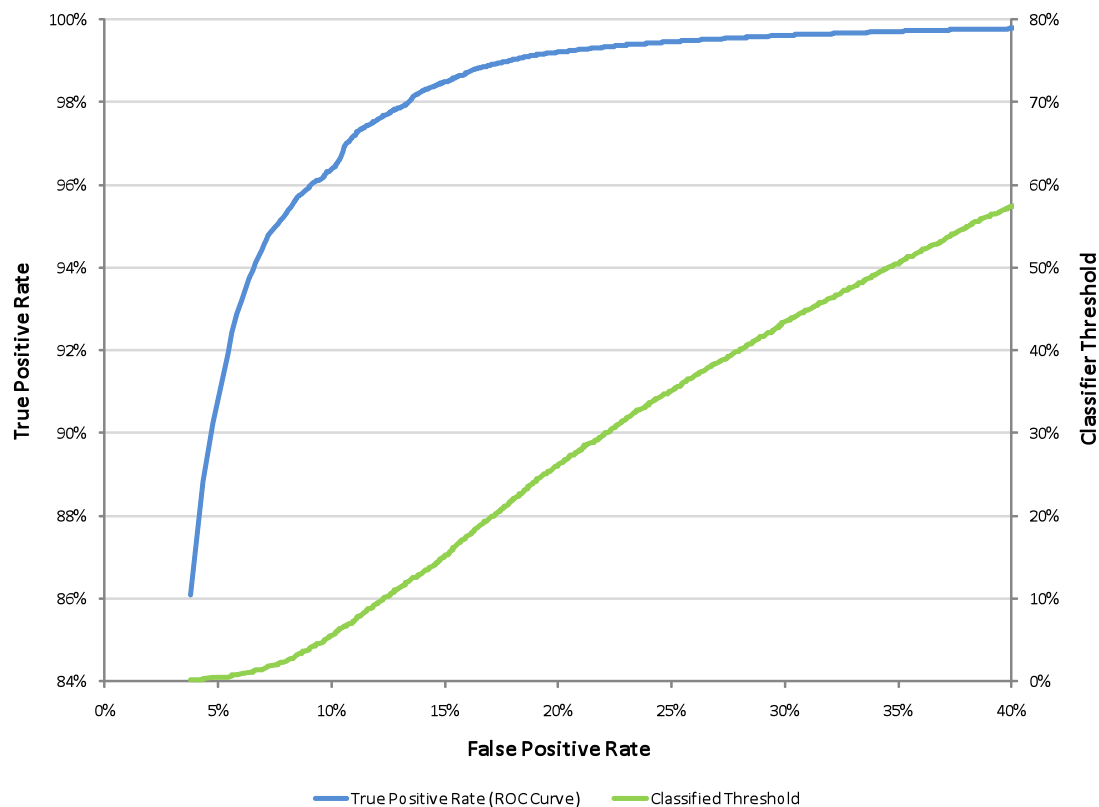


Figure 16: A Receiver Operating Characteristic (ROC) curve and corresponding classification threshold value of December 2008 machine-learning forecasts of 90-days-or-more delinquencies over the 3-month forecast horizon from January to March 2009.

Panel A: Performance Metrics for April 2008 to September 2008

Model for the Prediction Date of	Apr-08	May-08	Jun-08	Jul-08	Aug-08	Sep-08
Correctly Classified Instances (Rate)	0.989	0.991	0.991	0.991	0.991	0.991
Incorrectly Classified Instances (Rate)	0.011	0.009	0.009	0.009	0.009	0.009
Kappa statistic	0.751	0.753	0.735	0.751	0.751	0.749
Mean Absolute Error	0.006	0.004	0.004	0.004	0.004	0.004
Root Mean Squared Error	0.075	0.061	0.062	0.061	0.062	0.065
TP Rate	0.997	0.998	0.998	0.998	0.997	0.997
FP Rate	0.312	0.345	0.348	0.322	0.319	0.309
Precision	0.839	0.896	0.856	0.853	0.849	0.828
Recall	0.688	0.656	0.652	0.678	0.682	0.692
F-Measure	0.756	0.757	0.740	0.755	0.756	0.754
ROC Area	0.952	0.927	0.933	0.937	0.940	0.935

Panel B: Performance Metrics for October 2008 to March 2009

Model for the Prediction Date of	Oct-08	Nov-08	Dec-08	Jan-09	Feb-09	Mar-09
Correctly Classified Instances (Rate)	0.990	0.989	0.988	0.989	0.992	0.992
Incorrectly Classified Instances (Rate)	0.010	0.011	0.012	0.011	0.008	0.008
Kappa statistic	0.741	0.726	0.735	0.745	0.832	0.830
Mean Absolute Error	0.004	0.005	0.005	0.005	0.006	0.005
Root Mean Squared Error	0.065	0.067	0.069	0.070	0.075	0.073
TP Rate	0.997	0.997	0.997	0.997	0.995	0.993
FP Rate	0.320	0.356	0.346	0.320	0.129	0.036
Precision	0.827	0.848	0.854	0.839	0.803	0.734
Recall	0.680	0.644	0.654	0.680	0.871	0.964
F-Measure	0.746	0.732	0.741	0.751	0.836	0.834
ROC Area	0.937	0.921	0.929	0.924	0.944	0.895

Table 9: Standard performance statistics for machine-learning classification models of consumer credit risk. Each model is calibrated based on the prior 3-months of data (*Training Period*) and applied to the input data available on the *Prediction Date*. The results of classifications versus actual outcomes over the following 3-months (*Evaluation Period*) are used to calculate these performance metrics. See also Table 5 for the list of Training and Evaluation Periods.

high recall.

Finally, the area under the ROC curve is a widely used measure in the machine-learning literature for comparing models (see Figure 16), and can be interpreted as the probability of the classifier assigning a higher score, i.e., a higher probability of being of type-1, to a type-1 observation than to a type-0 observation. The ROC area of our model ranges from 0.89 to 0.95, which again shows that our machine-learning classifiers have strong predictive power in separating the two classes.

5.2 Robustness

Although the dataset used in our analysis is from a large consumer bank in the U.S., it does not cover the entirety of a consumer’s spending and saving activities. Consumers usually have business relationships with multiple banks and credit-card companies, hence it is rare that all elements of the feature vectors we described in Section 4.2 are available for a given account. To check the robustness of our model to missing features, we divide our data set into equally sized samples separated by the availability of features. More sophisticated analysis can be performed—using feature-selection techniques widely used in machine learning—to assign weights on features by their importance, however in this study, we focus on the simpler approach of rank-ordering the data by simply counting number of missing values, without differentiating among the features that are missing.

After dividing the data set in this manner, for each group we perform 10-fold cross validation (CV), i.e., stratifying the dataset into 10 bins, using 9 of them for training of the model and the remaining one for testing, and repeating this step 9 times using different set of bins for testing and training. Figure 17 contains the average confusion matrix (see Section 5.1) across the 10-fold CV results for the 90-days-or-more delinquency forecasts over a 6-month forecast horizon. The panels are labeled Group I through Group IV where Group I contains accounts with the largest number of missing features, and Group IV contains accounts with the fewest.

This analysis shows a significant improvement in both precision and recall values from Group I to Group II. Group I customers are typically single-account holders, i.e., individuals with only a checking account and a low number of transactions, also known as “thin-file” customers because their files contain few records. However, as a customer’s file becomes “thicker” with more transactions and accounts, the ability of machine-learning models to

forecast future delinquencies improves, presumably because of the additional features available for such customers.

To get a sense to the economic significance of the improvement possible with his model, figure 18 plots the Value Added (VA) measure (A.4) that is developed in Appendix A.2,¹⁷ estimated via the the 10-fold CV performed on four different forecasts of 90-days-or-more delinquencies over forecast horizons of 3, 6, 9, and 12 months. As expected, the model's forecast accuracy becomes better as the prediction horizon lengthens because the forecast problem is easier, e.g., predicting 90-days-or-more delinquency over a 12-month horizon is considerably easier than over a 3-month horizon. Also, across all four models, there is a significant improvement in VA from Group I to Group II, whereas the performance of the model in Groups III and IV are comparable to that of Group II, with Group IV yielding slightly lower performance.

¹⁷We assume an amortization period $N = 3$ years, a run-up before default of 30% ($B_d = 1.3B_r$), and an interest rate differential ($r = 5\%$).

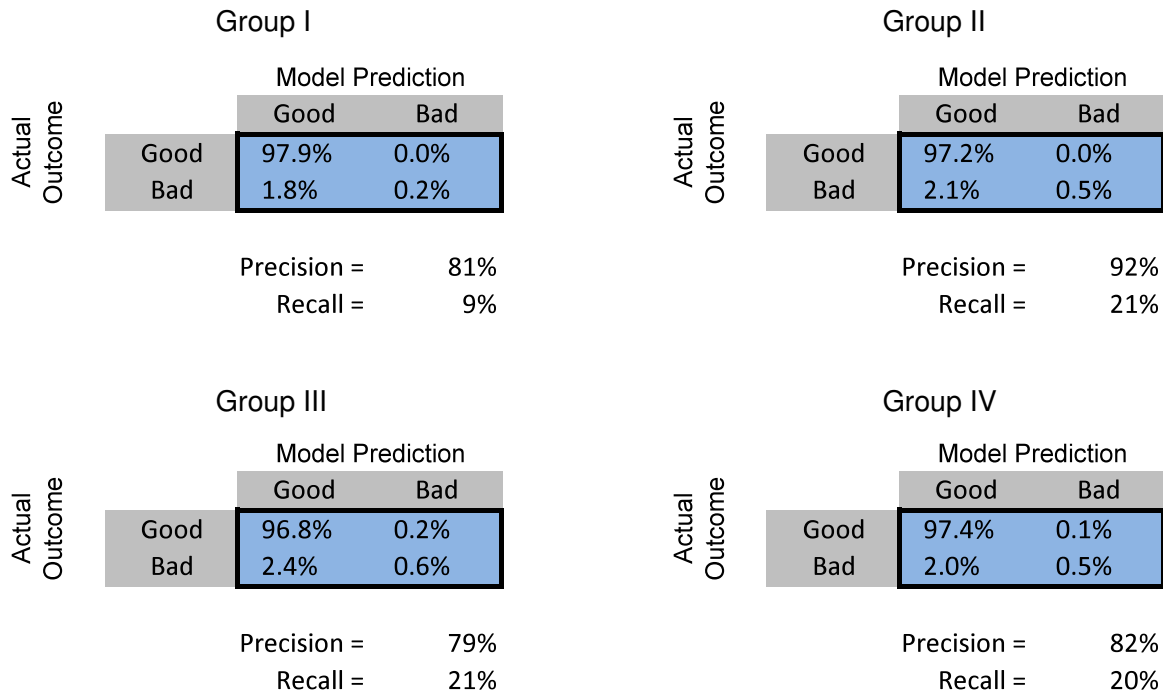


Figure 17: Average confusion matrices, averaged over 10 trials of machine-learning forecasts of 90-days-or-more delinquency over a 6-month forecast horizon on data from January 2005 through April 2008, with each trial based on a randomly chosen 10% subset of the entire dataset. For each trial, four models are estimated, one for each of four stratified sub-samples of equal size, stratified by on the number of non-missing features. Group I is the sample of accounts with the fewest available features (“thinnest”), and Group IV is the sample with the most (“thickest”).

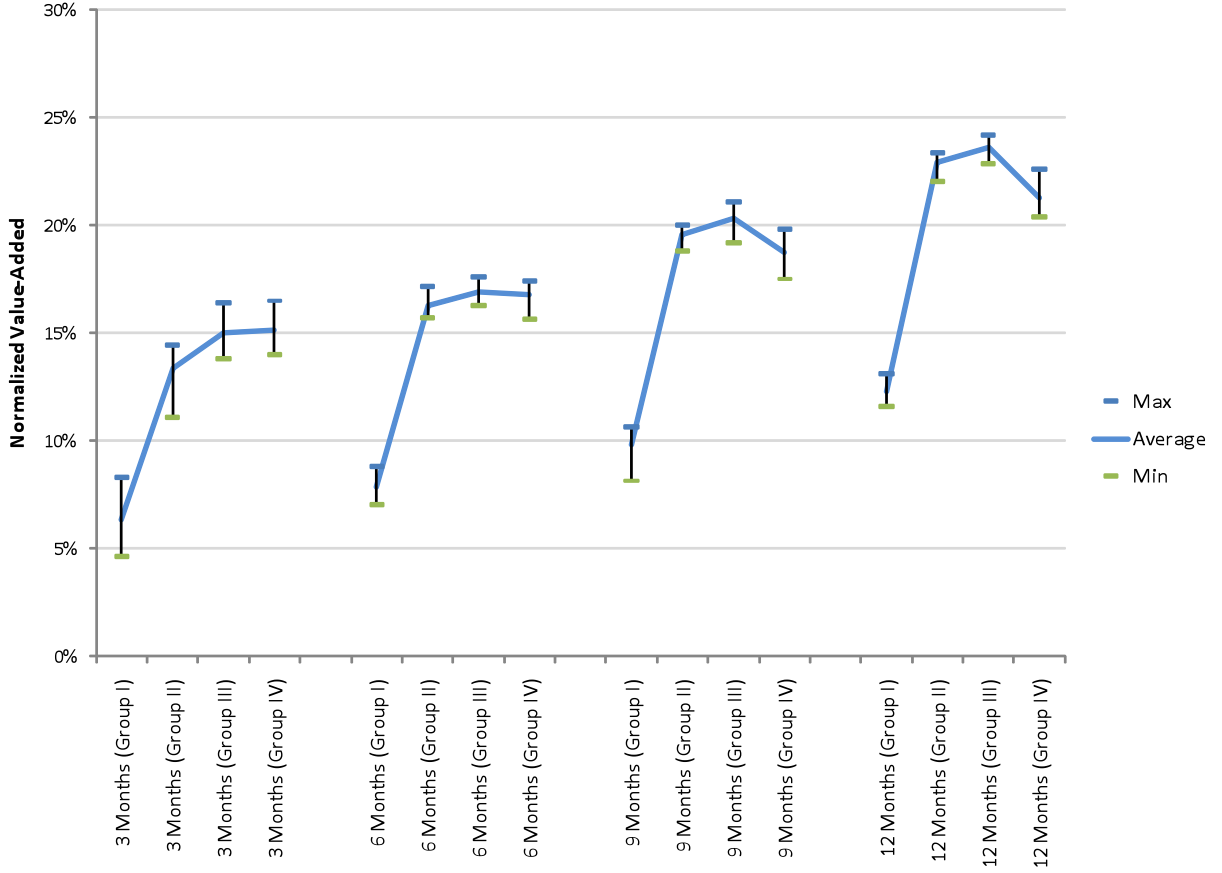


Figure 18: Value-added of machine-learning forecasts of 90-days-or-more delinquency over 3-, 6-, 9-, and 12-month forecast horizons on data from January 2005 through April 2008, for four equal stratified sub-samples of a randomly chosen 10%-subset of the data, stratified by on the number of non-missing features. Group I is the sample of accounts with the fewest available features (“thinnest”), and Group IV is the sample with the most (“thickest”). Minimum and maximum VA values are calculated using 10-fold cross validation from the entire data set using equation (A.4) with amortization period $N = 3$ years, run-up before default of 30% ($B_d = 1.3B_r$), $r = 5\%$, and the ROC-tangency classification threshold.

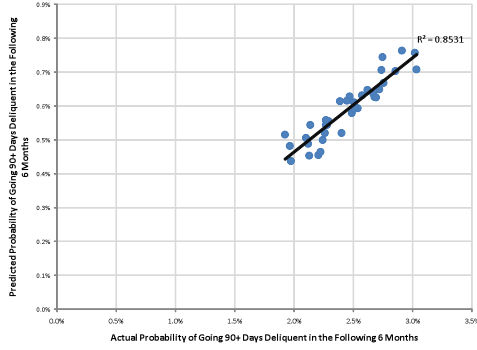
5.3 Macroprudential Risk Management

Machine-learning forecasts for individual accounts can easily be aggregated to generate macroeconomic forecasts of credit risk in the consumer lending business by simply tabulating the number of accounts forecasted to be delinquent over a given forecast period and for a specific classification threshold. The proportion of forecasted delinquencies in the population may then be used as an indicator of systemic risk for consumer lending, and its properties

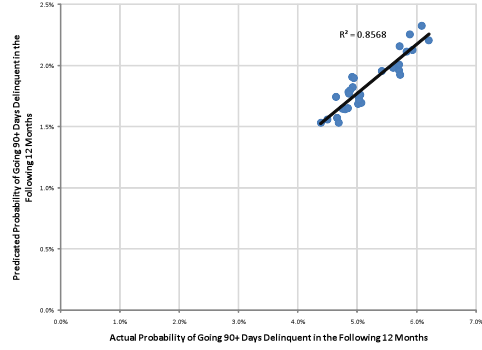
can also be summarized with confusion matrices with different classification thresholds. In this section, we construct such aggregate delinquency probabilities in the year prior to the credit crisis of 2007–2009, and show that this approach yielded increasingly higher forecasted delinquency probabilities, even during the boom years. This estimated trend suggests that machine-learning techniques applied to combined transactions and credit scores are capable of generating leading indicators of deterioration in consumer creditworthiness.

For purposes of aggregate risk measurement, a longer forecast horizon may be more appropriate, so we apply the same machine-learning algorithm of Section 4 to data from January 2005 to December 2008 and define the target variable as 90-days-or-more delinquency over a 6- or 12-month forecast horizon. Figure 19(a) and 19(b) show the aggregate results the machine-learning forecasts as compared to realized delinquencies for 6- and 12-month forecast horizons using the ROC-tangency classification threshold (in which the trade-off between true and false positive rates is one-to-one; see Section 5.1). The forecasted delinquencies are highly correlated with realized delinquencies—with linear regression R^2 's of 85% for both horizons—hence our forecasts are capturing the dynamics of the consumer credit cycle over the sample.

However, note that the forecasts consistently under-estimate the absolute level of delinquencies by a scalar multiple, which may be an artifact of the particular classification threshold selected. To correct for this underestimation, and to reduce the effects of overfitting, we use 10-fold cross validation to estimate the model's recall value (the number of accounts classified as bad as a percentage of the total number of bad accounts). Then we simply multiply the forecasted default rate by the reciprocal of this estimated recall rate. Figures 20(a) and 20(b), which contain time series plots of the adjusted forecasts as well as the realized aggregate delinquency rates, show that the adjustment factor is reasonably successful in correcting the underestimation bias, yielding forecasts that are accurate both in terms of level and dynamics.



(a) Predicted and actual 90-days-or-more delinquency rates (6-month)

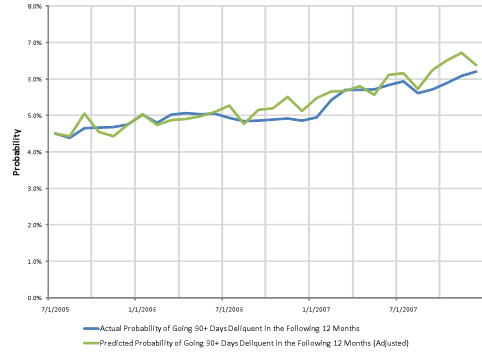


(b) Predicted and actual 90-days-or-more delinquency rates (12-month)

Figure 19: Forecasted and realized 90-days-or-more delinquency rates for 6- and 12-month forecast horizons, with ROC-tangency classifier threshold, based on data from January 2005 to December 2008. The horizontal axis starts in July 2005 since some features require 6 months of training data, and ends June 2008 or December 2007 since either 6 or 12 months of data are needed to calculate the realized delinquencies events to be compared again the predicted values.



(a) Time series of actual and predicted 90-days-or-more delinquency rates (6-month)



(b) Time series of actual and predicted 90-days-or-more delinquency rates (12-month)

Figure 20: Time series of predicted and realized 90-days-or-more delinquency rates for 6- and 12-month forecast horizons, with ROC-tangency classifier threshold, based on data from January 2005 to December 2008. The horizontal axis starts in July 2005 since some features require 6 months of training data, and ends June 2008 or December 2007 since either 6 or 12 months of data are needed to calculate the realized delinquencies events to be compared again the predicted values.

6 Conclusion

In the aftermath of one of the worst financial crises in modern history, it has become clear that consumer behavior has played a central role at every stage—in sowing the seeds of crisis, causing cascades of firesale liquidations, and bearing the brunt of the economic consequences. Therefore, any prospective insights regarding consumer credit that can be gleaned from historical data has become a national priority.

In this study, we develop a machine-learning model for consumer credit default and delinquency that is surprisingly accurate in forecasting credit events 3 to 12 months in advance. Although our sample is only a small percentage of the Bank’s total customer base, the results are promising. Our out-of-sample forecasts are highly correlated with realized delinquencies, with linear regression R^2 ’s of 85% for monthly forecasts over 6- and 12-month horizons. Moreover, crude estimates of the value-added of these forecasts yield cost savings ranging from 6% to 23% of total losses. Given the scale of industry wide charge-offs during 2008, even a 6% cost-savings would amount to hundreds of millions of dollars.

From a macroprudential risk management perspective, the aggregation of machine-learning forecasts of individuals may have much to contribute to the management of enterprise and systemic risk. Current credit-bureau analytics such as credit scores are based on slowly varying consumer characteristics, and therefore not as relevant for tactical risk management decisions by chief risk officers and policymakers. We find that machine-learning forecasts are considerably more adaptive, and are able to pick up the dynamics of changing credit cycles as well as the absolute levels of default rates.

We believe that our results are indicative of considerably more powerful models of consumer behavior that can be developed via machine-learning techniques, and are exploring further refinements and broader datasets in ongoing research.

A Appendix

In this appendix, we provide the definition for the performance metrics reported in Table 9 of Section 5.1 (Appendix A.1), and the details of the framework used to calculate the cost savings reported in Section 5.2 (Appendix A.2).

A.1 Definitions of Performance Metrics

Suppose a binary classifier has been applied to a sample of N observations. For each observation i , the model produces the probability that the observation with feature vector x_i belongs to class 1. This predicted probability, $f(x_i)$ is then compared to a threshold and observations are classified into class 1 or 0 based on the result of that comparison. For a given level of threshold, let True Positive (TP) be the number of instances that are actually of type 0 that were correctly classified as type 0 by the classifier, False Negative (FN) be the number of instances that are actually of type 0 but incorrectly classified as type 1, False Positive (FP) be the number of instances that are of type 1 but incorrectly classified as type 0 and, finally, True Negative (TN) be the number of instances that are of type 1 but incorrectly classified as type 1 (see Figure 14). Then one can define the following metrics to evaluate the accuracy of the classifier:

$$\text{Correctly Classified Instances (Rate)} \equiv (\text{TP} + \text{TN})/N$$

$$\text{Incorrectly Classified Instances (Rate)} \equiv (\text{FP} + \text{FN})/N$$

$$\text{kappa statistic} \equiv (P_a - P_e)/(1 - P_e) \quad , \quad P_a \equiv (\text{TP} + \text{TN})/N \quad ,$$

$$P_e \equiv [(\text{TP} + \text{FN})/N] \times [(\text{TP} + \text{FN})/N]$$

$$\text{Mean Absolute Error} \equiv 1/N \sum_{i=1}^N |f(x_i) - I_i|$$

$$\text{Root Mean Squared Error} \equiv 1/N \sum_{i=1}^N (f(x_i) - y_i)^2$$

$$\text{True Positive (TP) Rate} \equiv \text{TP}/(\text{TP} + \text{FN})$$

$$\text{False Positive (FP) Rate} \equiv \text{FP}/(\text{FP} + \text{TN})$$

$$\text{Precision} \equiv \text{TN}/(\text{TN} + \text{FN})$$

$$\text{Recall} \equiv \text{TN}/(\text{TN} + \text{FP})$$

$$F\text{-Measure} \equiv (2 \times \text{Recall} \times \text{Precision})/(\text{Recall} + \text{Precision})$$

where $f(x_i)$ is the model’s predicted probability that a customer with feature vector x_i is of type 1, and I_i is an indicator that is equal to 1 if customer i is type-1 and 0 otherwise.

A.2 Framework for Calculating Value Added

In this section, we provide a simple framework for analyzing the value added of classification-based algorithms for credit-line reduction. This framework requires some simplifying assumptions regarding the revenues and costs of the consumer lending business. As a starting point, suppose that in the absence of any forecasts, a lender will take no action with respect to credit risk, implying that customers who default or are delinquent will generate losses for the lender, and customers who are current in their payments will generate financing fees on their running balances. To make the analysis simpler, assume that all non-defaulting customers have a running balance of B_r , and defaulting customers also start with a balance of B_r but will increase their balance to B_d between the point at which the lender’s credit-risk management decision is made and the point when they default. For example, an empirically plausible set of parameters are $B_r = \$1,000$ and $B_d = \$1,200$. As it will be clear shortly, only the ratio of run-up balance, $B_d - B_r$, and the running balance, B_r , enters into the potential savings calculation. In this particular example, the critical factor is the 20% run-up in the balance before default.

While the lender’s loss from default is simply B_d , calculating the profit due to a customer in good-standing requires further discussion. For now, let the present value of the profit from such a customer be given by a fraction of the running balance, denoted by P_m for “Profit Margin Rate”, which can be a function of the interest rate and the expected length of time the account holder will use this credit line.¹⁸ Given these assumptions, we can calculate the profit with and without a forecast model as follows:

$$\begin{aligned} \text{Profit without Forecast} &= (TP + FN) \cdot B_r \cdot P_m - (FP + TN) \cdot B_d \\ \text{Profit with Forecast} &= TP \cdot B_r \cdot P_m - FP \cdot B_d - TN \cdot B_r . \end{aligned}$$

¹⁸For simplicity, we make the assumption that all “good” customers who are subjected to a line reduction immediately pay off their balances and close their accounts. To the extent that most customers do not behave in this manner, our estimates of the potential cost-savings are conservative.

Then the dollar savings is simply given by the difference:

$$\text{Savings} = \text{TN}(B_d - B_r) - \text{FN} \cdot B_r \cdot P_m \quad (\text{A.1})$$

where

$$\text{TN}(B_d - B_r) = \text{Savings due to correct decision}$$

$$\text{FN} \cdot B_r \cdot P_m = \text{Opportunity cost due to incorrect decision} .$$

We can simplify this even further by dividing the savings by the savings that would have been possible under the perfect-foresight case, i.e., the case in which all bad customers, and only bad customers, are identified and their credit was reduced. We will call this ratio the “value added” (VA) of the forecast, which is given by:

$$\text{Value Added} \equiv \frac{\text{TN} - \text{FN} \cdot P_m \cdot \frac{B_r}{B_d - B_r}}{\text{TN} + \text{FP}} \quad (\text{A.2})$$

Note that the final expression depends on the running and default balance only through the ratio of the running balance to the difference between the default and running balance. For example, it is clear from (A.1) that if this difference is zero, no value added can be achieved, which is expected because it is only the difference between B_d and B_r that can potentially be captured by a correct and timely decision by the lender.

Before attempting to evaluate VA, we must first specify how the profit margin rate, P_m , is calibrated. This factor is meant to capture the total financing charge that customers pay to the lender on their outstanding balances. This value depends on the credit-card interest rate as well as the speed with which customers pay off their balances. For simplicity, we assume that the difference between the credit-card interest rate and the lender’s funding cost is a fixed rate given by r , and customers pay back their entire credit card debt over an N -year period. In this simple setting, the financing charge is an annuity hence P_m is given

by: ¹⁹

$$P_m = r \cdot \frac{1 - (1 + r)^{-N}}{r} = 1 - (1 + r)^{-N}. \quad (\text{A.3})$$

Combining (A.2) and (A.3) yields the following expression for VA:

$$\text{VA}(r, N, TN, FN, FP) = \frac{TN - FN \cdot [1 - (1 + r)^{-N}] \cdot [\frac{B_d}{B_r} - 1]^{-1}}{TN + FP} \quad (\text{A.4})$$

¹⁹In the interest of parsimony and simplicity, we have implicitly assumed that the lender's funding cost is also r . For example, if $r=5\%$ implies that the lender pays 5% for funds and charges an interest rate of 10% on credit-card balances.

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Explainable Machine Learning Models of Consumer Credit Risk*

Randall Davis,[†] Andrew W. Lo,[‡] Sudhanshu Mishra,[§] Arash Nourian,[¶]
Manish Singh,^{||} Nicholas Wu,[§] Ruixun Zhang[§]

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Abstract

In this paper, we create machine learning (ML) models to forecast home equity credit risk for individuals using a real-world dataset, and demonstrate methods to explain the output of these ML models to make them more accessible to the end user. We analyze the explainability of these models for various stakeholders: loan companies, regulators, loan applicants, and data scientists, incorporating their different requirements with respect to explanations. For loan companies, we generate explanations for every model prediction of creditworthiness. For regulators, we perform a stress test for extreme scenarios. For loan applicants, we generate diverse counterfactuals to guide them with steps to reverse the model's classification. Finally, for data scientists, we generate simple rules that accurately explain 70-72% of the dataset. Our work is intended to accelerate the adoption of ML techniques in domains that would benefit from explanations of their predictions.

Keywords: Machine Learning; Interpretability; Explainable AI; Credit Lending; Inductive Logic Programming; Optimal Trees; LIME; SHAP; Counterfactual.

JEL codes: C55; C45; G0; G2; G51; C69.

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[†]MIT Computer Science and AI Laboratory;

[‡]Corresponding author; MIT Sloan School of Management; MIT Laboratory for Financial Engineering; MIT Computer Science and Artificial Intelligence Laboratory; 100 Main Street, E62-618, Cambridge, MA 02142, alo-admin@mit.edu.

[§]MIT Laboratory for Financial Engineering

[¶]Fair Isaac Corporation

^{||}smanish@mit.edu, MIT Computer Science and AI Laboratory; MIT Laboratory for Financial Engineering

1 Introduction

The total U.S. household debt at the end of the fourth quarter of 2020 is estimated to be \$14.56 trillion, with 189.6 million new credit accounts opened within the prior 12 months.¹ Given this massive amount of household debt, even a low delinquency rate can significantly affect the operation of the financial system. This potential impact makes the study of credit default risk an important real-world classification task.

Lenders generally use consumer credit ratings to grant and structure the terms of credit to consumers. To compute these credit ratings, a variety of factors that gauge the credit-worthiness of individuals have been described in the literature, which extends as far back as the 1940s (Chapman, 1940).

Recent advances in computing, innovative algorithms, and an explosion in the quantity of data have contributed to the growing success of complex nonlinear machine learning models, sometimes known as “deep” learning models. Unlike traditional machine learning techniques such as logistic regression and decision trees, deep learning models may have an extraordinary number of parameters. They are able to automatically learn nonlinear representations and interactions of input features from large datasets, a process that helps them to achieve superior performance compared to other machine learning methods. ML models have been explored for diverse applications in economics and finance (Gogas and Papadimitriou, 2021). They are widely used for a variety of different credit risk applications, including peer-to-peer lending (Ma et al., 2018; Duan, 2019), mortgage risk (Sirignano, Sadhwani, and Giesecke, 2016; Kvamme et al., 2018; Chen, Guo, and Zhao, 2021), credit card risk (Butaru et al., 2016), consumer credit risk (Jiang et al., 2021), and fair credit allocation (Tantri, 2021).

However, regulatory compliance is a major obstacle in the adoption of black-box models for credit risk modeling. In the United States, the Fair Credit Reporting Act of 1970 mandates that lenders must be able to disclose up to four key factors that adversely affected the credit score of a rejected consumer. More recently, the European Union’s General Data Protect Regulation (2018) created a right to explanation, whereby a user may ask for an explanation of an algorithmic decision that was made about them (Goodman and Flaxman, 2017).

¹Source: New York Fed Consumer Credit Panel/Equifax

Apart from regulatory compliance, different stakeholders have different requirements for explanation and transparency. For example, a loan applicant might like to know the possible steps that she could follow to make her creditworthy. Similarly, loan companies may need to provide explanations for their decisions regarding creditworthiness of applicants, while system developers may need to understand the specific features and relationships that underpin their models.

In this work, we first model a credit risk forecast on a real-world home equity line of credit (HELOC) dataset released by the Fair Isaac Corporation (FICO). We developed a wide range of ML models, including interpretable rule-based models (e.g., inductive logic programming and optimal trees) and black-box ML models (e.g., neural networks and random forests). We compare these models and find that neural networks outperform other linear and nonlinear models, reaching 74.75% accuracy. We also find that simple rules can explain 70-72% of the dataset.

The “explainability” of a model means the ability to give answers to the different stakeholders involved in the decision-making process (Croxxson, Bracke, and Jung, 2019). In this work, we identify the different stakeholders involved in credit risk management—loan companies, regulators, loan applicants, and data scientists—and provide explanations of the models according to their needs.

For loan companies, we generate interpretable explanations for every prediction using two post hoc explainability tools, Local Interpretable Model-Agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP), that are able to provide reasons for a loan denial. We modify the methodology of the LIME algorithm to incorporate the constraints on our data. The ability to generate explanations makes a model transparent about its functionality, in this case, informing loan companies about the relationships it has learned. For example, we identify that having a higher average age of line of credit leads to a decrease in the probability of default. These explanations also assist in finding representative borrowers from the past, for example, to find an accurate applicant from past to aid loan officers.

For regulators, we suggest a method of investigating potential regarding fairness of these models. Regulators may also be involved in investigating the model’s functionality in extreme scenarios. In anticipation of this issue, we provide a methodology to perform stress tests to help the regulators analyze the model’s behavior in extreme scenarios.

Since loan applicants are interested in getting approval for a loan, we generate a diverse set of suggestions that can help individuals deemed non-creditworthy become creditworthy. For individuals who already have been approved for loans, we are able to generate suggestions that can help them remain creditworthy. These counterfactual suggestions are generated by incorporating constraints based on specific items of domain knowledge, which makes these suggestions more usable in practice. We generated suggestions successfully for 99% of the individuals in our testing dataset.

Finally, for data scientists, we are able to summarize the dataset using a few simple rules that can help them understand the relationships and structure in the dataset. Understanding these relationships may provide a data scientist with the insights to develop better models in the future. Using inductive logic programming, we are able to explain 70% of the dataset using a single rule, and can explain up to 72% of the dataset using two simple rules.

Our work demonstrates that the functionality of black-box ML models can be explained to a range of different stakeholders, if the right tools are applied to the task, unlocking the future potential of applying AI to improve credit modeling. New explainable AI tools will be able to help stakeholders make counterfactual predictions and explain a model's output, leading to the ability to answer many "what-if" questions. This includes stress tests for extreme scenarios, and informing applicants about the factors which will improve their creditworthiness. More generally, it demonstrates the importance and potential of explainable AI to affect traditional fields like credit modeling significantly.

The remainder of this paper is structured as follows. Section 2 discusses the relevant literature. Section 3 describes the different models used in our paper, while Section 4 describes the dataset used in the analysis. Section 5 evaluates and compares different models. The explainability for various stakeholders is discussed in Section 6. Section 7 summarizes our findings and concludes the paper.

2 Literature Review

A range of statistical and operational research methods has been used over the long history of credit scoring models (Thomas, 1999). Most recently, machine learning techniques have been adopted for credit risk models. Lessmann et al. (2015), Thomas (1999), Breeden (2020),

and Gogas and Papadimitriou (2021) give detailed surveys of ML methods for credit risk forecasting. However, Gogas and Papadimitriou (2021) also highlight the shortcomings of ML with regard to the explainability of the models for fintech applications.

An extensive literature has been developed about explainable AI for models in health-care, computer vision, and natural language processing (Molnar, 2020). Even though these techniques were developed for other domains, some have been adopted for explaining credit risk forecast models.

A variety of solutions have been proposed to deal with the shortcomings of ML models used in credit risk forecasting. Interpretable ML models have been used by Khandani, Kim, and Lo (2010) (decision trees for a consumer credit risk model), and Obermann and Waack (2016) (a multiclass rule-based model for corporate credit ratings). Similarly, Dumitrescu et al. (2021) propose an interpretable penalized logistic tree regression model for credit scoring, while Chen et al. (2018) use an interpretable two-layer additive risk model for a home equity line of credit dataset. A two-layer additive risk model is the award-winning model in the recent FICO data science challenge. For these models, interpretability generally comes at the cost of performance compared to black-box models. For example, Caruana and Niculescu-Mizil (2006) found that random forests outperform decision tree classifiers. In this paper, we show that a neural network outperforms the two-layer additive risk model proposed in Chen et al. (2018).

Another category of solutions includes post hoc methods. Bussmann et al. (2020) use Shapley values to explain tree-based ensemble models and apply correlation networks to group the borrowing companies using the derived explanations. Similarly, Hadji Misheva et al. (2021) use the post hoc methods of Local Interpretable Model-agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP) to obtain local and global explanations for models trained on a Lending Club dataset. Albanesi and Vamossy (2019) propose a deep learning-based approach that combines the outputs of tree-based ensemble models and neural networks to predict consumer default, and use SHAP to provide model explanations. Other methods that use SHAP for explaining credit risk models include Ariza-Garzón et al. (2020) and Bracke et al. (2019). More recently, Qadi et al. (2021) propose a human-in-the-loop ML method that combines a post hoc explanation from SHAP with explanations from credit risk experts. Other methods of post hoc interpretability include layerwise relevance

propagation and activation analysis of hidden units of a neural network (Ponomareva and Caenazzo, 2019). Rudin and Shaposhnik (2019) propose a minimum set cover problem to generate a rule-based summary of a machine learning model on the home equity line of credit dataset. Similarly, Martens et al. (2007) propose a rule extraction method from a trained support-vector machine (SVM) model for credit scoring.

While these methods generate explanations of model outcomes, they are not tailored specifically for the different stakeholders involved in the credit risk pipeline. In addition, they do not embed the specific constraints of the dataset in their methodology of generating explanations, which may lead to explanations that are not practical or useful. Interpretability for different stakeholders has been introduced in a demo by IBM², but it remains incomplete, and does not use the state-of-the-art methods that we employ in this work.

In this paper, we present a methodology of generating explanations of black-box models for different stakeholders, a direction that has not been well explored previously. In the process of generating these explanations, we embed constraints to ensure that explanations and suggestions are meaningful and can be adopted for real-world use.

3 Machine Learning Methods

The fundamental goal of credit scoring is to determine the creditworthiness of an individual. Simply put, it is a binary classification task that labels credit applicants as creditworthy or non-creditworthy. A creditworthy applicant is likely to repay their financial obligation, while a non-creditworthy applicant is not.

We frame the consumer credit risk classification problem as predicting the probability of default for a borrower. More specifically, given a set of features, $x_1, x_2, \dots, x_k$, for a borrower i , the task is to predict a variable y_i , that is, the probability of default, $Pr(Default)$. The features x_i describes the borrower's credit history, for example, the number of lines of credit and the number of times the borrower defaulted in the past, among others. In practice, y_i is determined by long-standing credit scoring models, which are characterized by decisions such as whether the applicant had 90+ days delinquency in the two years after the opening of a new line of credit.

²<https://aix360.mybluemix.net/data>

We frame the classification as a supervised learning problem in which we train a function f that can approximate the relationship $y_i = f(x_1, x_2 \dots x_k)$. To train the function f , we use variety of ML models, including optimal classification trees, random forests, inductive logic programming, and neural networks. We describe these models next.

3.1 Random Forests

Random forests are an ensemble supervised learning technique. This technique aggregates multiple outputs from a set of predictors, in this case, multiple decision trees, in the belief that this will produce a more accurate classifier.

The key idea behind random forests is that a high-performing classifier can be constructed from a set of non-expert classifiers which are decision trees. A single decision tree is a supervised learning method that predicts the value of a target variable by learning simple if-then decision rules. It is constructed using the Classification And Regression Tree (CART) algorithm (Breiman et al., 1984). Each node in the decision tree is a condition on the value of a single feature that splits the data into two subsequent branches. CART recursively identifies the feature-value pair that best minimizes the tree's Gini impurity, a metric of the disorderliness of the labels of a set of data points.

Random forests are trained via a method called bootstrap aggregation, or bagging. The training data points are first randomly assigned into n groups with replacement, where n corresponds to the number of decision trees. Individual decision trees are fitted to a randomly chosen set of features in each group. To classify a new data point, the random forest aggregates the predictions from each of its constituent decision trees, and uses the majority vote as its classification. The random sampling of data points and features ensures that the resulting decision trees are uncorrelated. Thus, by aggregating their independent predictions, random forests are able to reduce variance and improve generalizability.

Random forests are difficult to interpret. For individual data points, each decision tree gives its if-else conditions that lead to a classification, but when these conditions are combined over the many trees of an ensemble, interpreting these conditions is impossible. This makes random forests effectively a black-box model.

3.2 Inductive Logic Programming

Inductive logic programming (ILP) (Muggleton, 1991) involves using first-order logic to represent and explain data. The dataset can be represented by a finite set of rules or clauses.

ILP requires that we specify the number of rules N , a given maximum rule size R , and the dimensional n binary input vector X . We construct Π , a $N \times R \times 2n$ tensor, and interpret $\text{softmax}(\Pi[i, j])$ as a probability distribution for the j th term in the i th rule; that is, we obtain a $2n$ -sized discrete probability distribution over the n features and their negations.

The rules are learned in a disjunctive normal form that consists of clauses with $\hat{\wedge}$ (AND) and $\hat{\vee}$ (OR) conditions. The AND and OR operators require binary operands. However, these operators must be extended to continuous operands to learn clauses from data. We use the product as a continuous extension of $\hat{\wedge}$, while a continuous extension of $\hat{\vee}$ is obtained from DeMorgan's Law, $\hat{\vee}(x) = 1 - A(1 - x)$, where A is the continuous extension of $\hat{\wedge}$. From parametrization and using these continuous extensions of $\hat{\vee}$ and $\hat{\wedge}$ on $[0, 1]$, we can compute an approximated label $\hat{y}$ for an input vector X by concatenating X with its $1 - X$ to get a $2n$ -vector X^* :

$$\hat{y} = \widehat{\bigvee_i} \widehat{\bigwedge_j} (X^* \cdot \text{softmax}(\Pi[i, j]))$$

where i ranges over the number of rules and j ranges over the size of a rule.

The binary cross-entropy loss between $\hat{y}$ and ground truth y can be minimized using stochastic gradient descent in order to learn the probability distributions in Π . Once the model is trained, we use the probability distribution Π to obtain a set of logical rules in disjunctive normal form, for all n from 1 to N and r from 1 to R , using the following expression:

$$\text{target} \leftarrow \bigvee_{n=1}^N \left(\bigwedge_{r=1}^R \text{argmax}_k (\Pi[n, r, k]) \right)$$

In our work, X is a numerical vector with each dimension corresponding to a different feature value. We first rank-normalize it to change its range to $[0, 1]$, and use the transformation

T to binarize it: $T(x) = \sigma(a(X - b))$, where σ is the sigmoid function, a is the fixed scale parameter, and b is the parameter learned during optimization.

Using the inductive logic programming model, we can thus learn simple interpretable rules that can be used for classification and summarizing datasets.

3.3 Optimal Classification Trees

Decision trees are constructed in a top-down greedy way using the CART algorithm (Breiman et al., 1984) as described in Section 3.1. At every node, the split is decided locally without the knowledge of other nodes. This makes decision trees only one-step optimal, leading to poor performance when classifying unseen points.

Optimal trees (Bertsimas and Dunn, 2017) are a variant of decision trees that are learned in a globally optimal manner. Optimal trees decide their split in one step, with knowledge of all other splits. The tree learning process is modeled as a mixed-integer optimization problem, which can be solved using fast available optimizers. Because optimal trees are constructed in a globally optimal fashion, they perform better than decision trees, and have all the advantages of other decision trees in terms of explainability. However, optimal trees become difficult to explain if they are very deep, and the rules learned at every node are complex. In addition, the number of variables involved in optimization for creation of optimal tree model is a linear function of dataset size and an exponential function of maximum depth. In general, mixed integer optimization does not scale well with a large number of variables. As a result, deeper optimal trees take a longer time to train on large datasets compared to decision trees, which limits their use for Big Data problems.

3.4 Neural Networks

Neural networks (NN) are supervised learning models loosely inspired by the biological networks of the human brain.

A NN is composed of three types of layers: an input layer, a number of hidden layers, and an output layer. An input layer relays the input features into the model, the hidden layers act as the computational engine, and the output layer generates the final model prediction. The input to each hidden unit is a linear combination of the units of the preceding layer. The hidden unit then computes its output by mapping its input through an activation function.

A nonlinear activation function is commonly used to create nonlinear interactions between the units of the neural network. It is worth noting that logistic regression is a special case of a NN, with one hidden layer containing one hidden unit with a sigmoid activation function.

In NNs, the value of each hidden unit can be computed as:

$$h_j(x) = f(w_j + \sum_{i=0}^n w_{ij} \cdot x_i).$$

Here, w_{ij} is the weight from input x_i to hidden unit h_j . The weights w_{ij} can be learned by minimizing the loss function using optimizers like stochastic gradient descent.

The weights of the neural network create complex and nonlinear interactions between input features, which do not have human-interpretable meanings. However, the complex nonlinear interactions learned by NNs lead in general to better model performance. Moreover, the decision boundaries learned by NNs can be both high-dimensional and extremely nonlinear. Thus, learned features may have varying significance at different points of the feature space.

4 Data

For our task, we used a home equity line of credit (HELOC) dataset provided by FICO. A HELOC is a revolving loan in which the collateral is the borrower's equity in their house. Like a credit card, a HELOC is available for a set time frame during which a borrower can withdraw money as needed.

The FICO dataset contains 10,459 borrowers who were granted HELOCs during a two-year application window from March 2000 to March 2002. In March 2003, a year after the application window had closed, a performance snapshot was captured, and the risks of the borrowers were evaluated. In this dataset, an applicant might have used their HELOC for a duration between one and three years, depending on the time of their approval. The dataset contained 5,459 non-creditworthy records, while the remaining 5,000 records were deemed creditworthy. In addition to the binary target variable of credit risk classification, each credit applicant is characterized by 23 predictor features, 21 continuous and 2 categorical. Further information about these features, and important terms relevant to the dataset, are provided

in Appendix B.2.

The records in our dataset contain special values, which require a careful approach. We deal with special values by discretizing continuous features into bins. The advantage of binning is that special values can be treated as a separate bin, and any outliers can be consolidated. Once a binning schema has been decided, a feature can be represented using one-hot encoding and weight of evidence (WoE) encoding.

In one-hot encoding, a feature that contains n bins can be represented as an n -dimensional vector f . If a feature value belongs to bin_i then the value of its k^{th} dimension $f_k = 1$ if $k = i$ and 0 otherwise, for $k \in [0, n)$. The drawbacks of one-hot encoding are that bins are treated as unordered categories, and sparsity is introduced. Sparse features can result in overfitting and biased parameters in a model if the training dataset is small. In addition, for continuous variables, there is no clear-cut formula to define the binning schema, and choosing it manually can be sub-optimal.

Weight of evidence (WoE) encoding is a popular statistical technique used in the credit rating industry (Siddiqi, 2012). It is used to automatically recode the values of continuous and categorical predictor variables into discrete bins, and to assign each bin a WoE value. The bins are determined such that they will produce the largest differences with respect to the WoE values. Additionally, monotonicity constraints can be specified to ensure that WoE values are strictly increasing or decreasing in feature values.

The formula for WoE encoding is derived from entropy theory and the information value. For bin_i , the WoE value can be computed as follows:

$$WoE_i = [\ln(\frac{\text{Relative Frequency of Goods}}{\text{Relative Frequency of Bads}})] * 100 \quad (1)$$

where the relative frequency of goods is defined as the ratio of the number of creditworthy individuals in bin_i to the total number of creditworthy individuals, and the relative frequency of bads is defined as the ratio of the number of non-creditworthy individuals in bin_i to the total number of non-creditworthy individuals.

Intuitively, the WoE value of a bin provides a measure of its predictive ability to separate creditworthy and non-creditworthy applicants. An important benefit of WoE encoding is

that it can be used to treat missing values and outliers without introducing sparsity. As WoE values are on the same scale, we can use them to compare the univariate effects of bins on the target variable within a feature or across all features. Its drawback, like most binning techniques, is that it results in a loss of information. As we will see in Section 5, models trained on WoE encoded data have a better performance than other methods.

5 Evaluation

In this section, we evaluate the different models described in Section 3. These machine learning models, including optimal trees, random forests, and neural networks, are trained by recoding all the features using weight of evidence encoding. An inductive logic programming model is trained using the methodology described in section 3.2. We also evaluate the neural network model trained on one-hot encoding version of the dataset. The details of these implementations and best-performing models are included in Appendix C for reproducibility.

We include the two-layer additive risk model in our evaluation (Chen et al., 2018), which was the best performing model of the FICO Data Challenge³. It partitions features into different subgroups, combining scores from different subgroups into a global model score. Its feature subgroups are generally interpretable, as they are created through the intervention of an expert in the field. The model resembles a two-layer sparse neural network. This model serves as a baseline for the other black-box models.

We evaluate these models using K-fold cross-validation. In K-fold cross-validation, the input dataset is randomly partitioned into K equal subsets. In each run, one of the subsets is chosen as the test set and the remaining as the training set. For this analysis, we choose $K = 5$, i.e., in each run of cross-validation, the training set contains 80% of the dataset (7,898 points), and the test set contains the remaining 20% (1,973 points). The class distributions of the train-test sets are included in Table C.1 in the appendix.

We evaluate the models using different metrics: accuracy, area under the curve (AUC), false positive rate (Type I error) and false negative rate (Type II error). The metrics are averaged across all five cross-validation datasets. The AUC measures the ability of a classifier to distinguish between the classes. Our dataset is fairly balanced, as is illustrated in Table

³<http://dukedatasciencefico.cs.duke.edu/>

C.1, hence the accuracy is also a good measure of goodness of fit along with the AUC. All models predict the $Pr(Default)$ values for every sample. To classify the different samples, we used a threshold of 0.5.

As Table 1 illustrates, we find that the NN model trained on WoE data has the best performance, with an accuracy of 74.75%. It outperforms the NN trained on one-hot encoded data, implying that the WoE encoded features have more information than one-hot encoded features. The two-layer additive risk model performs second best, with an accuracy of 74.12%. This model is easy to interpret; however, it requires the input and involvement of experts who know about the constraints in the dataset and the relationship between its different features (in order to divide features into different subgroups) that might not always be available for various applications.

On the other hand, the rule-based models, optimal trees (black box) and random forests suffer from the problem of explainability. For random forests, analyzing the ensemble of 140 trees at a time is intractable. Similarly, the optimal trees (black box) model has a single deep tree with multiple features deciding every split, which is difficult to interpret. If we train a shallow optimal tree, its accuracy drops to 72.28%, but the model is easy to interpret and analyze. We also obtain a small set (1 or 2) of simple rules from inductive logic programming that are easy to understand, but they come at the cost of a decrease in accuracy. Even though interpretable rule-based models do not perform as well as NN model, we will see in Section 6 that they are useful for explainability for the different stakeholders involved.

Using ILP, we obtain a single simple rule, that is, “if *ExternalRiskEstimate* is smaller than 72, then classify borrowers as non-creditworthy,” achieves an accuracy of 70.65%. *ExternalRiskEstimate* is a condensed version of the borrower’s credit risk, a metric similar to the FICO score. In general, companies use the credit score and a threshold associated with it to decide an individual’s creditworthiness; in our case, it is the *ExternalRiskEstimate* with a threshold of 72. We observe that the use of machine learning models using different features about a borrower’s credit history can increase the accuracy of the delinquency forecast by 4 percentage points compared to the naïve use of credit scores.

Credit companies are naturally interested in using models with the best performance. If credit companies give loans to borrowers by misclassifying defaulters as non-defaulters

(that is, false negatives), they will suffer losses. Similarly, if credit companies deny loans to borrowers who can repay (that is, false positives), they will lose business. Given the enormous size of the mortgage business in the United States, even a small increase in model performance can create a substantial impact. However, regulatory practice and the black-box nature of the models prevent them from harnessing these benefits. This motivates us to analyze the explainability of these different models.

Models	Test Accuracy	Test AUC	False Positive Rate	False Negative Rate
ILP (1 rule)	70.65	70.60	30.37	28.41
ILP (2 rules)	71.01	70.84	33.02	25.28
Optimal Trees (Interpretable)	72.28	74.18	28.25	27.21
Optimal Trees (Black Box)	74.12	74.46	29.16	22.85
Random Forest (140 Trees)	73.77	79.82	29.71	23.01
Two Layer Additive Risk	74.12	81.01	30.31	21.77
NN (one-hot)	74.10	80.59	25.16	26.60
NN (WoE)	74.75	81.40	28.31	22.42

Table 1: 5-Fold cross-validation performance of ML models. The NN trained on WoE data performs the best with a mean test accuracy of 74.7% and AUC of 81.4%. A higher AUC implies the NN model is better able to distinguish between the creditworthy and non-creditworthy classes compared to other models. All rule-based classifiers have smaller values of AUC compared to the others. The NN (one-hot) has the minimum false positive rate, while the two-layer additive risk model has the minimum false negative rate when the threshold used to classify the model’s output is 0.5. All the reported values are in percentages.

6 Explaining Machine Learning Models

We have shown in Section 5 that the neural network model outperforms every other model tested. However, the non-interpretable nature of neural networks limits its adoption. In this section, we analyze the explainability of models in order to help their wider adoption for

different applications.

The multiple parties involved in credit risk management require different explanations for different purposes. For instance, loan applicants who are denied loans are interested in finding the reason for the denials and suggestions that can make them more creditworthy. Data scientists, on the other hand, are more interested in understanding the data, while regulators demand fairness from the models and analyze the model's behavior in extreme scenarios. We characterize the kinds of explanation that are appropriate for the following end users: loan companies, data scientists, loan applicants, and regulators.

6.1 Interpretability for Loan Companies: Opening the Black Box

Loan companies use machine learning models to evaluate the creditworthiness of a borrower. Regulations require the loan companies to give a set of reasons for every denial of the application. Consequently, loan companies are interested in finding the factors that contribute to the creditworthiness of the individual. In addition, they require explanations for every prediction of the model. These explanations make the model transparent, and they assist in finding the most representative samples (borrowers from the past) for a data point (a new borrower). We use state-of-the-art methods to generate explanations for our model predictions, including LIME (Ribeiro, Singh, and Guestrin, 2016) and SHAP (Lundberg and Lee, 2017). We generate explanations for the best performing NN model.

LIME

LIME is a model-agnostic technique that approximates the decision boundary of a model at a particular data point using a linear approximation⁴. This linearity makes the LIME model interpretable. The approximation is constructed by training a locally-weighted linear regression model in the neighborhood of the data point of interest. The coefficients of the regression can be used to justify the data point's classification.

Constructing a locally-weighted linear regression model involves sampling and perturbing the data points that are used for training. One of the shortcomings of LIME is that the perturbed data points sampled by LIME may be invalid. For example, imagine a dataset

⁴The key idea behind LIME is that every segment of the decision boundary begins to look linear at increasingly smaller scales.

with two features A and B, with a constraint that $A < B$. Sampling each feature’s value independently, as is done in LIME’s original algorithm, may produce perturbed data points where this constraint is violated. We address this shortcoming of LIME by modifying the algorithm to take into account potential interdependencies of different input features.

This modification changes the methodology of sampling a data point of the LIME algorithm. In its original implementation, a data point with n features is sampled by independently sampling each of its n features from univariate normal distributions. In our modified implementation, a data point is sampled directly from a joint multivariate normal distribution across all features. This allows perturbations to be informed by the correlation of features. The multivariate normal distribution is still centered on the mean of each feature value, but the standard deviation along each feature is determined by the correlation matrix of the input data.

We first evaluate our modified LIME with respect to the validity of the perturbed data points. As noted, LIME samples perturbed points in the neighborhood of the input data point. We generate a set of 5,000 perturbed data points from several univariate normal distributions centered on the feature means and a single multivariate normal distribution centered similarly. We measure the quality of the sampled points by two metrics.

Correlation. The correlation between features of the perturbed data points should be similar to that of the features in the training dataset. We compute the mean-squared error (MSE) between the correlation matrix of the training data and that of the perturbed points generated by both implementations of LIME. Let us call these values $MSE^{original}$ and $MSE^{modified}$, respectively. We find that $MSE^{modified}$ is much smaller than $MSE^{original}$ (0.000 vs 0.053). This is expected because the correlation matrix of the training data is an input to the multivariate normal distribution that samples perturbed points in our modified implementation. Our perturbed data more closely resembles the characteristics of the original dataset.

Constraint Violation. There are 12 constraints relevant to the HELOC dataset (6 relational constraints and 6 value constraints). We determined these via discussion with FICO personnel. Some examples of these constraints are, “All feature values with percentage units

should be smaller than 100,” and “The number of lines of credit not delinquent must be less than the total number of lines of credits.” The 5,000 perturbed data points are scanned for violations of these constraints. The results for all constraints are shown in Table 2. We see that the modified implementation of LIME produces fewer violations than the original implementation in 7 out of the 12 constraints. Moreover, a perturbed point sampled by the modified implementation has on average 1.954 constraint violations, versus 2.627 for the original implementation. Hence, we conclude that we use more realistic data in our modified LIME for generating LIME approximations.

In the remainder of the paper, we use this modified LIME for our analysis. Also, whenever we use the term LIME, we refer to our modified LIME.

Index	Constraint	Original LIME Violations	Modified LIME Violations
1	All feature values interpreted quantitatively must be non-negative	4383	4024
2	$\text{PercentLOCNeverDelq} \leq 100$	1198	1183
3	$\text{PercentInstLOC} \leq 100$	1	2
4	$\text{PercentLOCWBalance} \leq 100$	340	296
5	$\text{FracRevLOCLimitUse} \leq 100$	62	75
6	$\text{FracInstLOCUse} \leq 100$	509	506
7	$\text{NumLOC90PlusDaysDelq} \leq \text{Num-LOC60PlusDaysDelq}$	1656	655
8	$\text{NumLOCReqLast6MExPastWeek} \leq \text{Num-LOCReqLast6M}$	2095	388
9	$\text{NumLOC60PlusDaysDelq} \leq \text{NumTotalLOC}$	191	239
10	$\text{NumLOC90PlusDaysDelq} \leq \text{NumTotalLOC}$	192	233
11	$\text{NumLOCNotDelq} \leq \text{NumTotalLOC}$	2257	1915
12	$\text{NumLOCInLast12M} \leq \text{NumTotalLOC}$	249	255

Table 2: Comparison of constraint violations among 5,000 perturbed points sampled from the original and modified implementations of LIME. The points sampled from the modified implementation of LIME have fewer violations in 7 out of the 12 constraints.

Explaining Model Predictions (LIME)

LIME learns a linear surrogate model. As a result, for each data point, the coefficients of its linear regression can be interpreted as the change in the output produced by a unit change

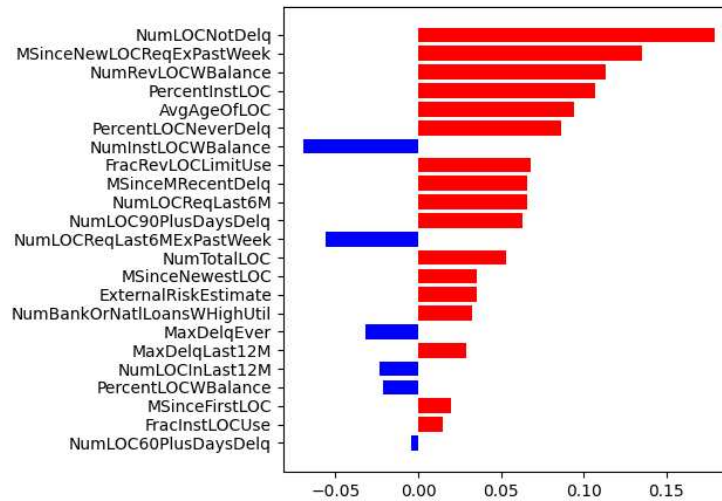


Figure 1: The LIME model approximation for a random data point. A unit increase in features with a red horizontal line will increase the probability of default. Similarly, a unit increase in features with a blue horizontal line will decrease the probability of default. The contribution of each feature in the model explanation can be obtained by multiplying each feature coefficient by the feature value.

in the corresponding feature value, given that other feature values are held constant. Figure 1 shows an example of a LIME explanation. A unit increase in the values of features with coefficients shown in red lines will increase the probability of default, while a unit increase in the values of features with coefficients shown in blue lines will decrease the probability of default.

After obtaining the feature importance for all the data points, they can be aggregated to find the overall feature importance. In Figure 2, we look at the global feature importance for the model. The top three most significant features are the months since the newest request for a new line of credit (excluding those requested in the past week), the external risk estimate, and the fraction of all revolving line of credit limits in use. From Figure 2a, we can observe the nonlinear nature of the classifier. For example, for the feature “months since the newest request for a new line of credit (excluding those requested in the past week),” having small and large values both contribute to a decrease in the probability of default, while having values around the median contributes to an increase in the probability of default. Hence, by aggregating the feature importance, we are able to find the relationship learned by the model. These generated explanations and the overall feature importance also

aid in discovering biases in the model.

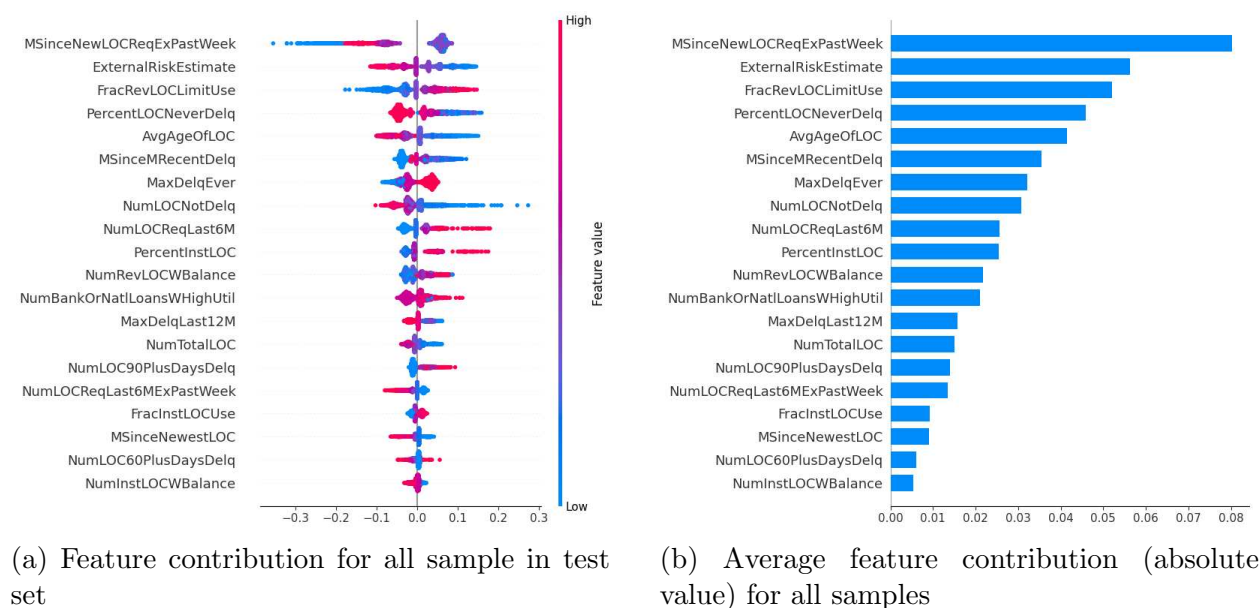


Figure 2: Feature importance obtained by aggregating LIME explanations for all samples in the test set. In Figure (a), the color-coding (blue-red) represents the value of the feature. For example, for external risk estimate, the larger values (in red) contribute to the decreasing probability of default, while the smaller values (in blue) contribute to the increasing probability of default. Figure (b) is obtained by aggregating the feature importance as obtained for all points. Figure (b) illustrates that the three foremost significant features are months since the newest request for a new line of credit (excluding those requested in the past week), the external risk estimate, and the fraction of all revolving line of credit limits in use.

The quality of LIME explanations depends on the fidelity of the LIME model. We evaluate the fidelity of the approximations produced by our modified implementation of LIME for the best-performing NN. We do this by constructing an approximation at each of the 1973 points in the test dataset and computing the fraction of times the NN and the LIME approximations produce the same classification. We find that out of 1973 data points, 1935 points have the same classification for the linear LIME approximation and the NN model being analyzed. The high fidelity of the LIME approximations shows that LIME is able to generate valid linear approximations that can be trusted for a large number of data points.

However, for the few cases where the classification of the NN model and LIME do not match and are different significantly, the linear approximation cannot be trusted. In order to generate explanations for these points, we discuss another method.

SHapley Additive exPlanations (SHAP)

SHAP is an explainable AI method with an economic foundation. It performs the Shapley value decomposition of the model output, giving the contributions of every feature at a data point. Like LIME, it is also a model-agnostic method.

Shapley values have several good properties that satisfy a number of important criteria, including local accuracy (i.e., the model explanation matches the original prediction), handling of missing data (i.e., if the feature is absent, its contribution to the model prediction will be zero), and consistency (i.e., if the model changes in a way that leads to larger marginal contributions for a feature, the Shapley values also increase). Shapley values also incorporate the interactions between features in the process of calculating feature importance, making SHAP a more reliable method for interpretability than LIME.

We use the Kernel SHAP implementation from Lundberg and Lee (2017). The Kernel SHAP procedure computes the Shapley values by running a weighted-least-squares regression whose solution is the Shapley values of features. To explain a point x_i , the different points used in the linear regression are obtained by selecting a subset of features from x_i , and the remaining subset of features not selected are replaced with values from background data points (that is, from training data). The weights of the linear regression are decided by the size of the sampled subset. For example, a subset of one feature has the maximum weight because it provides the maximum information about that feature contribution. Using the described weighted-least-squares regression, we obtain Shapley values for the data point x_i . It is worth highlighting that the Kernel SHAP implementation relies on taking subsets of features, that is, $2^{\text{Number of features}}$. As a result, Kernel SHAP does not scale well.

Using SHAP, we obtain explanations for the classification of all the data points in the test set. Similar to LIME, the Shapley values for features can be aggregated for all test points to find the overall impact of a feature on the model. We present the feature importance for all the points in the test sample in Figure 3, which shows that the months since the newest request for a new line of credit (excluding those requested in the past week), the fraction of all revolving lines of credit limits in use, and the average age of lines of credit are the top three most important features for the model. We observe a minor difference in the feature contribution obtained from the LIME and SHAP. This may be due to the fact that not all

LIME approximations are locally accurate.

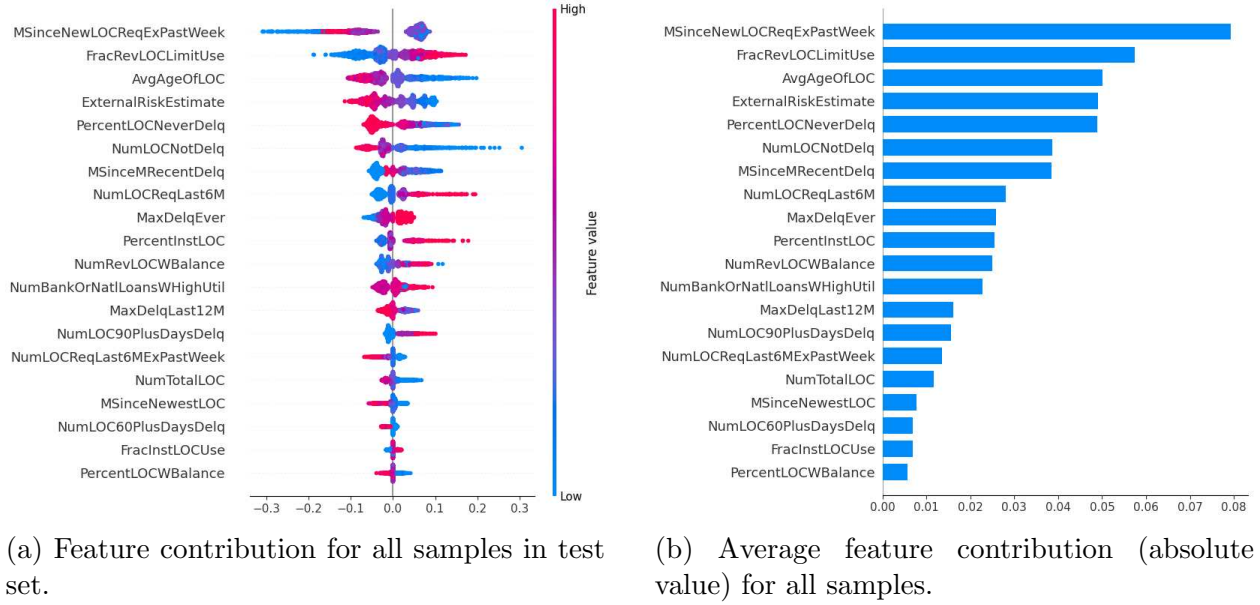


Figure 3: Feature importance (Shapley values) obtained by aggregating Kernel SHAP explanations for all the samples in the test set. In Figure (a), the color coding (blue-red) represents the value of the feature. For instance, the larger values (in red) for the external risk estimate contribute to the decreasing probability of default, while the smaller values (in blue) contribute to the increasing probability of default. We obtain the overall feature importance (as illustrated in Figure (b)) by aggregating Shapley values for all test points. The most important features are the months since the newest request for a new line of credit (excluding those requested in the past week), the fraction of all revolving lines of credit limits in use, and the average age of lines of credit. These are largely consistent with the LIME feature importance. Small discrepancies can be attributed to the inaccurate local approximations of LIME.

Both LIME and SHAP have their weaknesses. For example, in addition to feature importance, we can obtain locally linear approximations using LIME (as obtained in Figure 1), which are not available in SHAP. These local approximations of the decision boundary are necessary for other applications, as will become clear in the next section. On the other hand, the explanations using LIME might not satisfy important properties like local accuracy and consistency that Shapley values allow.

From the computational perspective, different runs of LIME produce slightly different explanations because of the randomness in sampling local data points. Consequently, it can potentially suffer from instability issues. On the other hand, computing Shapley values is not

scalable for datasets with a large number of features. In such scenarios, LIME approximations can be computed efficiently. Therefore, both methods (SHAP and LIME) should be used in practice depending on the use case, carefully and judiciously.

Finding Representative Samples

Another application of model explanations is to find the most representative data point for a particular data point analyzed. It can be useful in scenarios where loan companies might require representative data points from the past to explain the model prediction of a particular loan candidate. For example, the borrower x_i might default in the future because it is similar to x_j and x_k who defaulted in past.

The k-nearest neighbors algorithm (kNN) applied to the data naively can generate the most similar points in the sample. However, if we apply kNN to the original feature space, the less significant features will end up contributing to the distance calculations between the data points. To avoid this problem, every data point can be represented by a feature importance vector (for instance, a vector of Shapley values), and kNN can be applied to the feature importance vector. Using SHAP will ensure that less important features have smaller feature contributions (that is, smaller Shapley values). Hence, the contributions of less important features to the distance calculation in the kNN will be minimal. This modification leads to finding a more representative data point for each new data point being analyzed.

6.2 Interpretability for Regulators: Model Fairness and Stress Testing

In the previous subsection, we generated explanations for model predictions. Apart from concerns about the transparency of the credit approval algorithm, government regulations mandate that these algorithms must be fair. Concerns about the fairness of a model arise when the model gives importance to features like race, religion, or gender that are specified in law. Our HELOC dataset does not contain any such features overtly. Nevertheless, the fairness of a black box model can be examined (if features were present) using the individual and overall feature contributions that are obtained from LIME and SHAP.

Another important aspect of interpretability for regulators is stress-testing the models.

The credit risk models used by companies need to withstand appropriate stress testing. This involves simulating extreme scenarios and analyzing the model's behavior in response to generally extreme macroeconomic conditions.

Extreme macroeconomic conditions cannot be simulated in our model due to its lack of macroeconomic features. However, we demonstrate how the interpretability methods can help in stress testing a black-box model for extreme customers. The same methodology can be generalized for extreme macroeconomic conditions. We use two extreme customers in our stress testing. In the first case, an extreme customer is obtained by sampling points from a multivariate normal distribution with extreme values. While sampling, some feature values may no longer satisfy the feature-specific constraints; for them, we manually truncated the feature values. In the second case, we use a customer whose feature values are all -9, that is, there is no bureau record or investigation on file about this customer. We have multiple such data points in the dataset that are not included for training or testing purposes. The extreme points used in our analysis are included in Table 3.

Regulators are interested in verifying the validity of the model's output and analyzing its behavior. We preprocessed features for these extreme customers using weight of evidence encoding, which leads to extreme values being assigned to one of the bins. Due to this binning, the model was able to produce valid predictions. For case 1, the output is $Pr(default) = 0.53$, which implies that the person may or may not default with almost equal probability. Since this data point is sampled randomly, we cannot comment on the accuracy of the prediction. For case 2, the model's prediction is $Pr(default) = 0.75$, which implies that, when there is no information about the person on file, the person is likely to default, hence preventing lenders from approving credit for such individuals with no information in their credit file.

To understand the behavior of the model in the proximity of extreme points, we use the LIME model approximation. For cases 1 and 2, we present the LIME approximation in Figure 4. In both cases, the LIME model approximation prediction is close to the original model prediction. Hence, we conclude that the approximations can be trusted. This model approximation can be used by regulators to learn more about the model's behavior in an extreme scenario.

In this section, we demonstrated the use of explainable AI methods for generating expla-

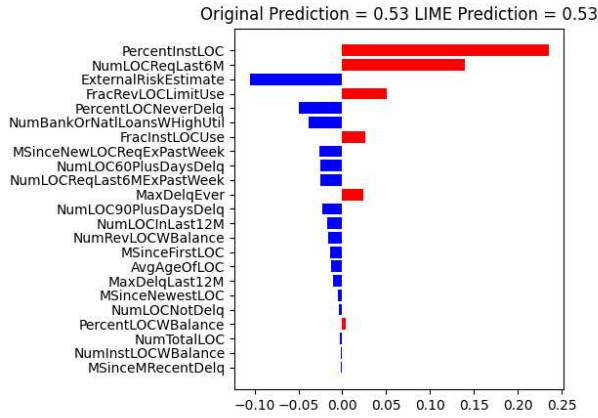
Feature	Data Pt 1	Data Pt 2
ExternalRiskEstimate	131	-9
MSinceFirstLOC	935	-9
MSinceNewestLOC	99	-9
AvgAgeOfLOC	467	-9
NumLOCNotDelq	89	-9
NumLOC60PlusDaysDelq	6	-9
NumLOC90PlusDaysDelq	5	-9
PercentLOCNeverDelq	100	-9
MSinceMRecentDelq	115	-9
MaxDelqLast12M	16	-9
MaxDelqEver	13	-9
NumTotalLOC	125	-9
NumLOCInLast12M	6	-9
PercentInstLOC	88	-9
MSinceNewLOCReqExPastWeek	35	-9
NumLOCReqLast6M	13	-9
NumLOCReqLast6MExPastWeek	12	-9
FracRevLOCLimitUse	71	-9
FracInstLOCUse	100	-9
NumRevLOCWBalance	23	-9
NumInstLOCWBalance	25	-9
NumBankOrNatlLoansWHighUtil	21	-9
PercentLOCWBalance	100	-9
Model Prediction	0.53	0.75

Table 3: Extreme Customers (data points). The first point is sampled, assuming features are multivariate Gaussian. The second point is obtained by setting all feature values to -9 (no bureau record or investigation).

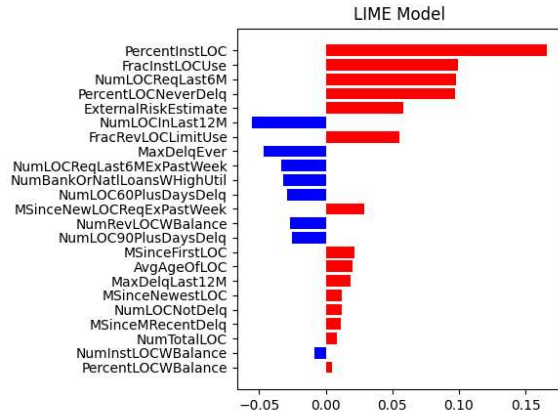
nations in model fairness testing and stress testing that may be able to assist regulators in their duties. Next, we discuss the use of model interpretability for loan applicants.

6.3 Interpretability for Loan Applicants: Counterfactual Suggestions

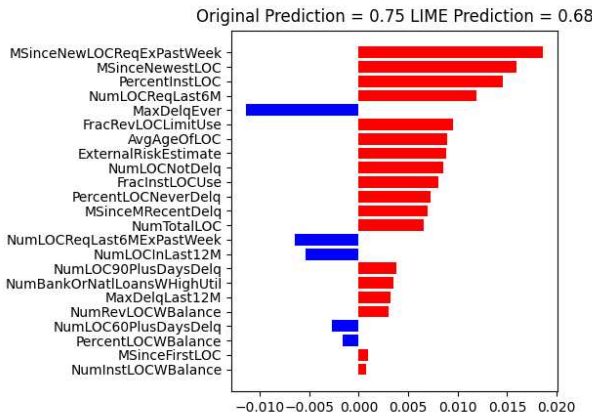
Loan applicants are interested in two major aspects of model interpretability. The first is to learn the reason behind their denial or approval, and the second is how to modify the features that might change their classification in the future. Explanations for the reason behind a denial or approval can be obtained from the techniques discussed in the previous



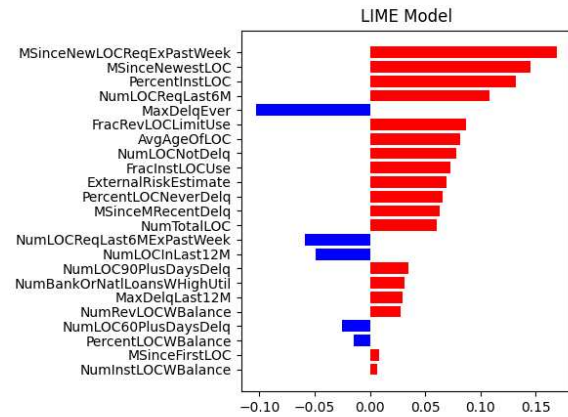
(a) Case 1: Feature Contribution.



(b) Case 1: LIME Model



(c) Case 2: Feature Contribution



(d) Case 2: LIME Model

Figure 4: LIME explanations for the two extreme scenarios (case 1: an extreme sample, case 2: no information in file). The regulators obtain the linear approximation using the LIME model in the proximity of an extreme sample. In both cases, the LIME approximation is close to the model's output. Hence, we conclude the linear model approximations are reliable. The LIME models are $\Sigma(\text{FeatureValue}) * (\text{LIME} - \text{Coefficient})$. The LIME coefficients are represented by horizontal lines. We can observe that for extreme case 1, the model gives its maximum importance to the percentage of lines of credit that are installment lines of credit, while for extreme case 2, the model gives its highest importance to the months since the newest request for a new line of credit excluding those requested in the past week. The above analysis can be extended to different extreme conditions.

sections. In this section, we discuss the methodology of generating instructions for reversing the model’s classification.

Applicants are interested in counterfactuals that provide information about the steps that might change the decision of the model. Consequently, the counterfactuals being generated should have the following properties: reverse classification (that is, the model prediction for the counterfactual should be reversed from the original decision), proximity (that is, the counterfactual should be close to the original data point), and diversity (that is, there should be multiple different counterfactuals from which an individual be able to choose).

We generate counterfactuals using the state-of-the-art method, Diverse Counterfactual Explanations (DiCE) (Mothilal, Sharma, and Tan, 2020). This method learns the counterfactuals c_i for a data point y optimizing over the requirements of the counterfactual. In particular, it learns the c_i ’s by minimizing the following loss:

$$L = \sum_{i=1}^k \text{loss}(f(c_i), y) + \lambda_1 \sum_{i=1}^k \text{dist}(c_i, y) - \lambda_2 * \text{diversity}(c_1, c_2 \dots c_k)$$

where the first part is the reverse classification loss, the second part is the proximity to the original data point, and the third part is the diversity component. λ_1 and λ_2 are the weights for proximity and diversity components, respectively.

We study a few illustrative examples and analyze the properties of the counterfactuals generated using DiCE.⁵

Applicants classified as non-creditworthy are interested in the steps that can make them creditworthy. The steps suggested must be practical enough for an applicant to implement. Some features are inherently impossible for individuals who are deemed non-creditworthy to improve. For example, the maximum delinquency ever in days cannot be decreased: it is an event that happened in the past, and cannot be changed. Likewise, the total number of lines of credit established cannot be increased, since the person has been turned down from opening a new line of credit.

In our system, the features that are impossible to modify are incorporated as constraints in the optimization of loss L . Table 4 shows an example of the set of suggestions to become

⁵We use the publicly available implementation of the DiCE algorithm, found at <https://github.com/interpretml/DiCE>.

creditworthy for an individual who is deemed non-creditworthy. All of the suggestions are possible to implement, though some might be difficult. In cases where it is hard for an individual to implement certain changes, he or she may be able to choose from a diverse set of counterfactuals to reverse the classification.

In Table 4, the first counterfactual suggests increasing the months since the applicant's most recent delinquency to more than four years. However, this requires an individual to wait for four years while making payments for all his lines of credit. In this case, the individual may instead choose to follow another set of steps, as suggested by the third counterfactual: increasing the percent of lines of credit never delinquent (by decreasing the total lines of credit), along with decreasing the number of lines of credit requests made in the six months prior to applying for a new line of credit. Similarly, diversity-constrained counterfactual suggestions can be made for all individuals who are deemed non-creditworthy, and individuals can follow the steps which are most convenient to their circumstances.

Likewise, the people who are classified as creditworthy would like to maintain their creditworthy status. Hence, they would like to know which actions to avoid that might make them non-creditworthy. Table 5 presents one such example. The applicant in Table 5 might face a credit denial if he or she takes steps such as opening up new lines of credit, decreasing the lines of credit that are not currently delinquent, increasing the fraction of revolving lines of credit, and making new requests for line of credit in the six months prior to applying for a line of credit. These steps result in the probability of default as suggested by the model increasing from 0.09 to 0.58, which may lead to a denial of credit. From these counterfactuals, the applicant thus knows the steps not to take that may decrease their creditworthiness.

To gauge the ability of DiCE to generate counterfactuals, we run it for all test data points (total: 1973) and count the number of points for which the counterfactuals were successfully found. In Table 6, we show that increasing the proximity constraint leads to a decrease in the average number of feature changes in the counterfactuals obtained. We also observe that increasing the proximity constraint leads to a decrease in the number of successfully generated counterfactuals. An optimal algorithm to generate counterfactuals for an individual thus is to start with a high value for the proximity constraint and slowly decrease it until a counterfactual is found.

Using DiCE and related counterfactual generating methods, we can generate suggestions

Feature	Original Value	New Value
MSinceMRecentDelq	0 – 4	48–
NumLOCReqLast6M	2 – 4	1 – 2
$Pr(Default)$	0.63	0.44
AvgAgeOfLOC	75 – 98	98–
NumTotalLOC	9 – 14	0 – 1
NumLOCReqLast6M	2 – 4	1 – 2
$Pr(Default)$	0.63	0.43
PercentLOCNeverDelq	0 – 82	98–
NumLOCReqLast6M	2 – 4	1 – 2
$Pr(Default)$	0.63	0.31

Table 4: The steps toward creditworthiness suggested to an individual deemed non-creditworthy. Following these steps, an individual should be able to decrease the probability of default predicted by the machine learning model. We present three diverse counterfactuals. In the first, it suggests changing the months since most recent delinquency from the value of 0-4 months to greater than 48 months, which means that person should not be delinquent for the next 4 years. In the second counterfactual, one of the suggestions is to decrease the total number of lines of credit to 0-1 (by closing accounts). Because some suggestions may be difficult for an individual to follow, we provide multiple diverse counterfactuals. The individual can follow the third counterfactual, which suggests decreasing the number of lines of credits to increase the percentage of lines of credit never delinquent, and decrease the number of lines of credit requests in the most recent six months prior to applying for a new line of credit. Providing multiple diverse counterfactuals allows the option of selecting the most convenient option to an individual.

for the loan applicants, as discussed above. One limitation of the counterfactual generation algorithm is the inability to take feature relationships into consideration. For example, if one of the generated counterfactuals suggests decreasing the number of the total lines of credit, then decreasing number of lines of credit also leads to the changes in the features that depend on the total number of lines of credit. However, this type of relationship is ignored in the present method. Solving this problem would involve the intervention of an expert who is familiar with the relationship between the model’s features, and can encode them as constraints in the counterfactual generation optimization process. We leave this for future studies.

Feature	Original Value	New Value
AvgAgeOfLOC	60 – 75	4 – 29
NumLOCNotDelq	12 – 17	4 – 6
NumLOCReqLast6M	0 – 1	1 – 2
FracRevLOCLimitUse	13 – 29	77–
$Pr(Default)$	0.09	0.58

Table 5: The steps that can make a creditworthy-deemed individual non-creditworthy, as suggested by the algorithm. In simple English, if the individual decreases the number of lines of credit not currently delinquent, opens up multiple new lines of credit leading to a decrease in the average age of their lines of credit, made multiple requests for lines of credit in the past six months, and increases the fraction of their revolving line of credit, then the individual will be deemed non-creditworthy.

Proximity Constraint(λ_1)	Mean # of Feature Changes	Loss Value (Distance)	# of success counterfactual found
0.5	3.96	0.054	1961
1.5	3.35	0.045	1909
5.0	2.66	0.035	1854

Table 6: Proximity Analysis. On increasing the proximity constraint (λ_1), the number of changes required for generating a successful counterfactual decreases, and the number of data points for which successful counterfactuals are generated also decreases. An optimal algorithm to generate counterfactuals for an individual is to start with a larger proximity constraint and slowly decrease it until a counterfactual is found.

6.4 Interpretability for Researchers and Data Scientists: Simple Rules to Summarize the Dataset

One property of interpretability of particular interest to data scientists is the summarization of the data and the model. A global view of the data and the model is able to give the researcher an idea about any possible problems with the model. It can additionally help them to present a summary of the model to managers or regulators. In this section, we discuss methods that may help data scientists to demystify these black box models for their particular needs.

We have already discussed the first aspect of interpretability, the summarization of the model. A good summary of the model includes determining the most important features that contribute to the model’s prediction. Using the methods described in the previous section

(LIME and SHAP), data scientists can obtain the most important features and know their contribution to the model’s overall prediction. For example, Figure 3 illustrates that the feature, “months since the newest request for a new line of credit (excluding those requested in past week),” is the most important feature in the model. The model summary also includes information about the relationship between the input and output of the model. Additionally, from Figure 3a, we can observe that higher values of the fraction of all revolving line of credit limits in use increases the probability of default.

Another important aspect for researchers is the summarization of the dataset. It involves discovering the intrinsic relationships within the dataset. To this end, we aim to learn a set of simple rules that can summarize the dataset. Tree-based classification approaches can be used to learn such rules. A tree classifier is a set of if-else statements that determines the classification of a data point. As illustrated in Table 1, the best optimal tree classifier with our dataset achieves an accuracy of 74.12%. However, the complexity of rules obtained from the best optimal trees classifier makes these rules difficult to analyze and interpret.

To obtain rules that are simple to understand, the decision tree must be constrained to a smaller depth, with fewer features at every node. Consequently, we use a simpler and interpretable optimal tree to obtain more easily analyzed rules. This simple optimal tree is shown in Figure 5. Here we show two simple rules that can achieve an accuracy of 72.28%. The rules state that a person will default on a loan (that is, be classified as non-creditworthy) if the number of months since a new line of credit has been requested (excluding those requested in the past week) is greater than 1, and the external risk estimate is less than 75, or if the number of months since a new line of credit has been requested (excluding those requested in the past week) is less than or equal to 1, and the external risk estimate is less than 68.

We use inductive logic programming to learn an even simpler set of rules with a small decrease in accuracy compared to the rules generated from the optimal trees model. As discussed earlier, ILP generates rules that are composed of a single condition: for example, using the rule *ExternalRiskEstimate* < 72 to classify people into non-creditworthy and creditworthy groups is able to achieve an accuracy of 70.65%.

In addition to ILP, there exist multiple other rule-finding methods in the literature. However, the ILP approach stands out in terms of its simplicity and ability to learn more effective

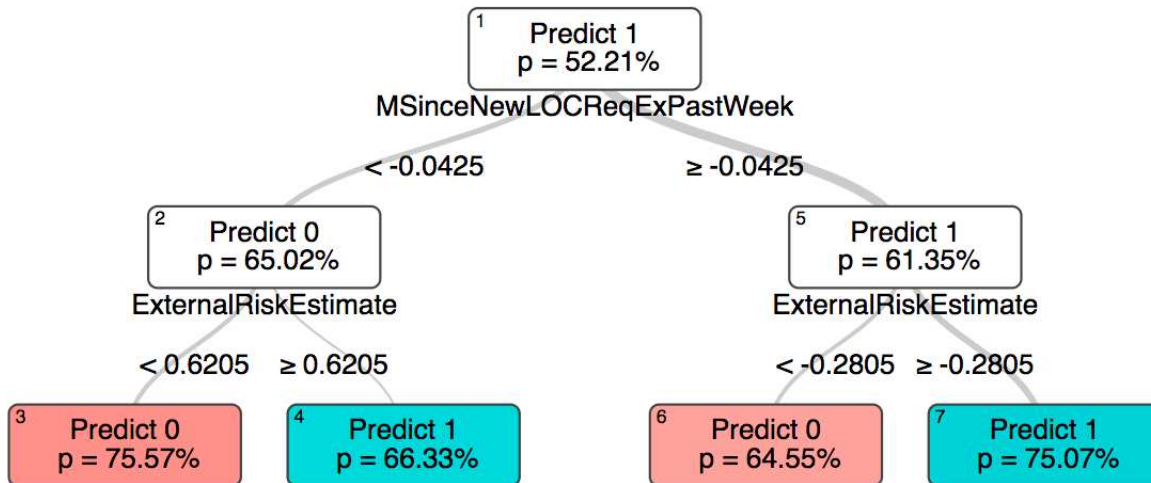


Figure 5: Explainable Optimal Tree Model used to learn simple rules summarizing the dataset. The rules states that a person will default on a loan (that is, be classified as non-creditworthy) if the number of months since a new line of credit has been requested (excluding the past week) is greater than 1 and the external risk estimate is less than 75, or the number of months since a new line of credit has been requested (excluding the past week) is less than or equal to 1 and the external risk estimate is less than 68.

rules. For example, the logistic rule regression/generalized linear rule model described in Wei et al. (2019) is another rule-based model, but it leads to multiple rules that are difficult to analyze together. Similarly, the Boolean rule column generation (BRCG) method described in Dash, Günlük, and Wei (2018) gives a set of rules to describe a dataset, but it requires dividing feature values into different bins before learning the rules. Data-binning limits the quality of the generated rules, since the algorithm might not select the threshold for the binning of features that is the optimal threshold for the rule.

Using the methods discussed above, data scientists should be more able to summarize datasets and demystify models. Obtaining a good summary of the dataset and understanding of the model will help in the creation of better classifiers.

7 Conclusion

In this paper, we have examined several different machine learning models, and used suitable tools to create explanations of their function according to the different needs of stakeholders involved in credit risk management. We have used state-of-the-art interpretable machine learning techniques, including Local Interpretable Model-Agnostic Explanations (LIME), SHapley Additive exPlanations (SHAP), and Diverse Counterfactual Explanations (DiCE), adapting them to our use case. We demonstrate the importance of domain-specific knowledge in order to explain these black-box models. These domain-specific constraints must be obtained from experts in the field, but can produce pragmatically valid suggestions and explanations.

In our results, we demonstrated that with the right tools, even black-box machine learning models are able to answer a series of important questions for credit risk modeling. These questions include: why does the model classify a data point in a certain way? What small changes in feature values could reverse the model's classification of an individual? How does the model behave in extreme scenarios? What relationships did the model learn? Are the models biased? What is the minimal summary of the dataset?

Answering these questions not only fulfills the legal requirements specified by regulators for the use of machine learning models in credit risk management, but also provides borrowers, lenders, and data scientists with answers to questions that they may desire from ML models.

Interpretable models have the additional benefit of being convincing and easy to accept. Interpretable models may also help domain experts troubleshoot the inner workings of a complex model, which in turn will make it more accurate and tailored to its domain.

Many problems in finance and economics have a common mathematical representation and internal statistical structure, and may therefore benefit from our framework to interpret the black-box machine learning models used to analyze them. These include loan defaults, mortgage prepayments, Federal Reserve rate decisions, corporate merger and acquisition decisions, asset return maximization, and insurance claims, among others. Our work is a step in the direction of bridging the gap between these black-box models and their use in a real world setting.

Future work along these lines should extend this explainability analysis by incorporating second-order effects into the explainability algorithms (LIME and DiCE), in order to generate more practical counterfactual suggestions and explanations. It may also be of interest to compare the insights derived from these explainability tools to the traditional factors used for credit risk forecast by involving an expert in the process. Finally, large real-world datasets, in particular those including macroeconomic and demographic features and the size of loan requests, should be used to extend this model's capabilities, in order to help quantify the model's overall fairness, response to stress testing, and the monetary impact due to the superior performance of black-box machine learning models.

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A Household Debt Statistics

Figure A.1a shows the total household debt balance in the United States and its composition over time. We observe that the household debt balance reached an all-time high at the end of the fourth quarter of 2020, with a value of \$14.56 trillion. The overall trend line of the debt balance is increasing. Figure A.1b shows the total number of new accounts opened over time, along with the number of inquiries and number of accounts closed. The total number of accounts opened in the last single-year period measured is 189.6 million. This enormous debt balance and number of new accounts opened shows the importance of credit risk management. The data and the Figure A.1 were obtained from the New York Fed Consumer Credit Panel/Equifax.

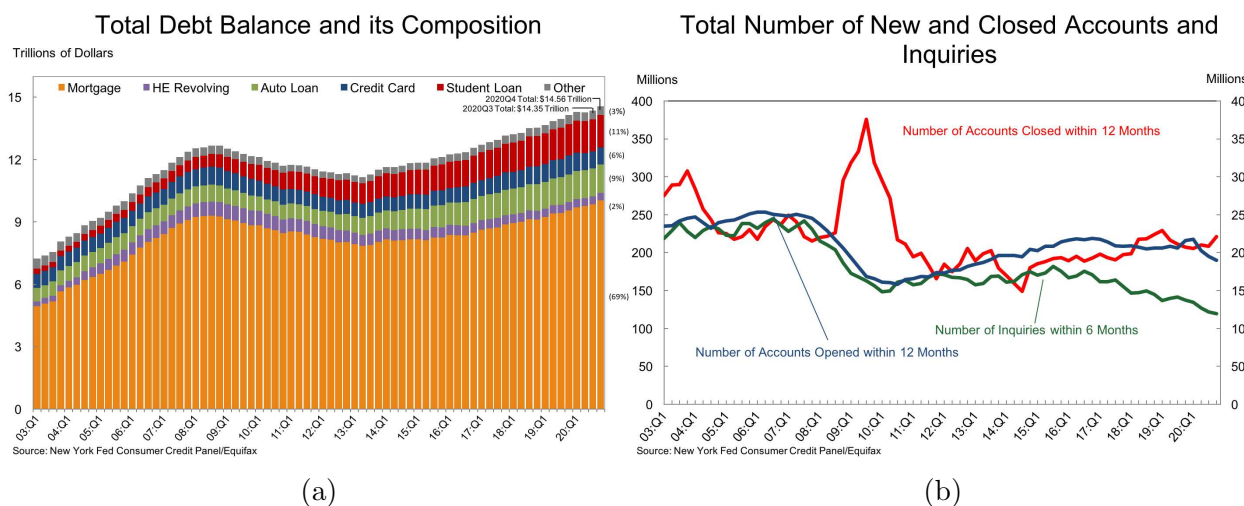


Figure A.1: Household Debt. The plots were obtained from the quarterly report on household debt and credit released by the New York Fed.

B Dataset

B.1 Glossary of Relevant Credit Modeling Terms

The following definitions provide helpful context for the description of the dataset's features.

1. Line of Credit: An agreement to provide credit.
2. Revolving Line of Credit: A line of credit with a maximum amount that the borrower can choose to use each month. The most common example is a credit card.

3. **Installment Line of Credit:** A line of credit with a fixed loan amount and a fixed monthly payment. A mortgage is a common example.
4. **Delinquent:** A line of credit is delinquent if its payments are not made in a timely manner.
5. **Utilization:** The amount still owed divided by the total amount borrowed; the fraction of available credit currently in use.

B.2 Explanation of Predictor Features

In addition to the binary target variable (Risk Classification), each credit applicant is characterized by 23 predictor features, 21 continuous and 2 categorical. These are:

1. A condensed version of the borrower's credit risk computed by FICO using all credit bureau information (ExternalRiskEstimate)
2. Months since the very first line of credit was established (MSinceFirstLOC)
3. Months since the newest line of credit was established (MSinceNewestLOC)
4. Average age in months of all existing lines of credit (AvgAgeOfLOC)
5. Number of lines of credit not currently delinquent (NumLOCNotDelq)
6. Number of lines of credit ever been 60 or more days delinquent (NumLOC60PlusDaysDelq)
7. Number of lines of credit ever been 90 or more days delinquent (NumLOC90PlusDaysDelq)
8. Percentage of lines of credit never been delinquent (PercentLOCNeverDelq)
9. Number of months since the most recent delinquency (MSinceMRecentDelq)
10. Maximum delinquency in days in the past year (MaxDelqLast12M)
11. Maximum delinquency ever in days (MaxDelqEver)
12. Total number of lines of credit established (NumTotalLOC)
13. Number of lines of credit established in the past year (NumLOCInLast12M)

14. Percentage of lines of credit that are installment lines of credit (PercentInstLOC)
15. Months since the newest request for a new line of credit excluding those requested in the past week (MSinceNewLOCReqExPastWeek)
16. Number of requests for new lines of credit in the last 6 months (NumLOCReqLast6M)
17. Number of requests for new lines of credit in the last 6 months excluding those requested in the past week (NumLOCReqLast6MExPastWeek)
18. Fraction of all revolving credit limits in use (FracRevLOCLimitUse)
19. Fraction of all installment lines of credit in use (FracInstLOCUse)
20. Number of revolving lines of credit with outstanding balances (NumRevLOCWBalance)
21. Number of installment lines of credit with outstanding balances (NumInstLOCWBalance)
22. Number of bank loans and national loans (a subset of all revolving trades) with an outstanding balance of at least 75% of the credit limit (NumBank/NatlLoansWHighUtil)
23. Percentage of lines of credit with outstanding balances (PercentLOCWBalance)

B.3 Data Cleaning

Our dataset contains special values, negative integers that are interpreted symbolically and do not hold any numeric significance. As a result, we cannot directly feed them into our machine learning models. We either have to drop them or encode them appropriately. A large fraction of the data points in our HELOC dataset contains at least one special value (7,957 of the 10,459 data points). Hence, dropping all such data points is infeasible.

In addition, the dataset has 588 records that solely contain the special value -9 for all feature values. 331 of these data points are labeled as non-creditworthy, and 266 as creditworthy. This is a problem for any model because the same input vector will produce opposite target labels. This happens because a borrower will receive a special value if they

are either a VIP and do not need to be investigated, or if they have no bureau record at all, i.e., they have no credit history. Such data points are dropped in our analysis.

We find special values are concentrated in 9 of our 23 input features. A standard technique for dealing with special values is to replace them with the mean values of the respective feature. A simple example illustrates that this is not a meaningful approach for our dataset. If a borrower has never had a delinquency, she will have the -7 (Condition not met) special value for the feature “months since most recent delinquency.” Clearly, replacing the feature value with the mean is not correct, since she will be moved from a desirable value of the feature to a less desirable one. To handle these special values, we used binning techniques.

B.4 Data Visualization

Before using the dataset to train machine learning models, we analyze its properties using a few exploratory data visualization techniques.

Figure B.1 visualizes the 23x23 correlation matrix of our dataset, identifying higher correlation values with lighter shades. We find three pairs of features with correlations greater than 0.8.

1. The total number of lines of credit (NumTotalLOC) and the number of lines of credit that are not currently delinquent (NumLOCNotDelq)
2. The number of lines of credit that have been 60+ days delinquent (NumLOC60PlusDaysDelq) and the number of lines of credit that have been 90+ days delinquent (NumLOC90PlusDaysDelq)
3. The number of request for new lines of credit in the past 6 months (NumLOCReqLast6M) and the number of request for new lines of credit in the past 6 months excluding the past week (NumLOCReqLast6MExPastWeek)

C Model Implementation Details

The class distribution of the dataset in the five-fold cross-validation is presented in Table C.1. By visual inspection, it illustrates that the dataset is reasonably balanced.

We give details of the implementation for the different ML models used in our evaluation.

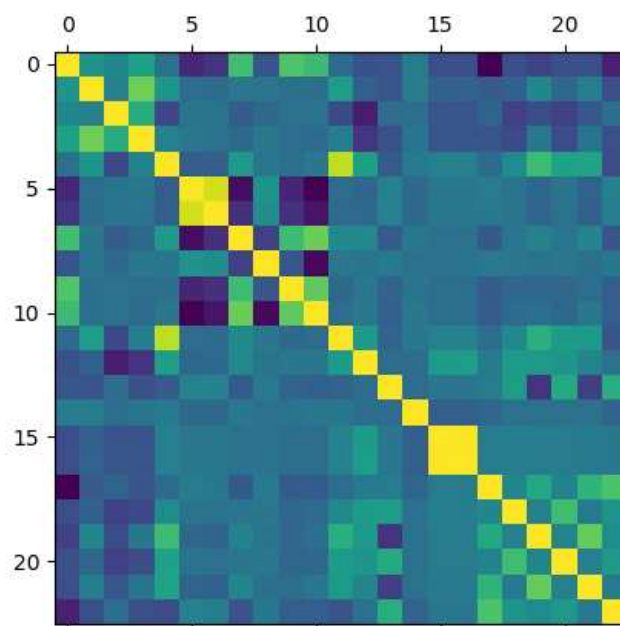


Figure B.1: 23 x 23 Correlation Matrix. The lighter shades represents highly correlated feature pairs. As expected, the diagonal has the lightest shade because it represents the correlation of a feature with itself.

For optimal trees, we used the Julia implementation available from the Interpretable AI website ⁶. We performed a grid search over a depth from 1 to 10, and the number of features used for the deciding split at each node from the set 1,2,3,4,5,10,15,20,23. The optimal tree (black box) is the best-performing model of all the parameter combinations. Its parameters are depth=2 and the number of features=10. The optimal tree (interpretable) model corresponds to a model depth=2 and a number of features=1.

We used the scikit-learn implementation of the random forest classifier⁷. A grid search was performed on the number of estimators (trees) from 1 to 150. The best performing model consisted of 140 trees. Other parameters were set to the provided default values.

The neural network (WoE) model was implemented using the scikit-learn implementation of MLPClassifier. We performed a grid search over the different architectures and the L2-

⁶<https://docs.interpretable.ai/v1.0/>

⁷<https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html>

Fold Number	Count of $y_i = 1$ (Train)	Count of $y_i = 0$ (Train)	Count of $y_i = 1$ (Test)	Count of $y_i = 0$ (Test)
1	4118	3770	1010	963
2	4088	3801	1040	932
3	4094	3795	1034	938
4	4111	3778	1017	955
5	4101	3788	1027	945

Table C.1: Five-fold cross-validation data distribution. $y_i = 1$ corresponds to the data point (i.e., individual) who defaulted. The dataset is fairly balanced.

regularization constant. The best-performing model had three hidden layers consisting of 5 units each, and a L2-regularization penalty constant of 0.5. Other parameters were set to the default values.

The neural network (one-hot) model was implemented using tensorflow-keras. This was done to ensure the availability of gradients from the model for the counterfactual generating algorithm (DiCE). The best-performing model had one hidden layer, with the number of nodes=20. It was trained using the Adam optimizer with a learning rate of 0.01. The activation type used was the Rectified Linear Unit (ReLU), and a L1-regularization penalty of 0.001 was used.

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PREDICTING CONSUMER DEFAULT:
A DEEP LEARNING APPROACH

Stefania Albanesi
Domonkos F. Vamossy

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Predicting Consumer Default: A Deep Learning Approach
Stefania Albanesi and Domonkos F. Vamossy
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ABSTRACT

We develop a model to predict consumer default based on deep learning. We show that the model consistently outperforms standard credit scoring models, even though it uses the same data. Our model is interpretable and is able to provide a score to a larger class of borrowers relative to standard credit scoring models while accurately tracking variations in systemic risk. We argue that these properties can provide valuable insights for the design of policies targeted at reducing consumer default and alleviating its burden on borrowers and lenders, as well as macroprudential regulation.

Stefania Albanesi
Department of Economics
University of Pittsburgh
4901 Wesley W. Posvar Hall
Pittsburgh, PA 15260
and CEPR
and also NBER
stefania.albanesi@gmail.com

Domonkos F. Vamossy
Department of Economics
University of Pittsburgh
230 S Bouquet St
Pittsburgh, PA 15260
d.vamossy@pitt.edu

1 Introduction

The dramatic growth in household borrowing since the early 1980s has increased the macroeconomic impact of consumer default. Figure 1 displays total consumer credit balances in millions of 2018 USD and the delinquency rate on consumer loans starting in 1985. The delinquency rate mostly fluctuates between 3 and 4%, except at the height of the Great Recession when it reached a peak of over 5%, and in its aftermath when it dropped to a low of 2%. With the rise in consumer debt, variations in the delinquency rate have an ever larger impact on household and financial firm balances sheets. Understanding the determinants of consumer default and predicting its variation over time and across types of consumers can not only improve the allocation of credit, but also lead to important insights for the design of policies aimed at preventing consumer default or alleviating its effects on borrowers and lenders. They are also critical for macroprudential policies, as they can assist with the assessment of the impact of consumer credit on the fragility of the financial system.

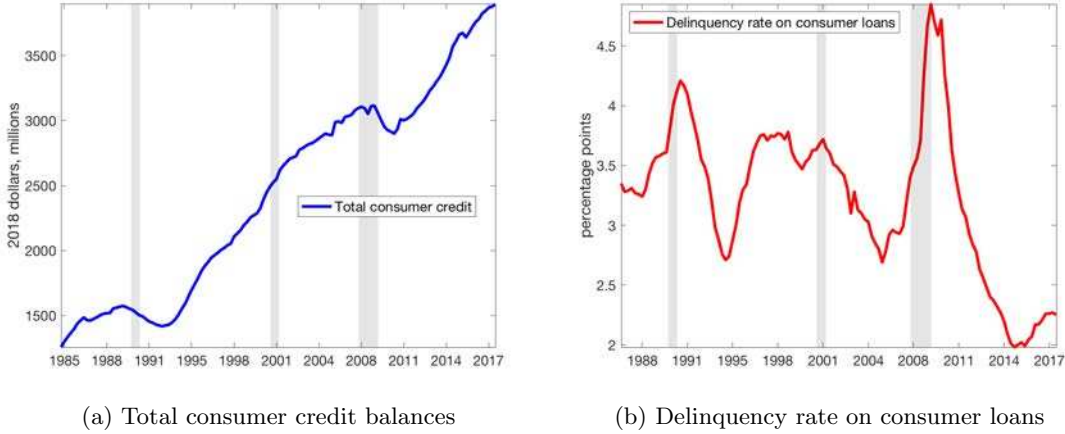


Figure 1: Source: Author’s calculations based on Federal Reserve Board data.

This paper proposes a novel approach to predicting consumer default based on deep learning. We rely on deep learning as this methodology is specifically designed for prediction in environments with high dimensional data and complicated non-linear patterns of interaction among factors affecting the outcome of interest, for which standard regression approaches perform poorly. Our methodology uses the same information as standard credit scoring models, which are one of the most important factors in the allocation of consumer credit. We show that our model substantially improves the accuracy of default predictions while increasing transparency and accountability. It is also able to track variations in systemic risk, and is able to identify the most important factors driving defaults and how they change over time. Finally, we show that adopting our model can accrue substantial savings to borrowers and lenders.

Credit scores constitute one of the most important factors in the allocation of consumer credit

in the United States. They are proprietary measures designed to rank borrowers based on their probability of future default. Specifically, they target the probability of a 90days past due delinquency in the next 24 months.¹ Despite their ubiquitous use in the financial industry, there is very little information on credit scores, and emerging evidence suggests that as currently formulated credit scores have severe limitations. For example, Albanesi, De Giorgi, and Nosal (2017) show that during the 2007-2009 housing crisis there was a marked rise in mortgage delinquencies and foreclosures among high credit score borrowers, suggesting that credit scoring models at the time did not accurately reflect the probability of default for these borrowers. Additionally, it is well known that credit scores are indiscriminately low for young borrowers, and a substantial fraction of borrowers are unscored, which prevents them from accessing conventional forms of consumer credit.

The Fair Credit Reporting Act, a legislation passed in 1970, and the Equal Opportunity in Credit Access Act of 1984 regulate credit scores and in particular determine which information can be included and must be excluded in credit scoring models. Such models can incorporate information in a borrower’s credit report, except age and location. These restrictions are intended to prevent discrimination by age and factors related to location, such as race.² The law also mandates that entities that provide credit scores make public the four most important factors affecting scores. In marketing information, these are reported to be length of credit history, which is stated to explain about 15% of variation in credit scores, and credit utilization, number and variety of the debt products and inquiries, each stated to explain about 25-30% of the variation in credit scores. Other than this, there is very little public information on credit scoring models, though several services are now available that allow consumers to simulate how various scenarios, such as paying off balances or taking out new loans, will affect their scores.

The purpose of our analysis is to propose a model to predict consumer default that uses the same data as conventional credit scoring models, improves on their performance, benefiting both lenders and borrowers, and provides more transparency and accountability. To do so, we resort to deep learning, a type of machine learning ideally suited to high dimensional data, such as that available in consumer credit reports.³ Our model uses inputs as features, such as debt balances and number of trades, delinquency information, and attributes related to the length of a borrower’s credit history, to produce an individualized estimate that can be interpreted as a probability of default. We target the same default outcome as conventional credit scoring models, namely a 90+ days delinquency in the subsequent 8 quarters. For most of the analysis, we train the model on data

¹The most commonly known is the FICO score, developed by the FICO corporation and launched in 1989. The three credit reporting companies or CRCs, Equifax, Experian and TransUnion have also partnered to produce VantageScore, an alternative score, which was launched in 2006. Credit scoring models are updated regularly. More information on credit scores is reported in Section 6.1 and Appendix E.

²Credit scoring models are also restricted by law from using information on race, color, gender, religion, marital status, salary, occupation, title, employer, employment history, nationality.

³For excellent reviews of how machine learning can be applied in economics, see Mullainathan and Spiess (2017) and Athey and Imbens (2019).

for one quarter and test it on data 8 quarters ahead, in keeping with the default outcome we are considering, so that our predictions are truly out of sample. We present a variety of performance metrics suggesting that our model has very strong predictive ability. Accuracy, that is percent of observations correctly classified, is above 86% for all periods in our sample, and the AUC-Score, a commonly used metric in machine learning, is always above 92%.

To better assess the validity of our approach, we compare our deep learning model to logistic regression and a number of other machine learning models. Deep learning models feature multiple hidden layers, designed to capture multi-dimensional feature interactions. By contrast, logistic regression can be interpreted as a neural network without any hidden layers. Our results suggest that deep learning is necessary to capture the complexity associated with default behavior, since all deep models perform substantially better than logistic regression. The importance of feature interaction reflects the complexity associated with default behavior. Additionally, our optimized model combines a deep neural network and gradient boosting and outperforms other machine learning models, such as random forests and decision trees, as well as deep neural networks and gradient boosting in isolation. However, all approaches show much stronger performance than logistic regression, suggesting that the main advantage is the adoption of a deep framework.

We also compare the performance of our model to a conventional credit score. By construction, credit scores only provide an ordinal ranking of consumers based on their default risk, and are not associated to a specific default probability. Yet, it is still possible to compare performance by assessing whether borrowers fall in different points of the distribution with the credit score compared to our model predictions. We find that our model performs significantly better than conventional credit scores. The rank correlation between realized default rates and the credit score is about 98%, where it is close to 1 for our model. Additionally, the Gini coefficient for the credit score, a measure of the ability to differentiate borrowers based on their credit score is approximately 81% and drops during the 2007-2009 crisis, while the Gini coefficient for our model is approximately 86% and stable over time. Perhaps most importantly, the credit score generates large disparities between the implied predicted probability of default and the realized default rate for large groups of customers, particularly at the low end of the credit score distribution. As an illustration, among Subprime borrowers, 17% display default behavior which is consistent with Near Prime borrowers and 15% display default behavior consistent with Deep Subprime. The default rates for Deep Subprime, Subprime and Near Prime borrowers are respectively 95%, 79% and 44%, so this misclassification is large, and it would imply large losses for lenders and borrowers in terms of missed revenues or higher interest rates. By contrast, the discrepancy between predicted and realized default rates for our model is never more than 4 percentage points for categories with at least a percent share of default risk.

Another advantage of our approach when compared to conventional credit scoring models is that we can generate a predicted probability of default for a much larger class of borrowers. Borrowers may be unscored because they do not have sufficient information in their credit report or because

the information is stale, and approximately 8% of borrowers fall into this category.⁴ The absence of a credit score implies that these borrowers do not qualify for most types of credit and is very consequential. Our model can generate a predicted probability of default for all borrowers with a non-empty credit record. We achieve this in part by not including lags in our specification, which implies that only current information in a borrower’s credit report is used. This is not costly from a performance standpoint as many attributes used as inputs in the model are temporal in nature and capture lagged behavior, such as ”worst status on all trades in the last 6 months.”

We also examine the ability of our model to capture the evolution of aggregate default risk. Since our data set is nationally representative and we can score all borrowers with a non-empty credit record, the average predicted probability of default in the population based on our model corresponds to an estimate of aggregate default risk. We find that our model tracks the behavior of aggregate default rates remarkably well. It is able to capture the sharp rise in aggregate default rates in the run up and during the 2007-2009 crisis and also captures the inversion point and the subsequent drastic reduction in this variable. With the growth in consumer credit, household balance sheets have become very important for macroeconomic performance. Having an accurate assessment of the financial fragility of the household sector, as captured by the predicted probability of default on consumer credit has become crucially important and can aid in macro prudential regulation, as well as for designing fiscal and monetary policy responses to adverse aggregate economic shocks. This is another advantage of our model compared to credit scores, since the latter only provides an ordinal ranking of consumers with respect to their probability of default. Our model can provide such a ranking but in addition also provides an individual prediction of the default rate which can be aggregated into a systemic measure of default risk for the household sector.

As a final application, we compute the value to borrowers and lenders of using our model. For consumers, the comparison is made relative to the credit score. Specifically, we compute the credit card interest rate savings of being classified according to our model relative to the credit score. Being placed in a higher default risk category substantially increases the interest rates charged on credit cards at origination and increasingly so as more time lapses since origination, whereas being placed in a lower risk category reduces interest rate costs. We choose credit cards as they are a very popular form of unsecured debt, with 73% of consumers holding at least one credit or bank card. In percentage of credit cards balances, average net interest rate expense savings are approximately 5% for low credit score borrowers. These values constitute lower bounds as they do not include the higher fees and more stringent restrictions associated with credit cards targeted to low credit score borrowers and the increased borrowing limits available to higher credit score borrowers. For lenders, we calculated the value added by using our model in comparison to not having a prediction of default risk or having a prediction based on logistic regression. We use logistic regression for this exercise as it is understood to be the main methodology for conventional credit scoring models. Over a loan with a three year amortization period, we find that the gains relative to no forecast

⁴See Brevoort, Grimm, and Kambara (2016). For more information, see Section 6.1.1.

are in the order of 75% with a 15% interest rate, while the gains for relative to a model based on logistic regression are approximately 5%. These results suggest that both borrowers and lenders would experience substantial gains from switching to our model.

Our analysis contributes to the literature on consumer default in a variety of ways. We are the first to develop a prediction model of consumer default using credit bureau data that complies with all of the restrictions mandated by U.S. legislation in this area, and we do so using a large and temporally extended panel of data. This enables us to evaluate model performance in a setting that is closer to the one prevailing in the industry and to train and test our model in a variety of different macroeconomic conditions. Previous contributions either focus on particular types of default or use transaction data that is not admissible in conventional credit scoring models. The closest contributions to our work are Khandani, Kim, and Lo (2010), Butaru et al. (2016) and Sirignano, Sadhwani, and Giesecke (2018). Khandani, Kim, and Lo (2010) apply a decision tree approach to forecast credit card delinquencies with data for 2005-2009. They estimate cost savings of cutting credit lines based on their forecasts and calculate implied time series patterns of estimated delinquency rates. Butaru et al. (2016) apply machine learning techniques to combined consumer trade line, credit bureau, and macroeconomic variables for 2009-2013 to predict delinquency. They find substantial heterogeneity in risk factors, sensitivities, and predictability of delinquency across lenders, implying that no single model applies to all institutions in their data. Sirignano, Sadhwani, and Giesecke (2018) examine over 120 million mortgages between 1995 to 2014 to develop prediction models of multiple states, such as probabilities of prepayment, foreclosure and various types of delinquency. They use loan level and zip code level aggregate information. They also provide a review of the literature using machine learning and deep learning in financial economics. Kvamme et al. (2018) also predict mortgage default using use convolutional neural networks and emphasize the advantages of deep learning. Both Sirignano, Sadhwani, and Giesecke (2018) and Kvamme et al. (2018) do not evaluate their models out of sample. Finally, Lessmann et al. (2015) reviews the recent literature on credit scoring, which is based on substantially smaller datasets than the one we have access to, and recommends random forests as a possible benchmark. However, we find that our hybrid model as well as our model components, a deep neural network and gradient boosted trees, improves substantially over random forests, possibly owing to recent methodological advances in deep learning, including the use of dropout, the introduction of new activation functions and the ability to train larger models.

Our model is interpretable, which implies that we are able to assess the most important factors associated with default behavior and how they vary over time. This information is important for lenders, and can be used to comply with legislation that requires lenders and credit score providers to notify borrowers of the most important factors affecting their credit score. Additionally, it can be used to formulate economic models of consumer default. The literature on consumer default⁵

⁵ Some notable contributions include Chatterjee et al. (2007), Livshits, MacGee, and Tertilt (2007), and Athreya, Tam, and Young (2012).

suggests that the determinants of default are related to preferences, such as impatience which increases the propensity to borrow, or adverse expenditure of income shocks. Based on these theories, it is then possible to construct theoretical models of credit scoring, of which Chatterjee et al. (2016) is a leading example. We find that the number of trades and the balance on outstanding loans are the most important factors associated with an increase in the probability of default, in addition to outstanding delinquencies and length of the credit history. This information can be used to improve models of consumer default risk and enhance their ability to be used for policy analysis and design.

We also identify and quantify a variety of limitations of conventional credit scoring models, particularly their tendency to misclassify borrowers by default risk, especially for relatively risky borrowers. This implies that our default predictions could help improve the allocation of credit in a way that benefits both lenders, in the form of lower losses, and borrowers, in the form of lower interest rates. Our results also speak to the perils associated with using conventional credit scores outside on the consumer credit sphere. As it is well known, credit scores are used to screen job applicants, in insurance applications, and a variety of additional settings. Economic theory would suggest that this is helpful, as long as credit score provide information which is correlated with characteristics that are of interest for the party using the score (Corbae and Glover (2018)). However, as we show, conventional credit scores misclassify borrowers by a very large degree based on their default risk, which implies that they may not be accurate and may not include appropriate information or use adequate methodologies. The broadening use of credit scores would amplify the impact of these limitations.

The paper is structured as follows. Section 2 describes our data. Section 3 discusses the patterns of consumer default that motivate our adoption of deep learning. Section 4 describes our prediction problem and our model. Section 5 provides a comprehensive performance assessment of our model, compares it to other approaches, and uses a variety of interpretability techniques to understand which factors are strongly associated with default behavior. Section 6 compares our model to conventional credit scores, illustrates its performance in predicting and quantifying aggregate default risk and calculates the value added of adopting our model over alternatives for lenders and borrowers.

2 Data

We use anonymized credit file data from the Experian credit bureau. The data is quarterly, it starts in 2004Q1 and ends in 2015Q4. The data comprises over 200 variables for an anonymized panel of 1 million households. The panel is nationally representative, constructed from a random draw for the universe of borrowers with an Experian credit report. The attributes available comprise information on credit cards, bank cards, other revolving credit, auto loans, installment loans, business loans, first and second mortgages, home equity lines of credit, student loans and collections. There is

information on the number of trades for each type of loan, the outstanding balance and available credit, the monthly payment, and whether any of the accounts are delinquent, specifically 30, 60, 90, 180 days past due, derogatory or charged off. All balances are adjusted for joint accounts to avoid double counting. Additionally, we have the number of hard inquiries by type of product, and public record items, such as bankruptcy by chapter, foreclosure and liens and court judgments. For each quarter in the sample, we also have each borrowers’s credit score. The data also includes an estimate of individual and household labor income based on IRS data. Because this is data drawn from credit reports, we do not know gender, marital status or any other demographic characteristic, though we do know a borrower’s address at the zip code level. We also do not have any information on asset holdings.

Table 1 reports basic demographic information on our sample, including age, household income, credit score and incidence of default, which here is defined as the fraction of households who report a 90 or more days past due delinquency on any trade. This will be our baseline definition of default, as this is the outcome targeted by credit scoring models. Approximately 34% of consumers display such a delinquency.

Table 1: Descriptive Statistics

Feature	Mean	Std. Dev.	Min	25%	50%	75%	Max
Age	45.8	16.3	18	32.2	45.1	57.8	83
Household Income	77.1	55.0	15	42	64	90	325
Credit Score	678.4	111.0	300	588	692	780	839
Default within 8Q	0.339	0.473	0	0	0	1	1

Credit score corresponds to Vantage Score 3. Household income is in USD thousands, trimmed at the 99th percentile. Source: Authors’ calculations based on Experian Data.

3 Patterns in Consumer Default

We now illustrate the complexity of the relation between the various factors that are considered important drivers of consumer default. Our point of departure are standard credit scoring models. While these models are proprietary, the Fair Credit Reporting Act of 1970 and the Equal Opportunity in Credit Access Act of 1984 mandate that the 4 most important factors determining the credit scores be disclosed, together with their importance in determining variation in credit scores. These include credit utilization and number of hard inquiries, which are supposed to capture a consumer’s demand for credit, the variety of debt products, which capture the consumer’s experience in managing credit, and the number and severity of delinquencies. Each of these factors is stated to account for 25-30% of the variation in credit scores. The length of the credit history is also seen

as a proxy on a consumer’s experience in managing credit, and this is reported as accounting for 10-15% of the variation in credit scores.⁶ The models used to determine credit scores as a function of these attributes are not disclosed, but they are widely believed to be based on linear and logistic regression as well as score cards. Additionally, available credit scoring algorithms typically do not score all borrowers.

Subsequently, we illustrate the properties of consumer default that suggest deep learning might be a good candidate for developing a prediction model. Specifically, we show that default is a relatively rare but very persistent outcome, there are substantial non-linearities in the relation between default and plausible covariates, as well as high order interactions between covariates and default outcomes.

3.1 Default Transitions

The default outcome we consider is a 90+ days delinquency, which occurs if the borrower has missed scheduled payments on any product for 90 days or more.⁷ This is the default outcome targeted by the most widely used credit scoring models, which rank consumers based on their probability of becoming 90+ days delinquent in the subsequent 8 quarters. We refer to borrowers who are either current or up to 60 days delinquent on their payments as *current*.

The transition matrix from current to 90+ days past due in the subsequent 8 quarters is given in Table 2. Clearly, the two states are both highly persistent, with a 77% of current customers remaining current in the next 8 quarters, and 93% of customers in default remaining in that state over the same time period. The probability of transition from current to default is 23%, while the probability of curing a delinquency with a transition from default to current is only 7%. These results suggest that default is a particularly persistent state, and predicting a transition into default is very valuable from the lender’s perspective, since they are unlikely to be able to recuperate their losses. But it is also quite difficult, as the current state is also very persistent.

3.2 Non-linearities

Our model includes a relatively large list of features, which is presented in Table 17. The summary statistics for these features are reported in Table 18 in the Appendix. As is demonstrated in the table, there is a wide dispersion in the distribution of these variables. For example, the average balance on credit card trades is approximately \$4,500, but the standard deviation, at \$9,800, is more than twice as large. Similarly, average total debt balances are approximately \$77,000, while the standard deviation is \$170,000 and the 75th percentile \$95,000, suggesting a high upper tail

⁶For an overview of the information available to borrowers about the determinants for their credit score, see <https://ficoscore.com>.

⁷For instance, if no payment has been made by the last day of the month within the past three months and the payment was due on the first day of the month three months ago. For credit cards, this occurs if the borrower does not make at least their minimum payment.

Table 2: Default Transitions

Current/Next 8Q	No default	Default
No default	0.776	0.224
Default	0.073	0.927

Quarterly frequency of transition from current to default. Current corresponds to 0-89 day past due on any trade, Default corresponds to 90+ day past due on any trade in the subsequent 8 quarters. Source: Authors' calculations based on Experian Data.

dispersion of this variable. The other features display similar patterns.

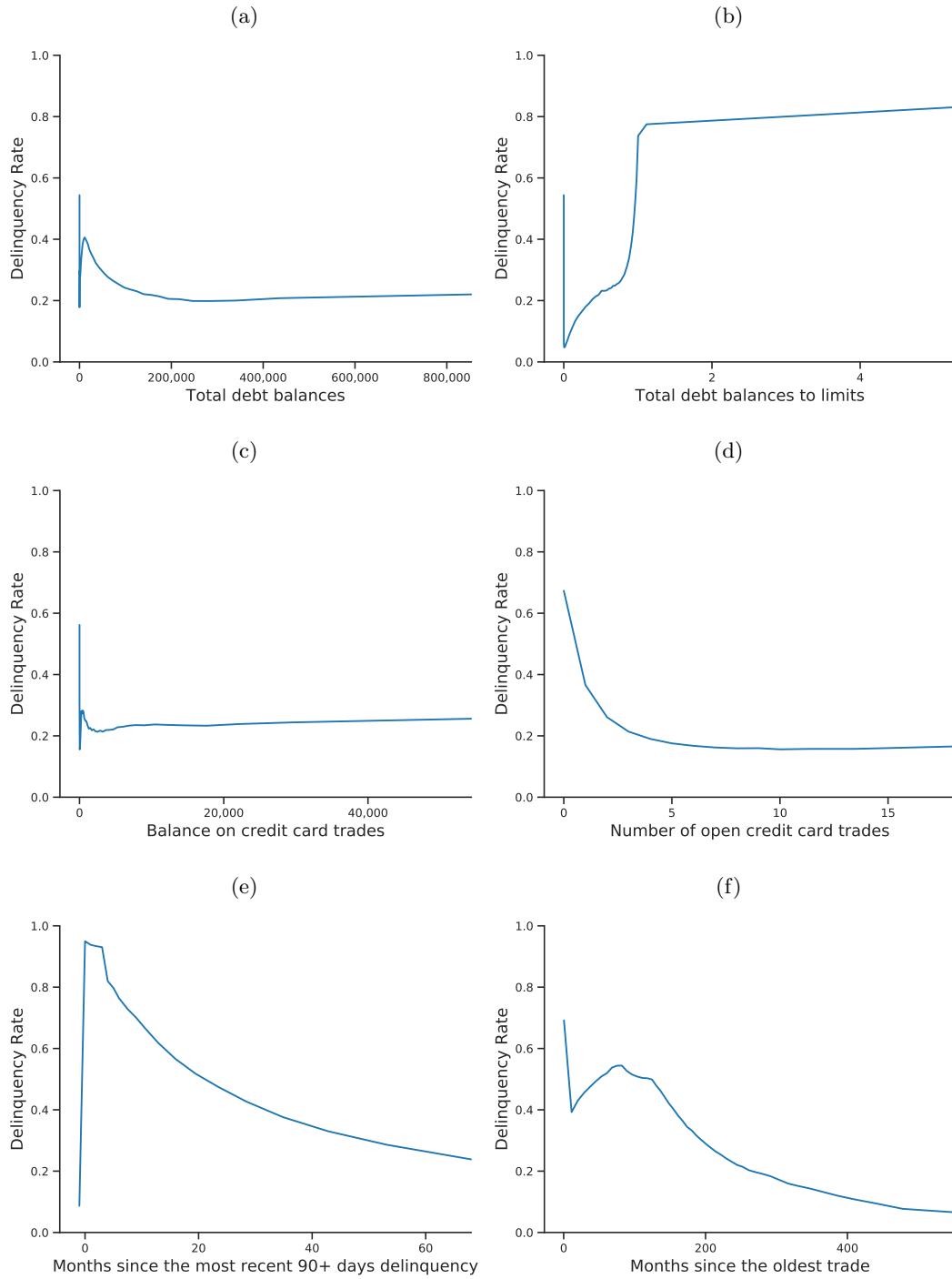
The features are used to predict the probability of default. We now illustrate the highly non-linear relation between the features and the incidence of default. Figure 2 shows how the default rate, defined as the fraction of borrowers with a 90+ day past due delinquency in the subsequent 8 quarters, varies with total debt balances, credit utilization, the credit limit on credit cards, the number of open credit card trades, the number of months since the most recent 90+ day past due delinquency and the months since the oldest trade was opened. The figures show that while the relation between the features and the incidence of default is mostly monotone, it is highly nonlinear, with very little variation in the incidence of default for most intermediate values of the variable and much higher or lower values at the extremes of the range of each covariate. The variables in the figure are just illustrative, a similar pattern holds for most plausible features.

3.3 High Order Interactions

Multidimensional interactions are another feature of the relation between default and plausible covariates, that is default behavior is simultaneously related with multiple variables. To see this, Figure 3 presents contour plots of the relation between the incidence of default and couples of covariates. The covariates reported here are chosen since they are important driving factors in default decisions, based on our model, as discussed in Section 5.2. Panels (a) and (b) explore the joint variation in the incidence of default with total debt balances, credit utilization (total debt balances to limits), and credit history. Blue values correspond to high delinquency rates while red values to low delinquency rates. As can be seen from both panels, higher credit utilization corresponds to higher delinquency rate, but for given credit utilization, an increase in total debt balances first decreases then increases the delinquency rate, where the switch in sign depends on the utilization rate. For given utilization rates, a longer credit history first increases then decreases the delinquency rate, provided the utilization rate is smaller than 1.⁸ Panels (c) and (d) explore the relation between default and credit card borrowing. Default rates decline with the number of

⁸Utilization rates above 1 can arise for a delinquent borrower if fees and other penalty add to their balances for given credit limits.

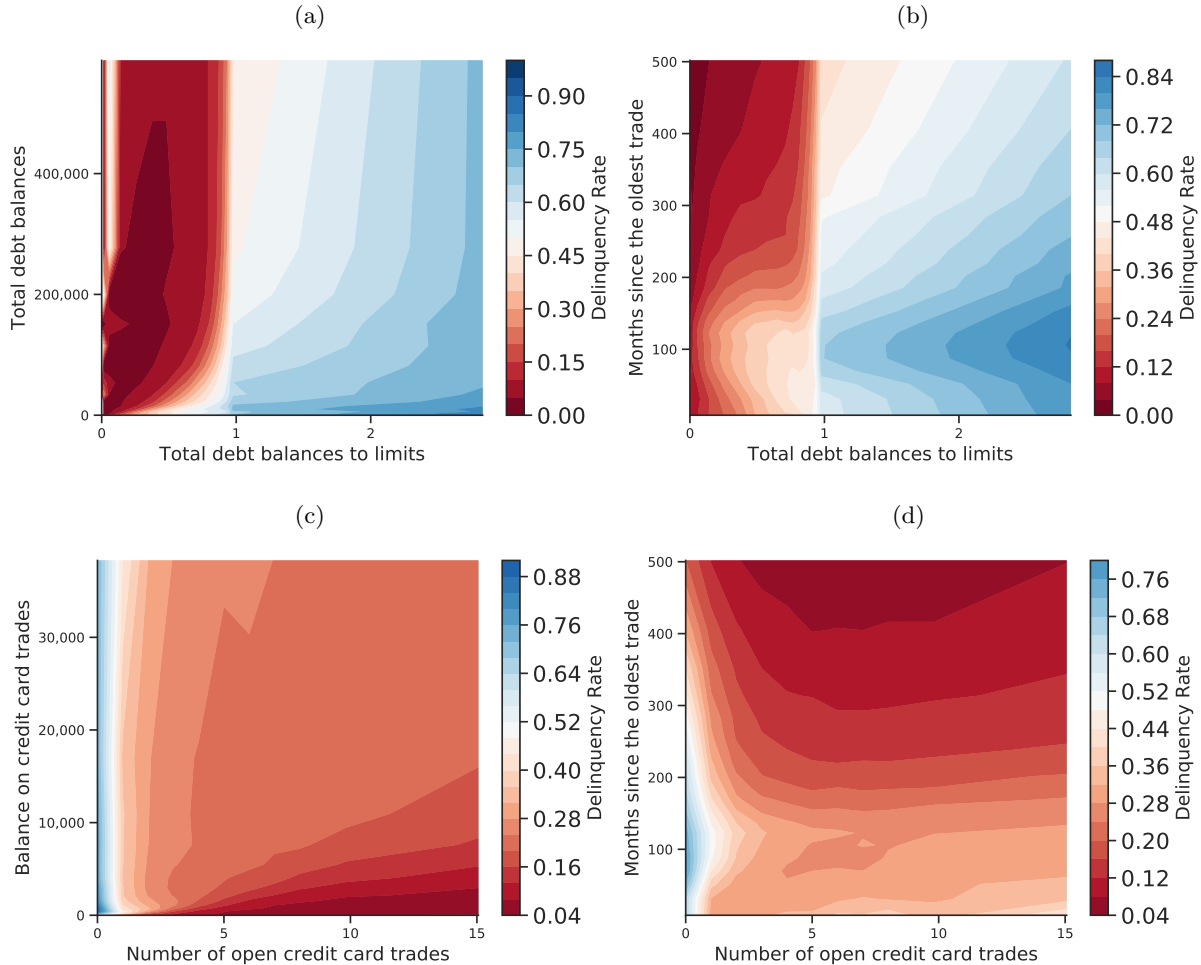
Figure 2: Nonlinear Relation Between Default and Covariates



Delinquency rate is the fraction with 90+ days past due trades in subsequent 8 quarters. In panel (e) and (f), -1 implies no past delinquency. Source: Authors' calculations based on Experian Data.

credit cards, though for a given number of credit card trades, they mostly increase with credit card balances. This relation, however varies with the level of both variables. An increase in the length of credit history is typically associated with lower default rates, however, if the number of open credit cards is low, this relation is non-monotone. The variables reported in the figures are illustrative of a general pattern in the joint relation between couples of covariates and default rates.

Figure 3: Multidimensional Relation Between Default and Covariates



Relationship between 90+ days past due delinquency rate and pairs of covariates. Source: Authors' calculations based on Experian Data.

This pattern of multidimensional non-linear interactions across covariates is fairly difficult to model using standard econometric approaches. For this reason, we propose a deep learning approach to be explained below.

4 Model

Predicting consumer default maps well into a supervised learning framework, which is one of the most widely used techniques in the machine learning literature. In supervised learning, a learner takes in pairs of input/output data. The input data, which is typically a vector, represent pre-identified attributes, also known as features, that are used to determine the output value. Depending on the learning algorithm, the input data can contain continuous and/or discrete values with or without missing data. The supervised learning problem is referred to as a "regression problem" when the output is continuous, and as a "classification problem" when the output is discrete. Once the learner is presented with input/output data, its task is to find a function that maps the input vectors to the output values. A brute force way of solving this task is to memorize all previous values of input/output pairs. Though this perfectly maps the input data to the output values in the training data set, it is unlikely to succeed in forecasting the output values if (1) the input values are different from the ones in the training data set or (2) when the training data set contains noise. Consequently, the goal of supervised learning is to find a function that generalizes beyond the training set, so that it correctly forecasts out-of-sample outcomes. Adopting this machine-learning methodology, we build a model that predicts defaults for individual consumers. We define default as a 90+ days delinquency on any debt in the subsequent 8 quarters, which is the outcome targeted by conventional credit scoring models. Our model outputs a continuous variable between 0 and 1 that can be interpreted under certain conditions as an estimate of the probability of default for a particular borrower at a given point in time, given input variables from their credit reports.

We start by formalizing our prediction problem. We adopt a discrete-time formulation for periods $0, 1, \dots, T$, each corresponding to a quarter. We let the variable D_t^i prescribe the state at time t for individual i with $D \subset \mathbb{N}$ denoting the set of states. We define $D_1^i = 1$ if a consumer is 90+ days past due on any trade and $D_1^i = 0$ otherwise. Consumers will transition between these two states over their lifetime.

Our target outcome is 90+ days past due in the subsequent 8 quarters, defined as:

$$Y_t^i = \begin{cases} 0 & \text{if } \sum_{n=t}^{t+7} D_n^i = 0 \\ 1 & \text{otherwise} \end{cases} \quad (1)$$

We allow the dynamics of the state process to be influenced by a vector of explanatory variables $X_{t-1}^i \in \mathbb{R}^{d_x}$, which includes the state D_{t-1}^i . In our empirical implementation, X_{t-1}^i represents the features in Table 17. We fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and an information filtration $(\mathcal{F}_t)_{(t=0,1,\dots,T)}$. Then, we specify a probability transition function $h_\theta : \mathbb{R}^{d_x} \rightarrow [0, 1]$ satisfying

$$\mathbb{P}[Y_t^i = y | \mathcal{F}_{t-1}] = h_\theta(X_{t-1}^i), y \in D \quad (2)$$

where θ is a parameter to be estimated. Equation 2 gives the marginal conditional probability for

the transition of individual i 's debt from its state D_{t-1}^i at time $t - 1$ to state y at time t given the explanatory variables X_{t-1}^i .⁹ Let g denote the standard *softmax* function:

$$g(z) = \left(\frac{1}{1 + e^{-z}} \right), z \in \mathbb{R}^K, \quad (3)$$

where $K = |D|$. The vector output of the function g is a probability distribution on D .

The marginal probability defined in equation 2 is the theoretical counterpart of the empirical transition matrix reported in Table 2. We propose to model the transition function h_θ with a hybrid deep neural network/gradient boosting model, which combines the predictions of a deep neural network and an extreme gradient boosting model. We explain each of the component models and their properties and the rationale for combining them below.

4.1 Deep Neural Network

One component of our model is based on deep learning, in the class used by Sirignano, Sadhwani, and Giesecke (2018). We restrict attention to feed-forward neural networks, composed of an input layer, which corresponds to the data, one or more interacting hidden layers that non-linearly transform the data, and an output layer that aggregates the hidden layers into a prediction. Layers of the networks consist of neurons with each layer connected by synapses that transmit signals among neurons of subsequent layers. A neural network is in essence a sequence of nonlinear relationships. Each layer in the network takes the output from the previous layer and applies a linear transformation followed by an element-wise non-linear transformation.

Figure 4 illustrates an example of a two layer neural network. This neural network has 3 input units (denoted x_1, x_2, x_3), 4 hidden units, and 1 output unit. Let n_l denote the number of layers in this network ($n_l = 2$). We label layer l as L_l , where layer L_0 is the input layer, and layer $L_{L=2}$ is the output layer. The layers between the input ($l = 0$) and the output layer ($l = L$) are called hidden layers. Given this notation, there are $L - 1$ hidden layers, 1 in this specific example. A neural network without any hidden layers ($L = 1$) is a logistic regression model.

There are two ways to increase the complexity a neural network: (1) increase the number of hidden layers and (2) increase the number of units in a given layer. Lower tier layers in the neural network learn simpler patterns, from which higher tier layers learn to produce more complex patterns. Given a sufficient number of neurons, neural networks can approximate continuous functions on compact sets arbitrarily well (see Hornik, Stinchcombe, and White (1989) and Hornik (1991)). This includes approximating interactions (i.e., the product and division of features). There are two main advantages of adding more layers over increasing the number of units to existing layers; (1) later layers build on early layers to learn features of greater complexity and (2) deep neural networks—those with three or more hidden layers—need exponentially fewer neurons than shallow

⁹The state y encompasses realizations of the state between time t and $t + 7$.

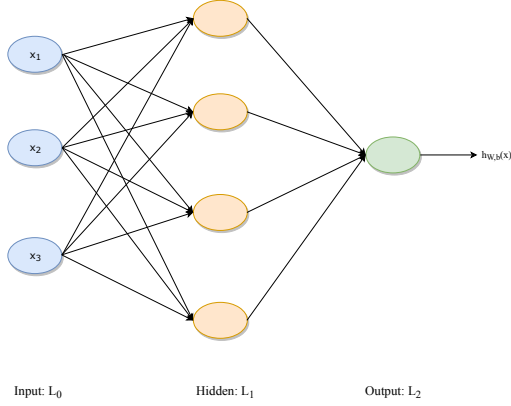


Figure 4: Two Layer Neural Network Example

networks (Bengio and LeCun (2007) and Montúfar et al. (2014)).

In the neural network represented in Figure 4, the parameters to be estimated are $(W, b) = (W^{(0)}, b^{(0)}, W^{(1)}, b^{(1)})$, where $W_{ij}^{(l)}$ denotes the weight associated with the connection between unit j in layer l and unit i in layer $l + 1$, and $b_i^{(l)}$ is the bias associated with unit i in layer $l + 1$. Thus, in this example $W^{(0)} \in \mathbb{R}^{3 \times 4}$, $b^{(0)} \in \mathbb{R}^{4 \times 1}$ and $W^{(1)} \in \mathbb{R}^{1 \times 4}$, $b^{(1)} \in \mathbb{R}$. This implies that there are a total of $21 = (3+1) \cdot 4 + 5$ parameters (four parameters to reach each neuron and five weights to aggregate the neurons into a single output). In general, the number of weight parameters in each hidden layer l is $N^{(l)}(1 + N^{(l-1)})$, plus $1 + N^{(L-1)}$ for the output layer, where $N^{(l)}$ denotes the number of neurons in each layer $l = 1, \dots, L$.

Let $a_i^{(l)}$ denote the activation (e.g., output value) of unit i in layer l . Fix W and b , our neural network defines a hypothesis $h_{W,b}(x)$ that outputs a real number between 0 and 1.¹⁰ Let $f(\cdot)$ denote the activation function that applies to vectors in an element-wise fashion. The computation this neural network represents, often referred to as forward propagation, can be written as:

$$z^{(1)} = W^{(0),T} x + b^{(0)}$$

$$a^{(1)} = f(z^{(1)})$$

$$z^{(2)} = W^{(1),T} a^{(1)} + b^{(1)}$$

$$h_{W,b}(x) = a^{(2)} = f(z^{(2)})$$

¹⁰This is a property of the sigmoid activation function.

There are many choices to make when structuring a neural network, including the number of hidden layers, the number of neurons in each layer, and the activation functions. We built a number of network architectures having up to fifteen hidden layers.¹¹ All architectures are fully connected so each unit receives an input from all units in the previous layer.

Neural networks tend to be low-bias, high-variance models, which imparts them a tendency to over-fit the data. We apply dropout to each of the layers to avoid over-fitting (see Srivastava et al. (2014)). During training, neurons are randomly dropped (along with their connections) from the neural network with probability p (referred to as the dropout rate), which prevents complex co-adaptations on training data.

We apply the same activation function (scaled exponential linear unit or SELU) at all nodes, which is obtained via hyperparameter optimization,¹² and defined as:

$$\text{SELU}(x) = \lambda \begin{cases} \alpha e^x - \alpha & \text{if } x \leq 0 \\ x & \text{otherwise} \end{cases} \quad (4)$$

where $\alpha \approx 1.6733$, and $\lambda \approx 1.0507$.

Let $N^{(l)}$ denote the number of neurons in each layer $l = 1, \dots, L$. Define the output of neuron k in layer l as $z_k^{(l)}$. Then, define the vector of outputs (including the bias term $z_0^{(l)}$) for this layer as $z^{(l)} = (z_0^{(l)}, z_1^{(l)}, \dots, z_{N^{(l)}}^{(l)})'$. For the input layer, define $z^{(0)} = (x_0^{(l)}, x_1^{(l)}, \dots, x_{N^{(l)}}^{(l)})'$. Formally, the recursive output of the l -th layer of the neural network is:

$$z^{(l)} = \text{SELU}(W^{(l-1),T} z^{(l-1)} + b^{(l-1)}), \quad (5)$$

with final output:

$$h_\theta(x) = g(W^{(L-1),T} z^{(L-1)} + b^{(L-1)}). \quad (6)$$

The parameter specifying the neural network is:

$$\theta = (W_0, b_0, \dots, W_{L-1}, b_{L-1}) \quad (7)$$

4.2 Decision Tree Models

The second component of our model is Extreme Gradient Boosting, which builds on decision tree models. Tree-based models split the data several times based on certain cutoff values in the explanatory variables.¹³ A number of such models have become quite prevalent in the literature, most notably random forests (see Breiman (2001) and Butaru et al. (2016)) and Classification and Regression Trees, known as CART. We briefly review CART and then explain gradient boosting.

¹¹The number of layers and the number of neurons in each layer, along with other hyperparameters of the model, are chosen by Tree-structured Parzen Estimator (TPE) approach. See Appendix C for more details.

¹²There are many potential choices for the nonlinear activation function, including the sigmoid, relu, and tanh.

¹³Splitting means that different subsets of the dataset are created, where each observation belongs to one subset. For a review on decision trees, see Khandani, Kim, and Lo (2010).

4.2.1 CART

There are a number of different decision tree-based algorithms. As an illustration of the approach, we describe Classification and Regression Trees or CART. CART models an outcome y_i for an instance i as follows:

$$\hat{y}_i = \hat{f}(x_i) = \sum_{m=1}^M c_m I\{x_i \in R_m\}, \quad (8)$$

where each observation x_i belongs to exactly one subset R_m . The identity function I returns 1 if x_i is in R_m and 0 otherwise. If x_i falls into R_l , the predicted outcome is $\hat{y} = c_l$, where c_l is the mean of all training observations in R_l .

The estimation procedure takes a feature and computes the cut-off point that minimizes the Gini index of the class distribution of $\mathbf{y}$, which makes the two resulting subsets as different as possible. Once this is done for each feature, the algorithm uses the best feature to split the data into two subsets. The algorithm is then repeated until a stopping criterium is reached.

Tree-based models have a number of advantages that make them popular in applications. They are invariant to monotonic feature transformations and can handle categorical and continuous data in the same model. Like deep neural networks, they are well suited to capturing interactions between variables in the data. Specifically, a tree of depth L can capture $(L - 1)$ interactions. The interpretation is straightforward, and provides immediate counterfactuals: "If feature x_j had been bigger / smaller than the split point, the prediction would have been $\bar{y}_0$ instead of $\bar{y}_1$." However, these models also have a number of limitations. They are poor at handling linear relationships, since tree algorithms rely on splitting the data using step functions, an intrinsically non-linear transformation. Trees also tend to be unstable, so that small changes in the training dataset might generate a different tree. They are also prone to overfitting to the training data. For more information on tree-based models see Molnar (2019).

4.2.2 eXtreme Gradient Boosting (XGBoost)

Gradient Boosted Trees (GBT) are an ensemble learning method that corrects for tree-based models' tendency to overfit to training data by recursively combining the forecasts of many over-simplified trees. Though shallow trees are "weak learners" on their own with little predictive power, the theory behind boosting proposes that a collection of weak learners, as an ensemble, creates a single strong learner with improved stability over a single complex tree.

At each step m , $1 \leq m \leq M$, of gradient boosting, an estimator, h_m , is computed on the residuals from the previous models predictions. A critical part of gradient boosting method is regularization by shrinkage as proposed by Friedman (2001). This consists in modifying the update rule as follows:

$$F_m(x) = F_{m-1}(x) + \nu \gamma_m h_m(x), \quad (9)$$

where $h_m(x)$ represents a weak learner of fixed depth, γ_m is the step length and ν is the learning

rate or shrinkage factor.

XGBoost is a fast implementation of Gradient Boosting, which has the advantages of fast speed and high accuracy. For classification, XGBoost combines the principles of decision trees and logistic regression, so that the output of our XGBoost model is a number between 0 and 1. For the remainder of the paper we refer to XGBoost as GBT.¹⁴

4.3 Hybrid DNN-GBT Model

We examined two techniques to create a hybrid DNN-GBT ensemble model. Ensemble models combine multiple learning algorithms to generate superior predictive performance than could be obtained from any of the constituent learning algorithms alone. The first method combines the two models by replacing the final layer of the neural network with a gradient boosted trees model. Examples of this approach are Chen, Lundberg, and Lee (2018) and Ren et al. (2017). The second, uses both models separately and then averages out the final predicted probabilities of the two models. We found the latter to perform better on our dataset. This method is similar to Kvamme et al. (2018), who combined a convolutional neural network with a random forest by averaging. Thus, our methodology relies on combining the output of the deep neural network with the output of a gradient boosted trees model. This is achieved in two steps:

1. For each observation, run DNN and GBT separately and obtain predicted probabilities for each of the models;
2. Take the arithmetic mean of the predicted probabilities.

5 Implementation

Table 17 lists the features from the credit report data we use as inputs in the model. These covariates are chosen based on economic theory (see for example Chatterjee et al. (2007)) as well as based on information from currently used credit scoring models. They include information on balances and credit limits for different types of consumer debt, severity and number of delinquencies, credit utilization by type of product, public record items such as bankruptcy filings by chapter and foreclosure, collection items, and length of the credit history. In order to be consistent with the restrictions of the Fair Credit Reporting Act on 1970 and the Equal Opportunity in Credit Access Act of 1984 we do not include information on age or zip code, and we do not include any information on income, to be consistent with current credit scoring models. Table 17 lists the full set of features used in our machine learning models. It is important to note that we do not use any lagged features. This is because many of the features have a temporal dimension, for example "worst present status

¹⁴For more on XGBoost, see Chen, Lundberg, and Lee (2018) and Ren et al. (2017).

on any traded in the last 6 months.” Importantly, excluding lags enables us to provide a default prediction to any borrower with a non-empty credit record, which implies that we can score virtually all consumers.

5.1 Classifier Performance

In this section, we describe the performance of our hybrid model under various training and testing windows. First, we evaluate our model on the pooled sample (2004Q1-2013Q4), where we apply a random 60%-20%-20% split to our training, validation, and testing sets. Then, to account for look-ahead bias, we train and test our models based on 8 quarter windows that were observable at the time of forecast. In particular, we require our training and testing sets to be separated by 8 quarters to avoid overlap. For instance, the second out-of-sample model was calibrated using input data from 2004Q2, from which the parameter estimates were applied to the input data in 2006Q2 to generate forecasts of delinquencies over the 8 quarter window from 2006Q3-2008Q2. This gives us a total of 32+1 calibration and testing periods reported in Table 3. The percentage of 90+ days past due accounts within 8 quarters varies from 32.5% to 35.9%.

The hybrid model outputs a continuous variable that, under certain circumstances, can be interpreted as an estimate of the probability of an account becoming 90+ days delinquent during the subsequent 8 quarters. One measure of the model’s success is its ability to differentiate between accounts that did become delinquent and those that did not; if these two groups have the same forecasts, the model provides no value. Table 3 presents the average forecast for accounts that did and did not fall into the 90+ days delinquency category over the 32+1 evaluation periods. For instance, during the testing period for 2010Q4, the model’s average prediction among the 35.44% of accounts that became 90+ days delinquent was 73.35%, while the average prediction among the 64.56% of accounts that did not was 15.86%. We should highlight that these are truly out-of-sample predictions, since the model is calibrated using input data from 2008Q4. This shows the forecasting power of our model in distinguishing between accounts that will and will not become delinquent within 8 quarters. Furthermore, this forecasting power seems to be stable over the 32+1 calibration and evaluation periods, partly driven by the frequent re-calibration of the model that captures some of the changing dynamics of consumer behavior.

We also look at accounts that are current as of the forecast date but become 90+ days delinquent within the subsequent 8 quarters. In particular, we contrast the model’s average prediction among individuals who were current on their accounts but became 90+ days delinquent with the average prediction among customers who were current and did not become delinquent. Given the difficulty of predicting default among individuals that currently show no sign of delinquency, we anticipate the model’s performance to be less impressive than the values reported in Table 3. Nonetheless, the values reported in Table 4 indicate that the model is able to distinguish between these two populations. For instance, using input data from 2008Q4, the average model prediction for individuals who were current on their debts and became 90+ days delinquent is 45.13%, contrasted with

Table 3: 1 Quarter Ahead Predictions, Full Sample– Hybrid DNN-GBT

Training Window	Testing Window	Data	Predicted	Delinquents	Non-Delinquents
2004Q1-2013Q4	2004Q1-2013Q4	0.3396	0.3363	0.7583	0.1181
2004Q1	2006Q1	0.3248	0.2924	0.6499	0.1204
2004Q2	2006Q2	0.3274	0.3041	0.6759	0.123
2004Q3	2006Q3	0.3306	0.3111	0.6852	0.1263
2004Q4	2006Q4	0.3347	0.3158	0.6861	0.1295
2005Q1	2007Q1	0.341	0.316	0.6854	0.1249
2005Q2	2007Q2	0.3444	0.3202	0.6872	0.1274
2005Q3	2007Q3	0.3469	0.3197	0.6871	0.1246
2005Q4	2007Q4	0.3505	0.3287	0.6958	0.1306
2006Q1	2008Q1	0.3535	0.3373	0.7098	0.1336
2006Q2	2008Q2	0.3545	0.3338	0.7012	0.1321
2006Q3	2008Q3	0.3558	0.336	0.7048	0.1324
2006Q4	2008Q4	0.3587	0.3407	0.7089	0.1348
2007Q1	2009Q1	0.3588	0.3488	0.7213	0.1403
2007Q2	2009Q2	0.358	0.3486	0.7226	0.1401
2007Q3	2009Q3	0.3573	0.3542	0.7296	0.1455
2007Q4	2009Q4	0.3589	0.3552	0.7307	0.145
2008Q1	2010Q1	0.3589	0.3563	0.7293	0.1474
2008Q2	2010Q2	0.3568	0.362	0.7351	0.155
2008Q3	2010Q3	0.3559	0.3644	0.7367	0.1587
2008Q4	2010Q4	0.3544	0.3623	0.7335	0.1586
2009Q1	2011Q1	0.3541	0.3582	0.7286	0.1552
2009Q2	2011Q2	0.3511	0.3589	0.7271	0.1596
2009Q3	2011Q3	0.35	0.3543	0.7217	0.1564
2009Q4	2011Q4	0.3484	0.3544	0.7255	0.1561
2010Q1	2012Q1	0.3467	0.3539	0.7306	0.154
2010Q2	2012Q2	0.3434	0.3504	0.7254	0.1543
2010Q3	2012Q3	0.3396	0.3488	0.7297	0.1529
2010Q4	2012Q4	0.3358	0.3463	0.7286	0.1531
2011Q1	2013Q1	0.3341	0.3475	0.7343	0.1534
2011Q2	2013Q2	0.3317	0.3441	0.7277	0.1537
2011Q3	2013Q3	0.3298	0.3425	0.7328	0.1504
2011Q4	2013Q4	0.3275	0.3399	0.7328	0.1486

Performance metrics for our model of default risk over 32+1 testing windows. For each testing window, the model is calibrated on data over the period specified in the training window, and predictions are based on the data available as of the data in the training window. For example, the fourth row reports the performance of the model calibrated using input data available in 2004Q3, and applied to 2006Q3 data to generate forecasts of delinquencies for within 8 quarter delinquencies. Average model forecasts over all customers, and customers that (ex-post) did and did not become 90+ days delinquent over the testing window are also reported. Source: Authors' calculations based on Experian Data.

12.34% for those who did not. As in Table 3, the model’s ability to distinguish between these two classes is consistent across the 32+1 evaluation periods listed in Table 4.

Under certain conditions, the forecasts generated by our model can be converted to binary decisions by comparing the forecast to a specified threshold and classifying accounts with scores exceeding that threshold as high-risk. Setting the threshold level comes with a trade-off. A low level threshold leads to many accounts being classified as high risk, and even though this approach may accurately capture customers who are actually high-risk and about to default on their payments, it can also give rise to many low-risk accounts incorrectly classified as high-risk. By contrast, a high threshold can result in too many high-risk accounts being classified as low-risk.

This type of trade-off is inherent in any classification problem, and involves trading off Type-I (false positives) and Type-II (false negatives) errors in a classical hypothesis testing context. In the credit risk management context, a cost/benefit analysis can be formulated contrasting false positives to false negatives to make this trade-off explicit, and applying the threshold that will optimize an objective function in which costs and benefits associated with false positives and false negatives are inputs.

A commonly used performance metric in the machine learning and statistics literature is a 2×2 contingency table, often referred to as the *confusion matrix*, that describes the statistical behavior of any classification algorithm. In our application, the two rows correspond to ex post realizations of the two types of accounts in our sample, *no default* and *default*. We define *no default* accounts as those who do not become 90+ days delinquent during the forecast period, and *default* accounts as those who do. The two columns correspond to ex ante classifications of the accounts into these categories. If a predictive model is applied to a set of accounts, each account falls into one of the four cells in the confusion matrix, thus the performance of the model can be assessed by the relative frequencies of the entries. In the Neymann-Pearson hypothesis-testing framework, the lower-left entry is defined as Type-I error and the upper right as Type-II error, while the objective of the researcher is to minimize Type-II error (i.e., maximize "power") subject to a fixed level of Type-I error (i.e., "size").

As an illustration, figure 5 shows the confusion matrix for our hybrid DNN-GBT model calibrated using 2011Q4 data and evaluated on 2013Q4 data and a threshold of 50%. This means that accounts with estimated delinquency probabilities greater than 50% are classified as default and 50% or below as no default. For this quarter, the model classified $61.29\% + 7.19\% = 68.48\%$ of the accounts as no default, of which 61.29% did indeed not default and 7.19% actually defaulted, that is, they were 90+ days delinquent in the subsequent 8 quarters. By the same token, of the $5.96\% + 25.56\% = 31.52\%$ borrowers who defaulted, the model accurately classified 25.56%. Thus, the model’s accuracy, defined as the percent of instances correctly classified, is the sum of the entries on the diagonal of the confusion matrix, that is, $61.29\% + 25.56\% = 86.85\%$.

We can compute three additional performance metrics from the entries of the confusion matrix, which we describe heuristically here and define formally in the appendix. *Precision* measures the

Table 4: 1 Quarter Ahead Predictions, Current– Hybrid DNN-GBT

Training Window	Testing Window	Data	Predicted	Delinquent	Non-delinquent
2004Q1-2013Q4	2004Q1-2013Q4	0.1676	0.1618	0.5421	0.085
2004Q1	2006Q1	0.1844	0.1558	0.4254	0.0948
2004Q2	2006Q2	0.1702	0.1452	0.3967	0.0937
2004Q3	2006Q3	0.1695	0.1487	0.4019	0.097
2004Q4	2006Q4	0.1727	0.151	0.4029	0.0984
2005Q1	2007Q1	0.1805	0.151	0.4002	0.0961
2005Q2	2007Q2	0.1813	0.1527	0.3947	0.0991
2005Q3	2007Q3	0.1831	0.1494	0.3899	0.0955
2005Q4	2007Q4	0.1847	0.1541	0.3995	0.0985
2006Q1	2008Q1	0.189	0.162	0.4172	0.1025
2006Q2	2008Q2	0.1896	0.16	0.4073	0.1022
2006Q3	2008Q3	0.1872	0.1589	0.4055	0.1021
2006Q4	2008Q4	0.1817	0.1574	0.401	0.1034
2007Q1	2009Q1	0.1781	0.1634	0.4182	0.1082
2007Q2	2009Q2	0.1752	0.1626	0.4177	0.1085
2007Q3	2009Q3	0.1713	0.1673	0.4317	0.1127
2007Q4	2009Q4	0.1661	0.1641	0.4232	0.1125
2008Q1	2010Q1	0.1683	0.1689	0.433	0.1154
2008Q2	2010Q2	0.1668	0.1768	0.449	0.1223
2008Q3	2010Q3	0.1661	0.1806	0.4577	0.1254
2008Q4	2010Q4	0.1644	0.1773	0.4513	0.1234
2009Q1	2011Q1	0.1674	0.1789	0.4594	0.1226
2009Q2	2011Q2	0.1668	0.1798	0.4593	0.1239
2009Q3	2011Q3	0.1669	0.1765	0.452	0.1214
2009Q4	2011Q4	0.1597	0.1726	0.4462	0.1206
2010Q1	2012Q1	0.1604	0.1701	0.4458	0.1174
2010Q2	2012Q2	0.1622	0.1702	0.4465	0.1167
2010Q3	2012Q3	0.1598	0.167	0.4435	0.1143
2010Q4	2012Q4	0.1575	0.165	0.4416	0.1133
2011Q1	2013Q1	0.1606	0.1703	0.4591	0.115
2011Q2	2013Q2	0.1603	0.1707	0.4556	0.1163
2011Q3	2013Q3	0.1578	0.1652	0.4531	0.1113
2011Q4	2013Q4	0.1548	0.1615	0.4457	0.1095

Performance metrics for our model of default risk over 32+1 testing windows for customers who are current as of the forecast date but become 90+ days delinquent in the following 8 quarters. For each testing window, the model is calibrated on data over the period specified in the training window columns, and predictions are based on the data available as of the data in the training window. For example, the fourth row reports the performance of the model calibrated using input data available in 2004Q3, and applied to 2006Q3 data to generate forecasts of delinquencies for within 8 quarter delinquencies. Average model forecasts over all current customers, and all current customers that did and did not become 90+ days delinquent over the testing window are also reported. Source: Authors' calculations based on Experian Data.

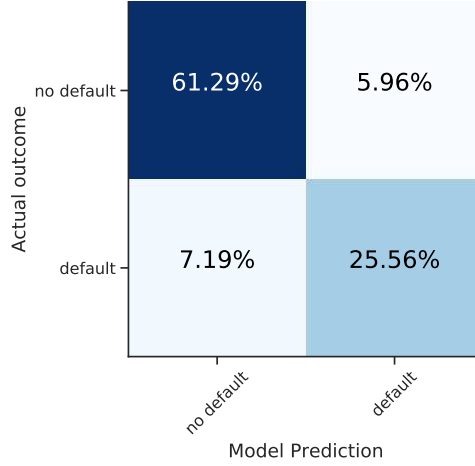


Figure 5: Confusion matrix for our model of default risk. Rows correspond to actual states, with default defined as 90+ days delinquent, no default otherwise. Classifier threshold: 50%. The numerical example is based on the model calibrated on 2011Q4 data and applied to 2013Q4 to generate out-of-sample predictions. Source: Authors' calculations based on Experian Data.

model's accuracy in instances that are classified as default. *Recall* refers to the number of accounts that defaulted as identified by the model divided by the actual number of defaulting accounts. Finally, the *F-measure* is simply the harmonic mean of precision and recall. In an ideal scenario, we would have very high precision and recall.

We can track the trade-off between true and false positives by varying the classification threshold of our model, and this trade-off is plotted in Figure 6. The blue line, called the Receiver Operating Characteristic (ROC) curve, is the pairwise plot of true and false positive rates for different classification thresholds (green line), and as the threshold decreases, the figure shows that the true positive rate increases, but so does the false positive rate. The ROC curve illustrates the non-linear nature of the trade-offs, implying that increase in true positive rates is not always proportionate with the increase in false positive rates. The optimal threshold then considers the cost of false positives with respect to the gain of true positives. If these are equal, the optimal threshold will correspond to the tangent point of the ROC curve with the 45 degree line.

The last performance metric we consider is the area under the ROC curve, known as AUC score, which is a widely used measure in the machine-learning literature for comparing models. It can be interpreted as the probability of the classifier assigning a higher probability of being in default to an account that is actually in default. The ROC area of our model ranges from 0.9241 to 0.9306, demonstrating that our machine-learning classifiers have strong predictive power in separating the two classes.

Table 5 reports the performance metrics widely used in the machine-learning literature for

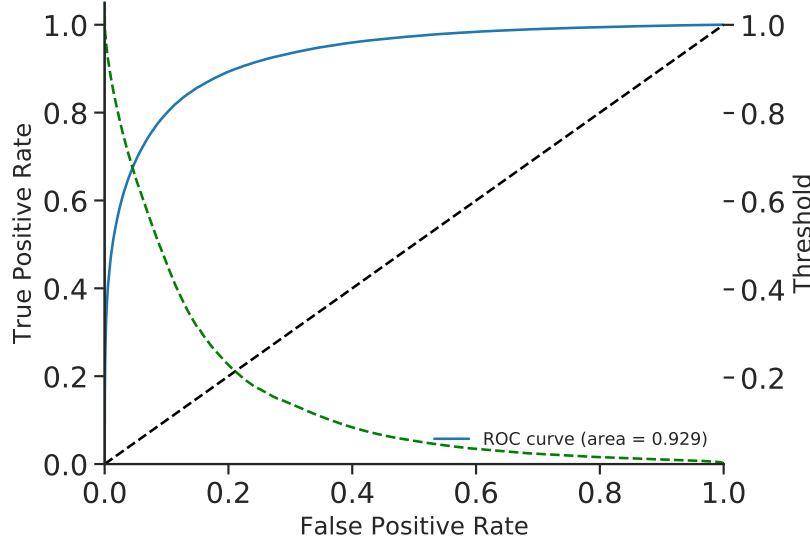


Figure 6: Receiver Operating Characteristic (ROC) curve and corresponding classification threshold value of out-of-sample forecasts of 90+ days delinquencies over the 8Q forecast horizon based on our model of default risk. The model is trained on 2011Q4 input data and evaluated on data from 2013Q4. Source: Authors' calculations based on Experian Data.

each of the 32+1 models discussed. Our models exhibit strong predictive power across the various performance metrics. For instance, the 85.31% precision implies that when our classifier predicts that someone is going to default, there is an 85.31% chance this person will actually default; while the 70.16% recall means that we accurately identified 70.16% of all the defaulters. Our approach of using only one quarter of data to train the model is rather restrictive. Using more quarters usually increases model performance, so since most credit scoring applications will use a training data that exceeds one quarter, performance metrics are likely to improve relative to what we report in our exercise.

Table 6 reports the same performance metrics for the population of borrowers who are current, that is, they do not have any delinquencies in the quarter they are assessed. As previously noted, this is a smaller population with a lower probability of default. Performance metrics drop marginally relative to those for the model applied to the population of all borrowers but they are still very strong. For example, the AUC score drops from 92-93% to 86-88%, accuracy mostly remain in the same range and the loss increases by 1-2 percentage points.

5.2 Model Interpretation

We use our hybrid DNN-GBT model to uncover associations between the explanatory variables and default behavior. Since we do not identify causal relationships, our goal is simply to find covariates that have an important impact on default outcomes. Our findings can be used to better

Table 5: Performance Metrics using Hybrid DNN-GBT

Training Window	Testing Window	AUC score	Precision	Recall	F-measure	Accuracy	Loss
2004Q1-2013Q4	2004Q1-2013Q4	0.9554	0.8703	0.8160	0.8470	0.8962	0.2535
2004Q1	2006Q1	0.9246	0.8531	0.7016	0.7700	0.8638	0.3233
2004Q2	2006Q2	0.9255	0.8509	0.7151	0.7771	0.8657	0.3178
2004Q3	2006Q3	0.9264	0.8488	0.7273	0.7834	0.8670	0.3156
2004Q4	2006Q4	0.9255	0.8499	0.7258	0.7829	0.8653	0.3187
2005Q1	2007Q1	0.9260	0.8599	0.7191	0.7832	0.8643	0.3205
2005Q2	2007Q2	0.9260	0.8617	0.7202	0.7846	0.8638	0.3211
2005Q3	2007Q3	0.9253	0.8606	0.7211	0.7847	0.8627	0.3246
2005Q4	2007Q4	0.9241	0.8554	0.7280	0.7866	0.8615	0.3276
2006Q1	2008Q1	0.9251	0.8513	0.7390	0.7912	0.8621	0.3262
2006Q2	2008Q2	0.9249	0.8569	0.7295	0.7881	0.8609	0.3271
2006Q3	2008Q3	0.9259	0.8563	0.7347	0.7909	0.8618	0.3248
2006Q4	2008Q4	0.9263	0.8582	0.7347	0.7916	0.8613	0.3247
2007Q1	2009Q1	0.9279	0.8512	0.7509	0.7979	0.8635	0.3202
2007Q2	2009Q2	0.9283	0.8513	0.7538	0.7996	0.8647	0.3193
2007Q3	2009Q3	0.9289	0.8439	0.7665	0.8034	0.8659	0.3166
2007Q4	2009Q4	0.9306	0.8474	0.7710	0.8074	0.8680	0.3124
2008Q1	2010Q1	0.9304	0.8462	0.7721	0.8074	0.8678	0.3137
2008Q2	2010Q2	0.9301	0.8367	0.7820	0.8084	0.8678	0.3143
2008Q3	2010Q3	0.9298	0.8340	0.7849	0.8087	0.8678	0.3144
2008Q4	2010Q4	0.9295	0.8326	0.7823	0.8066	0.8671	0.3157
2009Q1	2011Q1	0.9304	0.8379	0.7806	0.8082	0.8689	0.3133
2009Q2	2011Q2	0.9289	0.8329	0.7796	0.8054	0.8677	0.3161
2009Q3	2011Q3	0.9294	0.8416	0.7694	0.8039	0.8686	0.3145
2009Q4	2011Q4	0.9296	0.8354	0.7756	0.8044	0.8686	0.3135
2010Q1	2012Q1	0.9302	0.8332	0.7785	0.8049	0.8692	0.3115
2010Q2	2012Q2	0.9291	0.8319	0.7718	0.8007	0.8681	0.3134
2010Q3	2012Q3	0.9289	0.8278	0.7735	0.7997	0.8684	0.3119
2010Q4	2012Q4	0.9282	0.8199	0.7749	0.7968	0.8673	0.3136
2011Q1	2013Q1	0.9288	0.8156	0.7800	0.7974	0.8676	0.3121
2011Q2	2013Q2	0.9284	0.8169	0.7758	0.7958	0.8679	0.3117
2011Q3	2013Q3	0.9290	0.8101	0.7834	0.7965	0.8680	0.3103
2011Q4	2013Q4	0.9293	0.8108	0.7805	0.7954	0.8685	0.3083

Performance metrics for our model of default risk. The model calibrations are specified by the training and testing windows. The results of classifications versus actual outcomes over the following 8Q are used to calculate these performance metrics for 90+ days delinquencies within 8Q. Source: Authors' calculations based on Experian Data.

Table 6: Performance Metrics using Hybrid DNN-GBT

Training Window	Testing Window	AUC score	Precision	Recall	F-measure	Accuracy	Loss
2004Q1	2006Q1	0.8776	0.7688	0.4035	0.5292	0.8676	0.3167
2004Q2	2006Q2	0.8658	0.7315	0.3543	0.4774	0.8680	0.3159
2004Q3	2006Q3	0.8644	0.7189	0.3647	0.4839	0.8681	0.3159
2004Q4	2006Q4	0.8649	0.7236	0.3640	0.4843	0.8661	0.3189
2005Q1	2007Q1	0.8648	0.7425	0.3571	0.4823	0.8616	0.3284
2005Q2	2007Q2	0.8624	0.7364	0.3477	0.4724	0.8592	0.3320
2005Q3	2007Q3	0.8613	0.7376	0.3457	0.4708	0.8577	0.3368
2005Q4	2007Q4	0.8606	0.7273	0.3579	0.4797	0.8566	0.3386
2006Q1	2008Q1	0.8621	0.7206	0.3815	0.4989	0.8551	0.3405
2006Q2	2008Q2	0.8618	0.7267	0.3631	0.4842	0.8533	0.3420
2006Q3	2008Q3	0.8612	0.7194	0.3627	0.4823	0.8542	0.3401
2006Q4	2008Q4	0.8583	0.7130	0.3484	0.4681	0.8562	0.3363
2007Q1	2009Q1	0.8599	0.7001	0.3771	0.4902	0.8603	0.3285
2007Q2	2009Q2	0.8600	0.6978	0.3786	0.4909	0.8624	0.3251
2007Q3	2009Q3	0.8599	0.6802	0.4002	0.5039	0.8650	0.3204
2007Q4	2009Q4	0.8581	0.6740	0.3905	0.4945	0.8674	0.3162
2008Q1	2010Q1	0.8613	0.6858	0.4065	0.5105	0.8687	0.3146
2008Q2	2010Q2	0.8622	0.6687	0.4334	0.5259	0.8696	0.3129
2008Q3	2010Q3	0.8622	0.6622	0.4454	0.5326	0.8701	0.3125
2008Q4	2010Q4	0.8637	0.6675	0.4384	0.5292	0.8718	0.3091
2009Q1	2011Q1	0.8669	0.6785	0.4517	0.5424	0.8724	0.3084
2009Q2	2011Q2	0.8670	0.6770	0.4538	0.5434	0.8728	0.3085
2009Q3	2011Q3	0.8685	0.6923	0.4341	0.5336	0.8734	0.3069
2009Q4	2011Q4	0.8657	0.6773	0.4278	0.5244	0.8760	0.3027
2010Q1	2012Q1	0.8672	0.6774	0.4289	0.5253	0.8756	0.3019
2010Q2	2012Q2	0.8693	0.6895	0.4234	0.5247	0.8755	0.3015
2010Q3	2012Q3	0.8682	0.6839	0.4216	0.5217	0.8764	0.3000
2010Q4	2012Q4	0.8676	0.6811	0.4133	0.5144	0.8771	0.2984
2011Q1	2013Q1	0.8700	0.6754	0.4450	0.5365	0.8765	0.2996
2011Q2	2013Q2	0.8705	0.6789	0.4401	0.5340	0.8769	0.2986
2011Q3	2013Q3	0.8698	0.6711	0.4424	0.5333	0.8778	0.2967
2011Q4	2013Q4	0.8681	0.6681	0.4317	0.5245	0.8789	0.2950

Performance metrics for our model of default risk for the current population. Borrowers who are current do not have any delinquencies. The model calibrations are specified by the training and testing windows. The results of classifications versus actual outcomes over the following 8Q are used to calculate these performance metrics for 90+ days delinquencies within 8Q. Source: Authors' calculations based on Experian Data.

understand default behavior, further refine model specification and possibly aid in the formulation of theoretical models of consumer default. For this exercise, we mainly use the pooled model, which uses all available data. This allows us to assess factors that are critical in default behavior throughout the sample period with the best performing model. We also consider time variation in the factors influencing the default decision in subsets of our sample.

5.2.1 Explanatory Power of Variables

We start by examining the explanatory power of each of our features. We follow an approach similar to Sirignano, Sadhwani, and Giesecke (2018), which amounts to a perturbation analysis on the pooled sample using our hybrid model. First, we draw a random sample of 100,000 observations from the testing sample. Then, for each variable, we re-shuffle the feature, keeping the distribution intact and the model’s loss function is evaluated with the changed covariate. We repeat this step 10 times, and report the average of the loss and accuracy. Then, the variable is replaced to its original values, and a perturbation test is performed on a new variable. Perturbing the variable of course reduces the accuracy of the model, and the test loss becomes larger. If a particular variable has strong explanatory power, the test loss will significantly increase. The test loss for the complete model when no variables are perturbed is the Baseline value. Features that have large explanatory power, and whose information is not contained in the other remaining variables will increase the loss significantly if they are altered. Table 7 reports the results. Features relating to the number and credit available on revolving trades and features pertaining to delinquencies dominate the list. Specifically, credit amount on revolving trades, months since the most recent 90+ days delinquency and months since the oldest trade was opened each increase the loss by 15%, the number of open bankcard and credit card trades increase the loss by 13%, while the balance on first mortgage trades and monthly payment on first mortgages increase the loss by 11%. These results suggest that revolving debt, length of credit history and temporal proximity to a delinquency are all important factors in default behavior. Based on publicly available information, length of the credit history is also an important determinant of standard credit scoring models, though credit utilization rather than balances or number of trades is understood as the most critical. This approach to assessing the importance of different features for the predicted probability of default has two major shortcomings. First, when features are highly correlated, the interpretation of feature importance can be biased by unrealistic data instances. To illustrate this problem, consider two highly correlated features. As we perturb one of the features, we create instances that are unlikely or even impossible. For example, mortgage balances are highly correlated with and lower than total debt balances, yet this perturbation approach could create instances in which total debt balances are smaller than mortgage balances. Since many of the features are strongly correlated, care must be taken with interpretation of feature importance. We list the highly correlated features in Appendix C. An additional concern with this perturbation approach is that the distribution of some features are highly skewed, which implies that the probability of their value being different than where the mass

of their distribution is concentrated is quite low. Moreover, skewness varies substantially across features, therefore the informativeness of the perturbation may differ across variables. In the next section, we examine a more robust approach that is less susceptible to these limitations.

5.2.2 Economic Significance of Variables

We now turn to analyzing the economic significance of our features for default behavior. We adopt SHapley Additive exPlanations (SHAP), a unified framework for interpreting predictions, to explain the output of our hybrid deep learning model (for a detailed description of the approach see Lundberg and Lee (2017)). SHAP uses a game theoretical concept to assign each feature a local importance value for a given prediction.

To obtain the Shapley values for our hybrid model, we first compute the Shapley values for the Deep Neural Network model and the Gradient Boosted Trees model separately, then simply take the average of the Shapley values for these two models for each individual and for each feature.¹⁵ We use a random sample of 1 million observations for explaining the model. Features that are highly correlated can decrease the importance of the associated feature by splitting the importance between both features. To account for the effect of feature correlation on interpretability, we group features with a correlation larger than 0.7, and sum the SHAP values within each groups. We denote these groups with an asterisk for the rest of the analysis and report the composition of feature groups in Table 19 in the appendix.

Figure 7 sorts features by the sum of SHAP value magnitudes, and plots the distribution of the impact each feature has on the model output for the twelve most important features or groups of correlated features. The color represents the feature value (red: high, blue: low), whereas the position on the horizontal axis denotes whether the feature is above or below the reference value and by how much. The charts plot the distribution of SHAP values for individual instances in the 1 million testing sample. The most important feature in terms of SHAP value magnitude is the worst status on any trades. High values of this variable tend to be highly associated with a higher value of the default rate compared to the baseline or reference value, whereas low values to a lower than baseline default rate, though the distribution of instances is much more dispersed. Features capturing credit history, such as length of credit history and recent delinquencies, also have high SHAP values, specifically, higher than baseline values of this variable are associated mostly with lower than baseline values of the predicted default probability, while lower than baseline values of the credit history are associated with a higher than baseline values of the default rate, with a much more dispersed distribution. Additionally, credit card utilization, credit amount on derogatory trades and outstanding collections are typically associated with an increase in the predicted probability of

¹⁵We implement Deep SHAP, a high-speed approximation algorithm for SHAP values in deep learning models to compute the Shapley values for our 5 hidden layer neural network. Because our dataset is fairly large with many features, we do not use the entire training dataset for the prediction. We pass a random sample of 1,000 observations of the training dataset. For GBT, we implement TreeExplainer, a high-speed exact algorithm for tree ensemble methods. For this, we do not need background or training observations.

Table 7: Explanatory Power of Variables

Feature	Accuracy	Loss
Credit amount on revolving trades	0.8792	0.2939
Months since the most recent 90+ days delinquency	0.8795	0.2938
Months since the oldest trade was opened	0.8770	0.2932
Open credit card trades	0.8793	0.2889
Open bankcard revolving, and charge trades	0.8803	0.2854
Balance on first mortgage trades	0.8772	0.2829
Monthly payment on open first mortgage trades	0.8752	0.2821
Total credit amount on open trades	0.8830	0.2821
Credit amount on open credit card trades	0.8829	0.2815
Worst ever status on any trades in the last 24 months	0.8869	0.2805
Monthly payment on all debt	0.8825	0.2803
Total debt balances	0.8846	0.2801
Balance on bankcard revolving and charge trades	0.8860	0.2747
Balance on collections	0.8842	0.2743
Months since the most recently opened first mortgage	0.8856	0.2743
Monthly payment on open auto loan trades	0.8857	0.2739
Monthly payment on credit card trades	0.8848	0.2733
Balance on credit card trades	0.8867	0.2732
Months since the most recent 30-180 days delinquency	0.8865	0.2731
Credit amount on open mortgage trades	0.8868	0.2724
Credit amount on unsatisfied derogatory trades	0.8848	0.2717
Balance on revolving trades	0.8860	0.2717
Credit amount on open installment trades	0.8869	0.2713
Months since the most recently opened credit card trade	0.8869	0.2709
Worst present status on any trades	0.8888	0.2706
Months since the most recently opened auto loan trade	0.8870	0.2700
Early payoff trades	0.8879	0.2695
Credit amount paid down on open first mortgage trades	0.8898	0.2685
Mortgage type	0.8885	0.2679
Months since the most recently closed, transferred, or refinanced first mortgage	0.8887	0.2671
Fraction of mortgage to total debt	0.8896	0.2669
Credit card utilization ratio	0.8881	0.2665
Balance on open auto loan trades	0.8887	0.2663
Credit amount on non-deferred student trades	0.8901	0.2662
:	:	:
:	:	:
Mortgage inquiries made in the last 3 months	0.8948	0.2552
Bankcard revolving and charge inquiries made in the last 3 months	0.8947	0.2551
Auto loan or lease inquiries made in the last 3 months	0.8946	0.2551
Balance on open bankcard revolving, and charge trades with credit line suspended	0.8947	0.2551
Baseline	0.8947	0.2551

This table reports a perturbation analysis on the pooled sample using our hybrid model. For each variable, we re-shuffle the feature, keeping the distribution intact in the test dataset and the model’s loss function is evaluated on the test dataset with the changed covariate. We repeat this step 10 times, and report the average of the loss and accuracy. Then, the variable is replaced to its original values, and a perturbation test is performed on a new variable. Perturbing the variable of course reduces the accuracy of the model, and the test loss becomes larger. If a particular variable has strong explanatory power, the test loss will significantly increase. The test loss for the complete model when no variables are perturbed is the Baseline value.

default relative to the baseline, as are relatively high numbers of inquiries and monthly payment on debt. Higher total debt balances are also associated with a lower than baseline predicted probability of default, reiterating the notion that the borrowers with the most credit are also associated with lower predicted probability of default, which suggests that credit allocation decisions are made to minimize default probabilities. As in the perturbation exercise, we find that number of trades and balances seem to have the strongest association with variation in the predicted probability of default.

These results only point to correlations between the features and the predicted outcome and should not be interpreted causally. Yet, they can be used as a point of departure for a causal analysis of default and theoretical modeling. They are also important to comply with legal disclosure requirements. Both the Fair Credit Reporting Act and the Equal Opportunity in Credit Access Act require lenders and developers of credit scoring models to reveal the most important factors leading to a denial of a credit application and for credit scores. The SHAP value provides an individualized assessment of such factors that can be used for making credit allocation decisions and communicating them to the borrower.

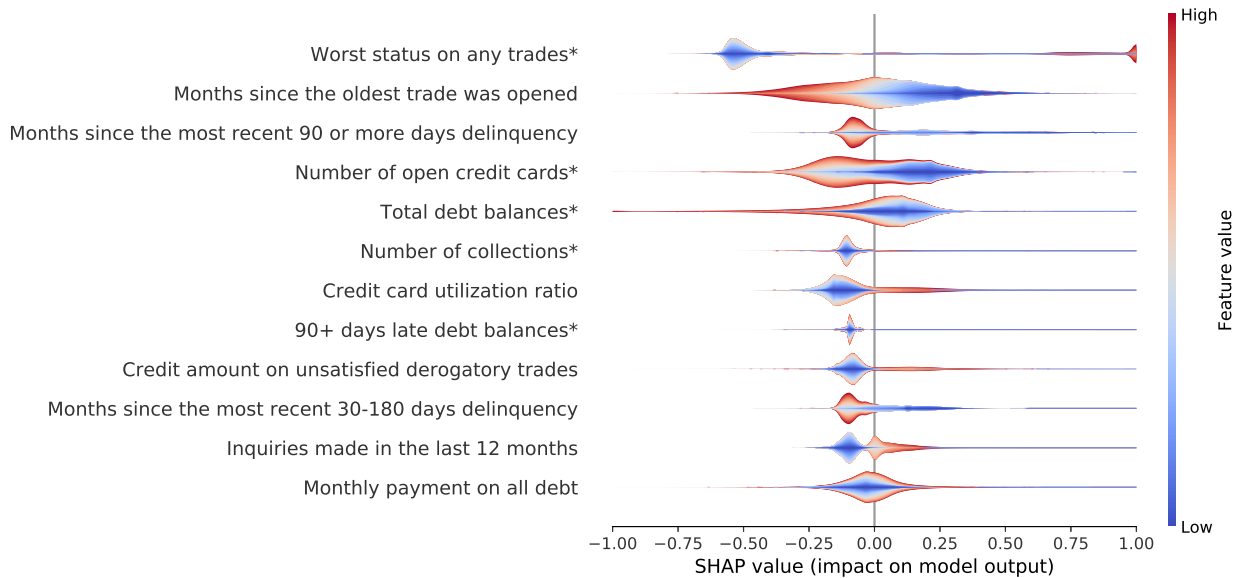


Figure 7: SHAP applied to predicted 90+ days delinquency within 8Q. Source: Authors' calculations based on Experian Data.

5.2.3 Temporal Determinants of Default

We next look at the changing dynamics of default behavior by comparing models that are trained in different periods of time. For this analysis, we use our hybrid model. Specifically, we target

the following time periods: 2006Q1, 2008Q1 and 2011Q1 as time periods before, during and after the 2007-2009 crisis, and compute default predictions for them with data trained in the same quarter and two years prior, that is in 2004Q1, 2006Q1, 2007Q1 and 2009Q1, respectively. We then calculate Shapley values for the two models.¹⁶ The first exercise provides an in-sample assessment for feature importance, while the second exercise can be used to assess feature importance out-of-sample. In both exercises, the model is the same, so comparing the results from the two exercises can help uncover which features are important for default prediction for a given period from an ex ante perspective and from an in-sample perspective.

Table 8 reports the results. The features are sorted by the sum of absolute SHAP value magnitudes over the first period. For each period, it is interesting to compare the variation in SHAP values from an ex ante and contemporaneous perspective, and additionally we are interested in comparing variation in SHAP values for given features in the different time periods. In all testing windows, the worst status on a mortgage trade has the first or second highest SHAP value. Total debt balances are the feature with the first or second highest SHAP value in all testing periods except for 2008Q1 where for the within same quarter prediction they are only the fifth in rank. Student debt is the third ranked feature in terms of SHAP values in all testing periods, and trades every more 90+ days delinquent or derogatory is either fourth or fifth. In 2006Q1 and 2011Q1, number of open credit card trades come in fifth, but ranks lower in 2008Q1, while features pertaining to credit card debt rank in a range for fourth to seventh depending on the period. Fraction of 60 days delinquent debt to total debt ranks sixth to tenth depending on the time period. The SHAP value is quite stable over time for most features, but there are some variables for which it changes substantially. One example is foreclosed first mortgages which ranks below fifteenth for 2006Q1 and 2008Q1 but moves up to eighth out of sample and ninth in sample for 2008Q1. The length of the credit history does not register high SHAP value in any of the time periods, nor do features related to credit utilization or inquiries. Overall these results confirm our findings from the pooled model, suggesting the balances and number of trades, in addition to delinquency status, have a strong association with default risk according to our model.

5.3 Model Comparisons

We examine the out-of-sample behavior of a collection of alternative machine learning techniques in this section. To motivate our choice of deep learning, leading to our hybrid DNN-GBT model, we begin by illustrating the importance of hidden layers that enable us to capture multi-level interactions between features by comparing how neural networks of different depth perform on the pooled sample. For this exercise, we fix the number of neurons per layer at 512, and build neural

¹⁶We do this for both the Deep Neural Network and the Gradient Boosted Trees and similarly to how we obtain the output, we simply take the average of the Shapley values. For our DNN explanations, we use a random sample of 1,000 observations of the testing data scaled by the mean and standard deviation of the corresponding testing data for reference value.

Table 8: SHAP Values over Time

Features	2006Q1			Prediction Date 2008Q1 Model			2011Q1		
	2004Q1	2006Q1	2008Q1	2006Q1	2008Q1	2009Q1	2009Q1	2011Q1	2011Q1
Worst present status on a mortgage trade	0.446 (1)	0.485 (2)	0.48 (2)	0.439 (1)	0.439 (1)	0.472 (1)	0.472 (1)	0.395 (2)	0.395 (2)
Total debt balances*	0.301 (2)	0.539 (1)	0.497 (1)	0.206 (5)	0.206 (5)	0.416 (2)	0.416 (2)	0.473 (1)	0.473 (1)
Student debt*	0.282 (3)	0.271 (3)	0.242 (3)	0.247 (3)	0.247 (3)	0.302 (3)	0.302 (3)	0.317 (3)	0.317 (3)
Trades ever 90 or more days delinquent or derogatory*	0.22 (4)	0.236 (5)	0.235 (4)	0.237 (4)	0.237 (4)	0.283 (4)	0.283 (4)	0.245 (4)	0.245 (4)
Number of open credit cards*	0.215 (5)	0.038 (19)	0.033 (22)	0.16 (7)	0.16 (7)	0.042 (20)	0.042 (20)	0.202 (5)	0.202 (5)
Fraction of 60 days delinquent debt to total debt	0.122 (6)	0.185 (6)	0.186 (6)	0.119 (8)	0.119 (8)	0.075 (10)	0.075 (10)	0.099 (10)	0.099 (10)
Credit card debt*	0.105 (7)	0.26 (4)	0.216 (5)	0.18 (6)	0.18 (6)	0.09 (7)	0.09 (7)	0.163 (6)	0.163 (6)
Months since the most recently opened home equity line of credit trade	0.089 (8)	0.036 (21)	0.035 (21)	0.08 (11)	0.08 (11)	0.06 (14)	0.06 (14)	0.148 (8)	0.148 (8)
Credit amount on unsatisfied derogatory trades	0.085 (9)	0.106 (8)	0.087 (9)	0.1 (10)	0.1 (10)	0.094 (6)	0.094 (6)	0.078 (13)	0.078 (13)
Balance on HELOC loans*	0.077 (10)	0.179 (7)	0.171 (7)	0.023 (32)	0.023 (32)	0.016 (41)	0.016 (41)	0.088 (11)	0.088 (11)
Months since the most recent 30-180 days delinquency	0.076 (12)	0.075 (11)	0.076 (11)	0.039 (21)	0.039 (21)	0.044 (19)	0.044 (19)	0.064 (16)	0.064 (16)
Auto loan*	0.076 (11)	0.037 (20)	0.04 (20)	0.267 (2)	0.267 (2)	0.098 (5)	0.098 (5)	0.036 (23)	0.036 (23)
Worst status on any trades*	0.071 (13)	0.091 (9)	0.09 (8)	0.063 (15)	0.063 (15)	0.058 (15)	0.058 (15)	0.052 (19)	0.052 (19)
90+ days late debt balances*	0.062 (14)	0.055 (15)	0.061 (13)	0.023 (33)	0.023 (33)	0.029 (26)	0.029 (26)	0.045 (20)	0.045 (20)
Foreclosed first mortgages*	0.048 (15)	0.04 (18)	0.041 (19)	0.033 (24)	0.033 (24)	0.082 (8)	0.082 (8)	0.108 (9)	0.108 (9)
Balance on open 60 days late revolving trades	0.044 (16)	0.06 (13)	0.058 (14)	0.038 (22)	0.038 (22)	0.038 (23)	0.038 (23)	0.088 (12)	0.088 (12)
Number of mortgages*	0.043 (17)	0.065 (12)	0.067 (12)	0.05 (17)	0.05 (17)	0.052 (17)	0.052 (17)	0.074 (14)	0.074 (14)
Fraction of credit card debt to total debt	0.034 (18)	0.015 (38)	0.016 (37)	0.026 (30)	0.026 (30)	0.032 (25)	0.032 (25)	0.054 (17)	0.054 (17)
Charge-off amount on unsatisfied charge-off trades	0.031 (19)	0.058 (14)	0.051 (16)	0.04 (20)	0.04 (20)	0.007 (68)	0.007 (68)	0.033 (25)	0.033 (25)
Home equity line of credit trades ever 90 or more days delinquent or derogatory	0.028 (20)	0.018 (33)	0.016 (36)	0.013 (49)	0.013 (49)	0.014 (47)	0.014 (47)	0.027 (28)	0.027 (28)

This table reports the Shapley values for a selected 20 features for three out-of-sample models. For each prediction window, we compute the Shapley value for each of the observations and for each feature. We then calculate the average of the absolute value for each feature, and report the results for the selected features. Finally, we rank the results based on the feature's relative rank in the given prediction window in parentheses. Source: Authors' calculations based on Experian data.

networks up to 8 hidden layers.¹⁷

We benchmark our results against the logistic regression, which is a commonly used technique in credit scoring and can be interpreted as a neural network with no hidden layers. Table 9 reports the in- and out-of-sample behavior for neural networks with 0-8 hidden layers. The number of hidden layers measure the complexity of the network, and we found that the marginal improvements in performance beyond 3 layers are small. Depth, however, is not directly proportional to out-of-sample performance due to higher model capacity. Table 9 also shows that applying dropout substantially improves the out-of-sample fit for networks of higher depths. This demonstrates that dropout serves as an effective regularization tool and addresses over-fitting for networks of greater depths.

Table 9: Neural networks comparison: Loss & Accuracy

Model	In-sample Loss		Out-of-sample Loss	
	w/o Dropout	Dropout	w/o Dropout	Dropout
Logistic Regression	0.3451	0.3452	0.3444	0.3446
1 layer	0.3097	0.3113	0.3123	0.3116
2 layers	0.2987	0.2983	0.3089	0.3041
3 layers	0.2878	0.2888	0.3074	0.2987
4 layers	0.2850	0.2862	0.3078	0.2972
5 layers	0.2856	0.2867	0.3076	0.2975
6 layers	0.2850	0.2852	0.3075	0.2967
7 layers	0.2825	0.2838	0.3066	0.2958
8 layers	0.2901	0.2853	0.3062	0.2966

Model	In-sample Accuracy		Out-of-sample Accuracy	
	w/o Dropout	Dropout	w/o Dropout	Dropout
Logistic Regression	0.8564	0.8564	0.8569	0.8570
1 layer	0.8693	0.8684	0.8683	0.8685
2 layers	0.8745	0.8748	0.8701	0.8722
3 layers	0.8798	0.8789	0.8714	0.8742
4 layers	0.8811	0.8799	0.8715	0.8747
5 layers	0.8807	0.8801	0.8721	0.8747
6 layers	0.8813	0.8803	0.8721	0.8749
7 layers	0.8821	0.8809	0.8723	0.8753
8 layers	0.8785	0.8807	0.8719	0.8750

In-sample and out-of-sample loss (categorical cross-entropy) and accuracy for neural networks of different depth and for logistic regression. Models are calibrated and evaluated on the pooled sample (2004Q1 - 2013Q4). Source: Authors' calculations based on Experian Data.

The results in Table 9 suggest that there are complex non-linear relationships among the fea-

¹⁷The architecture reported in Table 9 was not optimized. We picked 512 for each layer randomly. Another variant of this exercise is where we use the optimal 5-layer neural network architecture and we remove the neurons layer by layer. We report the results in Table 23.

tures used as inputs in the model. This is further supported by the fact that permitting non-linear relationships between default behavior and explanatory variables produces the largest model improvement. Going from a linear model (0 layers) to the simplest non-linear model (1 layer) generates the most sizable reduction in out-of-sample loss. To see this from another angle, we plot the ROC curves for our neural networks considered in Figure 8. We can see that the logistic regression is dominated by all models that allow for non-linear relationships, while the improvements for deeper models are marginal.

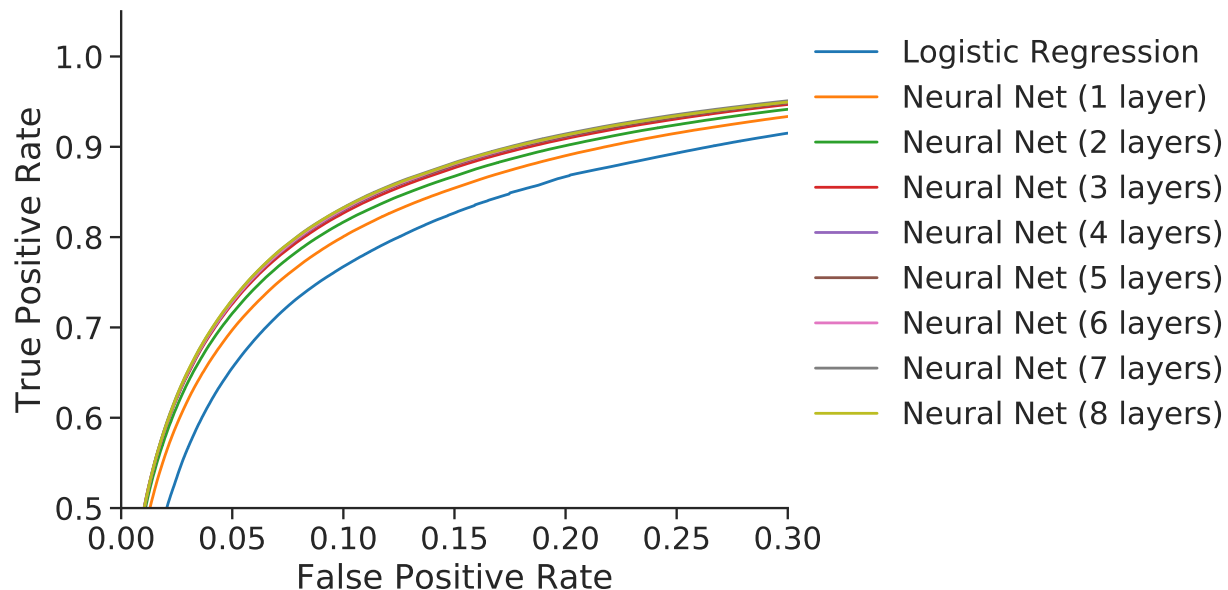


Figure 8: Out-of-sample ROC curves for various models. Models are calibrated and evaluated on the pooled sample (2004Q1 - 2013Q4). Source: Authors' calculations based on Experian Data.

We next analyze a number of machine learning techniques that are possible alternatives to our hybrid model. These algorithms have been used in other credit scoring applications, and include decision trees (CART, see Khandani, Kim, and Lo (2010)), random forests (RF, see Butaru et al. (2016)), neural networks (see West (2000)), gradient boosting (GBT, see Xia et al. (2017)) and logistic regression. We use the out of sample loss as our main comparison metric, with lower loss values corresponding to better model performance. We tune the hyper-parameters for each model and present the results in Table 10 for our baseline 1 quarter training/validation samples. Our hybrid model performs the best, with gradient boosting coming second, while deep neural network outperforms both random forest and decision trees in all samples. We repeat this comparison with an expanding training window and find that deep neural networks benefit the most from more training data.

Empirically, ensembles perform better when there is a significant diversity among the models

(see Kuncheva and Whitaker (2003)). Table 22 in Appendix D shows the SHAP values for our hybrid DNN-GBT model in comparison to GBT and DNN models for the pooled sample. The results suggest there are significant differences between the DNN and GBT. For instance, total debt balances is the third most important feature for GBT, while only tenth for DNN. Even more striking perhaps is that monthly payment on all debt is ranked as tenth most important for GBT, while only 45th for DNN. The ensemble approach can thus be thought of as providing diversification, which can reduce the variance of any of the two models. If one of the models puts too high of a weight on a feature, the other model may mitigate this effect.

It is important to emphasize that these results do not imply that there does not exist a random forest or CART model that cannot outperform our hybrid model. The best model will depend on the specific sample. The exercise is intended to illustrate that the complexity of the model is proportional to its accuracy to a certain degree, and that deep neural networks improve substantially on shallow models, such as logistic regression.

6 Applications

In this section, we use our model in a number of applications. We first provide a comparison between the performance of our model and conventional credit score. Second, we show that our model can score more borrowers. Finally, we show that our model is able to accurately predict variations in aggregate default risk.

6.1 Comparison with Credit Score

In this section, we compare the performance of our deep neural networks to a conventional credit score.¹⁸ The credit score is a summary indicator intended to predict the risk of default by the borrower and it is widely used by the financial industry. For most unsecured debt, lenders typically verify a prospective borrower’s credit score at the time of application and sometimes a short recent sample of their credit history. For larger unsecured debts, lenders also typically require some form of income verification, as they do for secured debts, such as mortgages and auto loans. Still, the credit score is often a key determinant of crucial terms of the borrowing contract, such as the interest rate, the downpayment or the credit limit. We have access to a widely used conventional credit score that uses information from the three credit bureaus.

Table 11 shows the relationship between credit score, predicted probability and realized default rate, where default is defined as usual as 90+ days delinquency in the subsequent 8 quarters. The calculation proceeds as follows. We first compute the number of unique credit scores in the data. We create the same number of bins of equal size in our predicted probability distribution, and calculate the realized frequency of 90+ days delinquencies in the subsequent 8 quarters for each

¹⁸The deep-learning forecasts for each quarter are obtained using out-of-sample input data, as reported in Table 3.

Table 10: Model Comparison: Out-of-Sample Loss

Training Window	Testing Window	Combined	GBT	DNN	RF	CART	Logistic
2004Q1	2006Q1	0.3233	0.3235	0.3276	0.3274	0.3432	0.3499
2004Q2	2006Q2	0.3178	0.3185	0.3220	0.3231	0.3372	0.3465
2004Q3	2006Q3	0.3156	0.3161	0.3202	0.3210	0.3361	0.3445
2004Q4	2006Q4	0.3187	0.3198	0.3223	0.3245	0.3396	0.3464
2005Q1	2007Q1	0.3205	0.3209	0.3249	0.3259	0.3427	0.3491
2005Q2	2007Q2	0.3211	0.3215	0.3252	0.3272	0.3440	0.3512
2005Q3	2007Q3	0.3246	0.3246	0.3294	0.3301	0.3455	0.3525
2005Q4	2007Q4	0.3276	0.3275	0.3323	0.3327	0.3498	0.3560
2006Q1	2008Q1	0.3262	0.3264	0.3308	0.3316	0.3477	0.3558
2006Q2	2008Q2	0.3271	0.3277	0.3316	0.3321	0.3495	0.3563
2006Q3	2008Q3	0.3248	0.3256	0.3285	0.3299	0.3474	0.3565
2006Q4	2008Q4	0.3247	0.3259	0.3282	0.3293	0.3462	0.3547
2007Q1	2009Q1	0.3202	0.3211	0.3248	0.3244	0.3425	0.3503
2007Q2	2009Q2	0.3193	0.3202	0.3241	0.3236	0.3427	0.3496
2007Q3	2009Q3	0.3166	0.3174	0.3213	0.3205	0.3383	0.3464
2007Q4	2009Q4	0.3124	0.3127	0.3172	0.3170	0.3348	0.3425
2008Q1	2010Q1	0.3137	0.3141	0.3185	0.3182	0.3360	0.3453
2008Q2	2010Q2	0.3143	0.3144	0.3193	0.3188	0.3355	0.3463
2008Q3	2010Q3	0.3144	0.3148	0.3195	0.3187	0.3353	0.3468
2008Q4	2010Q4	0.3157	0.3156	0.3214	0.3196	0.3365	0.3459
2009Q1	2011Q1	0.3133	0.3138	0.3179	0.3174	0.3320	0.3453
2009Q2	2011Q2	0.3161	0.3163	0.3211	0.3202	0.3346	0.3472
2009Q3	2011Q3	0.3145	0.3148	0.3192	0.3186	0.3322	0.3457
2009Q4	2011Q4	0.3135	0.3137	0.3181	0.3176	0.3313	0.3446
2010Q1	2012Q1	0.3115	0.3118	0.3160	0.3166	0.3298	0.3440
2010Q2	2012Q2	0.3134	0.3140	0.3177	0.3186	0.3313	0.3453
2010Q3	2012Q3	0.3119	0.3127	0.3160	0.3176	0.3310	0.3440
2010Q4	2012Q4	0.3136	0.3142	0.3180	0.3195	0.3325	0.3455
2011Q1	2013Q1	0.3121	0.3124	0.3168	0.3167	0.3313	0.3440
2011Q2	2013Q2	0.3117	0.3123	0.3158	0.3169	0.3320	0.3432
2011Q3	2013Q3	0.3103	0.3108	0.3148	0.3152	0.3315	0.3416
2011Q4	2013Q4	0.3083	0.3088	0.3126	0.3136	0.3281	0.3406
Average		0.3171	0.3177	0.3216	0.3220	0.3377	0.3476

Performance comparison of machine learning classification models of consumer default risk. The model calibrations are specified by the training and testing windows. The results of predicted probabilities versus actual outcomes over the following 8Q testing period are used to calculate the loss metric for 90+ days delinquencies within 8Q. Combined refers to the hybrid DNN-GBT model, DNN refers to deep neural network, RF refers to random forest, GBT refers to gradient boosted trees, while CART refers to decision tree. Source: Authors' calculations based on Experian Data.

of these bins. Since higher credit scores correspond to lower probability of default, we present the negative of the rank correlation with realized defaults for the credit score. The results indicate that even though credit score is successful in rank-ordering customers by their future default rates, with rank correlations between 0.983 and 0.990, our deep neural network performs better, with rank correlations always at 0.999. Figure 14 in Appendix E plots the time series of these rank correlations by quarter for the entire sample period. The figure shows that the rank correlation for the predicted probability of default generated by our model is remarkably stable over time, while for the credit score it fluctuates from lows of around 0.975 before 2012 to a peak go 0.995 in 2013Q2 with notable quarter by quarter variation. This property of credit scores may be due to the fact that the credit score is an ordinal ranking and its distribution is designed to be stable over time, even if default risk at an individual or aggregate level may change substantially. Figure 13 in Appendix E displays the histogram of credit score distributions in our sample for selected years, and show that these distributions are virtually identical over time.

A common way to measure the accuracy of conventional credit scoring models is the Gini coefficient, which measures the dispersion of the credit score distribution and therefore its ability to separate borrowers by their default risk. The Gini coefficient is related to a key performance metric for machine learning algorithm, the AUC score, with $Gini = 2 * AUC - 1$, so we can compare the performance of the credit score to our model along this dimension. Figure 14 plots the Gini coefficient for the credit score and our predicted default probability by quarter. The Gini coefficient for our model is about 0.85 between 2006Q1 and 2008Q3, and then rises to 0.86. For the credit score, the Gini coefficient is close to 0.81 until 2012Q3 when it drops to approximately 0.79 until the end of the sample, suggesting a drop in performance of the credit score in the aftermath of the Great Recession.

Table 11: Credit Score-Predicted Probability of Default Comparison

Metric	2006	2007	2008	2009	2010	2011	2012	2013
Rank Correlation								
Credit Score	0.9881	0.9804	0.9882	0.9816	0.9825	0.9861	0.9906	0.9944
Predicted Probability	0.9991	0.9992	0.9992	0.9992	0.9992	0.9991	0.9993	0.9992
GINI Coefficient								
Credit Score	0.8108	0.8142	0.8137	0.8143	0.8078	0.8008	0.7942	0.7898
Predicted Probability	0.8538	0.8535	0.8536	0.8606	0.8616	0.8608	0.8601	0.8580

Rank correlation between credit score, predicted probability of default according to our model and subsequent realized default frequency by year. For the credit score, report the rank correlation between each unique value of the score and the default frequency. For predicted probability of default based on our hybrid DNN-GBT model, we first generate a number of bins equal to the number of unique credit score realizations in the data and then calculate the realized default frequency for each bin. Source: Authors' calculations based on Experian Data.

Figure 9 plots a scatterplot of the realized default rate against the credit score (left panel) and

our predicted probability (right panel) for all quarters in the year 2008. In addition to the raw data, we also plot second-order polynomial-fitted curves to approximate the relationship. The scatter plots of realized default rates against the predictions from our hybrid model lay mostly on the 45 degree line, consistent with the very high rank correlations reported in Table 11. By contrast, the relation between realized default rates and credit scores has an inverted S-shape, with the realized default rate equal to one for a large range of low credit scores and equal to zero for a large range of high credit scores, and a large variation only for intermediate credit scores.

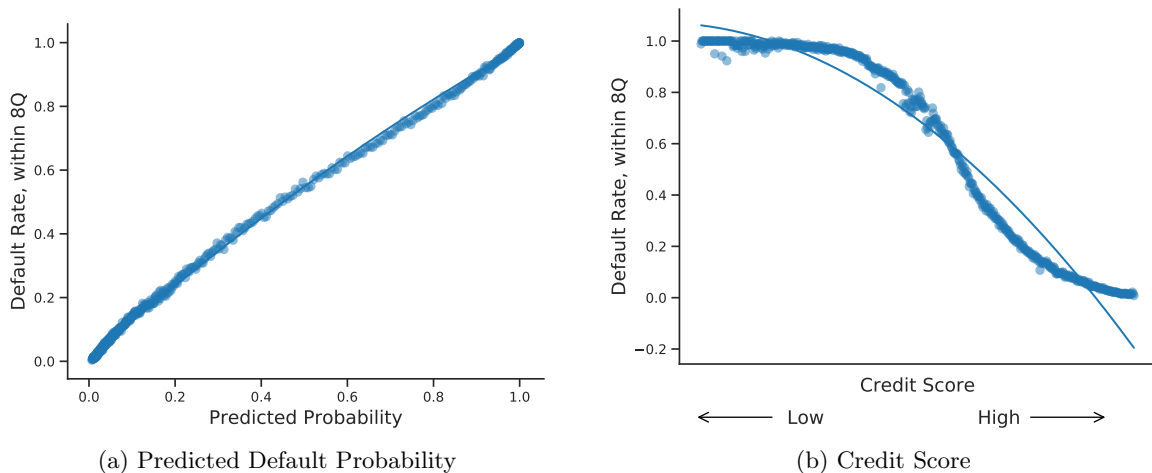


Figure 9: Scatter plot of realized default rates against model predicted default probability (a) and the credit score (b), with associated second-order polynomial fitted approximations for the year 2008. Source: Authors' calculations based on Experian Data.

Figure 10 plots second-order polynomial-fitted curves approximating the relation between realized default rates and those predicted by our model and the credit score for all years in which our model prediction is available, starting in 2006 until 2013, to examine how the relation between realized and predicted defaults varies with aggregate economic conditions. For the years at the height of the Great Recession, the default rate seems to be somewhat higher than our model prediction, but in all years the relation is very close to a 45 degree line. By contrast, there is virtually no change in the relation between the realized default rate and the credit score. This is by construction, since the distribution of credit scores is designed to only provide a relative ranking of default risk across borrowers.¹⁹ This property of the credit score implies that it is unable to forecast variations in aggregate default risk. In Section 6.2, we will show that our model is able to capture variations in aggregate default risk while retaining a consistent ability to separate borrowers by their individual default risk.

Table 12 reports the rank correlation with the realized default rate and the Gini coefficients by

¹⁹Credit scores are specifically designed to provide a stable ranking by using multiple years of data.

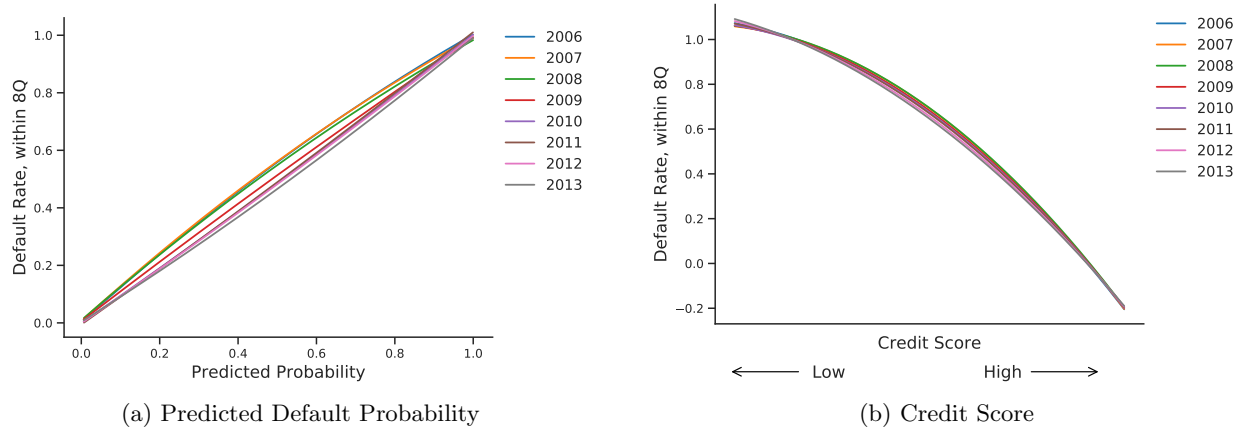


Figure 10: Second-order polynomial approximation of the relationship between realized default rates against model predicted default probability (a) and the credit score (b) for selected years. Source: Authors' calculations based on Experian Data.

year for the credit score and the probability of default predicted by our model restricting attention to the current population, that is those borrowers who do not have any outstanding delinquencies in the quarter of interest. The rank correlation between the credit score and the realized default rate drops by 1-3 percentage points for these borrowers, whereas for our model it drops by less than a quarter of 1 percent. The Gini coefficient drops from 80-81% to 68-69% for the credit score and from 85-86% to 72-74%. These results suggest that when measured on the population of current borrowers, the performance advantage of our model relative to a conventional credit score grows.²⁰

To understand the differences in performance across the credit score and our model, we examine how the ranking of borrowers varies under the two approaches. To do so, we consider the industry classification of borrowers into five risk categories Deep Subprime, Subprime, Near Prime, Prime and Super Prime.²¹ As shown in Table 13, these categories account for respectively, 6.5%, 21.2%, 14.1%, 33.3% and 24.9% of all borrowers. We then create 5 correspondingly sized bins in our predicted probability of default with bin 1 corresponding to the 6.5% of borrowers with the highest predicted default risk and bin 5 to the 24.9% of all borrowers with the lowest predicted default risk. Finally, we calculate the fraction of borrowers in each credit score category that is in each of the 5 predicted default risk categories and their realized and predicted default rate. The results are displayed in Table 13. We also report the average realized and predicted default rate for each

²⁰Appendix E also plots the time series of the rank correlation and the Gini coefficient for the credit score and our model for the current population. The credit score shows a large drop in these statistics for the credit score during the Great Recession whereas for our model they are both stable over time. This is consistent with the notion that the performance of the credit score dropped during the 2007-2009 period.

²¹The threshold levels for these categories are: 1) Deep Subprime: up to 499 credit score; 2) Subprime: 500-600 credit score; 3) Near Prime: 601-660 credit score; 4) Prime: 661-780 credit score; 5) Super Prime: higher than 781 credit score.

Table 12: Credit Score-Predicted Probability of Default Comparison– Current Borrowers

Metric	2006	2007	2008	2009	2010	2011	2012	2013
Rank Correlation								
Credit Score	0.9670	0.9613	0.9445	0.9653	0.9683	0.9585	0.9806	0.9559
Predicted Probability	0.9978	0.9979	0.9979	0.9975	0.9975	0.9973	0.9971	0.9974
Credit Score	0.6933	0.6935	0.6908	0.6807	0.6777	0.6833	0.6795	0.6810
Predicted Probability	0.7366	0.7244	0.7216	0.7187	0.7238	0.7335	0.7360	0.7391

Rank correlation between the credit score, predicted probability of default according to our model and subsequent realized default frequency by year for the population of current borrowers. Current borrowers do not have any delinquencies. For the credit score, report the rank correlation between each unique value of the score and the default frequency. For predicted probability of default based on our hybrid model, we first generate a number of bins equal to the number of unique credit scores in the data and then calculate the realized default frequency for each bin. Source: Authors' calculations based on Experian Data.

credit score category overall (columns 7 and 8) and for each predicted default risk category for all credit score (last 5 rows).

These results suggest that our model does well in predicting the default probability of borrowers in all categories, with a slight tendency to under-predict the probability of default by 1-2 percentage points for the Deep Subprime and Subprime borrowers. The majority of Deep Subprime borrowers fall in the two lowest categories of predicted default risk. For Subprime borrowers, 64% fall into the corresponding second category of default risk, while 15% fall in the first and 17% into the third. This corresponds to a sizable discrepancy as the average realized default probability for Subprime borrowers is 79%, whereas it is 99% for those in the first category and only 44% for those in the third. By contrast, the predicted default risk is very close to the realized default risk for Subprime borrowers in all categories of the predicted default risk distribution, with a discrepancy under 1 percentage point for predicted default risk category 1 and 2, and 5 percentage points for category 3 and 4. Near Prime borrowers also display a wide dispersion across predicted default risk categories with only 43% falling into the corresponding third category, 24% falling in category 2 (higher default risk) and 31% falling into category 4 (lower default risk). Again, the realized default rates vary substantially for Near Prime borrowers by predicted default risk category, from 77% in category 2, to 41% and 20% in category 3 and 4, respectively, while the predicted default risk is much closer to the realized, with a maximum 4 percentage point discrepancy. The discrepancy in classification for the credit score are lower for Prime and Super Prime borrowers. 13% of Prime borrowers fall into category 3 (higher default risk), 13% in category 5 (lower default risk) and 71% in the corresponding category 4. The realized default rates are 11% for Prime borrowers in category 4, and 34% and 3% respectively for Prime borrowers in category 3 and 5. Only 18% of Super Prime borrowers fall in category 4 of predicted default risk (higher risk) and 82% fall in the corresponding category 5. Moreover, the differences in realized default risk between these categories are minor,

with a realized default rate of 5% and 2% for categories 4 and 5, respectively. These results suggest that credit scores misclassify borrowers across risk categories with very different realized default rates. By contrast, as shown in the bottom 5 rows of Table 13 and by columns (6) and (8), our model is very successful at predicting the default rate for borrowers irrespective of their credit score.

6.1.1 Coverage

Consumers with limited credit histories encounter severe difficulties in accessing credit markets. As explained earlier in the paper, lenders often rely on credit scores to make lending decisions. If a borrower’s credit report does not have sufficient information to evaluate their default risk, lenders are unlikely to grant credit. Consequently, consumers with limited credit histories have a hard time accessing credit markets. These consumers can be divided into two groups. The first group consists of individuals without credit records, often referred to in the industry as “credit invisibles.” The second group are those consumers who do have credit records, but are considered “unscorable,” that is they have insufficient credit histories to generate a credit score. Brevoort, Grimm, and Kambara (2016) find that 11% of the US population lacks credit records and an additional 8.3% have a credit record but do not have a credit score.

A credit record may be considered unscorable for two reasons: (1) it contains insufficient information to generate a score, meaning the record either has too few accounts or has accounts that are too new to contain sufficient payment history to calculate a reliable credit score; or (2) it has become stale in that it contains no recently reported activity. The exact definition of what constitutes insufficient or stale information differs across credit scoring models, as each model uses its own proprietary definition.²²

The challenges that credit invisibles and unscored consumers face in accessing credit markets has generated considerable attention from researchers and industry participants. Brevoort, Grimm, and Kambara (2016) show that young, minority and low income borrowers are disproportionately represented among the unscored. Several studies have explored the potential of various types of alternative data to supplement the information contained in credit reports and allow credit scores to be generated for these consumers.²³ Our model generates a predicted probability of default for every individual in our sample without an empty credit record, so effectively there are no active borrowers that we do not score. Our ability to score every active borrower is partly due to the fact that our model does not use any lagged observations. As previously discussed, many of the features in our model have a temporal dimension which renders the use of lags unnecessary. This

²²The FICO score has the most restrictive requirements and does not score borrowers who show no updates or reports on credit file in past 6 months, or no accounts at least 6 months old. The rest of the industry has been trying to expand the universe of scorable borrowers and typically adopts a more flexible approach to increase the ranks of borrowers who are scored. Vantage Score has been particularly pushing the need to expand the universe of scorable customers and they have implemented several successful changes to the most recent version of their scoring model that have indeed substantially decreased the number of unscored consumers. For more information, see <https://www.vantagescore.com/resource/174/scoring-credit-invisibles-using-machine-learning-techniques>.

²³See for example Jagtiani and Lemieux (2018). For an industry perspective, see Carroll and Rehmani (2017).

Table 13: Comparison with Credit Score

(1)	Credit Score	Predicted Default Probability		Default Rate		Average Default Rate	
	Share	(3)	Share	Realized	Predicted	Realized	Predicted
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Deep Subprime	6.48%	1	47.83%	99.55%	99.49%	95.45%	94.59%
		2	51.21%	92.24%	90.85%		
		3	0.90%	63.66%	52.25%		
		4	0.04%	43.72%	12.97%		
		5	0.01%	34.04%	1.95%		
Subprime	21.22%	1	14.90%	99.35%	99.35%	78.64%	77.27%
		2	64.41%	84.32%	83.85%		
		3	16.67%	51.93%	46.41%		
		4	4.00%	21.78%	18.07%		
		5	0.02%	16.18%	2.13%		
Near Prime	14.09%	1	1.34%	99.11%	99.21%	43.71%	42.61%
		2	24.19%	76.56%	78.58%		
		3	42.94%	41.39%	40.38%		
		4	30.90%	19.63%	15.91%		
		5	0.63%	3.13%	2.56%		
Prime	33.31%	1	0.08%	98.85%	99.18%	14.31%	14.37%
		2	2.42%	74.86%	78.37%		
		3	13.17%	33.54%	36.31%		
		4	70.92%	10.67%	10.27%		
		5	13.40%	3.18%	2.46%		
Super Prime	24.90%	1	0.00%	100.00%	99.31%	2.56%	2.66%
		2	0.09%	81.36%	81.77%		
		3	0.21%	32.13%	34.30%		
		4	18.00%	5.43%	5.95%		
		5	81.69%	1.76%	1.76%		
All	100	1	99.44%	99.41%			
		2	83.95%	83.88%			
		3	41.65%	40.65%			
		4	11.42%	10.62%			
		5	2.02%	1.89%			

Borrowers are classified into 5 categories of default risk standard in the industry named in column (1). The threshold levels for these categories are: 1) Deep Subprime: up to 499 credit score; 2) Subprime: 500-600 credit score; 3) Near Prime: 601-660 credit score; 4) Prime: 661-780 credit score; 5) Super Prime: higher than 781 credit score. The fraction of borrowers in each category is reported in column (2). Borrowers are also assigned to 5 categories of predicted default risk based on our hybrid model from the highest default risk (1) to the lowest (5), where the share of borrowers in each predicted default risk category is the same as for the credit score categories Deep Subprime to Super Prime. For each credit score risk category the share of borrowers in each predicted default risk category is reported in column (4). Columns (5) and (6) report the corresponding realized and predicted default probability for each credit score category interacted with predicted default risk category. Columns (7) and (8) report the average realized and predicted default probability for each credit score category. All rates, fractions and shares in percentage. Total # of observations: 17,732,772. Time period 2006Q1-2013Q4. Source: Authors' calculations based on Experian Data.

constitutes an additional advantage of our model when compared to traditional credit scores.

6.2 Predicting Systemic Risk

We next turn to analyze the aggregate forecasting power of our hybrid model. We aggregate the deep-learning forecasts for individual accounts to generate macroeconomic forecasts of credit risk by taking the average of the predicted probabilities over a given forecast period. Since our sample of consumers is nationally representative in each quarter, this will provide an unbiased estimate of the aggregate default risk predicted by our model. We calculate the aggregate default probability for 2006Q1-2013Q4, and show that our model is able to predict the spike in delinquencies during the 2007-2009 financial crisis and also the reduction in delinquencies since then. This estimate of aggregate default risk could be used as a proxy of systemic risk in the household sector. The results are displayed in Figure 11. Panel (a) plots the aggregate predicted default rate from our hybrid model and compares it to the aggregate realized default rate. While our predicted aggregate default rate is approximately 2 percentage points lower than the realized in 2006 and 2007, it rises at a similar speed as the realized default rate. It peaks in 2010Q2, approximately 2 quarters after the peak in the realized rate and then declines in the ensuing period, again reflecting the behavior of the realized rate, though it overestimates it by about 1 percentage point. Panel (b) shows a scatter plot of the predicted aggregate default rate against the realized for the different quarters in our sample period. The correlation between the predicted and realized aggregate default rate is 36%.

6.3 Value Added

We assess the economic salience of our hybrid DNN-GBT model by analyzing its value added for lenders and borrowers. For lenders, we examine the role our model can play in minimizing the losses from default. For borrowers, we calculate the interest savings for borrowers who are misclassified as having an excessively high probability of default based on the credit score compared to our model.

6.3.1 Lenders

We follow the framework proposed by Khandani, Kim, and Lo (2010), which compares the value of having a prediction of default risk to having none, and we make the same simplifying assumptions with respect to the revenues and costs of the consumer lending business. Specifically, in absence of any forecasts, it is assumed a lender will take no action regarding credit risk, implying that customers who default will generate losses for the lender, and customers who are current on their payments will generate positive revenues from financing fees on their running balances. To simplify, we assume that all defaulting and non-defaulting customers have the same running balance, B_r , but defaulting customers increase their balance to B_d prior to default. We refer to the ratio between B_d and B_r as "run-up." It is assumed that with a model to predict default risk, a lender can avoid losses of defaulting customers by cutting their credit line and avoiding run-up. Then, the value

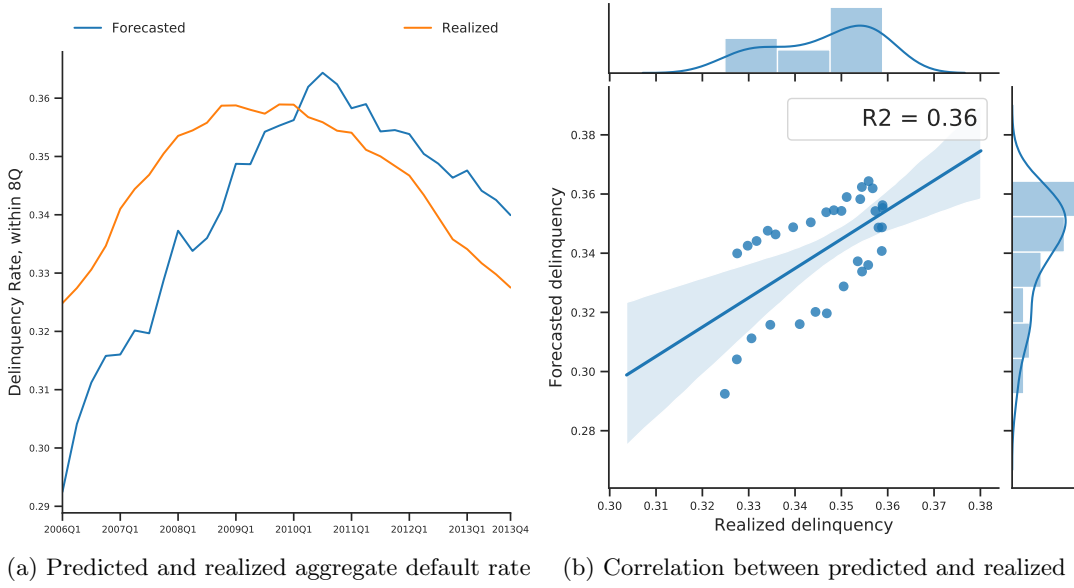


Figure 11: Fraction of consumers with 90+ days delinquency within the subsequent 8 quarters, predicted by our hybrid DNN-GBT model and realized. Aggregate default rates are obtained by averaging across all consumers in each period. Source: Authors' calculations based on Experian Data.

added as proposed by Khandani, Kim, and Lo (2010) can be written as follows:

$$VA(r, N, TN, FN, FP) = \frac{TN - FN[1 - (1 + r)^{-N}][\frac{B_d}{B_r} - 1]^{-1}}{TN + FP} \quad (10)$$

where r refers to the interest rate, N the loan's amortization period, and TN, FN, FP refer to true negatives, false negatives and false positives respectively. Panel (a) of Figure 12 plots the Value Added (VA) as a function of interest rate and the ratio of run-up balance for our out-of-sample forecasts of 90+ days delinquencies over the subsequent 8 quarters for 2012Q4. These estimates imply cost savings of over 60% of total losses when compared to having no forecast model for a run-up of 1.2 at a 10% interest rate for an amortization period for 3 years.

We next compare the value added of our hybrid model with default predictions generated by a logistic regression. This exercise illustrates the gains from adopting a better technology for credit allocation. Panel (b) of Figure 12 shows more modest, but substantial cost savings in the range of 1-9% and approximately 5% for a 1.2 run-up at a 10% interest rate with a 3 year amortization period. This exercise then confirms the advantages of using deep learning over other technologies in predicting default.

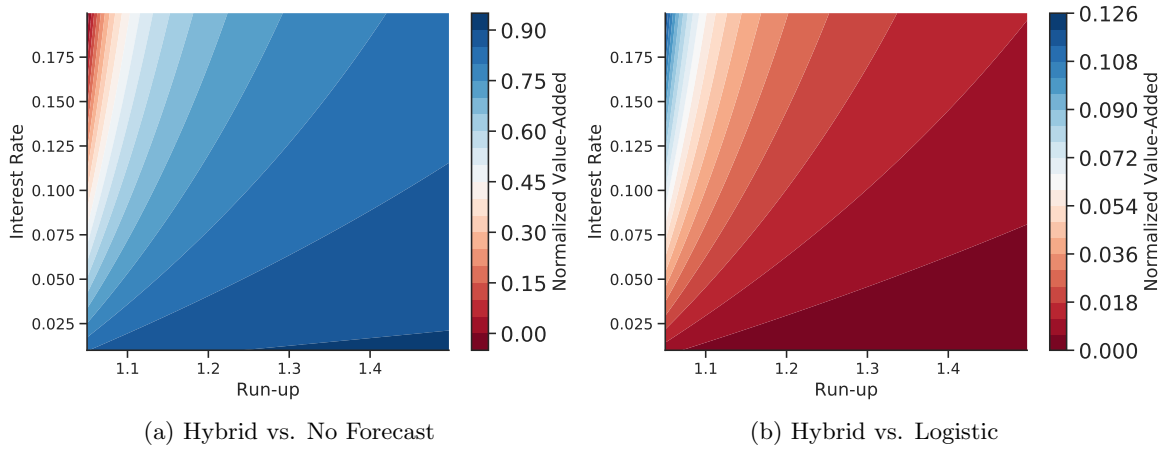


Figure 12: Value-added of machine-learning forecasts of 90+ days delinquency over 8Q forecast horizons on data from 2012Q4. VA values are calculated with amortization period $N = 3$ years and a 50% classification threshold. Source: Authors' calculations based on Experian Data.

6.3.2 Borrowers

We now examine the potential cost savings for consumers who would be offered credit according to the predicted default probability implied by our model instead of a conventional credit score. Following our approach in Section 6.1, we create credit score categories based on common industry standards and corresponding predicted probability bins with the same number of observations, and we place customers in these bins based on their credit score at account origination for each of their credit cards.²⁴ The distribution of customers is summarized in Table 24 in Appendix F. We then follow the information on interest rates by credit score category in Table 2 in Agarwal et al. (2015) to obtain the cost of credit on credit card balances.²⁵ To obtain the cost savings for consumers, we use the difference in interest rates by credit score category based on how they would be classified according to our model. For customers who are placed in higher risk categories by the credit score compared to our predicted probability of default, interest rates on credit cards are higher than they would have been if they had been classified according to our model. Thus, using our model to score consumers rather than the credit score would generate the cost savings for them. For customers placed in risk categories by the credit score that are too low relative to the default risk predicted by our model, interest rates will be higher under our model. The calculation is made for each individual consumer. The average for each credit score category is then computed and reported in Table 25. The information on interest rates and balances, and the dollar value of cost savings for

²⁴We look at the months since most recently opened credit card account to infer account origination. We drop customers with months since the most recently opened credit card greater than 72.

²⁵Credit card interest rates are notoriously invariant to overall changes in interest rates, so the calculations reported in this section apply irrespective of the time period. See Ausubel (1991) and Calem and Mester (1995).

different credit card categories is reported in Table 14. We report this in percentage of credit card balances in the top panel and in current USD terms in the bottom panel. The largest gains accrue to customers with Subprime and Near Prime credit scores. As we showed in Section 6.1, they are more likely to be attributed a probability of default by the credit score that is too low compared to our model predictions. Additionally, the biggest variation in credit card interest rates occurs across Subprime and Near Prime borrowers in comparison to Prime based on Agarwal et al. (2015). The cost savings for these borrowers average out to 4-5% of total credit card balances of \$1,085-1426. Gains for Prime and Super Prime borrowers who are attributed a lower default probability by our model are very modest, as credit card interest rates vary little by credit score for Prime and Superprime borrowers. On the other hand, Prime and Superprime borrowers who are placed in group 1, corresponding to the highest predicted default probability based on our model face, substantial losses in the order of 4-5% of total credit card balances or \$274-423. The cumulated interest rate cost savings across all consumers in our sample is \$723,636,560, which amounts to \$40 per capita.

This calculation provide us with a lower bound for the cost savings of being classified according to our model in comparison to the credit score, as they do not take into account the higher credit limits and potential behavioral responses of customers faced with higher borrowing capacity and lower interest rates. As shown in Agarwal et al. (2015), changes in the cost of funds for lenders mainly translate into changes in credit limits and exclusively for higher credit score borrowers. Therefore, being placed in a higher risk category for consumers also inhibits their ability to benefit from expansionary monetary policy. Additionally, we do not take into account the fact that more expensive credit in the form of higher interest rate costs makes it more likely that the consumer will incur missed payments in response to temporary changes in income. Fees for missed payments constitute a substantial component of credit card costs for consumers, and the ability to avoid these fees would contribute to substantial cost savings for consumers (see Agarwal et al. (2014)).

7 Conclusion

We have proposed to use deep learning to develop a model to predict consumer default. Our model uses the same data used by conventional scoring models and abides with all legislative restrictions in the United States. We show that our model compares favorably to conventional credit scoring models in ranking individual consumers by their default risk, and is also able to capture variations in aggregate default risk. Our model is interpretable and allows to identify the factors that are most strongly associated with default. Whereas conventional credit scoring models emphasize utilization rates, our analysis suggests that the number and balances on open trades are the factors which associate more strongly to higher default probabilities. Our model is able to provide a default prediction for all consumers with a non-empty credit record. Additionally, we show that our hybrid DNN-GBT model performs better than standard machine learning models of default based on

Table 14: Cost of Credit Risk Misclassification

		Credit Score			
		Subprime, Near Prime	Prime Low	Prime Mid	Prime High, Superprime
Annual Interest Rate Savings					
Predicted Default Bin	1	0.00%	-5.13%	-4.28%	-4.93%
	2	5.13%	0.00%	0.85%	0.20%
	3	4.28%	-0.85%	0.00%	-0.65%
	4	4.93%	-0.20%	0.65%	0.00%
Annual Average Cost Saving (\$)					
Predicted Default Bin	1	0	-423	-274	-309
	2	1085	0	83	15
	3	1426	-179	0	-54
	4	1239	-44	118	0

This table reports the average cost savings for consumers across credit score and predicted default probability bins for our sample. The cumulative savings for consumers on both credit card and bankcard debt adds up to \$723,636,560. Time period 2006Q1-2013Q4. Source: Authors' calculations based on Experian Data.

logistic regression and can accrue cost saving to lenders in the order of 1-9% compared to default predictions based on logistic regression, as well as interest rate cost savings for consumers of up to \$1,426 per year.

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Appendix

A Performance Metrics

Suppose a binary classifier is given and applied to a sample of N observations. For each instance i , let y_i denote the true outcome. For each observation, the model generates a probability that an observation with feature vector x_i belongs to class 1. This predicted probability, $f(x_i)$ is then evaluated based on a threshold to classify observations into class 1 or 0. Given a threshold level (c), let True Positive (TP) denote the number of observations that are correctly classified as type 0, True Negative (TN) be the number of observations that are correctly classified as type 1, False Positive (FP) be the number of observations that are type 1 but incorrectly classified as type 0, and, finally, False Negative (FN) be the number of observations that are actually of type 0 but incorrectly classified as type 0. Based on these definitions, one can define the following metrics to assess the performance of the classifier:

$$\text{True Positive Rate (TPR)} \equiv \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (11)$$

$$\text{False Positive Rate (FPR)} \equiv \frac{\text{FP}}{\text{FP} + \text{TN}} \quad (12)$$

$$\text{Precision} \equiv \frac{\text{TN}}{\text{TN} + \text{FN}} \quad (13)$$

$$\text{Recall} \equiv \frac{\text{TN}}{\text{TN} + \text{FP}} \quad (14)$$

$$\text{F-measure} \equiv \frac{2 \times \text{Recall} \times \text{Precision}}{\text{Precision} + \text{Recall}} \quad (15)$$

$$\text{Accuracy} \equiv \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (16)$$

$$\text{Youden's J statistic} \equiv \frac{\text{TP}}{\text{TP} + \text{FN}} + \frac{\text{TN}}{\text{TN} + \text{FP}} - 1 \quad (17)$$

$$\text{ROC AUC} = \int_{-\infty}^{\infty} \text{TPR}(c) \text{FPR}'(c) dc \quad (18)$$

$$\text{Cross-entropy loss} = -\frac{1}{N} \sum_{i=1}^N (y_i \cdot \log(f(x_i)) + (1 - y_i) \cdot \log(1 - f(x_i))) \quad (19)$$

$$\text{Brier score} = \frac{1}{N} \sum_{i=1}^N (f(x_i) - y_i)^2 \quad (20)$$

B Data Pre-Processing

Our original dataset contains 33,600,000 observations. We discard observations of individuals with missing birth information, deceased individuals and restrict our analysis to individuals aged between 18 and 85, residing in one of the 50 states or the District of Columbia, with 8 consecutive quarters of non-missing default behavior. This leaves us with 22,004,753 data points. Our itemized sample

restrictions are summarized in Table 15 below.

Table 15: Itemized Sample Restrictions

	Observations
Credit Report Data	33,600,000
Remove	
Deceased	- 513,270
Age	- 4,718,804
Residence	- 953,215
Prediction Window	- 5,409,958
Prediction Sample	22,004,753

Feature Scaling

We normalize all explanatory variables by their means and standard deviations:

$$z_i = \frac{x_i - \mu_x}{\sigma_x} \quad (21)$$

where $\mathbf{x} = (x_1, x_2, \dots, x_k)$, and $\mathbf{z}_i$ is the i^{th} normalized data.

Train-Test Split

For most of our analysis we split the data to account for look-ahead bias, i.e., the training set consists of data 8Q prior to the testing data. Then, we scale the testing data by the mean and standard deviation of the training data. In an alternative specification, we split our pooled data into three chunks: training set (60%), holdout set (20%), and testing set (20%). We report each specifications in Table 3 - Table 4. Except for parts of Section 5.2, we used the predictions generated by our models on the temporal splits.

In each specifications, we randomly shuffled the data to ensure that the mini-batch gradients are unbiased. If gradients are biased, training may not converge and accuracy may be lost.

C Model Estimation

Our estimation consists of four steps. First, we specify the loss function. Second, we choose the optimization algorithm. Third, we optimize the hyperparameters of the model. Fourth, we train our models.

Loss Function

Suppose $\mathbf{y}$ is the ground truth vector of default, and $\hat{\mathbf{y}}$ is the estimate obtained directly from the last layer given input vector $\mathbf{x} = (x_1, x_2, \dots, x_k)$. By construction, $y_i = \{0, 1\}$ and $\hat{y}_i \in [0, 1]$. We minimize the categorical cross-entropy loss function²⁶ to estimate the parameter specified in (7). We do this by choosing θ that minimizes the distance between the predicted $\hat{\mathbf{y}}$ and the actual $\mathbf{y}$ values. Given N training examples, the categorical cross-entropy loss can be written as:

$$L(\hat{y}, y) = -\frac{1}{N} \sum_{i=1}^N (y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i)) \quad (22)$$

We apply an iterative optimization algorithm to find the minimum of the categorical cross-entropy loss function. We next describe this algorithm.

DNN Optimization Algorithm

Deep learning models are computationally demanding due to their high degree of non-linearity, non-convexity and rich parameterization. Given the size of the data, gradient descent is impractical. We follow the standard approach of using stochastic gradient descent (SGD) to train our deep learning models (see Goodfellow et al. (2016)). Stochastic gradient descent is an iterative algorithm that uses small random subsets of the data to calculate the gradient of the objective function. Specifically, a subset of the data, referred to as a mini-batch (the size of the mini-batch is called the batch size), is loaded into memory and the gradient is computed on this subset. The gradient is then updated, and the process is repeated until convergence.

We adopt the Adaptive Moment Estimation (Adam), a computationally efficient variant of the SGD introduced by (see Kingma and Ba (2014)) to train our neural networks. The Adam optimization algorithm can be summarized as follows:

1. Fix the learning rate α , the exponential decay rates for the moment estimates: $\beta_1, \beta_2 \in [0, 1)$, and the objective function. Initialize the parameter vector θ_0 , the first and second moment vector m_0 and v_0 respectively, and the timestep t .
2. While θ_t does not converge, do the following:
 - (a) Compute the gradients with respect to the objective function at timestep t :

$$g_t = \nabla_{\theta} f_t(\theta_{t-1}) \quad (23)$$

- (b) Update the first and second moment estimates:

$$m_t = \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t \quad (24)$$

$$v_t = \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2 \quad (25)$$

²⁶Loss function measures the inconsistency between the predicted and the actual value. The performance of a model increases as the loss function decreases. There are several other types of loss functions, including mean squared error, hinge, and Poisson. The categorical cross-entropy is often used for classification problems.

(c) Compute the bias-corrected first and second moment estimates:

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t} \quad (26)$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t} \quad (27)$$

(d) Update the parameters:

$$\theta_t = \theta_{t-1} - \frac{\alpha \cdot \hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \quad (28)$$

The hyperparameters have intuitive interpretations and typically require little tuning. We apply the default setting suggested by the authors of Kingma and Ba (2014), these are $\alpha = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\epsilon = 10^{-7}$.

GBT Algorithm

Fit a shallow tree (e.g., with depth $L = 1$). Using the prediction residuals from the first tree, fit a second tree with the same shallow depth L . Weight the predictions of the second tree by $\nu \in (0, 1)$ to prevent the model from overfitting the residuals, and then aggregate the forecasts of these two trees. Until a total of K trees is reached in the ensemble, at each step k , fit a shallow tree to the residuals from the model with $k-1$ trees, and add its prediction to the forecast of the ensemble with a shrinkage weight of ν .

Regularization

Neural networks are low-bias, high-variance models (i.e., they tend to overfit to their training data). We implement three routines to mitigate this. First, we apply dropout to each of the layers (see Srivastava et al. (2014)). During training, neurons are randomly dropped (along with their connections) from the neural network with probability p (referred to as the dropout rate), which prevents complex co-adaptations on training data.

Second, we implement "early stopping", a general machine learning regularization tool. After each time the optimization algorithm passes through the training data (i.e., referred to as an epoch), the parameters are gradually updated to minimize the prediction errors in the training data, and predictions are generated for the validation sample. We terminate the optimization when the validation sample loss has not decreased in the past 50 epochs. Early stopping is a popular substitute to l2 regularization, since it achieves regularization at a substantially lower computational cost.

Last, we use batch normalization (see Ioffe and Szegedy (2015)), a technique for controlling the variability of features across different regions of the network and across different datasets. It is motivated by the internal covariate shift, a phenomenon in which inputs of hidden layers follow different distributions than their counterparts in the validation sample. This problem is frequently encountered when fitting deep neural networks that involve many parameters and rather complex structures. For each hidden unit in each training step, the algorithm cross-sectionally de-means and variance standardizes the batch inputs to restore the representation power of the unit.

Hyperparameter selection

Deep learning models require a number of hyperparameters to be selected. We follow the standard approach by cross-validating the hyperparameters via a validation set. We fix a training and validation set, and then train neural networks with different hyperparameters on the training set and compare the loss function on the validation set. We cross-validate the number of layers, the number of units per layer, the dropout rate, the batch size, and the activation function (i.e., the type of non-linearity) via Tree-structured Parzen Estimator (TPE) approach (see Bergstra et al. (2011)),²⁷ and select the hyperparameters with the lowest validation loss.

The training set for our out-of-sample hyperparameter optimization comes from 2004Q3, while the validation set is from 2006Q3. Table 16 summarizes our machine learning model hyperparameters. For our neural network, we used 5 hidden layers, with 150-600-1000-600-400 neurons per layer, SELU activation function, a batch size of 4096, a learning rate of 0.003, and a dropout rate of 50%. For our GBT, We found that a learning rate of 0.05, a max tree depth of 6, a max bin size of 64, with 1000 trees gave us good performance. All GBT models were run until their validation accuracy was non-improving for a hundred rounds and were trained on CPUs.

Table 16: Hyperparameters for Machine Learning Models: Out-of-sample Exercise

Model	Tree Depth	# of Trees
CART	7	
RF	20	900
GBT	6	1000

For the pooled sample prediction, we increased the number of neurons per layers to 512,1024,2048,1024,512 and decreased the dropout rate to 20%, keeping the activation function, the batch size, and the learning rate unchanged. We instituted early stopping with a patience of 1,000 for GBT, and trained a model of depth 6 with up to 10,000 trees and a learning rate of 0.3. We report the results of the best performing GBT.

Implementation

We include 139 features for each individual. Since we work with panel data, there is a sample for each quarter of data. We train roughly 20 million samples, which takes up around 20 gigabytes of data. Our deep learning models are made up of millions of free parameters. Since the estimation procedure relies on computing gradients via backpropagation, which tends to be time and memory intensive, using conventional computing resources (e.g., desktop) would be impractical (if not infeasible). We address the time and memory intensity with two methods. First, to save memory, we use single precision floating point operations, which halves the memory requirements and results in a substantial computational speedup. Second, to accelerate the learning, we parallelized our computations and trained all of our models on a GPU cluster²⁸. In our setting, GPU computations

²⁷We use TPE since it outperformed random search (see Bergstra et al. (2011)), which was shown to be both theoretically and empirically more efficient than standard techniques such as trials on a grid. Other widely used strategies are grid search and manual search.

²⁸1 node with 4 NVIDIA GeForceGTX1080 GPUs. The pooled model trains within 36 hours.

were over 40X faster than CPU for our deep neural networks. For a discussion on the impact of GPUs in deep learning see Schmidhuber (2015).

We conduct our analysis using Python 3.6.3 (Python Software Foundation), building on the packages numpy (Walt, Colbert, and Varoquaux (2011)), pandas (McKinney et al. (2010)) and matplotlib (Hunter (2007)). We develop our deep neural networks with keras (Chollet et al. (2015)) running on top of Google TensorFlow, a powerful library for large-scale machine learning on heterogeneous systems (Abadi et al. (2016)). We run our machine learning algorithms using sci-kit learn (Pedregosa et al. (2011)) and (Chen and Guestrin (2016)).

Features

Table 17 lists our model inputs. Table 18 provides summary statistics for selected features. For the SHAP value analysis, we grouped features that had a correlation higher than 0.7. These groups are presented in Table 19.

Table 17: Model Inputs

90 day delinquencies in the last 36 months	Credit amount paid down on open first mortgage trades
90 days delinquencies in the last 12 months	Credit amount paid down on open second mortgage trades
90 days delinquencies in the last 24 months	Credit card trades opened in the last 12 months
90 days delinquencies in the last 6 months	Credit card utilization ratio
Auto loan or lease inquiries made in the last 3 months	Dismissed bankruptcies
Auto loan trades opened in the last 6 months	Early payoff trades
Balance on 30 days late bankcard trades	Fannie Mae first mortgage trades opened prior to June 2009
Balance on 30 days late mortgage trades	First mortgage trades opened in the last 6 months
Balance on 60 days late bankcard trades	Fraction of 30 days delinquent debt to total debt
Balance on 60 days late mortgage trades	Fraction of 60 days delinquent debt to total debt
Balance on 90-180 days late bankcard trades	Fraction of 90 days delinquent debt to total debt
Balance on 90-180 days late mortgage trades	Fraction of auto loan to total debt
Balance on authorized user trades	Fraction of credit card debt to total debt
Balance on bankcard trades	Fraction of HELOC to total debt
Balance on collections	Fraction of mortgage to total debt
Balance on credit card trades	Freddie Mac first mortgage trades opened prior to June 2009
Balance on derogatory bankcard trades	HELOC trades ever 90 or more days delinquent or derogatory
Balance on derogatory mortgage trades	HELOC utilization ratio
Balance on first mortgage trades	Inquiries made in the last 12 months
Balance on HELOC trades	Installment trades
Balance on mortgage trades	Installment utilization ratio
Balance on open 30 days late installment trades	Judgments with amount >\$1000
Balance on open 30 days late revolving trades	Monthly payment on all debt
Balance on open 60 days late installment trades	Monthly payment on credit card trades
Balance on open 60 days late revolving trades	Monthly payment on HELOC trades
Balance on open 90-180 days late installment trades	Monthly payment on open auto loan trades
Balance on open 90-180 days late revolving trades	Monthly payment on open first mortgage trades
Balance on open auto loan trades	Monthly payment on open second mortgage trades
Balance on open bankcard trades with credit line suspended	Monthly payment on student trades
Balance on open derogatory installment trades	Mortgage trades
Balance on open derogatory revolving trades	Mortgage type
Balance on open installment trades	Mortgage inquiries made in the last 3 months
Balance on open personal liable business loans	Open auto loan trades
Balance on open revolving trades	Open bankcard trades
Balance on revolving trades	Open bankcard trades opened in the last 6 months
Balance on second mortgage trades	Open credit card trades
Balance on student trades	Open first mortgage trades
Bankcard inquiries made in the last 3 months	Open HELOC trades
Bankruptcies filed within the last 24 months	Open mortgage trades
Chapter 13 bankruptcies	Open personal liable business loans
Chapter 7 bankruptcies	Open second mortgage trades
Charge-off amount on unsatisfied charge-off trades	Petitioned bankruptcies
Charge-off trades	Public record bankruptcies
Collections placed in the last 12 months	Public records filed in the last 24 months
Credit amount on deferred student trades	Total 30 days late debt balances
Credit amount on non-deferred student trades	Total 60 days late debt balances
Credit amount on open credit card trades	Total 90 or more days delinquent debt balances
Credit amount on open HELOC trades	Total 90-180 days late debt balances
Credit amount on open installment trades	Total credit amount on open trades
Credit amount on open mortgage trades	Total debt balances
Credit amount on revolving trades	Total derogatory debt balances
Credit amount on unsatisfied derogatory trades	Trades legally paid in full for less than the full balance
:	:

:
 Months since the most recently closed transferred or refinanced first mortgage
 Months since the most recently opened auto loan trade
 Months since the most recently opened credit card trade
 Months since the most recently opened first mortgage
 Months since the most recently opened HELOC trade
 Months since the most recently opened second mortgage
 Months since the most recent 30-180 days delinquency on auto loan or lease trades
 Months since the most recent 30-180 days delinquency on mortgage trade
 Months since the most recent 30-180days delinquency on credit card trades
 Months since the most recent foreclosure proceeding started on first mortgage
 Months since the most recent public record bankruptcy filed
 Months since the most recent 30-180 days delinquency
 Months since the oldest trade was opened
 Months since the most recent 90 or more days delinquency
 Presence of outstanding governmental agency debts
 Presently foreclosed first mortgage trades that occurred in the last 24 months
 Student trades ever 90 or more days delinquent or derogatory occurred in the last 24 months
 Trades ever 90 or more days delinquent or derogatory occurred in the last 24 months
 Presently foreclosed first mortgages
 Ratio of inquiries to trades opened in the last 6 months
 Utility trades
 Utilization ratio
 Unsatisfied collections
 Unsatisfied repossession trades
 Worst ever status on a credit card trade in the last 24 months
 Worst ever status on a mortgage trade in the last 24 months
 Worst ever status on an auto loan or lease trade in the last 24 months
 Worst ever status on any trades in the last 24 months
 Worst present status on a mortgage trade
 Worst present status on a revolving trade
 Worst present status on an auto loan or lease trade
 Worst present status on an installment trade
 Worst present status on an open trade
 Worst present status on any trades
 Worst present status on bankcard trades

List of features included in our model.

Table 18: Summary Statistics

Feature	Mean	Std. Dev	25%	Median	75%
Balance on collections	724.74	3951.88	0	0	0
Balance on credit card trades	4573.20	9824.31	0	834	4591
Balance on mortgage trades	63408.03	160889.35	0	0	76301
Balance on open auto loan trades	4472.88	11608.48	0	0	3915
Balance on open installment trades	8754.12	32554.07	0	0	10583
Balance on open personal liable business loan	290.94	17183.32	0	0	0
Balance on revolving trades	4532.12	9804.11	0	761	4478
Balance on student trades	3523.54	15537.59	0	0	0
Charge-off amount on unsatisfied charge-off trades	1264.56	83747.09	0	0	0
Collections placed in the last 12 months	0.38	1.24	0	0	0
Credit amount on open credit card trades	21475.90	30662.29	0	8600	31947
Credit amount on open installment trades	12178.34	37332.78	0	0	17286
Credit amount on revolving trades	21382.80	30396.75	0	8641	31890
Credit amount on unsatisfied derogatory trades	11048.38	70187.96	0	0	1500
Credit amount paid down on open first mortgage trades	6109.98	164527.99	0	0	2255
Credit card utilization ratio	0.51	2.31	0	0.06	0.37
Early payoff trades	0.94	1.64	0	0	1
Fraction of 30 days delinquent debt to total debt	0.02	0.11	0	0	0
Fraction of 60 days delinquent debt to total debt	0.01	0.08	0	0	0
Fraction of 90 days delinquent debt to total debt	0.03	0.15	0	0	0
Fraction of auto loan to total debt	0.12	0.27	0.00	0.00	0.04
Fraction of credit card debt to total debt	0.27	0.40	0.00	0.03	0.44
Fraction of home equity line of credit to total debt	0.03	0.13	0.00	0.00	0.00
Fraction of mortgage to total debt	0.30	0.42	0.00	0.00	0.82
Inquiries made in the last 12 months	1.38	2.28	0	1	2
Judgments with amount >\$1000	0.06	0.32	0	0	0
Monthly payment on all debt	907.95	11059.39	20	333	1232
Monthly payment on credit card trades	121.22	366.75	0	31	125
Monthly payment on open auto loan trades	132.70	313.24	0	0	227
Monthly payment on student trades	21.12	4628.97	0	0	0
Months since the most recently opened credit card trade	35.64	46.17	7	20	47
Months since the oldest trade was opened	196.45	126.35	98	178	271
Open auto loan trades	0.34	0.60	0	0	1
Open credit card trades	3.56	3.91	0	2	5
Open mortgage trades	0.50	0.83	0	0	1
Total 60 days late debt balances	619.09	13253.67	0	0	0
Total 90 or more days delinquent debt balances	3125.34	33186.26	0	0	0
Total credit amount on open trades	108480.31	259094.00	3000	33146	146535
Total debt balances	77126.36	170742.97	318	11738	95808
Total derogatory debt balances	1287.84	18422.40	0	0	0
Trades ever 90 or more days delinquent or derogatory, last 24 months	1.22	2.78	0	0	1
Utilization ratio	0.56	0.88	0.027	0.52	0.83

Table 19: Feature Groups

<u>Total debt balances*</u>	<u>Number of collections*</u>
Total debt balances	Collections placed in the last 12 months
Total credit amount on open trades	Trades ever 90 or more days delinquent
Balance on mortgage trades	Unsatisfied collections
Credit amount on open mortgage trades	
Balance on first mortgage trades	<u>Number of open credit cards*</u>
	Open credit card trades
<u>30 days late debt balances*</u>	Credit amount on open credit card trades
Total 30 days late debt balances	Open bankcard trades
Balance on 30 days late mortgage trades	Credit amount on revolving trades
<u>60 days late debt balances*</u>	<u>Number of HELOC loans*</u>
Total 60 days late debt balances	Open home equity line of credit trades
Balance on 60 days late mortgage trades	Home equity line of credit utilization ratio
<u>90+ days late debt balances*</u>	<u>Number of mortgages*</u>
Total 90 or more days delinquent debt balances	Fraction of mortgage to total debt
Total 90-180 days late debt balances	Mortgage trades
Balance on 90-180 days late mortgage trades	Open mortgage trades
Total derogatory debt balances	Mortgage type
Balance on derogatory mortgage trades	Open first mortgage trades
<u>Balance on installment loans*</u>	<u>Foreclosed first mortgages*</u>
Credit amount on open installment trades	Presently foreclosed first mortgages
Balance on open installment trades	Presently foreclosed first mortgage trades, last 24 months
<u>Balance on HELOC loans*</u>	<u>Auto loan*</u>
Credit amount on open home equity line of credit trades	Open auto loan trades
Balance on home equity line of credit trades	Monthly payment on open auto loan trades
Balance on open revolving trades	Balance on open auto loan trades
<u>Student debt*</u>	<u>Credit card debt*</u>
Balance on student trades	Balance on credit card trades
Credit amount on non-deferred student trades	Balance on revolving trades
	Balance on bankcard trades
<u>Worst status on any trades*</u>	<u>Trades ever 90 or more days delinquent or derogatory*</u>
Worst ever status on any trades in the last 24 months	90 days delinquencies in the last 6 months
Worst present status on any trades	90 days delinquencies in the last 12 months
	90 days delinquencies in the last 24 months
	90 days delinquencies in the last 36 months
<u>Worst status on credit cards*</u>	<u>Bankruptcy history*</u>
Worst present status on a revolving trade	Public record bankruptcies
Worst present status on bankcard trades	Chapter 7 bankruptcies
	Months since the most recent public record bankruptcy filed
<u>Fraction of 90 days late debt to total debt*</u>	
Worst present status on an open trade	
Fraction of 90 days delinquent debt to total debt	

D Model Comparison

We trained a GBT with up to 10,000 trees, a learning rate of 0.3, and an early-stopping parameter of 1,000 to compare the performance of gradient boosting with deep neural networks. We also built on Table 10 by keeping our models' architecture the same, but expanded the training data by including observations up till the date specified by the training window. This exercise illustrates that while the performance of GBT remains similar, DNN benefits significantly from having more data to train on.

We also looked at the performance of the two models when we allow only the most recent 4 quarters for training.

We also investigated the SHAP values across four different models on the pooled sample: (1) logistic, (2) DNN, (3) GBT, and (4) the hybrid model. Table 22 summarizes these results.

Table 20: Model Comparison: DNN vs. GBT

Training Window*	Testing Window	AUC-score		Loss	
		DNN	GBT	DNN	GBT
2004Q1-2013Q4	2004Q1-2013Q4	0.9561	0.9429	0.2506	0.2832
2004Q1	2006Q1	0.9228	0.9241	0.3272	0.3236
2004Q2	2006Q2	0.9243	0.9251	0.3212	0.3190
2004Q3	2006Q3	0.9246	0.9264	0.3208	0.3162
2004Q4	2006Q4	0.9247	0.9256	0.3223	0.3194
2005Q1	2007Q1	0.9254	0.9261	0.3230	0.3209
2005Q2	2007Q2	0.9249	0.9258	0.3255	0.3228
2005Q3	2007Q3	0.9245	0.9253	0.3272	0.3250
2005Q4	2007Q4	0.9232	0.9243	0.3327	0.3284
2006Q1	2008Q1	0.9247	0.9249	0.3298	0.3289
2006Q2	2008Q2	0.9235	0.9244	0.3331	0.3306
2006Q3	2008Q3	0.9250	0.9253	0.3310	0.3284
2006Q4	2008Q4	0.9251	0.9256	0.3289	0.3281
2007Q1	2009Q1	0.9267	0.9275	0.3263	0.3233
2007Q2	2009Q2	0.9268	0.9276	0.3244	0.3222
2007Q3	2009Q3	0.9279	0.9288	0.3197	0.3183
2007Q4	2009Q4	0.9300	0.9307	0.3149	0.3133
2008Q1	2010Q1	0.9302	0.9308	0.3155	0.3141
2008Q2	2010Q2	0.9302	0.9308	0.3146	0.3128
2008Q3	2010Q3	0.9302	0.9308	0.3137	0.3117
2008Q4	2010Q4	0.9301	0.9307	0.3141	0.3124
2009Q1	2011Q1	0.9312	0.9317	0.3112	0.3097
2009Q2	2011Q2	0.9297	0.9302	0.3137	0.3126
2009Q3	2011Q3	0.9298	0.9302	0.3130	0.3117
2009Q4	2011Q4	0.9300	0.9302	0.3122	0.3115
2010Q1	2012Q1	0.9305	0.9307	0.3106	0.3099
2010Q2	2012Q2	0.9291	0.9293	0.3130	0.3124
2010Q3	2012Q3	0.9288	0.9290	0.3119	0.3113
2010Q4	2012Q4	0.9282	0.9286	0.3128	0.3121
2011Q1	2013Q1	0.9290	0.9293	0.3109	0.3101
2011Q2	2013Q2	0.9286	0.9289	0.3109	0.3103
2011Q3	2013Q3	0.9292	0.9294	0.3091	0.3085
2011Q4	2013Q4	0.9293	0.9295	0.3080	0.3074

Performance comparison of the two best performing machine learning classification models of consumer default risk. The model calibrations are specified by the training and testing windows. * implies that all data was used up to the quarter specified. The results of predicted probabilities versus actual outcomes over the following 8Q (testing period) are used to calculate the loss metric and the AUC-score for 90+ days delinquencies within 8Q. DNN refers to deep neural network, GBT refers to gradient boosted trees. Source: Authors' calculations based on Experian Data.

Table 21: Model Comparison: DNN vs. GBT

Training Window Start	Training Window End	Testing Window	AUC-score		Loss	
			DNN	GBT	DNN	GBT
2004Q1	2004Q1	2006Q1	0.9229	0.9242	0.3273	0.3234
2004Q1	2004Q2	2006Q2	0.9239	0.9251	0.3226	0.3190
2004Q1	2004Q3	2006Q3	0.9251	0.9265	0.3209	0.3161
2004Q1	2004Q4	2006Q4	0.9244	0.9257	0.3218	0.3192
2004Q2	2005Q1	2007Q1	0.9253	0.9261	0.3236	0.3205
2004Q3	2005Q2	2007Q2	0.9245	0.9258	0.3252	0.3220
2004Q4	2005Q3	2007Q3	0.9244	0.9253	0.3271	0.3244
2005Q1	2005Q4	2007Q4	0.9237	0.9243	0.3282	0.3273
2005Q2	2006Q1	2008Q1	0.9238	0.9251	0.3300	0.3270
2005Q3	2006Q2	2008Q2	0.9237	0.9244	0.3313	0.3287
2005Q4	2006Q3	2008Q3	0.9247	0.9254	0.3286	0.3263
2006Q1	2006Q4	2008Q4	0.9249	0.9257	0.3277	0.3261
2006Q2	2007Q1	2009Q1	0.9266	0.9277	0.3246	0.3216
2006Q3	2007Q2	2009Q2	0.9268	0.9278	0.3235	0.3209
2006Q4	2007Q3	2009Q3	0.9278	0.9288	0.3196	0.3172
2007Q1	2007Q4	2009Q4	0.9297	0.9307	0.3149	0.3124
2007Q2	2008Q1	2010Q1	0.9293	0.9306	0.3169	0.3133
2007Q3	2008Q2	2010Q2	0.9292	0.9304	0.3164	0.3129
2007Q4	2008Q3	2010Q3	0.9288	0.9302	0.3163	0.3128
2008Q1	2008Q4	2010Q4	0.9285	0.9299	0.3179	0.3142
2008Q2	2009Q1	2011Q1	0.9292	0.9307	0.3157	0.3121
2008Q3	2009Q2	2011Q2	0.9276	0.9291	0.3189	0.3150
2008Q4	2009Q3	2011Q3	0.9276	0.9291	0.3185	0.3147
2009Q1	2009Q4	2011Q4	0.9276	0.9294	0.3180	0.3138
2009Q2	2010Q1	2012Q1	0.9291	0.9302	0.3144	0.3113
2009Q3	2010Q2	2012Q2	0.9277	0.9290	0.3164	0.3133
2009Q4	2010Q3	2012Q3	0.9279	0.9288	0.3140	0.3119
2010Q1	2010Q4	2012Q4	0.9273	0.9284	0.3154	0.3132
2010Q2	2011Q1	2013Q1	0.9281	0.9291	0.3137	0.3111
2010Q3	2011Q2	2013Q2	0.9275	0.9286	0.3142	0.3112
2010Q4	2011Q3	2013Q3	0.9280	0.9292	0.3127	0.3095
2011Q1	2011Q4	2013Q4	0.9282	0.9293	0.3116	0.3084
Average			0.9267	0.9278	0.3202	0.3172

Performance comparison of the two best performing machine learning classification models of consumer default risk. The model calibrations are specified by the training and testing windows. The results of predicted probabilities versus actual outcomes over the following 8Q (testing period) are used to calculate the loss metric and the AUC-score for 90+ days delinquencies within 8Q. DNN refers to deep neural network, GBT refers to gradient boosted trees. Source: Authors' calculations based on Experian Data.

Table 22: Shap Values across Models

Feature	Hybrid	Logistic	DNN	GBT
Worst status on any trades*	0.596 (1)	0.095 (1)	0.155 (1)	1.038 (1)
Months since the oldest trade was opened	0.181 (2)	0.036 (4)	0.077 (4)	0.303 (4)
Months since the most recent 90 or more days delinquency	0.177 (3)	0.018 (9)	0.068 (5)	0.329 (2)
Number of collections*	0.164 (4)	0.056 (2)	0.132 (3)	0.211 (9)
Number of open credit cards*	0.163 (5)	0.039 (3)	0.137 (2)	0.245 (6)
Total debt balances*	0.161 (6)	0.01 (13)	0.028 (10)	0.308 (3)
Credit card utilization ratio	0.144 (7)	0.002 (43)	0.015 (23)	0.275 (5)
90+ days late debt balances*	0.125 (8)	0.016 (11)	0.033 (7)	0.221 (8)
Credit amount on unsatisfied derogatory trades	0.117 (9)	0.001 (57)	0.018 (18)	0.221 (7)
Months since the most recent 30-180 days delinquency	0.101 (10)	0.003 (36)	0.045 (6)	0.175 (11)
Inquiries made in the last 12 months	0.097 (11)	0.018 (8)	0.023 (12)	0.174 (12)
Monthly payment on all debt	0.091 (12)	0.0 (88)	0.008 (45)	0.179 (10)
Utilization ratio	0.089 (13)	0.008 (18)	0.017 (20)	0.165 (13)
Months since the most recently opened credit card trade	0.07 (14)	0.01 (14)	0.023 (13)	0.127 (14)
Credit card debt*	0.066 (15)	0.018 (10)	0.02 (16)	0.125 (15)
Balance on collections	0.063 (17)	0.003 (35)	0.008 (41)	0.123 (17)
Balance on installment loans*	0.063 (16)	0.001 (54)	0.014 (25)	0.124 (16)
Credit amount paid down on open first mortgage trades	0.062 (18)	0.001 (51)	0.013 (31)	0.114 (19)
Months since the most recently opened first mortgage	0.057 (19)	0.001 (65)	0.015 (22)	0.114 (18)
Monthly payment on credit card trades	0.052 (20)	0.009 (16)	0.014 (30)	0.097 (22)
Auto loan*	0.051 (21)	0.003 (29)	0.017 (19)	0.1 (21)
Monthly payment on open first mortgage trades	0.05 (22)	0.001 (74)	0.006 (54)	0.1 (20)
Number of mortgages*	0.045 (23)	0.028 (6)	0.029 (9)	0.079 (26)
Balance on HELOC loans*	0.044 (25)	0.006 (22)	0.006 (53)	0.087 (23)
Installment utilization ratio	0.044 (24)	0.004 (27)	0.013 (33)	0.082 (24)
Student debt*	0.042 (26)	0.008 (17)	0.014 (26)	0.079 (25)
Months since the most recently opened auto loan trade	0.039 (27)	0.001 (67)	0.022 (14)	0.072 (28)
Charge-off amount on unsatisfied charge-off trades	0.038 (28)	0.001 (66)	0.002 (69)	0.076 (27)
Fraction of 90 days late debt to total debt*	0.035 (29)	0.029 (5)	0.024 (11)	0.051 (36)
Worst ever status on a credit card trade in the last 24 months	0.034 (30)	0.001 (55)	0.014 (24)	0.061 (30)

This table reports the Shapley values for four selected machine learning classification models of consumer default risk. We sorted the features based on the feature’s relative rank (in parentheses) using the hybrid model. Source: Authors’ calculations based on Experian Data.

Table 23: Neural networks comparison: Loss & Accuracy

Model	In-sample Loss		Out-of-sample Loss	
	w/o Dropout	Dropout	w/o Dropout	Dropout
Logistic Regression	0.3451	0.3451	0.3449	0.3450
1 layer	0.3109	0.3106	0.3122	0.3116
2 layers	0.2965	0.2900	0.3078	0.3003
3 layers	0.2804	0.2460	0.3047	0.2744
4 layers	0.2669	0.2142	0.3005	0.2575
5 layers	0.2534	0.2013	0.2978	0.2506

Model	In-sample Accuracy		Out-of-sample Accuracy	
	w/o Dropout	Dropout	w/o Dropout	Dropout
Logistic Regression	0.8564	0.8564	0.8566	0.8566
1 layer	0.8687	0.8688	0.8681	0.8684
2 layers	0.8751	0.8787	0.8705	0.8736
3 layers	0.8829	0.9017	0.8729	0.8862
4 layers	0.8897	0.9163	0.8755	0.8943
5 layers	0.8968	0.9230	0.8785	0.8981

In-sample and out-of-sample loss (categorical cross-entropy) and accuracy for neural networks of different depth and for logistic regression. Models are calibrated and evaluated on the pooled sample (2004Q1 - 2013Q4). Source: Authors' calculations based on Experian Data.

E Comparison with Credit Scores

The credit score is a summary indicator intended to predict the risk of default by the borrower and it is widely used by the financial industry. For most unsecured debt, lenders typically verify a perspective borrower's credit score at the time of application and sometimes a short recent sample of their credit history. For larger unsecured debts, lenders also typically require some form of income verification, as they do for secured debts, such as mortgages and auto loans. Still, the credit score is often a key determinant of crucial terms of the borrowing contract, such as the interest rate, the downpayment or the credit limit.

The most widely known credit score is the FICO score, a measure generated by the Fair Isaac Corporation, which has been in existence in its current form since 1989. Each of the three major credit reporting bureaus— Equifax, Experian and TransUnion— also have their own proprietary credit scores. Credit scoring models are not public, though they are restricted by the law, mainly the Fair Credit Reporting Act of 1970 and the Consumer Credit Reporting Reform Act of 1996. The legislation mandates that consumers be made aware of the 4 main factors that may affect their credit score. Based on available descriptive materials from FICO and the credit bureaus, these are payment history and outstanding debt, which account for more than 60% of the variation in credit scores, followed by credit history, or the age of existing accounts, which explains 15-20% of the variation, followed by new accounts and types of credit used (10-5%) and new "hard" inquiries, that is credit report inquiries coming from perspective lenders after a borrower initiated credit application.

U.S. law prohibits credit scoring models from considering a borrower's race, color, religion, national origin, sex and marital status, age, address, as well as any receipt of public assistance, or the exercise of any consumer right under the Consumer Credit Protection Act. The credit score cannot be based on information not found in a borrower's credit report, such as salary, occupation, title, employer, date employed or employment history, or interest rates being charged on particular accounts. Finally, any items in the credit report reported as child/family support obligations are not permitted, as well as "soft" inquiries²⁹ and any information that is not proven to be predictive of future credit performance.

²⁹These include "consumer-initiated" inquiries, such as requests to view one's own credit report, "promotional inquiries," requests made by lenders in order to make pre-approved credit offers, or "administrative inquiries," requests made by lenders to review open accounts. Requests that are marked as coming from employers are also not counted.

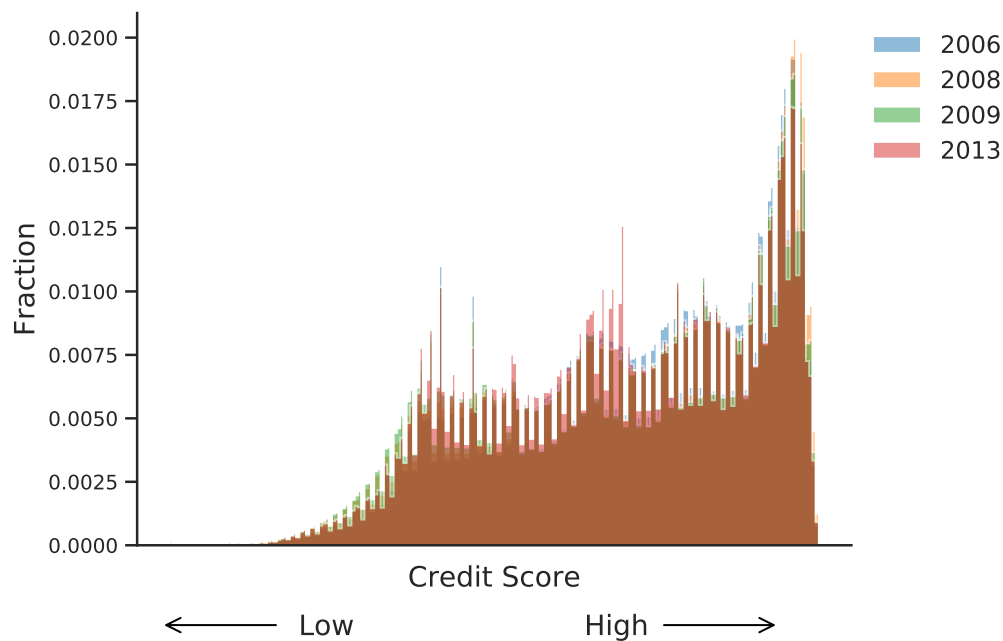


Figure 13: Histogram of the credit score in our data by year for selected years. Source: Source: Authors' calculations based on Experian data.

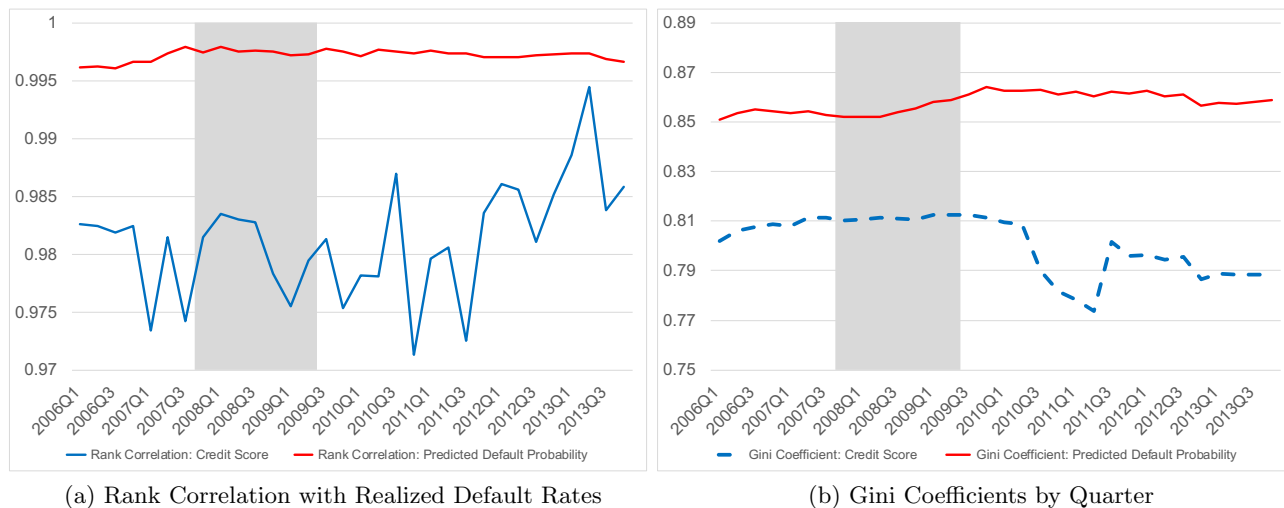


Figure 14: Absolute value of rank correlation with realized default rate for the credit score and model predicted default probability (a) and Gini coefficients for the credit score and model predicted default probability by quarter (b). Source: Authors' calculations based on Experian data.

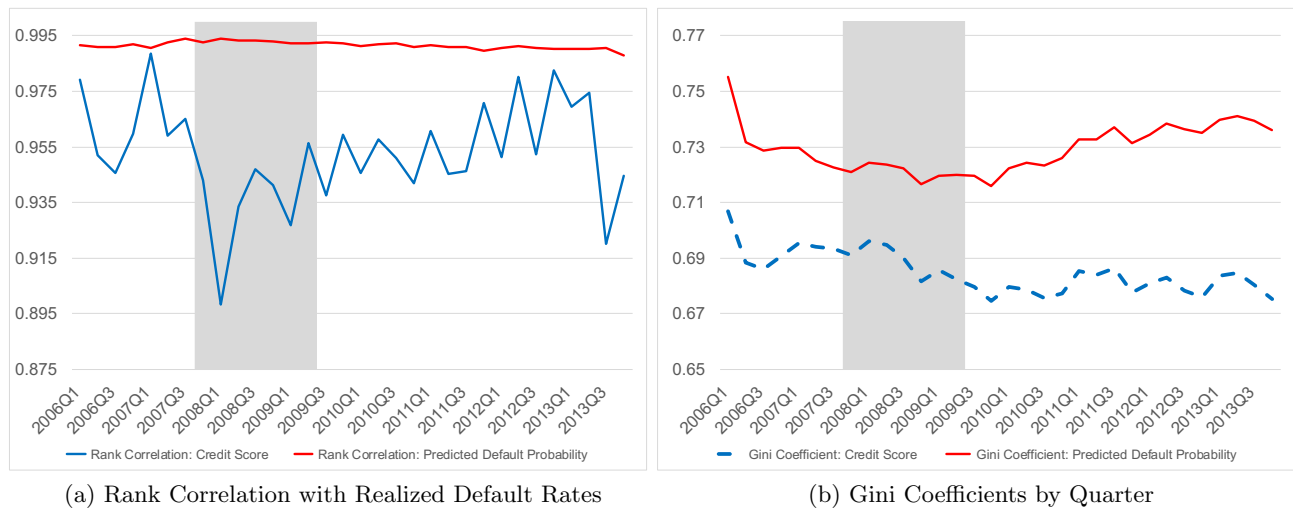


Figure 15: Absolute value of rank correlation with realized default rate for the credit score and model predicted default probability (a) and Gini coefficients for the credit score and model predicted default probability by quarter (b) for the population of current borrowers. Borrowers are current if they have not outstanding delinquencies. Source: Authors' calculations based on Experian data.

F Cost Savings for Consumers

Table 24: Distribution of Customers by Credit Score and Predicted Default

		Credit Score			
		Subprime, Near Prime	Prime Low	Prime Mid	Prime High, Superprime
Predicted Default Bin	1	36.22%	3.63%	1.24%	0.41%
	2	3.87%	3.41%	2.41%	1.08%
	3	1.09%	2.52%	4.01%	3.49%
	4	0.33%	1.21%	3.44%	31.64%

This table reports the share of customers in each predicted credit score categories and corresponding predicted default probability bins. Customers are classified based on credit scores and predicted default probabilities at account origination for each of their credit cards included in the balances. Source: Authors' calculations based on Experian data.

Table 25: Cost Savings for Consumers: Descriptive Statistics

				Credit Score		Credit Score
		Subprime, Near Prime	Prime Low	Prime Mid	Prime High, Superprime	
PP bin	Age					
	1	42	44	47		
	2	40	41	44		
	3	43	42	42		
	4	47	47	45		
PP bin	Household Income (\$)					
	1	57,673	65,929	74,326		
	2	65,291	66,527	74,474		
	3	83,843	77,536	75,962		
	4	107,025	101,535	97,465		
PP bin	Total Debt Balances (\$)					
	1	65,275	84,746	91,062		
	2	105,681	94,048	103,516		
	3	174,560	126,318	106,815		
	4	260,262	200,065	163,614		
PP bin	Total bankcard balances (\$)					
	1	3,822	4,332	3,968		
	2	10,480	6,955	5,502		
	3	16,330	10,606	6,794		
	4	20,660	14,464	9,824		
PP bin	Credit Card Limit (\$)					
	1	10,666	13,621	18,930		
	2	20,740	18,832	22,303		
	3	34,333	28,705	26,571		
	4	49,673	44,007	40,584		
PP bin	90+ DPD on All Accounts, Next 24 Months					
	1	0.62	0.44	0.37		
	2	0.36	0.29	0.25		
	3	0.29	0.23	0.18		
	4	0.24	0.19	0.13		
PP bin	90+ DPD on Bankcards, Next 24 Months					
	1	0.35		0.35		
	2	0.21		0.21		
	3	0.14		0.14		
	4	0.08		0.08		
PP bin	Credit Score, Current					
	1	606	661	694		
	2	644	671	701		
	3	659	684	711		
	4	673	697	724		
PP bin	Predicted Probability, Current					
	1	0.60	0.44	0.37		
	2	0.34	0.29	0.25		
	3	0.26	0.21	0.18		
	4	0.20	0.17	0.12		

This table reports the descriptive statistics on customers based on their credit score and predicted default probability based on our hybrid model. Credit score categories and predicted default probability bins are computed at account origination for credit cards. Source: Authors' calculations based on Experian data.



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IMF Working Paper

FinTech in Financial Inclusion Machine Learning Applications in Assessing Credit Risk

By Majid Bazarbash

***IMF Working Papers* describe research in progress by the author(s) and are published to elicit comments and to encourage debate.** The views expressed in IMF Working Papers are those of the author(s) and do not necessarily represent the views of the IMF, its Executive Board, or IMF management.

I N T E R N A T I O N A L M O N E T A R Y F U N D

IMF Working Paper

Monetary and Capital Markets Department

**FinTech in Financial Inclusion:
Machine Learning Applications in Assessing Credit Risk****Prepared by Majid Bazarbash***

Authorized for distribution by Ulric Eriksson von Allmen

May 2019

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Abstract

Recent advances in digital technology and big data have allowed FinTech (financial technology) lending to emerge as a potentially promising solution to reduce the cost of credit and increase financial inclusion. However, machine learning (ML) methods that lie at the heart of FinTech credit have remained largely a black box for the nontechnical audience. This paper contributes to the literature by discussing potential strengths and weaknesses of ML-based credit assessment through (1) presenting core ideas and the most common techniques in ML for the nontechnical audience; and (2) discussing the fundamental challenges in credit risk analysis. FinTech credit has the potential to enhance financial inclusion and outperform traditional credit scoring by (1) leveraging nontraditional data sources to improve the assessment of the borrower's track record; (2) appraising collateral value; (3) forecasting income prospects; and (4) predicting changes in general conditions. However, because of the central role of data in ML-based analysis, data relevance should be ensured, especially in situations when a deep structural change occurs, when borrowers could counterfeit certain indicators, and when agency problems arising from information asymmetry could not be resolved. To avoid digital financial exclusion and redlining, variables that trigger discrimination should not be used to assess credit rating.

JEL Classification Numbers: C52, C53, C55, G21, G23

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Author's Email Address: MBazarbash@IMF.org

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ACRONYMS

4GL	4 th Generation Programming Language	LTV	Loan-to-Value Ratio
BIS	Bank of International Settlements	ML	Machine Learning
DBMS	Database Management Systems	MSE	Mean Squared Error
DTI	Debt-to-Income Ratio	NN	Neural Network
FSB	Financial Stability Board	PD	Probability of Default
GBDT	Gradient Boosting Decision Tree	PDP	Partial Dependent Plot
IMF	International Monetary Fund	SQL	Structured Query Language
LASSO	Least Absolute Shrinkage Selection Operator	SVM	Support Vector Machine
LGD	Loss Given Default	VAR	Vector Autoregressive Model

GLOSSARY

Five Cs of Credit	A system to assess credit risk of business borrowers based on five features of the borrower: character, capacity, capital structure, coverage, and conditions
Cross-Validation	A model validation technique for assessing how the results of a statistical model will generalize to an independent data set and can make predictions
Estimation Error	Average difference between the estimated value of the target variable and realized values. Common measures are the mean squared errors or mean absolute errors.
FinTech	Technologically enabled financial innovation that could result in new business models, applications, processes or products with an associated material effect on financial markets, financial institutions and provision of financial services
Generalization	The ability of a model to offer comparable in-sample and out-of-sample predictive performance
Machine Learning	A method of designing a sequence of actions to solve a problem that optimize automatically through experience and with limited or no human intervention.
Nonparametric Model	A data analysis models whose structure is not specified a priori, but is instead determined from data
Overfitting	The problem of a statistical model that offers high in-sample, fit but poor out-of-sample predictive performance

I. INTRODUCTION

One of the major challenges of low-income and emerging economies is the high cost of credit. The high cost of borrowing and credit rationing gives rise to financial exclusion of especially small borrowers such as small- and medium-size enterprises and households that play a significant role in the macroeconomy ([Sahay et al. \(2015a\)](#)). As a promising solution, modern technological advances have enabled new business models to employ modern data analysis techniques on big data and automate tasks to make credit decisions more efficiently. FinTech credit¹ promises to offer loans at a higher speed and lower cost, and therefore grant loans to a larger fraction of the population, resulting in elevated financial inclusion².

Policymakers and economists have recognized FinTech's potential to transform the financial system but have also raised concerns about the opaque nature of the way modern technology could create value and generate financial stability risks ([The Bali FinTech Agenda \(2018\)](#)). In particular, for assessing credit risk, FinTech companies rely heavily on machine learning (ML) techniques that have been developed in the computer science field. Economists who want to improve their understanding of financial inclusion in the new FinTech landscape need to understand the technology, its impact on credit extension, and the benefits and risks it carries.

This paper's contribution to the literature in financial inclusion and FinTech is that it provides a nontechnical analysis of how ML could enhance assessing credit risk of borrowers while describing the main strengths and weaknesses of ML-based lending. To this end, the paper takes two important steps. First, to address the finance aspect of FinTech, it discusses the fundamental challenges that arise when assessing credit risk, confronted by both FinTech and traditional lenders. Second, the paper takes ML out of the black box by going over key concepts and ideas in ML and reviewing the most common techniques in ML in a nontechnical and illustrative way. Building on these two steps, the paper then discusses what ML-based lending can and cannot do to enhance credit assessment and ways in which FinTech credit can contribute to responsible financial inclusion. Therefore, it takes a step toward drawing policy implications for using FinTech credit to promote financial inclusion.

To describe the underlying challenges in prudent lending, the paper draws on the credit risk modeling literature. Credit risk assessment is seen as estimating a loan's probability of default and loss given default while taking agency problems arising from information asymmetry into consideration. Practitioners often seek information for five broad attributes (capacity, capital structure, coverage, character, and conditions), the well-known five Cs of credit ratings. ML has the potential to improve the assessments of some of the five Cs of the credit rating process, and thereby improve credit screening more broadly. For instance, [Jagtiani & Lemieux \(2017\)](#) document superior performance of loans originated by Lending

¹ The use of machine learning for assessing credit risk is not confined to FinTech credit companies, but has been incorporated by some large banks in their credit rating process at least in form of the challenger model. Nonetheless, FinTech credit companies, including peer-to-peer lending platforms and balance sheet lenders, use ML as their major method for assessing credit risk.

² For a discussion of macroeconomic gains of improving financial inclusion, see [Sahay et al. \(2015b\)](#).

Club in the US compared to traditional lenders and find a more accurate risk pricing through ML-based credit analysis.

The paper provides a survey of the recent ML literature and discusses the several methods that have gained the highest popularity in practice. These models include decision trees, random forest, gradient boosting trees, support vector machines, and neural networks. The paper presents the fundamental ideas underpinning these methods, and discusses their strengths, weaknesses, and extensions. While the discussion avoids mathematical detail and computational methods involved in applying the algorithms, it provides sufficient background for a nontechnical learner to understand the models.

The core philosophy of ML is to apply potentially complicated algorithms that could be run by machines to learn patterns in data with the primary goal of making predictions. ML models are designed to analyze a large amount of information contained in data from various sources. ML models could identify patterns in the data that standard econometric models cannot³. Nonetheless, a major shortcoming of using complex algorithms is that, in general, these patterns cannot be readily communicated with analysts and verified against expert knowledge. Therefore, while ML could make remarkable use of available data for making more accurate predictions than traditional methods, it could generate misleading results when data relevance is questioned. For example, in situations where there are rapid structural changes in the environment that are not fully reflected in the data, a naïve use of ML models may overweight information prior to the structural change and, for instance, result in the underestimation of the credit risk of borrowers.

An evolving body of literature has emerged in recent years that studies various aspects of FinTech, trends in FinTech development, its economic impact, potential risks, and possible policy and regulatory implications (see, for example, contributions by [Phillipon \(2016\)](#), [Berg et al. \(2018\)](#), [Buchak et al. \(2018\)](#), [Das \(2018\)](#), [Fuster et al. \(2018\)](#), [Gomber et al. \(2018\)](#)⁴, by IMF staff: [He et al. \(2017\)](#), [Lukonga \(2018\)](#), [Berkmen et al. \(2019\)](#), [McNeill et al. \(forthcoming\)](#) and [The Bali FinTech Agenda \(2018\)](#), FSB: [\(2017a, 2017b\)](#), [CGSF & FSB \(2017\)](#), BIS: [Claessens et al. \(2018\)](#), [BIS \(2018\)](#), and CGAP: [Kaffenberger et al. \(2018\)](#))⁵. These studies focus on the implications of FinTech, taking a rather broad view of ML. To facilitate a more in-depth understanding of the strengths and shortcomings of using ML for assessing credit risk and its implications for fostering financial inclusion, this paper provides a nontechnical primer to fundamental concepts in ML and some of the most common ML techniques.

A body of literature in economics presents studies of the applications of machine learning in various areas. For an extensive discussion of applications of machine learning models in economics, see NBER 2017 conference papers under “The Economics of Artificial

³ See for instance, [Lotterman et al. \(2012\)](#), among others, who show superior performance of machine learning models, including support vector machines and neural networks relative to the typical linear models in predicting loss given default. For a recent example, see [Bazarbash \(2019\)](#).

⁴ For a review of recent academic literature studying the use of digital technology in finance, see [Gomber et al. \(2017\)](#).

⁵ The list is highly selective and not intended to be comprehensive.

Intelligence: An Agenda,” particularly [Athey \(2018\)](#). Other similar studies are [Varian \(2014\)](#), [Athey & Imbens \(2015\)](#), and [Mullainathan & Spiess \(2017\)](#). This paper takes a less technical approach and discusses ML in the context of credit risk assessment, which is not the main focus of prior studies.

There has been a surge in recent years in the use of ML tools for estimating credit risk, especially since the establishment of Basel II, which called for development of internal credit rating models by banks, and since the global financial crisis. However, internal rating models based on the standard linear econometric approach have been generally shown to exhibit poor performance in forecasting loss given default ([Altman and Hotchkiss \(2010\)](#)). Studies of credit risk show that while ML models outperform traditional models, their performance depends on the specific ML model, the environment, and the sample used in the analysis (see [Bastos \(2010\)](#), [Qi and Zhao \(2011\)](#), [Lotterman et al. \(2012\)](#), [Tows \(2016\)](#), [Bazarbash \(2017\)](#), and [Nazemi et al. \(2018\)](#)). Motivated by these studies, this paper covers the most common and powerful ML methods applied for credit analysis.

The rest of the paper is organized as follows. [Section II](#) presents the main elements of prudent lending in the context of the five Cs of credit and agency problems. [Section III](#) discusses main ML tools and techniques. Building on these sections, [Section IV](#) analyzes strengths and weaknesses of ML-based lending in contrast to traditional methods. [Section V](#) includes concluding remarks.

II. THE FIVE CS OF CREDIT RATING, PD, AND LGD

Lending decisions involve evaluating a borrower’s credit risk, which consists of assessing the capacity and motivation of the borrower to repay the loan and the lender’s protection against losses if the borrower defaults. The credit rating is summarized by two important metrics: the probability of default (PD) and the expected loss given default (LGD). The PD rating shows the likelihood of failing to repay the loan, and therefore depends on the income-generating capacity of the borrower during the loan lifecycle. The LGD measures the fraction of lost value relative to the outstanding balance at default, which is linked to the lender’s protection against loss upon default.

Information to assess the PD and LGD of a loan comes from a wide range of sources. This section discusses the common framework the five Cs of credit, used by practitioners to group borrower information under five main features that are important for credit rating⁶. The five Cs of credit rating is primarily used for business borrowers, but shares common attributes with credit rating of households. The five Cs of credit include capacity, capital structure, coverage, character, and conditions. An important way in which ML could enhance assessing credit risk is to improve evaluation of each of these features.

Capacity

The borrower’s financial capacity shows the ability of the business to repay the loan, considering other senior obligations of the company and the company’s profitability. A key indicator of capacity is the debt-to-income (DTI) ratio. A lower DTI in good times is often

⁶ See [Tirole \(2010\)](#), [Investopedia](#), and websites of [Bank of America](#) and [Wells Fargo](#).

considered indicative of sufficient cash-generating capacity in downturns. However, the explanatory power of indicators like DTI for financial capacity depends crucially on the operational attributes of the underlying business.

Alternatively, a more fundamental analysis could directly model the future path of business income in terms of business performance incorporating factors such as:

- the business model;
- the infrastructure;
- key expertise;
- competitiveness;
- customer base;
- potential growth; and
- technological advantage.

Capital Structure

Capital structure reflects the composition of the company's liabilities. A higher capital ratio means a higher stake in the company by the business owners, which tends to provide a higher incentive for the managers to run the business profitably and delays default triggers due to a higher cushion provided by the capital.

Coverage

The effective value of pledged collateral and loan coverage indicates protection against loan losses in case of default. Loan coverage comes in two main forms: pledged collateral assets and loan guarantors. Like the capital ratio, higher security due to higher coverage provides the lender with more confidence in loan recovery in the event of default and reduces the LGD. An important challenge in evaluating coverage is the lack of a formal market for some assets, a situation that arises in particular for small business borrowers. Accordingly, the lender does not have a reference market price to use as a basis for estimating the price of such assets and project the future value evolution. In this case, a pricing model needs to be developed based on the specifics of each asset and possible implications of changes in the borrower's business on the asset's liquidation value.

Character

The borrower's character constitutes the track record for missed payments and default on previous debts, fraud, legal expenses, and basically any information implying questionable character of the business owner. Such information usually appears on the borrower's credit report and is used to generate a credit score (for example, the FICO score in the US) that banks predominantly use in credit assessment. A higher rating for character is often perceived as a manager with a higher sense of responsibility in running the business. The significance of character is particularly high in loans made to small companies, where the manager's role is pivotal in the success of the business.

Conditions

General conditions are beyond the company's control, but have cyclical influence on the loan performance. Conditions could include macroeconomic conditions (the business cycle) such as the growth rate, unemployment rate, inflation rate, interest rate, tariffs, regulations, the

financial cycle, industry-wide factors, and geography. Unlike borrower-specific features, such information remains the same for loans in the same industry and market. One way to think about the five Cs of credit rating is that “conditions” reflect systematic drivers while other Cs represent borrower-specific credit risk drivers. From a finance perspective, underlying conditions beyond the control of the business manager represent the nondiversifiable component of risk.

III. MACHINE LEARNING (ML) METHODS

This section presents some of the most prominent applications of machine learning that have proved successful in various applications and are used in credit risk analysis. It starts by reviewing fundamental concepts in machine learning and introducing the key concept of cross-validation in ML⁷. Next, it presents and discusses three classes of ML models, including tree-based models (decision trees, random forest, and gradient boosting trees), support vector machines, and neural networks⁸.

A. Fundamental Concepts

What is Machine Learning?

At its core, ML takes advantage of the high computational power of “machines” to run (mostly complicated) algorithms to “learn” from data. [Jordan and Mitchell \(2015\)](#) define a learning problem as “improving some measure of performance when executing some task, through some type of training experience.” The measure of performance is often defined as minimizing the overall error of estimation. The estimation error captures the average difference between the estimated value of the target variable and realized values. Common measures are the mean squared errors or mean absolute errors.

Machine learning models as nonparametric models

Machine learning models are largely nonparametric as opposed to econometric models that focus predominantly on parametric modeling⁹. Parametric models posit a functional form that links the outcome variable to explanatory variables based on a set of assumptions (or theory) that reflect the modeler’s understanding of the relationship. For example, it may be assumed that price elasticities of demand are invariant with respect to the volume of demand and accordingly posit a log-linear function that could be estimated by an ordinary least square approach. Another example is to assume variables follow a mean-reverting and Markovian structure with an autocorrelation parameter invariant to time, and therefore assume an autoregressive model as a standard vector autoregressive model.

⁷ The material in this section predominantly draws on standard ML textbooks by [James et al. \(2013\)](#) and [Hastie et al. \(2016\)](#) as well as a [scikit-learn user guide](#).

⁸ According to [Risk](#) magazine (July 3, 2018), the two most common ML techniques deployed in credit risk modeling are random forest and gradient boosting decision trees. Moreover, lenders are also experimenting with neural networks.

⁹ A more detailed comparison of ML and econometric modeling approaches is given in the next section. ML also includes parametric models such as Least Absolute Selection and Shrinkage Operator (LASSO) and ridge regressions, used for dimension reduction when many features are included in the model.

Estimation of model parameters allows for quantifying and possibly testing the modeler's understanding of the functional form that is assumed to hold between the features and output variable. Other common examples of econometric models that are parametric are the ordinary least squares, logistic model, and structural models in economics. The advantages of parametric models are parsimony, interpretability, fast estimation speed, and less need for a large training dataset. However, the main weaknesses of parametric models are specification risk, limited complexity, and generally poor predictive power.

On the other hand, nonparametric models make minimal functional assumptions to gain more flexibility in learning from the data. All machine learning models discussed in this section fall under the class of nonparametric models. The main strength of nonparametric models is their high flexibility and therefore their high predictive performance, which increases when a larger sample is used to train the model¹⁰. The main concern in using nonparametric models is the risk of overfitting.

Cross-validation

The problem of overfitting is usually addressed in machine learning by a model assessment technique called cross-validation. In cross-validation, data are randomly partitioned into two samples called train and test samples. The information of features and responses in the train sample are used to estimate the model parameters, that is, “train” the model. The prediction error of the estimated model is measured by feeding the features of the test sample into the model and comparing the model's predictions with the actual values of the outcome variable. Because the train and test samples are independent of each other, the prediction error measures out-of-sample predictive power of the model, that is, prediction of response on a new observation¹¹.

In this context, a model with the problem of overfitting yields a low train error but a high test error. As a result, the model does not generalize well to make accurate predictions on the test sample and is over-parametrized.

k-fold cross-validation

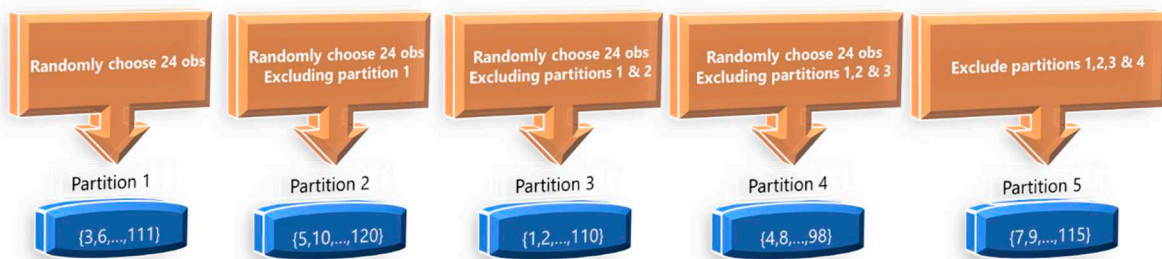
In practice, a version of cross-validation called *k-fold cross-validation* is often used. [Figure 1](#) illustrates a fivefold cross-validation for a sample with 120 observations. The objective is to randomly split the sample into five partitions with equal numbers of observations and use four partitions to train the model and the left-out partition to evaluate the prediction error based on, for instance, the mean squared error (MSE). Partitions 2 to 5 are used for training the model, and the prediction error is calculated on partition 1, call MSE_1 . Similarly, leaving out partition 2, training the model on partitions 1, 3, 4, and 5, and evaluating test error yields MSE_2 . The fivefold cross-validation prediction error is the average prediction errors on the five partitions, that is, $\frac{1}{5} \sum_{i=1}^5 MSE_i$.

¹⁰As [Russell and Norvig \(2016\)](#) put it, “Nonparametric models are good when you have a lot of data and no prior knowledge, and when you don't want to worry too much about choosing the right features.”

¹¹ For example, in case of regressions, it is possible to obtain a negative R^2 for the goodness of fit on the test sample, because of including noisy variables among the features. The effect of noisy variables would not be captured by the R^2 of the model on the train sample, because the model minimizes the mean squared error using the actual values of the outcome variable.

Machine learning models often have hyper-parameters that govern the shape of the algorithm. Examples include the size of trees in the tree-based models, the learning rate in a support vector machine, and the number of layers and nodes in a neural network (discussed in more detail later in this section). These hyper-parameters are often calibrated to minimize the cross-validation prediction error. However, this approach implies that the information in the validation set is used in model estimation. As an alternative, the sample could be independently split into three subsamples: the train, test, and prediction. The train sample is used for estimating model parameters, the test sample is used to calibrate the hyper-parameters, and the prediction sample is used to evaluate the prediction error.

Figure 1. Fivefold Cross-Validation of a Sample with 120 Observations.



Note: The numbers in brackets show the indexes of the observations in the data. The average value of MSE of predictions on each partition, where that partition is left out in model estimation, represents the fivefold cross-validation MSE.

Supervised and unsupervised learning models

Another distinction often made between machine learning models is based on the availability of information for the outcome variable. When data for the outcome measure are available and the model is built based on labeled data, it is called a *supervised learning* model. The objective of a supervised learning model is to make predictions based on modeling the relationship between features and the outcome variable. For example, suppose a lender is interested in predicting the probability of default of auto loan applicants based on borrower attributes (for example, income, age, career, tenure, debt, etcetera), car features (for example, market price, gas usage, seating capacity, etcetera), and economic indicators (for example, interest rate, employment rate, inflation rate, etcetera). If in addition to these features, the dataset includes information about loan repayment behavior of past borrowers and their behavior can be considered representative of new applicants, the ML model is a supervised learning model. In this case, observed payment performance is used to train the ML model. Most of the models developed for credit analysis are supervised learning models.

On the other hand, if the dataset does not include values for the outcome variable, the ML model is an *unsupervised learning* model. Because in an unsupervised learning model information for the outcome variable is not available, the goal is to find similarities (such as modes or associations) among features and group observations based on the identified characteristics. In the example of credit rating auto loan applicants, suppose the dataset did not include repayment outcomes (called “unlabeled” data). In this case, believing that the features were relevant for explaining the loan performance behavior, unsupervised learning

models could be applied to cluster borrowers of similar features. If repayment information were partially available, it is assumed that the applicants in a cluster would behave similarly.¹²

To predict the credit risk using supervised learning models, in addition to having a labeled dataset, the relevancy condition should be satisfied. The relevancy condition requires the labeled dataset to be representative of the types of loans whose credit risk the model is intending to assess. For example, if new loans are believed to belong to a new line of business with vastly different risk properties, a supervised learning model could yield misleading predictions and is therefore inappropriate to use, even if the model displays high performance measures on the original dataset. Therefore, expert judgment is needed to ensure the dataset's relevance to the analysis, for example, by ruling out deep structural changes in the environment.

Classification versus regression models

Supervised machine learning models are classified based on the type of the outcome variable. The outcome variable is a categorical variable when possible outcomes of the variable are groups or labels, and the values of the variable indicates belonging to a group rather than reflecting a measurement, for instance, taking on 1 for female and 0 for male. The modeling exercise in which the outcome variable is categorical is called *classification*. Because default outcome is binary (“default” or “no default”), PD models are classification models. On the other hand, if a quantitative outcome variable is modeled, for example, LGD, the modeling exercise is called *regression*.

B. Prominent Machine Learning Models

Some machine learning models in the literature have demonstrated good performance in a variety of predictive modeling exercises. The three most prominent classes of ML models presented in this section can be considered classic ML models with extensions that have attracted special interest in credit risk modeling. These models include: (1) tree-based models; (2) support vector machines; and (3) neural networks (NN). For easier exposition, I first explain the basic concept of each model and then provide an evaluation of the main strengths and weaknesses.

Tree-Based Models: Decision Trees, Random Forest, Gradient-Boosting Trees

Tree-based models in machine learning were inspired by if-then-else decision trees, which have been around for a long time in decision theory sciences. The if-then-else decision trees break down a large decision into a sequence of small decisions covering all possible outcomes in a hierarchical structure, where later decisions depend on former ones and the ultimate outcome is based on the sequence of all decisions in a branch. As a result, a clear connection is made between a set of possible decisions and their consequences, which have

¹² A third group of models is sometimes defined as a combination of supervised and unsupervised learning called *reinforcement learning*. Reinforcement learning starts with an unsupervised learning model based on unlabeled data, chooses an action with an impact on the environment, the effect of which could be evaluated as a reward function, and the model is improved based on the feedback effect ([FSB 2017](#)).

the advantage of being easily communicated. Decision trees are commonly used in medicine for diagnosing and treating.

In ML, decision tree models are both useful for classification and regression modeling. However, they are not often directly used in practice. Two successful extensions of decision trees that have demonstrated high predictive performance are among the most common machine learning tools used in practice: random forest and gradient boosting decision trees. After introducing the basics of decision tree models, these two methods will be presented.

Decision Trees¹³

A decision tree adopts a hierarchical structure like a flowchart that starts from a root node, progresses to lower nodes through possible states or decisions (represented as a branch), and ends at the terminal node that shows the consequence of the entire branch. Decision trees have the advantage of being used for both regression and classification models.

It is easier to demonstrate how a decision tree model works with a chart (see [Figure 2](#)). Consider a *regression tree* whose objective is to estimate the LGD using a dataset that contains two features of loans, including the loan-to-value (LTV) ratio and the DTI ratio, and the LGD for each loan. An estimated regression tree is displayed in [Figure 2](#). The tree splits the sample space into five regions. The first split is based on the LTV with a threshold of 0.80. If the LTV is less than 0.8, the DTI threshold is 0.50, but if the LTV is more than 0.8, then the DTI threshold is 0.36. Moreover, for a loan with an $LTV < 0.8$ and a $DTI > 0.5$, the tree further splits the LTV, with a threshold of 0.3. The numbers shown at the end of each branch of the tree are an average LGD of all the loans contained in each region. As a result, the model's prediction for a new loan with an $LTV < 0.30$ and an $DTI > 0.50$ is 4 percent. The tree's depth is 3, which measures the number of splits in the longest branch.

In its simplest form, a tree is constructed by making two decisions at each node until a given criterion is met¹⁴: (1) which feature to make the split upon; and (2) the associated threshold value for splitting. The objective is to minimize the mean square error of the LGD in the region, that is, find a region with average LGD (the predicted value) that is closest to LGD values of loans belonging to that region. Therefore, a decision tree finds loans that are similar to each other based on splitting the feature space.

A classification tree, that is, a tree to model PD, is developed in a similar way except that the assigned value at the terminal node is the fraction of observed defaults among loans in the region. The decision to split at each node is made to maximize a metric called *purity*, that is, a split to ensure a group of loans with similar default rates fall in the same region¹⁵.

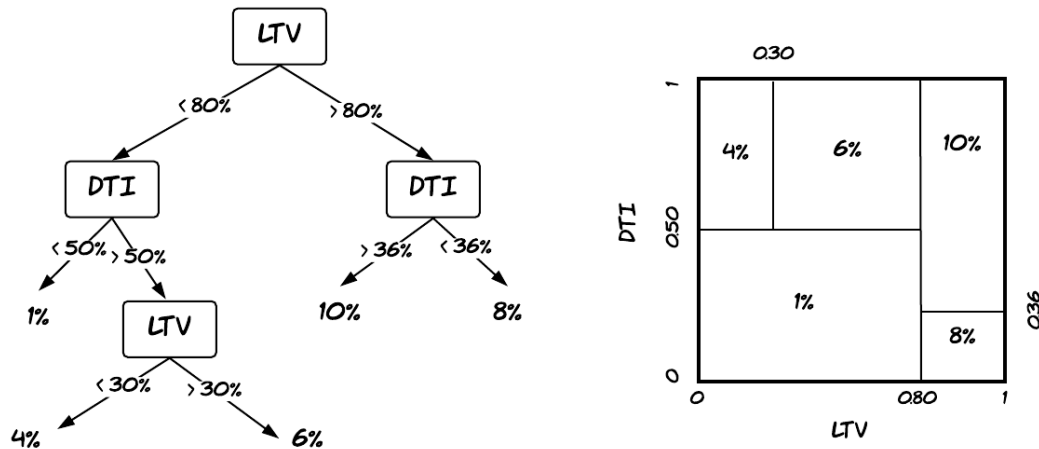
Decision trees should have a stopping threshold to control the size of the tree (for example, the maximum depth of the tree, the minimum number of observations remaining in the final node). Furthermore, trees are *pruned* to avoid overfitting.

¹³ [Breiman et al. \(1984\)](#).

¹⁴ Extended versions of trees allow for making multiple splits at each node.

¹⁵ Common measures of impurity are misclassification, Gini, and the Entropy.

Figure 2. Left: An Illustrative Decision Tree Model for Estimating LGD. Right: Partitioning of the Features Space Implied by the Estimated Tree.



Note: Labeled numbers represent average LGD values corresponding to the region. LTV is the Loan-To-Value ratio and DTI is the Debt-To-Income ratio.

Two main advantages of decision-tree models are:

- They are easy to interpret and illustrate for small trees. As a result, they can be easily used to offer clear guidelines in decision making, which is especially important in training credit raters. The guidelines could be further examined against intuition and expert opinion.
- In addition to single-output cases, decision trees can handle multi-output problems. For example, in addition to the LGD, the time to resolution of loans reflecting liquidity risk could be an important factor in risk management.

However, decision trees have certain disadvantages that don't make them the prime method to use, especially when dealing with large data, as is the practice in FinTech lending. Three main practical disadvantages of decision trees are:

- They can create complex trees with unintuitive criteria that are prone to overfitting and therefore do not generalize well to make robust predictions.
- An optimal decision tree is subject to the curse of dimensionality, and in practice they use locally optimal solutions (so-called "greedy" algorithms), which cannot guarantee globally optimal trees. Consequently, they can be unstable and noise sensitive where small variations in the data could sometimes lead to a completely different tree.
- Decision trees are biased when some classes dominate in the sample. This is particularly important in credit analysis, especially when evaluating the credit of underserved populations who don't have a long history or a reliable track record that reflects their creditworthiness.

Motivated by these limitations, extensions of decision trees have been developed that mitigate some of these constraints. These extensions take advantage of *ensemble learning*

methods. Ensemble methods combine several machine learning techniques (for example, trees) into one predictive model in order to decrease variance or bias, or to improve predictions. Random forest and gradient boosting decision trees are two prominent ensemble methods that are based on decision trees.

*Random forest*¹⁶

Random forest, or forest of randomized trees, is a technique that combines many decision trees based on two principles: (1) *bagging* technique; and (2) *decorrelating* underlying trees¹⁷. The first principle, *bagging*, is a general-purpose procedure in which the base model (a decision tree) is estimated on many randomly drawn subsamples with replacement from the train dataset where the subsamples have the same size as the train dataset (also called “bootstrapped” sampling). The predicted values by estimated trees are then averaged to yield the final prediction. In classification, a majority vote is taken to determine the final prediction¹⁸.

Based on the second principle, *decorrelation*, only a subset of features is considered at each node when the decision to split is made. For instance, if there are 16 features in the model, at each node only four are considered to find the optimal split. Furthermore, these four features are randomly selected. This way, each variable has the chance to be used in modeling the outcome variable without having to compete with a dominant feature. As a result, the underlying trees are “decorrelated.” A rule of thumb for choosing the size of the subset of features to select at each node is the square root of the number of features. For example, if the number of features (risk drivers) is p , then the number of features to be included at each node is about $\sqrt{p}$. Therefore, if there are 105 features, the random forest method randomly chooses $10(\approx \sqrt{105})$ features and uses them at each node to find the optimal feature to make the split upon.

[Figure 3](#) illustrates different steps of random forest. First, the entire sample is randomly split into train and test datasets. At the bagging stage, random subsamples of the train dataset are generated, and each subsample is fed into estimating the trees. To develop the trees, include m chosen features. For example, if a new loan has characteristics displayed in the figure by blue nodes, then its LGD is predicted according to the average values of predicted values by all trees of all subsamples.

The random forest model is widely used in many real-world applications, including banking, medical studies, stock markets, and e-commerce. The two most prominent applications of random forest in banking are (1) identifying loyal customers for lending—in line with this paper’s discussion of credit assessment—and (2) detecting fraudulent activities. Random forest is a powerful technique in identifying outliers, which is why they are used in applications such as fraud detection, identifying spam emails, rare diseases, etcetera.

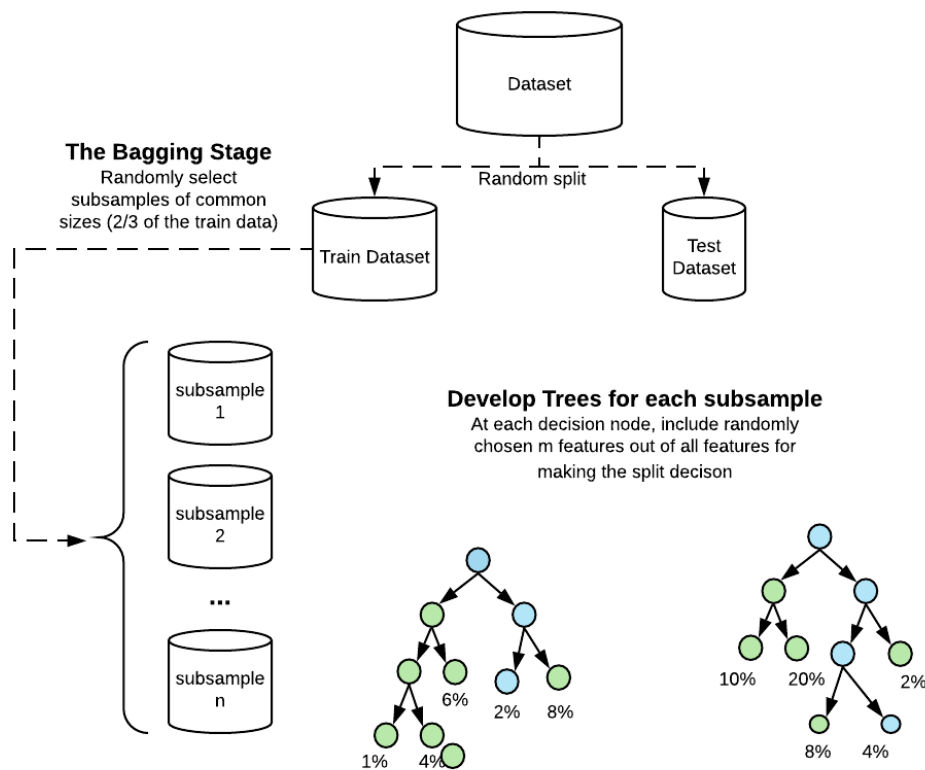
¹⁶ [Breiman \(1996, 2001\)](#).

¹⁷ For a more technical analysis of random forests, see [Biau \(2012\)](#).

¹⁸ The gain in lowering the variance of estimation based on bagging is based on the fact that the variance of the mean of n independent random variables with variance σ^2 is given by σ^2/n , hence a lower variance.

An extension of random forest is extremely randomized trees proposed by [Geurts, Ernst and Wehenkel \(2006\)](#), which makes two important modifications to the random forest algorithm. First, the criterion of splitting is not based on an optimized split, that is, one that minimizes prediction errors of implied partitions ([Figure 2](#)), but thresholds are randomly drawn for each candidate feature and the best of these randomly generated thresholds is picked as the splitting rule. Second, extremely randomized trees do not use bootstrapped samples but the train dataset. These two modifications lower the time of processing to a great extent while offering comparable performance to a random forest. Extremely randomized trees are applied in the same areas that random forests are. A well-known use in medicine is medical image analysis¹⁹.

Figure 3. The Random Forest Model



The random forest model improves upon the decision trees and has the following two advantages:

- Relative to decision trees, they are less subject to overfitting, have superior generalization properties, and can make more robust predictions, which is why they have attracted practical interest in various fields.
- Unlike decision trees, a dominant feature does not lead to the results in the random forest model due to the decorrelation stage, which provides the chance for various features to

¹⁹ See [Criminisi & Shotton \(2013\)](#).

contribute to the final prediction value. For the same reason, the random forest model can easily handle a large set of correlated features.

At the same time, the random forest model yields complicated rules, which can be hard to interpret and communicate, especially when the feature space and the number of generated trees is large. One way to finesse this problem is to rank features based on their contribution to the final prediction, a technique called “feature importance” in machine learning. For regression models, feature importance is based on the sum of reductions in residual sum of squares caused by nodes in which a specific feature was the basis for splitting. Similarly, in the context of classification, feature importance adds up to the reduction in the Gini impurity index²⁰ at each split resulting from a specific feature. The relative ranking of features allows for identifying features with the highest role in making predictions by the estimated model.

Feature importance could play a big role in assessing the results of credit analysis based on a random forest model against intuition. If the identified features are widely different from the expected set of features, the model or the data should be evaluated more closely.

Gradient-Boosting Decision Tree (GBDT)

GBDT is another successful tree-based ensemble machine learning technique that, like random forest, has attracted a lot of interest in real-life applications. A well-known example is finding high-quality user-generated content in social media through Yahoo²¹. Another example is applying GBDT to learn to rank for information retrieval, which is a key component in automated data analysis²².

Instead of building and combining parallel and independent trees as in random forest, GBDT builds a series of trees, each trained based on the mistakes of the previous tree²³. Specifically, at the first step, a simple tree is estimated on the entire train dataset. The second tree is estimated using prediction errors of the first tree as the output variable so as to decrease the prediction error of the first tree. Other trees are built sequentially in the same manner. The final prediction is the weighted sum of predictions of all trees. The weighting is governed by a key parameter specific to GBDT called the *learning rate*.

It has been empirically shown that using a small learning rate (for example, less than 0.1) with a large number of trees dramatically improves the model’s out-of-sample predictive power compared to a learning rate equaling one with a smaller number of trees. Clearly, GBDT with a small learning rate comes at the cost of higher computational time because of the need for more iteration to achieve a well-functioning GBDT.

The baseline trees in GBDT are generally shallow, with only a few branches, and are therefore considered weak learners, that is, each tree only adds a small amount to the

²⁰ The Gini impurity is computed by summing the probability of an item with a label being chosen times the probability of a mistake in categorizing that item.

²¹ See [Agichtein et al. \(2008\)](#).

²² See [Liu \(2009\)](#). Liu defines learning to rank for information retrieval as “a task to automatically construct a ranking model using training data, such that the model can sort new objects according to their degrees of relevance, preference, or importance.”

²³ [Friedman \(2001\)](#).

predictive power. However, many trees are added sequentially, with the goal of minimizing the error made by the previous tree. In calculating the final prediction, GBDT weights the predictions of all trees by the degree of correction that each tree can make to errors. This way, GBDT aggregates the predictive power of many weak learners to produce a strong predictor. In general, GBDT is considered to be superior to random forest, which performs better than basic trees.

Because the number of trees is usually very large, it is not feasible to display in a figure the underlying steps that an estimated GBDT takes in making predictions. However, like random forest, it is also possible to rank contributions of features in predicting the outcome, that is, generate feature importance. Another tool that summarizes the findings of an estimated GBDT is the partial dependence plot (PDP). PDPs show the dependence between the response variable and a feature by marginalizing over the values of all other features. PDPs can be interpreted as the expected effect on the response variable as a function of a feature while keeping other features fixed at their average.

As an extension of GBDT motivated by the bagging technique, [Friedman \(2002\)](#) proposes *stochastic gradient boosting* that improves upon GBDT by using a random subsample of the training data (like random forest) rather than the entire train data when estimating a tree at each iteration. A fraction between 0.5 and 0.8 of training data leads to best performance. Furthermore, stochastic gradient boosting decreases the computational time because it uses a smaller sample to estimate the trees at iteration.

Support Vector Machines (SVMs)

An SVM is a machine learning technique that is based on an intuitive approach that finds the best separation between observations that belong to different classes²⁴. While SVMs were developed primarily to solve classification problems, such as face detection, handwriting recognition, image classification, and bioinformatics, their extension allows for modeling regression problems as well.

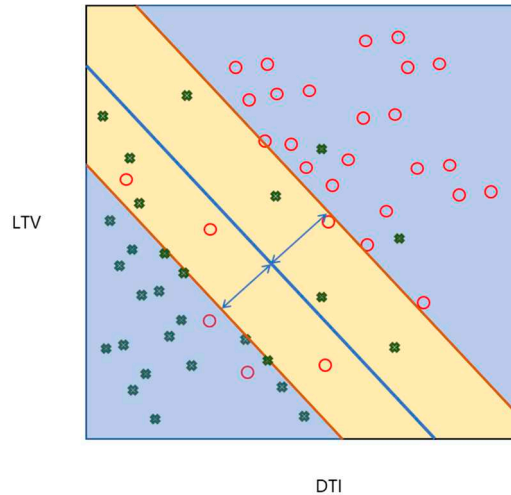
Consider the problem of predicting the default outcome of loans based on two features: the DTI and LTV ratios. [Figure 4](#) displays a sample SVM estimated for this model. The objective of an SVM is to find the best separating line²⁵ that splits the feature space based on default outcome. Unlike trees that make splits based on a single feature and therefore result in box-shape regions, SVMs find the border of separation based on a function of all of the features. To find the best separating line, the SVM tries to maximize the minimum distance between the line and closest points in each class, which is known as the *margin*. As shown in [Figure 4](#), the classes are not always completely separable. In this case, the SVM compromises some points (hence misclassifying those) and finds the separating line to generate the most homogenous observations in each region.

²⁴ [Cortes et al. \(1995\)](#) and [Ben-hur et al. \(2001\)](#).

²⁵ In higher dimensions, the separating line is called “separating hyperplane.”

The decision rule implied by a simple SVM, as displayed in [Figure 4](#), is quite intuitive. For instance, “if $0.5 \times \text{LTV} + 0.8 \times \text{DTI}$ is greater than 3, the loan is labeled as risky otherwise safe.”

Figure 4. An SVM Model for Predicting Default Outcome Based on Debt-To-Income (DTI) and Loan-To-Value (LTV) Ratios.



Note: Red circles represent “no default” and green crosses represent “default.” The thick blue line is the separating line estimated by SVM; red lines are support vectors distanced from the blue line by the margin.

The functional form that governs the shape of the separating line is linear. In this case, the SVM is based on a linear *kernel*. Natural extensions of the SVM when the data are not linearly separable is to use nonlinear kernels, which could therefore yield nonlinear decision boundaries. In effect, a nonlinear kernel is a transformation of the feature space, which could generate a sharper separation of outcome classes. Common nonlinear kernels used with an SVM are polynomials and radial basis function (RBF) kernels. Examples are displayed in [Figure 9](#). It is also possible to construct a customized kernel.

SVMs offer three main advantages:

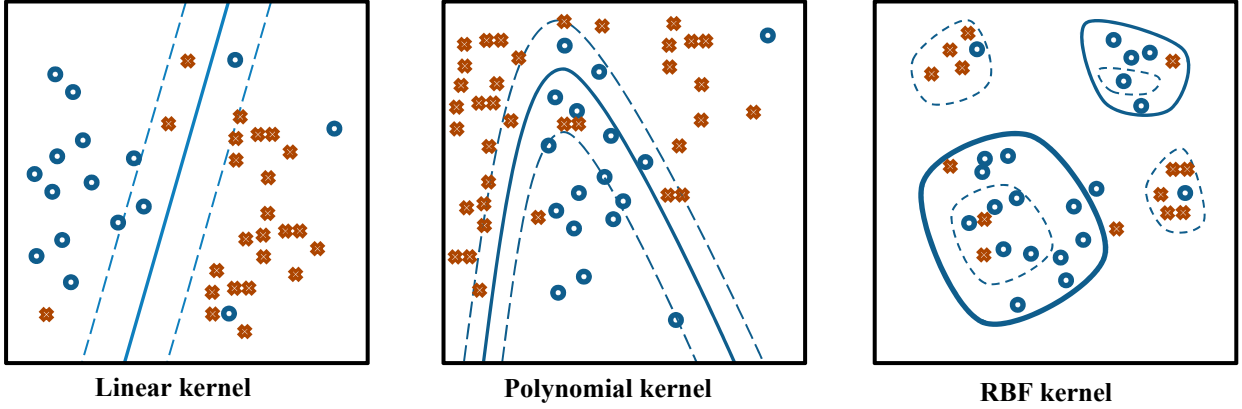
- Because they use a subset of the training dataset to evaluate the separating line and the support vectors, SVMs are memory efficient. Likewise, they scale well for high-dimensional problems and they are less prone to overfitting.
- The nonlinear kernel gives a high flexibility to SVMs, especially when the shape of the kernel is motivated by business knowledge and intuition about the problem. However, the performance of SVMs is sensitive to the choice of kernels.
- SVMs even work well with unstructured and semi-structured data like text and images.

However, their main disadvantage is:

- Estimated SVMs are difficult to interpret. For instance, it is hard to understand what variable weights represent and how individual features impact the outcome variable. In

this sense, they are more black-box models. Moreover, they do not directly provide probability estimates, which is particularly useful in estimating the probability of default²⁶.

Figure 5. SVM With Linear and Nonlinear Kernels



Neural Networks

NNs are another class of powerful predictive ML models that have become widely used, for example in the areas of computer vision and natural language processing. NNs were primarily motivated by the human brain (hence their name²⁷). An example neural network structure is shown in [Figure 6](#). NNs are built to learn the pattern between features and output through several inner layers. At the first layer, features (denoted by X_1 to X_4 in [Figure 6](#)) are used to evaluate the value of nodes Z_1 to Z_6 , which are then used as input for calculating nodes of the second layer. The calculation is based on a function on a weighted sum of inputs where the function's output range is between 0 and 1.²⁸ A common functional form is the sigmoid $\sigma(x) = \frac{1}{1+e^{-x}}$. For example,

$$Z_1 = \frac{1}{1+e^{-(\omega_1 X_1 + \dots + \omega_4 X_4 + b)}}$$

where ω_1 to ω_4 determine the contribution of each feature to the node and the constant b is intended to capture the *bias*. The above form resembles the logistic regression and therefore, in their basic form, NNs can be seen as comprising many logistic regressions, the outputs of one layer used as inputs in lower layers.

To evaluate each Z in layer 1, five coefficients (four weights and a bias) are needed. Adding up for five nodes, this means layer 1 has 25 parameters. Suppose layer n was the only other hidden layer of NN in [Figure 6](#). To estimate the three nodes in this layer, 18 more parameters should be estimated. In sum, the two-layer NN has 40 parameters to estimate (compare to five coefficients for an ordinary linear regression). It is immediately clear that a deeper NN

²⁶ It is possible to use workarounds such as fivefold cross-validation, but this approach is usually time-consuming when the sample size is large.

²⁷ This class of models is sometimes called “artificial neural networks,” particularly in the field of computational neurobiology, where the term “neural network” is reserved for the study of the human brain.

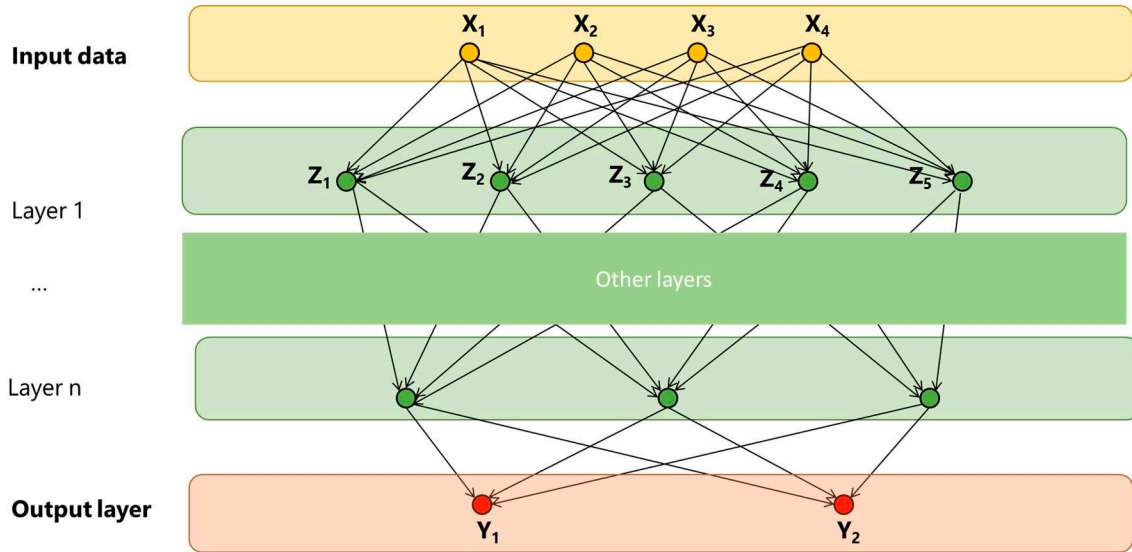
²⁸ The function is commonly referred to as *activation function*, which acts like a classification model, which combined with a threshold, leads to activate 1 or inactive 0 outcome from a node. Other activation functions include hyperbolic tangent, softmax, and rectifier functions.

would have many parameters, thereby requiring a large data to train, a more complicated optimization algorithm, and high computational power.

Extended versions of NNs, famously called *deep learning*, are one of the high-impact areas of ML models that have been successfully deployed in numerous fields due to higher efficiency and reasonable processing time²⁹. Examples include computer vision, speech recognition, natural language processing, audio recognition, social network filtering, machine translation, and bioinformatics, where they have produced results comparable to and in some cases superior to human experts³⁰.

The main advantage of NNs is their flexibility and ability to recognize complicated patterns in the data, which is particularly useful when the size of the data is large. As the sample size becomes larger, performance gains of traditional models remain low, whereas NNs are able to capture deeper connections between features and the output. However, those patterns are not understandable by human beings, making NNs black-box models. In effect, NNs learn by memorizing patterns in the data through the estimation of a large set of parameters. In addition, NNs use nonlinear structure via the activation function, which increases the model's flexibility.

Figure 6. A Neural Network with Four Input Variables (Features), Two Outcomes, and n Hidden Layers.



²⁹ Examples include deep NNs that use gradient-based optimization, convolutional networks, deep stacking networks, and long short-term memory (Bengio 2009).

³⁰ For a comprehensive historical review of deep learning and its applications, see Schmidhuber (2015). For financial applications, see Fu, Cheng, and Zhang (2016), who use convolutional NNs to detect fraudulent activities of credit cards of a large commercial bank. Sirignano et al. (2016) employ deep learning on a large dataset with 120 million observations for studying mortgage risk.

A challenging issue in using NNs is the way business knowledge should be incorporated in the model to improve the model's performance. This could only be done through modifying the feature's input to the model, but the structure of deep NNs could not be modified to reflect non-data information. This constraint of NNs has made it less attractive to be used as a standalone technique for lending purposes, especially when there is the risk of data contamination.

C. How Does Machine Learning Differ from Econometrics?

With this background, I turn to discuss some of the main differences between mainstream econometrics and ML approaches. While a detailed examination of the commonalities and differences among these two approaches is beyond the scope of this paper, the following discussion focuses on their most prominent distinctions in data analysis.

The literature in ML and econometrics significantly overlap, as the main objective of both is to learn from data. Both estimate models that explain the relationship between features (explanatory variables) and the output (response) variable that are intended to guide decision making. However, one broad way of making a distinction between the two approaches is that the main focus of ML models is to make the best out-of-sample prediction of the outcome variable without necessarily trying to explain the structure of the underlying relationship³¹, whereas mainstream econometrics aims at finding casual relationships between observed features and the outcome variable³².

The main challenge in data analysis that econometric studies attempt to address is how to deal with selection bias, which occurs when the relationship between features and the outcome variable differs between the observed data and the intended causal effect of a decision variable.³³ The famous result of the selection bias problem is that correlation does not imply causation, and therefore finding a high correspondence between the features and the outcome variable does not necessarily inform how the outcome will respond to a decision or a policy action in the presence of selection bias. The identification of the causal effect is guided by theory and restricting assumptions to infer how features influence the outcome variable.

However, econometric studies rarely look for maximizing the out-of-sample forecasting power of the model, which is the central focus of ML. The problem of selection bias relates to the ML modeling's assumption of generalizability of the observed data for making predictions for the unobserved sample. This means that the prediction problem should be

³¹ A recent literature in ML and artificial intelligence has tried to make ML models more interpretable. See [Gilpin et al. \(2018\)](#) for a review. Moreover, econometric tools such as time-series analysis are often used for forecasting purposes. Nonetheless, these methods are built to have some aspect of interpretability of the underlying parameters.

³² Both literatures share many common tools; for example, nonparametric models such as kernel regressions appear in econometrics, and some ML models such as linear discriminant analysis, naïve Bayes, and even a simple neural network could be considered parametric models. This section broadly highlights the main differences in the mainstream practice of ML and econometrics that are relevant in credit risk assessment.

³³ See, for example, standard econometrics textbooks like [Angrist and Pischke \(2008\)](#) and [Wooldridge \(2015\)](#) for elaboration.

carefully defined and the relevance of the sample for making prediction should be verified. For instance, in the case of assessing the credit risk, the sample should be representative of the borrowers who will be assessed using the model. For example, a model that is trained based on data that only contain large companies may not capture risk characteristics of loans to small- and medium-size enterprises because of underlying differences between the two loan classes.

Unlike ML, econometric studies pay close attention to the statistical significance of an estimated parameter. As long as a variable can increase the predictive performance of the model, it will be included in the ML modeling. In a similar vein, in ML model parameters do not often have useful interpretations to generate business insight. For instance, the number of trees used in random forest or GBDT, the margin parameter in SVM, and the depth and nodes of neural networks govern each model's structure and flexibility to learn from data, but do not add any insight in terms of credit risk assessment. Furthermore, while it is possible to draw decision rules from an estimated ML model, because of their high complexity, these rules could often only be applied by machines rather than by human agents. While this is useful for automating credit decisions, it has the shortcoming of making it difficult to evaluate the soundness of decisions against intuition and business knowledge.

IV. STRENGTHS AND WEAKNESSES OF ML-BASED LENDING FOR FINANCIAL INCLUSION

Building on previous sections, this section discusses potential advantages and shortcomings of using ML for credit rating and how ML-based lending could influence financial inclusion focusing on the credit aspect of financial inclusion. Because undue credit growth could lead to financial instability, the paper's focus is on promoting prudent lending in the economy by FinTech. Admittedly, there are other important enabling factors for prudent lending such as sound regulation and supervisory regimes, depth of the financial system, and high financial literacy of borrowers that are beyond the scope of this paper.

A. Strengths of ML-Based Lending

FinTech companies aim at changing the way traditional lending functions by automating the steps of credit assessment and lending decisions. This is carried out by training ML models through information gathered from a wide range of sources.

ML can make assessing credit risk of small borrowers feasible and economical.

Banks often refrain from assessing the credit risk of a small borrower because the small size of the loan and potentially high risk of the loan do not justify the cost of employing a professional examiner and meeting underwriting standards. The sunk cost of initiating a credit assessment is a major factor in financial exclusion of many households and small businesses ([*World Economic Forum 2016*](#)).

However, automated credit assessment allows FinTech companies to conduct credit scoring of small borrowers much more frequently than traditional lending. By automating the credit rating process, FinTech credit companies could better assess the creditworthiness of borrowers through making small amounts of loans at a high frequency and monitoring the repayment behavior of the borrower. This way, FinTech credit companies can assess

eligibility of borrowers for receiving loans rather than rationing those borrowers who lack sufficient standard data such as financial reports used in traditional credit risk assessment.³⁴

Another way that nonbank lenders could obtain massive information about potential borrowers is the experience of the so-called “BigTech” lending companies.³⁵ BigTech companies have the advantage of accessing information about customers’ business activity as a byproduct of offering e-commerce, payment and telecommunication platform services ([World Economic Forum 2015](#)). Big tech firms’ marketplace allows for observing the sales trend and cash flows of businesses active in their e-commerce space, thereby extremely reducing the cost of collecting information. Using ML analysis, they could therefore process the creditworthiness of borrowers without requiring professional financial reports, which is often cited as a big burden for small borrowers. Furthermore, they can make cashflow-based loans without requiring collateral, another major hindrance for small companies trying to obtain credit from banks ([BIS 2019](#)).

Similarly, big tech companies that offer payment services to consumers can leverage transaction data to acquire more accurate information about consumption behavior and income stream to assess creditworthiness of borrowers, therefore removing the need for an established credit registry to collect consumer information. This way, big tech lenders could provide financing to consumers who are unable to take personal loans from banks and improve financial inclusion.

For an empirical example, [Schweitzer and Barkley \(2017\)](#) study the Federal Reserve’s Small Business Survey and find that business borrowers who received funding from online platforms share similar characteristics with those who were denied credit, suggesting FinTech lending contributed to financial inclusion of these small borrowers.³⁶

ML can harden soft information

ML models are well known for performing particularly well in the presence of hard information ([Liberti and Peterson 2018](#)). According to them, hard information is characterized by two important properties: (1) it has a numeric value for data entries that measures a feature in a cardinal way; and (2) it is independent of the data collection process. Variables such as income, debt, number of employees, and sales are some examples of hard information that could be used in estimating the debt servicing capacity of the borrower. Similarly, among other Cs of credit rating, capital structure, coverage, and conditions are generally based on hard information. Price variables used for evaluating these attributes are hard information as long as they are inferred from market data.

However, a major challenge in evaluating asset values of small borrowers is that there is no liquid market for discovering the price. The common practice among traditional lenders in

³⁴ According to data by TransUnion, FinTech credit accounted for 38 percent of total personal loans in the US in 2018, rising from only 5 percent in 2013 (see [CNBC article](#) on Feb 21, 2019).

³⁵ The term “BigTech” refers to the direct provision of financial services or of products very similar to financial products by technology companies ([BIS 2019](#)).

³⁶ Using data of consumer credit in Germany, [de Roure et al \(2016\)](#) find that the risk-adjusted interest rates on P2P loans are lower than those on bank loans.

this case is to rely on a specific pricing model for the asset, which therefore contains many judgmental factors by the assessor. Similarly, the borrower's character is evaluated based on the rater's subjective assessment and may not easily generalize for other cases. Furthermore, information that a lender obtains through maintaining a long-term relationship is borrower-specific and judgmental, and therefore soft information.

Nevertheless, the challenge with processing soft information by ML-based analysis is a data limitation issue. Once sufficient data that are not predominantly noisy are collected, classification ML models could be used to turn soft information into hard information. Thanks to the high capacity of ML models to include large and diverse data, ML-based lenders can turn almost anything into data, ranging from social media activities to the physical location of applicants' activities. ML models offer the possibility of enhanced preprocessing and incorporating unstructured data in the analysis, which facilitates hardening soft information of data collected from a wide range of sources.

As a result, ML models could enhance pricing of assets pledged by small borrowers by finding common patterns in the nontraditional data—they could build reliable and powerful pricing models³⁷. Similarly, social media information could be used to sharpen evaluation of the borrower's character. For instance, [Jagatiani and Lemieux \(2017\)](#) show that using nontraditional data resulted in more informative credit scoring by FinTech lending firms in the US in comparison to the traditional FICO score system. Likewise, Berg et al. (2018) show that easily accessible information from digital footprints could lead to superior performance of credit rating relative to credit bureau scores.

ML can better capture nonlinearities

By searching for relationships in small partitioning of the sample (such as tree-based models, discussed in the previous section), ML models can capture local relationships between risk indicators and credit risk outcomes that traditional models or human agents fail to detect. As long as extra information collected and used by ML models is informative about borrowers, ML-based models can outperform traditional methods by capturing higher dimensions of nonlinearity. For example, through effective partitioning of borrower features, ML may find credit indications for those borrowers who have been identified as poor credit based on traditional indicators and denied credit. This way, by better capturing nonlinearities in the relationship between the credit risk outcome and features, ML is capable of better assessing the borrowers, particularly those with poor credit history, and avoiding being driven by the majority. Specifically, high-quality borrowers, due to their favorable historical record, dominate the sample and tend to drive the results in their favor if the model does not distinguish between different sections of the sample (for example, logistic regression). However, methods like random forest can focus on a particular class of borrowers by partitioning the sample and detecting possibly strong relationships for those borrowers. Therefore, while borrowers with a poor credit history based on traditional indicators of creditworthiness may be denied credit by traditional lenders, FinTech lenders may find them eligible to borrow funds.

³⁷ See, for instance, [Gu, Kelly, and Xiu \(2018\)](#).

ML could mitigate information asymmetry

Another layer of complication arises in lending due to information asymmetry between the lender and the borrower, which has created a large literature in corporate finance. Moral hazard and adverse selection problems could arise when the lender cannot reliably obtain information about the borrower, and the borrower has an incentive to misreport in order to obtain better loan terms, but taking this possibility into account, the lender would ration credit to avoid making losses. Such rationing behavior becomes extreme, as information asymmetry is more intensified ([Tirole 2010](#)).

ML could potentially mitigate financial exclusion due to information asymmetry by identifying indicators of borrower riskiness through analyzing larger data and identifying differences among borrowers. This could be accomplished by detecting nonlinear relations between features and credit risk that are otherwise unidentified by traditional approaches, leading to adverse selection and moral hazard problems. Modern technology allows for more enhanced monitoring of borrowers by screening the borrower more frequently and effectively. For instance, text, image, and video data could be collected on a company's business operations and could be used to assess compliance with loan covenants. Such enhanced monitoring could help to mitigate moral hazard problems.

While ML could in principle lead to better evaluation of the credit risk of borrowers, it could have heterogeneous impact on borrowers leading to the exclusion of a subset of borrowers, because the lender can make more informed credit decisions. Such financial exclusion is, however, consistent with prudent lending and sound risk management practices and could result in higher profitability of lenders and potentially superior financial stability.

B. Weaknesses of ML-Based Lending

ML-based lending bears risks of financial exclusion

ML models are trained using available data that may not necessarily be representative of all classes of borrowers that the creditor considers lending to—a situation that violates the principle of generalizability. As a result, the lack of sufficient relevant data for some classes would impose restrictions on the conclusions by ML analysis and could lead to redlining applicants belonging to a particular group. A case in point is a sample with a class of borrowers that have been historically denied credit for nonbusiness reasons (for example, gender, race, religion, ethnicity, residence in certain areas, etcetera). If the raw sample is used for making predictions regarding default behavior of underserved borrowers, the results will be contaminated by the effect of traditionally unfair financial exclusion of those borrowers, and the model will further predict poor credit rating. In such cases, features including gender, race, and religion that could cause bias should be excluded from the model. Such variables could, however, be used to assess the existence of bias by a scoring model and avoid discrimination.³⁸

³⁸ In this regard, a line of research studies discrimination-aware data mining. See [Berendt & Preibusch \(2014\)](#), [Lepri et al. \(2017\)](#) and [Žliobaitė \(2017\)](#).

ML-based credit rating could cause consumer protection, ethical, and data privacy issues

The use of information gathered from large pools of data (including social network and internet of things) not only raises an issue around the relevance of the information (the noise) but further compounds concerns on consumer protection, as exclusion could stem from using personal data to exclude consumers on the basis of unexplainable computer decisions³⁹. Therefore, in addition to the risk of unjustified exclusion, there is a risk to consumer protection because of the opaqueness of ML-based credit rating. As ML-based credit rating is often based on a large set of risk drivers⁴⁰, it may be hard to detect and prove the dominance of one factor in the final credit rating outcome, which creates space for unethical decisions and impedes supervision.

While this is a common problem between ML and traditional credit rating, it could be more difficult to detect with a large set of risk indicators. One way to mitigate such problems is to ensure that the set of features does not include discriminatory variables, and hence there is no role for such variables to drive the results. Supervisors could enforce this requirement by monitoring the set of input variables in the credit risk modeling process. Moreover, as discussed in the previous section, ML models could provide top variables that contribute most to the credit score, that is, by using “feature importance” of each algorithm. Lenders should actively monitor most significant drivers of credit rating of borrowers and assess against their business insight. Feature selection techniques such as LASSO could be used to reduce the number of attributes that drive the credit score, which is another way to make analysis of data with many features more tractable and thereby easier to monitor.

ML may not address structural changes

A similar challenge arises when a large change in the environment renders historical data irrelevant for assessing new customers, that is, analysis of the historical sample does not generalize for new applicants because of significant structural changes (for example, a change in financial development, macroeconomic policies, and industrial changes)⁴¹. If the modeler fails to identify such dramatic changes, the trained model’s risk assessment could have costly implications for the creditors. Of course, if there are historical observations of similar structural changes, ML could capture changes in the environment.

One shortcoming of ML models is that the underlying algorithm cannot be easily modified to reflect tacit information such as business knowledge, intuition, and anticipation of future events not realized in the data. Specifically, as discussed in the previous section, ensemble tree-based models such as random forest grow many trees and randomly select features, and therefore their underlying algorithm is too inflexible to incorporate tacit knowledge. Likewise, deep NNs have many parameters without a clear link to a particular piece of information and thus offer little modeling flexibility in this regard. SVMs have some limited flexibility in the choice of kernels, but it is not exactly clear how kernel choices could be

³⁹ For example, [Fuster et al. \(2018\)](#) propose a cross-category measure of disparity and find that ML models could worsen disparity relative to the logit model according to this measure in the US mortgage sample.

⁴⁰ According to [Claessens et al. \(2018\)](#), “the website of one Indian P2P platform claims that its credit assessment involves a review of more than 1,000 data points per borrower.”

⁴¹ See, for instance, [Rajan, Seru, and Vig \(2010\)](#).

modified to reflect complex knowledge. By contrast, in a Bayesian framework, it is sometimes possible to build intuition-type information into priors or the shape of the likelihood function.

Faking indicators by borrowers could damage ML-based scoring

If borrowers realize that an indicator is important for their credit rating, they are motivated to artificially change its value if they can. For example, the number of social media connections could be indicative of a high character score. However, considering it does not reflect the strength of relationships, applicants could expand the size of their network to achieve better credit rating. In that case, the information content of the variable changes over time and a model that is trained based on old data would yield erroneous credit scoring for new borrowers.

V. CONCLUDING REMARKS

I conclude by reviewing the main points of the paper and a brief discussion of the potential for FinTech credit in emerging economies with poor financial development. Achieving widespread financial inclusion is a major step in achieving sustainable development goals in many countries, and FinTech credit appears as a promising solution and a potential leapfrog for countries with low financial inclusion.

One of the core strengths of FinTech credit is the use of machine learning techniques and big data analytics to promote credit scoring. This paper reviews the underlying challenges in assessing credit risk of particularly small borrowers and discusses the most prominent ML techniques applied in assessing credit risk of borrowers for a nontechnical audience. It provides sufficient background to analyze general strengths and weaknesses of using FinTech for increasing financial inclusion.

Among the strengths of ML in promoting financial inclusion, it could particularly facilitate low-cost automated evaluation of small borrowers that would be otherwise left out of the traditional credit market. By turning soft information to hard information, ML allows for incorporating a wide range of information from various sources and structures. Moreover, ML can capture nonlinearities in the relationship between risk drivers and credit risk outcomes substantially beyond traditional models. These strengths could enable ML to mitigate the information asymmetry problem that is at the heart of lending, particularly to small borrowers with a short history of formal financial reports.

However, ML has certain weaknesses that should be taken into consideration when applying for credit risk assessment. Heavy reliance on learning from data, particularly in a context where the size of the sample is considerably larger than traditional ways of credit scoring, could result in noisy information playing a role in driving results of credit analysis and leading to financial exclusion of creditworthy applicants. The bias in the sample should be identified and avoided by analysts as much as possible to avoid digital financial exclusion. Along the same lines, ML may not capture structural changes in a timely manner, because

arrival of informative data may be slow due to the lengthy process of default observation.⁴² This could negatively impact FinTech lenders, who rely on ML to assess borrowers without evaluating the relevance of data used for training the model for new applicants. Furthermore, borrowers may realize and counterfeit certain indicators that drive their credit score, thereby decreasing the relevance of those features for new applicants. Finally, it should be noted that ML is exposed to some of the key concerns in econometrics, most importantly the endogeneity and selection bias problem. The analyst should check the sample to ensure proper treatment of these issues and avoid superfluous results. Nonetheless, proper choice of risk drivers makes these issues less of a concern for credit risk assessment and default outcome prediction.

Taking these strengths and weaknesses into consideration, emerging market economies (EMEs) that are challenged by low financial inclusion could highly benefit from FinTech lending. ML-based credit rating could have important consequences for the credit market in EMEs. By enhancing credit analysis, FinTech lending can reduce the time of credit decision and offer lower loan rates to existing borrowers. Lengthy credit decisions and high loan rates are among the main financial constraints in EMEs that confine the pace of the private sector growth. Banks and FinTech lenders benefit by appropriately pricing the risk, thereby lowering debt servicing costs for borrowers while improving their asset quality.

FinTech credit could increase financial inclusion by lending to underserved population relying on enhanced credit rating given that credit bureaus are either nonexistent or poorly developed in most EMEs. The asset quality of these borrowers will likely be lower than traditional borrowers, which implies the lenders take on more risk and should therefore offset the risk by holding sufficient buffer reserves and capital. ML evaluation of credit risk should allow for measuring the amount of buffer.

Nevertheless, EMEs may face some challenges in reaping these benefits while ensuring that the development of FinTech credit does not expose systemic risks to the financial system. First, given the central role of data in ML-based credit analysis, it should be legally and technologically possible to gather digitalized data reliably from various sources and avoid noisy and biased data as much as possible. The lower the noise of the data and the less need for data cleaning, the more effective the results of credit analysis. As a complement for high-quality data availability, cybersecurity measures should be in place because of the sensitivity of credit information.

Second, technological infrastructure for implementing big-data-based decision making should be developed and accessible for FinTech credit. Moreover, ML analysts should consistently review ML models for credit rating to avoid potential weaknesses of naïve application of ML. For example, analysts should conduct ongoing monitoring of the environment to identify structural changes, examine indicators that could be faked by borrowers, assess whether the most important drivers of credit risk implied by ML are justified based on business knowledge.

⁴² Specifically, data of loan performance at least over a full business cycle and a credit cycle is needed to appropriately train the ML model.

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